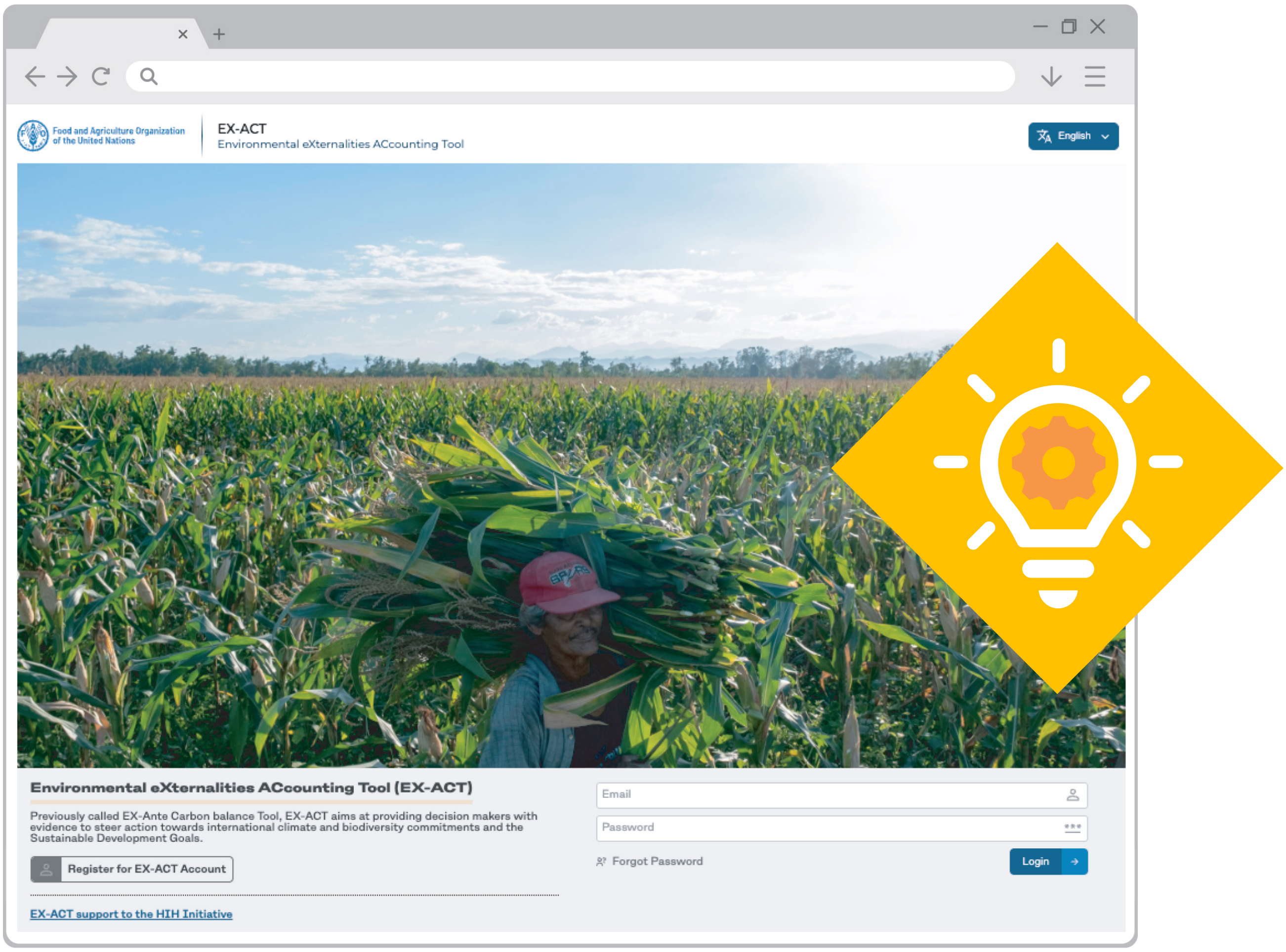
Food and Agriculture Organization of the United Nations

ENVIRONMENTAL EXTERNALITIES ACCOUNTING TOOL (EX-ACT) HANDBOOK

Guiding sustainable investments



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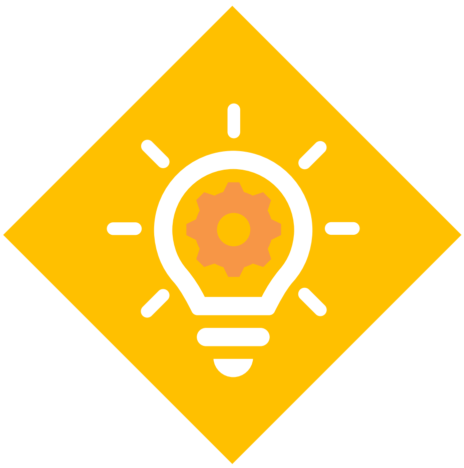

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Abbreviations

AGB	above-ground biomass
AFOLU	agriculture, forestry and other land use
AGDM	above-ground residue dry matter
AR4	Fourth Assessment Report of the Intergovernmental Panel on Climate Change
AR5	Fifth Assessment Report of the Intergovernmental Panel on Climate Change
AR6	Sixth Assessment Report of the Intergovernmental Panel on Climate Change
BGB	below-ground biomass
CBD	Convention on Biological Diversity
CH_4	methane
CO_2	carbon dioxide
CO_2eq	carbon dioxide equivalent
DIC	dissolved inorganic carbon
DOC	dissolved organic carbon
d.m.	dry matter
DW	deadwood
ETF	Enhanced Transparency Framework
EX-ACT	Environmental eXternalities Accounting Tool
FAO	Food and Agriculture Organization of the United Nations
FRA	Forest Resources Assessment
FUI	fuel use intensity
GHG	greenhouse gas
GWP	global warming potential
HCFC	hydrochlorofluorocarbon
HFC	hydrofluorocarbon
HWP	harvested wood product
IPCC	Intergovernmental Panel on Climate Change
IPPU	industrial processes and product use
LI	litter
MRV	measurement, reporting and verification
N	nitrogen
N_2O	nitrous oxide
NDCs	nationally determined contributions

NH_3	ammonia
NO_x	nitrogen oxides
POC	particulate organic carbon
SAR	Second Assessment Report of the Intergovernmental Panel on Climate Change
SDGs	Sustainable Development Goals
SOC	soil organic carbon
SPC	shadow price of carbon
t d.m.	tonnes of dry matter
UNCCD	United Nations Convention to Combat Desertification
UNFCCC	United Nations Framework Convention on Climate Change
USDA	United States Department of Agriculture
VS	volatile solids
WRB	World Reference Base

Rationale

Agrifood systems play a crucial role in global efforts to mitigate climate change and preserve biodiversity. They contribute a significant portion of global greenhouse gas (GHG) emissions but also display some of the largest mitigation potential at scale (IPCC, 2023). By influencing land use, habitat conservation and ecosystem health, they are important catalysts for effective biodiversity protection. As international commitments move towards common reporting frameworks to ensure comparability, reliability and completeness in environmental reporting (CBD, 2022; UNFCCC, 2022), providing tools that ensure alignment with these frameworks and streamline investment and policy analysis in the agrifood sector is of primary relevance.

In light of this, the Agrifood Economics and Policy (ESA) Division of the Food and Agriculture Organization of the United Nations (FAO) has developed a new online application, named Environmental eXternalities ACcounting Tool (EX-ACT), formerly known as EX-Ante Carbon balance Tool, to help private and public investors, as well as political stakeholders and countries, to align financial commitments and policies to requirements of the United Nations Framework Convention on Climate Change (UNFCCC) and its Paris Agreement, the Convention on Biological Diversity (CBD) and the United Nations Convention to Combat Desertification (UNCCD).

OBJECTIVES OF THE HANDBOOK

The handbook aims to document the methodology and use of the GHG accounting section of the new application, specifically introducing the methodology used to build the tool, the tool structure, and the equations and parameters used to calculate the carbon balance.

TARGET READERSHIP

The handbook is intended for technical users engaged in the design, implementation, and evaluation of agrifood system interventions with a focus on environmental externalities. This includes professionals working in government agencies, international organizations, research institutions, and private sector entities who require a robust and transparent framework for estimating GHG emissions and removals.

Users are expected to have a working knowledge of climate change mitigation in agriculture, land-use dynamics, and GHG accounting principles. Familiarity with international reporting frameworks such as those established under the UNFCCC, the CBD, and the UNCCD is also assumed.

STRUCTURE OF THE DOCUMENT

The present handbook is structured to provide a comprehensive and practical reference for users of the EX-ACT tool:

- It starts with an overview of the EX-ACT suite of tools, describing the objectives, uses, scope, and outputs of the application.
- This is followed by a detailed explanation of the methodology underpinning the tool, including the general approach to GHG accounting, the equations and parameters used, and the tiered assessment framework.
    - The methodology section is further developed through dedicated modules, each addressing a specific activity or sector. Each module provides both the scientific basis and the practical steps for data entry and calculation within the EX-ACT tool.
- The handbook also includes sections on the interpretation of results, the calculation of carbon balances, and the integration of economic valuation through the shadow price of carbon.
- The document concludes with annexes containing detailed tables of default values, emission factors, and parameters, as well as a comprehensive list of references.

This structure is designed to support both technical users seeking methodological detail and practitioners looking for clear, step-by-step guidance on applying the EX-ACT tool for robust and transparent GHG assessments in agrifood systems. The following section describes the tool’s design, scope, and outputs in greater detail, providing the necessary context before the methodology is presented in Section 2.

1. Introduction to the Environmental eXternalities ACcounting Tool

The EX-ACT tool, developed by FAO, is an accounting system to evaluate the environmental impacts of interventions in agrifood systems.

It is a Python-based tool with a geospatial component, aiming at providing reliable estimates of GHG emissions and removals at all stages of a project/policy/investment:

- Inception and design stage, through the use of Tier 1 Intergovernmental Panel on Climate Change (IPCC) factors and geospatial information.
- Implementation stage, through the use of yearly data on actual achievements and the integration of Tier 2 factors.
- Completion and *ex post*, through the integration of ground measurements and actual project implementation data.

The EX-ACT tool is designed to integrate environmental externalities into the assessment of agrifood system interventions. It is suitable for use throughout the intervention lifecycle. It includes features that support monitoring, reporting and verification (MRV) processes and is aligned with the Enhanced Transparency Framework (ETF) under the UNFCCC, ensuring compatibility with international reporting requirements.

The use of the tool is structured to fit within the time and resource constraints typically associated with project and policy design and evaluation processes, with scalable features depending on the level and quality of information available for the assessment.

The tool can be applied to a range of intervention types, including individual projects, sectoral investments and policies, and national programmes. Its outputs support decision-making by quantifying GHG impacts, thereby enabling alignment with climate mitigation targets such as those outlined in nationally determined contributions (NDCs).

1.1 Scope

The EX-ACT covers the full spectrum of agrifood system activities, including agriculture, forestry and other land use (AFOLU), fisheries and aquaculture, agricultural inputs, infrastructure, and downstream value chain operations.

Within the AFOLU sector, the tool accounts for both land-use change (e.g. deforestation, afforestation/reforestation, and other transitions) and land management practices. These include annual and perennial cropping systems, flooded rice cultivation, grassland management, livestock breeding, and the use of inland and coastal wetlands.

Agricultural inputs assessed include fertilizers, pesticides, machinery use, irrigation, and related components. Infrastructure activities include the construction of irrigation systems, on-farm buildings, and feeder roads. Downstream value chain activities covered by the tool include processing, packaging, storage, and transport.

The carbon balance is calculated based on changes in GHG emissions and carbon stock variations across five measurable carbon pools: above-ground biomass, below-ground biomass, litter, deadwood and soil.

In addition to carbon dioxide (CO_2), EX-ACT estimates emissions of other key GHGs, including nitrous oxide (N_2O), methane (CH_4), and hydrofluorocarbons (HCFCs), ensuring comprehensive coverage of emissions relevant to agrifood systems.

1.1.1 CATEGORIES AND REPRESENTATION OF AGRICULTURAL ACTIVITIES

The EX-ACT targets GHG accounting on a unit basis for all activities implemented in agrifood systems, including land-based, head-based, catch-based and quantity-based activities.

For land-based activities, the tool is based on the six broad categories and sub-categories proposed by IPCC for reporting GHG inventories. These are forest land, cropland, grassland, wetlands, settlements and other land.

For head-based activities, the tool is based on seventeen livestock categories proposed by IPCC for reporting GHG inventories. These are cattle (dairy/other), buffalo (dairy/other), sheep (dairy/other), swine (growing/breeding), goats, camels, horses, mules and asses, deer, ostrich, poultry (chicken layer/chicken broiler), llamas and alpacas.

For catch and quantity-based activities, the tool is based on categories proposed by IPCC (namely for fertilizers and energy sources) or other complementary scientific sources (for fisheries, aquaculture, inputs, irrigation and infrastructure).

Table 1 summarizes unit-based accounting categories according to how they are captured in EX-ACT.

Table 1. Accounting categories and Intergovernmental Panel on Climate Change reporting sector covered in the tool

Accounting category in line with IPCC	Accounting category in EX-ACT	Activity represented through the accounting category	IPCC reporting sector
Forest land	Forest management	Afforestation	AFOLU
		Deforestation	
		Rotational planting	
		Timber harvesting	
		Disturbance management	
Cropland	Annual cropland	All practices related to the management of annual and pluriannual crops	AFOLU
	Perennial cropland	Management of woody perennials and agroforestry systems	
	Flooded rice	All practices related to the management of rice crop under partial or total flooding	
	Set aside land	Temporary set-aside of annual cropland or another idle cropland	
Grassland	Grassland management	All practices related to the management of soil stocks of grasslands and pastures	AFOLU
Wetlands and flooded lands	Coastal wetland management	Extraction and excavation	AFOLU
		Drainage	
		Rewetting	
		Revegetation	
	Organic soils	Drainage	
		Rewetting	
		Soil fire management	
		Peat extraction	
	Waterbodies management	All practices related to the management of ponds, canals, ditches and other constructed water bodies	
Settlements	Settlement and infrastructure	Settlement	AFOLU IPPU
		Infrastructure creation	

Accounting category in line with IPCC	Accounting category in EX-ACT	Activity represented through the accounting category	IPCC reporting sector
Other land	Other land	All practices related to the management of bare soil, rock, ice and land areas that do not fall under any of the other five categories	AFOLU
Livestock	Livestock management	All practices related to the management of livestock, its productivity and its manure	AFOLU
Fisheries	Small fisheries	Practices linked to energy use for the fishing, refrigeration and landing of catch in small vessels	Energy
	Large fisheries	Practices linked to energy use for the fishing, refrigeration and landing of catch in small vessels	
Aquaculture	Aquaculture	Practices linked to fish rearing and energy use in aquaculture ponds	AFOLU Energy
Inputs	Inputs	Practices linked to production and field use of fertilizers, pesticides and animal feed	AFOLU IPPU
Energy	Energy	Practices linked to the use of solid, gaseous (including biogas), liquid fuels and electricity	Energy
Irrigation	Irrigation	New irrigation systems	Energy IPPU
		Operational phase of irrigation	

Note: IPCC = Intergovernmental Panel on Climate Change; EX-ACT = Environmental eXternalities ACcounting Tool; AFOLU = agriculture, forestry and other land use; IPPU = Industrial Processes and Product Use, Volume 3 of the IPCC 2019 Refinement.

Source: Authors' own elaboration.

1.1.2 TIERED ASSESSMENT

EX-ACT can accommodate different levels of assessment, or tiers, which represent a level of methodological complexity to estimate GHG emissions following the definition in IPCC (2006):

- **Tier 1** methods are designed to be the simplest to use, for which equations and default parameter values (i.e. emission and stock change factors) are provided in IPCC (2006, 2014, 2019) or other scientific sources for modules not sourced from IPCC (fisheries and aquaculture, for example). While users need to provide project specific activity data, Tier 1 parameters are available to compute a carbon balance based on default emission factors applicable globally or at regional level.
- **Tier 2** can use the same methodological approach as Tier 1 but applies emission and stock change factors that are based on country- or location-specific data. Country-defined emission factors are usually more specific to the climatic conditions, land-use systems and livestock categories in that country. Higher temporal and spatial resolution and more disaggregated activity data are typically used in Tier 2 to correspond with country-defined coefficients for specific regions and specialized land-use or livestock categories.
- **Tier 3** refers instead to the use of more complex methodologies, including GHG modelling techniques. They are tailored to address very specific local circumstances and are driven by high-resolution activity data.

The tool provides default values at Tier 1 level and can accommodate Tier 2 (and, in some cases, Tier 3) data through the Tier 2 parameters tab.

1.2 Outputs

The Environmental eXternalities ACcounting Tool was first developed to provide rapid assessments of GHG fluxes in the context of projects at design and inception stages. Through the years, it evolved to incorporate more detailed provisions, with an eye to allowing users to reliably estimate GHG fluxes of their interventions at all stages of implementation of an investment or policy.

The current version of the tool, and specifically its GHG section, allows for the assessment of the following GHGs, in line with international and private frameworks for climate-related reporting:

- **Emission inventory:** the tool automatically computes baseline emissions and stocks based on the choices made at the start situation. The inventory emissions and carbon stocks are reported in the results section and can be used to report on the baseline situation at the start of the project or policy.
- **Absolute emissions:** the tool automatically computes absolute emissions related to each scenario identified (with/without project). The absolute emissions estimates are available in the final result matrix and can be used to report on scenario-specific GHG fluxes.
- **Net emissions:** the tool computes net emissions through the carbon balance. Net emissions are always reported at activity level, under each activity section, as well as at the project level, through the final result matrix.
- **Unit emissions:** the tool provides results defined according to units targeted by the project. As an example, the carbon balance can be reported in tonnes of CO_2 equivalent per hectare, per unit of livestock, per ton of fertilizer, and so on. The tool does not provide results in emission intensity, but these can be obtained manually by dividing absolute emissions under each scenario by the units of production for the activities analyzed under said scenario.
- **Removals:** carbon removals can be estimated manually in the tool. Absolute emissions related to soils and biomass, whenever negative, represent a carbon removal. Users that are interested in reporting the absolute removals of their projects can look at absolute emissions under the with project scenario and use negative values for CO_2 stored in soils or in biomass.

2. Methodology

This section illustrates the methodology behind the GHG assessment performed by EX-ACT, including the general approach for estimating each GHG emission and the methodologies linked to each module of the tool. It also presents information on what results can be obtained from the application and how they are computed.

The EX-ACT is primarily based on the IPCC methodology for GHG assessment, complemented by other scientific research. Specifically, the main methodological sources are Volume 4 of the *2019 Refinement to the 2006 Guidelines for National Greenhouse Gas Inventories* (IPCC, 2019), the *2013 Supplement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories: Wetlands* (IPCC, 2014), and Volume 4 of the *2006 Guidelines for National Greenhouse Gas Inventories* (IPCC, 2006).

The GHG fluxes from farm operations, inputs, transport and irrigation systems implementation are based on Lal (2004). Emission factors for electricity use are based on United Nations Framework Convention on Climate Change (UNFCCC, 2021), and those for on-farm infrastructure are from Agence de l'Environnement et de la Maîtrise de l'Energie, ADEME (2020). Emission factors for the fishery sector are derived from Iribarren *et al.* (2011), Parker and Tyedmers (2015), Winther *et al.* (2009) as presented in Maestripieri *et al.* (2025). Soil carbon stock in mangroves is complemented by the review from Atwood *et al.* (2017). Yield information for the available annual cropland categories is obtained from FAOSTAT for the latest available year and continuously updated. Forest carbon reference values at the country level can also be drawn from the FAO Forest Resource Assessment (FAO, 2025) through the Tier 2 parameters tab of the forest management module.

Section 2 is organized as follows:

- Section 2.1 outlines the general computation methodology applied in EX-ACT.
- Section 2.2 then introduces the methodology for estimating carbon stock changes and associated CO_2 emissions and removals in biomass and soil.
- Section 2.3 introduced the methodology for estimating non-CO_2 emissions associated with biomass burning, mineralization of soil organic matter, and residue retention.
- Section 2.4 describes the detailed methodologies applied to each accounting module, covering land-use change, forest and cropland management, flooded rice and grassland systems, set-aside and other land categories, settlements and infrastructure, livestock, organic soils, wetlands, waterbodies, fisheries, aquaculture, inputs, energy, irrigation, and post-harvest emissions.
- Section 2.5 explains how EX-ACT calculates GHG results through carbon balance, dynamics of change, the shadow price of carbon and inventory emissions and removals.
- Section 2.6 focuses on building a robust analysis by defining project boundaries, scenario building and additionality, permanence, uncertainty, and transparency in carbon appraisals.

2.1 General methodology applied in the Environmental eXternalities ACcounting Tool

The EX-ACT tool computes climate impacts of agrifood system activities as carbon balances. A carbon balance is the difference between GHG fluxes generated with the activity and those linked to the most likely situation taking place in the absence of the investment determining the activity (scenario without the activity) as illustrated in Equation 1:

$$C_{Balance} = GHG_{with} - GHG_{without}$$

(Equation 1)

The carbon balance in EX-ACT is computed for every year of the **total accounting period** (the time between the starting year of activities and the last year of accounting) set by users.

The carbon balance is also computed differently depending on whether the GHG fluxes are generated during the **implementation phase** (i.e. the active phase of the project or investment, when activities are being implemented) or after it (i.e. a period where emissions and carbon stock changes continue to occur as a result of the implemented activities, also known as **capitalization phase**) (see Equation 2). During the implementation phase, a coefficient of change is applied to calculations to represent a dynamic reality in which activities are completed progressively through the years. For this reason, results of GHG fluxes linked to inventory emission factors and inventory coefficients are scaled by an additional factor between 0 and 1, depending on the dynamic of change selected[1]:

$$GHGimpl_{situation} \text{ in any given year} = EF_{activity} * Unit_{situation} * Coefficient_{change}$$

(Equation 2)

where:

- $GHGimpl_{situation}$ = gross fluxes linked to the implementation time and the specific project situation analyzed (without/with project);
- $EF_{activity}$ = emission factor of the considered activity. It can be replaced by carbon stock coefficients;
- $Unit_{situation}$ = number of units of the activity (i.e. hectares for land activities, heads for livestock-based activities, tonnes of catch for fishery-based activities);
- $Coefficient_{change}$ = coefficient associated with the selected dynamic of change for implementation of the activity (i.e. linear, exponential, immediate);

At the end of the implementation period, the tool assumes activities have been completed (e.g. all targeted hectares have been converted, all livestock heads have been improved); consequently, results of GHG fluxes are not scaled (Equation 3):

$$GHGcap_{situation} \text{ in any given year} = EF_{activity} * Unit_{situation}$$

(Equation 3)

where:

- $GHGcap_{situation}$ = gross fluxes linked to the capitalization time and the specific project situation analyzed (without/with project);
- $EF_{activity}$ = emission factor of the considered activity. It can be replaced by carbon stock coefficients;
- $Unit_{situation}$ = number of units of the activity (i.e. hectares for land activities, heads for livestock-based activities, tonnes of catch for fishery-based activities);

A number of GHG fluxes are also influenced by emission or carbon stock levels available in scientific literature, which means their accounting could stop based on reference data:

- For biomass, growth stops as soon as the forest or agroforestry system reaches reference levels provided by the FRA (FAO, 2025) or inserted in Tier 2;
- For dead organic matter (DOM) in forestry, growth on land units impacted in a given year will stop accruing after 20 years, due to the lack of growth rates at Tier 1 level of the IPCC;
- For soils, carbon stock changes on land units impacted in a given year will stop accruing after 20 years, in line with IPCC (2019) guidance on time dependency for soil organic carbon stock changes;
- For excavation in coastal wetland, soil organic carbon losses stop when the drainage depletes all available soil carbon according to reference levels available in the tool.

For all other activities, and in all cases where emission or carbon stock levels do not cause accounting to cease previously, the tool will stop accounting for GHG fluxes at the end of each activity's accounting period.

The duration of accounting can significantly influence the final carbon balance; thus, it is key to ensure that the data entered in the tool and the total duration of the activity chosen is in line with the specific recommendations for GHG accounting (such as those of IPCC), as applicable.

[1] The default assumption is that activities are implemented gradually, and thus achievements are obtained linearly throughout the implementation years.

The sections that follow detail how the general framework is applied to estimate CO_2 emissions and removals (Section 2.2) and non-CO_2 emissions (Section 2.3), before the module-by-module methodology presented in Section 2.4. More information on how the dynamics of change affect EX-ACT results are available in Section 2.5.2.

2.1.1 GAS CONVERSION: C TO CO_2 AND N TO N_2O

The formulas used in this handbook follow IPCC notation conventions. To display the results in EX-ACT, tonnes of carbon and tonnes of direct nitrogen emissions are converted into CO_2 and N_2O emissions based on their molecular weight ratios (see Equations 4 and 5):

$$\mathrm{CO2} = \mathrm{CO2} - \mathrm{C} * \frac{44}{12} \text{ or } \mathrm{CO2} = \Delta\mathrm{C} * \frac{44}{12}$$

(Equation 4)[2]

$$\mathrm{N2O} = \mathrm{N2O} - \mathrm{N} * \frac{44}{28}$$

(Equation 5)[3]

2.1.2 CONVERSION OF NON-CO_2 EMISSIONS INTO CO_2EQUIVALENTS: GLOBAL WARMING POTENTIAL

Results of computations related to non-CO_2 emissions are converted into tonnes of CO_2 equivalents (tCO_2eq) according to the Global Warming Potential (GWP) of each gas analyzed.

By default, EX-ACT converts emissions from non-CO_2 GHGs using their GWP over a 100-years horizon. Default GWP values provided are from the IPCC's *Fifth Assessment Report* (IPCC, 2013). However, users may choose to use other values as necessary, by changing the default settings of EX-ACT in the Description section. Reference sources for available GWP values are as follows:

- **100 yr SAR:** Global warming potential over 100 years from the Second Assessment Report (SAR), *Climate Change 1995 - The Science of Climate Change* (IPCC, 1995).
- **100 yr AR4:** Global warming potential over 100 years from the Fourth Assessment Report (AR4), *Climate Change 2007 - The Physical Science Basis* (IPCC, 2007).
- **100 yr AR5 with CC feedback (default):** Global warming potential over 100 years from the Fifth Assessment Report (AR5) with climate carbon feedback, *Climate Change 2013 - The Physical Science Basis* (IPCC, 2013).
- **100 yr AR5 w/out CC feedback:** Global warming potential over 100 years from the Fifth Assessment Report (AR5) without climate carbon feedback, *Climate Change 2013 - The Physical Science Basis* (IPCC, 2013)
- **100 yr AR6:** Global warming potential over 100 years from the Sixth Assessment Report (AR6), *Climate Change 2021 - The Physical Science Basis* (IPCC, 2021).

All GWP values used are available in Table A149 in the annex.

2.2 General methodology for estimating CO_2 emissions and removals

EX-ACT estimates annual carbon stock changes and associated CO_2 emissions and removals based on IPCC methodology outlined in Chapter 2 of IPCC 2019 refinement (IPCC, 2019).

For each land-use category, EX-ACT estimates annual carbon stock changes and related emissions and removals of CO_2 coming from five carbon pools, namely above-ground biomass (AGB), below-ground biomass (BGB), deadwood (DW), litter (LI), and soil (SO), according to Equation 6:

$$\Delta\mathrm{C_{LUi}} = \Delta\mathrm{C_{AGB}} + \Delta\mathrm{C_{BGB}} + \Delta\mathrm{C_{DW}} + \Delta\mathrm{C_{LI}} + \Delta\mathrm{C_{SO}} + \Delta\mathrm{C_{HWP}}$$

(Equation 6)[4]

[2] Based on Chapter 2 of IPCC (2019).
[3] Based on Chapter 11 of IPCC (2019).
[4] Corresponding to Equation 2.3 in Chapter 2 of IPCC (2019).

where:

- ΔC_{LUi}= carbon stock changes for a stratum of a land-use category;
- ΔC_{AGB} = carbon stock changes in above-ground biomass;
- ΔC_{BGB} = carbon stock changes in below-ground biomass;
- ΔC_{DW} = carbon stock changes in deadwood;
- ΔC_{LI}= carbon stock changes in litter;
- ΔC_{SO} = carbon stock changes in soils;
- ΔC_{HWP} = carbon stock changes in harvested wood products.[5]

Carbon stock changes are calculated for all land uses defined in EX-ACT, reported in Table 2. Carbon stock changes are computed for each carbon pool reported in Table 3 separately and depending on methodology explained in detail in Section 2.4.

Table 2. Overview of Intergovernmental Panel on Climate Change land uses adopted in the Environmental eXternalities ACcounting Tool

Land use	Definition
Forest land	This category includes all land with woody vegetation consistent with thresholds used to define Forest Land in the national greenhouse gas inventory. It also includes systems with a vegetation structure that currently fall below, but in situ could potentially reach the threshold values used by a country to define the Forest Land category.
Cropland	This category includes cropped land, including rice fields, and agroforestry systems where the vegetation structure falls below the thresholds used for the Forest Land category.
Grassland	This category includes rangelands and pastureland that are not considered Cropland. It also includes systems with woody vegetation and other non-grass vegetation such as herbs and brushes that fall below the threshold values used in the Forest Land category. The category also includes all grassland from wild lands to recreational areas as well as agricultural and silvi-pastural systems, consistent with national definitions.
Wetlands	This category includes areas of peat extraction and land that is covered or saturated by water for all or part of the year (e.g. peatlands and other wetland types) and that does not fall into the Forest Land, Cropland, Grassland or Settlements categories. It includes reservoirs as a managed sub-division and natural rivers and lakes as unmanaged sub-divisions. Further definitions of wetlands sub-divisions are provided in the IPCC Wetland Supplement (IPCC, 2014).
Settlements	This category includes all developed land, including transportation infrastructure and human settlements of any size, unless they are already included under other categories. This should be consistent with national definitions.
Other land	This category includes bare soil, rock, ice, and all land areas that do not fall into any of the other five categories.

Note: IPCC = Intergovernmental Panel on Climate Change.

Source: Land use definitions from Chapter 3.2 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories. Volume 4 – Agriculture, Forestry and Other Land Use.* Geneva, Switzerland. https://www.ipcc-nggip.iges.or.jp/public/2019rf/vol4.html

[5] As of 31/12/2025, carbon stock changes in HWP are still excluded from EX-ACT computations.

Table 3. Definition of carbon pools in agriculture, forestry and other land-use changes

Carbon pool	Definition
Above-ground biomass (AGB)	All biomass of living vegetation, both woody and herbaceous, above the soil including stems, stumps, branches, bark, seeds, and foliage.
Below-ground biomass (BGB)	All biomass of live roots. Fine roots of less than (suggested) 2mm diameter are often excluded because these often cannot be distinguished empirically from soil organic matter or litter.
Dead organic matter - deadwood	Includes all non-living woody biomass not contained in the litter, either standing, lying on the ground, or in the soil. Deadwood includes wood lying on the surface, dead roots, and stumps, larger than or equal to 10 cm in diameter (or the diameter specified by the country).
Dead organic matter (DOM) - litter	Includes all non-living biomass with a size greater than the limit for soil organic matter (suggested 2 mm) and less than the minimum diameter chosen for deadwood (e.g. 10 cm), lying dead, in various states of decomposition above or within the mineral or organic soil. This includes the litter layer as usually defined in soil typologies. Live fine roots above the mineral or organic soil (of less than the minimum diameter limit chosen for below-ground biomass) are included in litter where they cannot be distinguished from it empirically.
Soil organic matter	Includes organic carbon in mineral soils to a specified depth chosen by the country and applied consistently through the time series. Live and dead fine roots and DOM within the soil that are less than the minimum diameter limit (suggested 2 mm) for roots and DOM, are included with soil organic matter where they cannot be distinguished from it empirically. The default for soil depth is 30 cm and guidance on determining country-specific depths is given in Chapter 2.3.3.1 of IPCC (2019).

Source: Land-use definitions from Chapter 1, Table 1.1, Volume 4 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories. Volume 4 – Agriculture, Forestry and Other Land Use*. Geneva, Switzerland. https://www.ipcc-nggip.iges.or.jp/public/2019rf/vol4.html

By default, EX-ACT performs the calculations using Tier 1 values for AGB, BGB, DW, LI and soil carbon stocks provided by IPCC (2019) that represent average values differentiated by region, climate, moisture level and soil type, drawing from recent scientific literature. The default Tier 1 values for each pool by category and type of vegetation, are provided in the tables of Annex I.

The EX-ACT allows user-defined values for carbon stock levels, i.e. location-specific parameters, to overwrite default values available in the tool by using the Tier 2 parameters tab in each module.

The EX-ACT estimates the increase or decrease in biomass and soils carbon stocks due to i) changes in management practices on a given land, or due to ii) a land-use change and associated management practices, if any, on the converted area. Details on computations adopted are presented below.

2.2.1 BIOMASS CARBON STOCK CHANGES ACCORDING TO THE GAIN-LOSS METHOD

In computing carbon stock changes in biomass for **forest land, perennial cropland** and **mangrove forests in coastal wetlands**, EX-ACT adopts the gain-loss method, based on estimates of annual change in a given carbon pool, subtracting the pool's carbon losses from carbon gains. This method is depicted in Equation 7:

$$\Delta C\ Biomass = \Delta C_G - \Delta C_L$$

(Equation 7)[6]

where:

- $\Delta C\ Biomass$ = annual biomass carbon stock change, tC/yr;
- ΔC_G = annual biomass carbon gains, tC/yr;
- ΔC_L = annual biomass carbon losses, tC/yr.

[6] Corresponding to Equation 2.4 in Chapter 2 of IPCC (2019).

The annual increase in biomass carbon stock in tonnes of carbon (tC) per year is calculated using Equation 8:

$$\Delta C_G = \sum_{i,j} (A_{i,j} * G_{TOTAL\ i,j} * CF_{i,j})$$

(Equation 8)[7]

where:

- ΔC_G = annual change in biomass carbon gains, in tC/yr.
- $A_{i,j}$ = area concerned, in ha.
- $G_{TOTAL\ i,j}$ = average annual biomass growth, in tonnes of dry matter (t d.m.)/ha/year.
- $CF_{i,j}$ = carbon fraction, in tC/t d.m.

To get the average annual biomass growth ($G_{TOTAL\ i,j}$), the tool adopts by default the Tier 1 approach in Equation 9, based on the average annual AGB growth for a specific woody vegetation type (G_W) in t d.m./ha/year.

$$G_{TOTAL} = \sum \{ G_W * (1 + R)\}$$

(Equation 9)[8]

where:

- G_W = average annual above-ground biomass growth, in t d.m./ha/year;
- R = below-ground biomass to above-ground biomass ratio, in t d.m./t d.m.

Reductions in biomass carbon stocks are generated as a result of biomass losses caused by wood and fuelwood removal, and disturbances, and are calculated according to Equation 10:

$$\Delta C_L = L_{wood\ removals} + L_{fuelwood} + L_{disturbance}$$

(Equation 10)[9]

where:

- ΔC_L = annual change in biomass carbon losses, in tC/yr;
- $L_{wood\ removals}$ = annual biomass carbon losses due to wood removals, in tC/yr;
- $L_{fuelwood}$ = annual biomass carbon losses due to fuelwood removals, in tC/yr;
- $L_{disturbance}$= annual biomass carbon losses due to disturbances, in tC/yr.

Wood and fuelwood removals are calculated as in Equation 11:

$$L_{wood\ removals} + L_{fuelwood} = \{A_{removal} * B_W * (1 + R) * CF * fr\}$$

(Equation 11)[10]

where:

- $A_{removal}$ = area concerned, in ha (user input at the beginning of the module);
- B_W = average AGB of the area affected, in t d.m./ha. This is dependent on the type of forest, climate, moisture, and location selected;
- R = the ratio of BGB to AGB, in t d.m./t d.m;
- CF = the carbon fraction, in tC/t d.m;
- fr = the fraction of biomass removed, in percent

Disturbances may include pests, diseases, severe weather events, fire, and can be linked to anthropogenic activities. The tool estimates the annual carbon loss in biomass due to disturbances, in tonnes of carbon per year (tC/yr), according to Equation 12:

$$l_{disturbance} = \{A_{disturbance} * B_W * (1 + R) * CF * fd\}$$

[7] Corresponding to Equation 2.9 in Chapter 2 of IPCC (2006).
[8] Corresponding to Equation 2.10 in Chapter 2 of IPCC (2006).
[9] Corresponding to Equation 2.11 in Chapter 2 of IPCC (2006).
[10] Adapted from Equation 2.12 and Equation 2.13 in Chapter 2 of IPCC (2006).

(Equation 12)[11]

where:

- $A_{disturbance}$ = area impacted by the disturbance, in ha (user input at the beginning of the module);
- B_W = average AGB of the area affected, in t d.m./ha. This is dependent on the type of forest, climate, moisture, and location selected;
- R = the ratio of BGB to AGB (root-to-shoot ratio), in t d.m./t d.m;
- CF = the carbon fraction, in tC/t d.m. By default, in EX-ACT it is set to 0.47 in line with IPCC (2019);
- fd = the fraction of biomass destroyed by the disturbance, in percent (user input): if fd = 100 percent, all biomass is lost; if fd < 100 percent, the disturbance kills a portion of the biomass.

The forest, perennial cropland and coastal wetland modules provide more information on default factors used for biomass reference levels, growth rates and root-to-shoot ratios.

2.2.2 BIOMASS CARBON STOCK CHANGES ACCORDING TO THE STOCK-DIFFERENCE METHOD

In computing carbon stock changes in biomass for **all other land uses,** EX-ACT uses the stock-difference method. The stock-difference method is based on estimates of the difference in each pool's total carbon stock between time t2 and time t1. It requires the pool's carbon stock inventories for a given land area, at two points in time, according to Equation 13:

$$\Delta C = \frac{(C_{t2} - C_{t1})}{(t_2 - t_1)}$$

(Equation 13)[12]

where:

- ΔC = annual carbon stock change in the pool, tC/yr
- C_{t1} = carbon stock at time 1, tC/yr
- C_{t2} = carbon stock at time 2, tC/yr
- t_1 = time 1
- t_2 = time 2

At Tier 1, the tool only computes biomass stock changes for land converted to a new land-use category. The computation is performed according to Equation 14:

$$\Delta C \text{ Conversion} = \sum\{(B_{final} - B_{initial}) * \Delta A_{TO-OTHERS}\} * CF$$

(Equation 14)[13]

where:

- ΔC Conversion = initial change in biomass carbon stocks on land converted to another land category, tC/yr;
- B_{final} = biomass carbon stock on land-use category after conversion, t d.m./ha;
- $B_{initial}$ = biomass carbon stock on land-use category before conversion, t d.m./ha;
- $\Delta A_{TO-OTHERS}$ = area of land use converted to another land-use category in a certain year, ha/yr;
- CF = carbon fraction, tC/t d.m. By default, in EX-ACT it is set to 0.47.

Each respective module provides more information on default factors used for biomass reference levels and carbon stocks.

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[11] Corresponding to Equation 2.14 in Chapter 2 of IPCC (2006).
[12] Corresponding to Equation 2.5 in Chapter 2 of IPCC (2006).
[13] Corresponding to Equation 2.16 in Chapter 2 of IPCC (2006).

2.2.3 SOIL ORGANIC CARBON STOCK CHANGES

EX-ACT estimates the increase or decrease in soil organic carbon stocks in mineral soils following i) changes in management practices on a given land, or ii) following a land-use change and associated management practices, if any, on the converted area.

In line with IPCC (2019), EX-ACT assumes that:

i. Over time, SOC reaches a spatially averaged stable value, which is specific to the soil, climate, land use and management practices applied. All estimates of a change between SOC stocks are performed over 20 years, the default time period for transition between equilibrium SOC values.

ii. SOC stock changes during the transition to a new equilibrium SOC occur in a linear fashion. Although SOC changes in response to management changes may often be best described by non-linear functions, it is considered that this approach provides a good approximation of SOC stock changes for multi-year periods.

iii. SOC stocks for mineral soils are computed to a default depth of 30 cm because default reference SOC stocks and stock change factors are based on a 30 cm depth.

The methodology used for calculating the annual rates of SOC stock changes in mineral soils (in **both land remaining in the same land-use category and land converted to a new land-use category**) is based on the difference of SOC stocks between the initial and final situations, as described in Section 2.1 and depicted in Equation 15.

$$\Delta \mathrm{C\ Mineral} = \frac{(\mathrm{SOC_{final}} - \mathrm{SOC_{initial}})}{\mathrm{D}}$$

where:

$$\mathrm{SOC} = \sum (\mathrm{SOC_{REF}} * \mathrm{F_{LU}} * \mathrm{F_{MG}} * \mathrm{F_I} * \mathrm{A})$$

(Equation 15)[14]

where:

- $\Delta \mathrm{C\ Mineral}$ = annual change in mineral SOC stocks, tC/yr;
- $\mathrm{SOC_{final}}$ = SOC stock in the final situation assessed, tC;
- $\mathrm{SOC_{initial}}$ = SOC stock in the initial situation assessed, tC;
- D = time dependence of stock change factors, by default set to 20 years in EX-ACT in alignment with IPCC (2019);
- $\mathrm{SOC_{REF}}$ = SOC in the reference condition, tC/ha;
- $\mathrm{F_{LU}}$ = stock change factor for mineral SOC land-use system or sub-systems, dimensionless;
- $\mathrm{F_{MG}}$ = stock change factor for mineral SOC for management regime, dimensionless;
- $\mathrm{F_I}$ = stock change factor for mineral SOC for the inputs of organic amendments; dimensionless;
- A = land area under the different management, ha.

The SOC value depends on the SOC in the reference condition, and on the stock change factors, which include: i) a land-use factor ($\mathrm{F_{LU}}$) that reflects C stock changes associated with type of land use; ii) a management factor ($\mathrm{F_{MG}}$) representing the principal management practice specific to the land-use sector (e.g. different tillage practices in cropland); and iii) an input factor ($\mathrm{F_I}$) representing different levels of C input to soil (IPCC, 2019).

The EX-ACT provides default values for $\mathrm{SOC_{REF}}$ using the IPCC simplified soil classification ($\mathrm{SOC_{REF}}$ stocks for mineral soils to a depth of 30 cm; see Table A2 in the annex, and default values for stock change factors using the updated Table 5.5 of IPCC (2019) (see Table A31 in the annex).

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[14] Adapted from Equation 2.25 in Chapter 2 of IPCC (2019).

2.3 General methodology for estimating changes in non-CO_2 emissions

The EX-ACT estimates emissions for GHGs other than CO_2, specifically, N_2O, CH_4 and HFCs and HCFCs, based on the general IPCC methodology outlined in Chapter 2 of IPCC (2019).The main sources of these gases are energy use, refrigerant leakage, biomass burning, soil management, and livestock and manure management.

The generic approach for calculating non-CO_2 emissions considers multiplying an emission factor for a specific gas or source category with activity data related to the emission source (area, animal numbers or unit of weight, as shown in Equation 16):

$$\text{Emissions} = AD * EF$$

(Equation 16) [15]

where:

- AD = activity data;
- EF = emission factor.

The following sections illustrate the methodology for non-CO_2 emissions from a set of activities that are applicable to multiple accounting categories, and namely biomass burning, mineralization of soil organic matter, residue retention and refrigerant leakage. The methodology for estimating non-CO_2 emissions from livestock and manure management, being for a specific activity, is reported in the dedicated livestock module in Section 2.4.10. Analogously, the methodology for non-CO_2 energy emissions is reported in Section 2.4.18.

2.3.1 NITROUS OXIDE AND METHANE EMISSIONS ASSOCIATED WITH BIOMASS BURNING

The emissions of individual GHGs (N_2O and CH_4) from biomass burning (AGB, BGB, DW and LI) are calculated according to Equation 17, based on multiplying the total mass of fuel available for combustion by the combustion factor associated with the vegetation type used, and by the corresponding emission factor for each GHG. The amount of CH_4 and N_2O are calculated in kg per gas, and the sum expressed in tCO_2eq is determined using the GWP indicated in the Description module:

$$GHG_{fire} = A * M_B * C_f * G_{ef} * 10^{-3}$$

(Equation 17)[16]

where:

- $\mathrm{GHG_{fire}}$ = amount of GHG emissions from fire, tonnes of each GHG (CH_4, N_2O, CO_2);
- A = area burnt, in ha, inserted by users;
- $\mathrm{M_B}$ = mass of fuel available for combustion, tonnes/ha;
- $\mathrm{C_f}$= combustion factor, dimensionless;
- $\mathrm{G_{ef}}$= emission factor, g/kg d.m. burnt.

The EX-ACT provides default values of the factors $\mathrm{C_f}$ and $\mathrm{G_{ef}}$ respectively from Table 2.5 and Table 2.6 of IPCC (2019), which are detailed in Table A16 in the annex (for forest types) and Table A29 in the annex (for non-forest land uses).

Estimation methods and default values for defining $\mathrm{M_B}$vary according to each module (i.e. land-use category):

- When fire is occurring during a land-use change, $\mathrm{M_B}$ includes all the above-ground biomass available in the initial land use.
- When crop residues are burnt, i.e. in annual cropland**,** perennial cropland and flooded rice modules; $\mathrm{M_B}$ includes residue biomass as computed in the tool according to Equations 22 and 23 (see each module's default-values table for specific assumptions).

[15] Corresponding to Equation 2.6 in Chapter 2 of IPCC (2019).
[16] Corresponding to Equation 2.27 in Chapter 2 of IPCC (2019).

- When grassland is burnt, i.e. in the grassland module; M_B includes above-ground biomass as provided by IPCC.
- When forests are subject to disturbances or forest biomass is burnt during forest management practices (i.e. rotations, logging), i.e. in the forest management module; M_B includes all above and below-ground biomass.

More information and default values for M_B are contained in each land-use module.

2.3.2 NITROUS OXIDE EMISSIONS FROM MINERALIZATION OF SOIL ORGANIC MATTER

Estimation of direct N_2O emissions from soils is based on IPCC methodology outlined in Chapter 11 of IPCC (2019). N_2O emissions originate from N mineralization as a result of loss of soil organic carbon through land-use change or a change in management practices, and are calculated using Equation 18:

$$N_2O - N_{SOM} = F_{SOM} \times EF$$

(Equation 18)[17]

where:

- $N_2O - N_{SOM}$ = Direct N_2O emissions produced from mineralised soils, kilograms of N;
- F_{SOM} = the net annual amount of N mineralized in mineral soils as a result of loss of soil carbon through change in land-use or management practice, in kg N, calculated as in Equation 19.
- EF = emission factor for direct N_2O emissions from N, kg of N_2O/ kg N;

The total amount of N mineralized in mineral soils as a result of loss of soil carbon through change in land-use or management practice (the term F_{SOM} in Equation 18) is calculated using Equation 19:

$$F_{SOM} = \sum\left(\frac{\Delta C_{mineral}}{R} \times 1000\right)$$

(Equation 19)[18]

- $\Delta C_{mineral}$ = average annual loss of soil carbon for each land-use type, in tC, where:
  - $\Delta C_{mineral} = SOC_{final} - SOC_{initial}$, and $SOC_{final} < SOC_{initial}$
- R = carbon-to-nitrogen (C:N) ratio of the soil organic matter

In line with Chapter 11 of IPCC (2019), EX-ACT provides a default value of 15 for land-use changes from forest land or grassland to cropland (annual and perennial), a default value of 10 for cases involving management changes on cropland (remaining cropland).

The estimation of emissions from soil mineralisation applies to **all land-based modules**, whenever the changes in SOC stocks imply a loss of carbon.

2.3.3 NITROUS OXIDE EMISSIONS ASSOCIATED WITH RESIDUE RETENTION

Retention of agricultural residues can stimulate soil N_2O emissions via regulating soil carbon and nitrogen availability and microbial activities. Crop residues may play multiple roles, for example they can act as organic N fertilizer, organic carbon substrate and energy provider in regulating soil N_2O production and emission and therefore, N_2O emissions associated with agricultural residues are estimated. Estimation of N_2O emissions associated with residue retention is based on IPCC methodology outlined in Chapter 11 of IPCC (2019). Emissions from residue retention are computed under the annual cropland and flooded rice modules of the EX-ACT application, whenever the residue management practice is chosen accordingly. N_2O emissions from crop residue retention are estimated through Equation 20:

$$N_2O - N_{CR} = F_{CR} \times EF_1$$

(Equation 20)[19]

where:

[17] Adapted from Equation 11.2 of Chapter 11 of IPCC (2019).
[18] Corresponding to Equation 11.8 of Chapter 11 of IPCC (2019).
[19] Adapted from Equation 11.2 of Chapter 11 of IPCC (2019).

- $N_2O - N_{CR}$ = the annual direct N_2O emissions produced from managed soils through residue retention, in kg N/y;
- F_{CR} = the total amount of N in crop residues (above-ground and below-ground), including N-fixing crops, returned to soils, in kg N, calculated as in Equation 21.
- EF_1 = the emission factors to estimate direct N_2O emissions from managed soils due to N additions from crop residues, in kg N_2O–N/kg N.

The total amount of N in crop residues returned to the soil (the term F_{CR} in Equation 20) is calculated using Equation 21:

$$F_{CR} = AGR_{(T)} \times N_{AG(T)} + BGR_{(T)} \times N_{BG(T)}$$

(Equation 21)[20]

where:

- $AGR_{(T)}$ = the annual total amount of above-ground crop residue for crop T that is retained, in kg d.m./yr;
- $N_{AG(T)}$ = N content of above-ground residues for crop T, in kg N/kg d.m.;
- $BGR_{(T)}$ = annual total amount of below-ground crop residue for crop T that is retained, in kg d.m./yr;
- $N_{BG(T)}$ = the N content of below-ground residues for crop T, in kg N/kg d.m.

The amounts of above-ground and below-ground crop residues for crop T (the terms $AGR_{(T)}$ and $BGR_{(T)}$) are estimated as in Equation 22:

$$AGR_{(T)} = AG_{DM(T)} \times Area_{(T)}$$

$$BGR_{(T)} = \left(Crop_{(T)} + AG_{DM(T)} \right) \times RS_{(T)} \times Area_{(T)} \times Frac_{renew(T)}$$

(Equation 22)[21]

where:

- $AG_{DM(T)}$ = the above-ground residue dry matter for crop T that is retained, in kg d.m./ha (see Equation 23);
- $Crop_{(T)}$ = the harvested annual dry matter yield for crop T, in kg d.m./ha;
- $RS_{(T)}$ = the ratio of below-ground root biomass to above-ground shoot biomass for crop T, in kg d.m./ha;
- $Area_{(T)}$ = the total annual area harvested of crop T, ha/yr;
- $Frac_{renew(T)}$ = fraction of total area under crop T that is renewed annually, dimensionless.

And where $AG_{DM(T)}$ is calculated as in Equation 23, based on the methodology and equation illustrated in Table 11.2 of IPCC (2019):

$$AG_{DM(T)} = Crop_{(T)} \times Slope_{(T)} + Intercept_{(T)}$$

(Equation 23)[22]

where:

- $Crop_{(T)}$ = crop yield, in kg d.m./ha;
- $Slope_{(T)}$ = slope coefficient, dimensionless;
- $Intercept_{(T)}$ = intercept coefficient, dimensionless.

Default values for $Slope_{(T)}$, $Intercept_{(T)}$, $N_{AG(T)}$, $N_{BG(T)}$, and $RS_{(T)}$ are reported in Table A32 in the annex, while default crop yield values are obtained from FAO (2022a) and subject to constant updates.

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[20] Adapted from Equation 11.6 of Chapter 11 of IPCC (2019).
[21] Corresponding to Equation 11.6 in Chapter 11 of IPCC (2019).
[22] Adapted to methodology illustrated in Table 11.2 in Chapter 11 of IPCC (2019).

2.3.4 REFRIGERANT LEAKAGE EMISSIONS

Emissions from refrigerant leakages are based on(Equation 24 Equation 16 and Chapter 7, Volume 3 of IPCC (2019). All refrigerant systems leak refrigerants during their operation. EX-ACT quantifies these emissions based on common refrigerant gases. By default, conversion to CO_2eq is based on GWP values from IPCC assessment reports, reported in Table A149 in the annex. See the introduction of this Section 2.1.2 for more information on GWPs.

The annual refrigerant emissions are given by Equation 24:

$$\mathrm{E} = \mathrm{R} * \mathrm{EF}$$

(Equation 24)[23]

where:

- E = Annual emissions from refrigerant leakage, tCO_2eq/yr;
- R = Annual refrigerant gas leakage, kg/yr;
- EF = The emission factor for a given refrigerant type, tCO_2eq/kg refrigerant.

In turn, the EF for refrigerant gases follows from the GWP, and an adjustment factor for kilograms of refrigerant to tonnes of CO_2 emissions, in Equation 25:

$$\mathrm{EF} = \frac{\mathrm{GWP}}{1000}$$

(Equation 25)

where:

- GWP = The GWP value corresponding to a specific refrigerant and IPCC assessment report, dimensionless.

2.4 Methodology applied in the Environmental eXternalities ACcounting Tool by module

Building on the general methodology for CO_2 and non-CO_2 emissions outlined in Sections 2.2 and 2.3, this section details how those str applied within each module of EX-ACT. Each subsection covers the following:

- the **GHG(s)** concerned by the module;
- the **carbon pools** related to each GHG flux, or the emission source;
- the **methodology** for estimating the GHG flux(es). If the methodology is specific to a module, it is detailed in that module's section. Otherwise, it is described in Section 2.1 which outlines the general methodology, and is referenced here;
- the **default values** embedded in EX-ACT and corresponding sources.

Each assessment module includes a Tier 2 parameters tab which allows users to modify carbon pools and default values used for computations.

2.4.1 LAND-USE CHANGE

The land-use change module of EX-ACT covers GHG fluxes linked to any transitions from a specific IPCC-defined land-use category to a different one. The module covers the following computations:

1. Changes in biomass carbon stocks linked to the removal of biomass under the original land use:
   - CO_2 emissions and removals associated with biomass carbon stock changes are calculated according to the methodology illustrated in Section 2.2.
2. Emissions from fire (biomass burning) during conversion of land (e.g. slash and burn):
   - CH_4 and N_2O emissions from fire used during conversion of land (if applicable) are calculated according

[23] Based on Equation 2.6 of Chapter 2 of IPCC (2019).

to the methodology illustrated in Section 2.3.

The estimation of soil organic carbon stock changes linked to land-use changes (and the emissions from potential soil mineralization) cannot be separated from the influence of management factors of the initial and final land uses. For this reason, soil organic carbon stock changes are not accounted for under the land-use change modules but only provided in the final land-use module once all information linked to the land use, management and input factors are completed by users.

Table 4 illustrates the GHGs addressed by the module, with the corresponding level of assessment, i.e. Tier 1 or Tier 2 and includes in-text references to the equations used for the GHG fluxes calculations, as well as the corresponding IPCC equations for easier cross-reference.

Table 4. Overview of greenhouse gas fluxes estimated in the land-use change module of the Environmental eXternalities ACcounting Tool and corresponding equations

Activity		GHG	Flux description	Equation	IPCC equation
Land-use change	Tier 1/2	CO_2	Biomass carbon stock changes	14	2.16 of IPCC (2006)
	Tier 1	CH_4 and N_2O	Biomass burning in case of fire use during land conversion	17	2.27 of IPCC (2019)

Note: GHG = greenhouse gas; CO_2 = carbon dioxide; N_2O = nitrous oxide; CH_4 = methane; IPCC = Intergovernmental Panel on Climate Change.

Sources: Authors' own elaboration based on IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Volume 4 – Agriculture, Forestry and Other Land Use. Geneva, Switzerland. https://www.ipcc-nggip.iges.or.jp/public/2019rf/vol4.html; IPCC. 2006. *2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Volume 4 – Agriculture, Forestry and Other Land Use. Kanagawa, Japan. https://www.ipcc-nggip.iges.or.jp/public/2006gl.

The land-use categories concerned follow IPCC and are described in Table 2 in Section 2.2. Emissions from land-use changes are calculated whenever the initial land-use differs from the final land-use, in line with the definitions above.

Biomass carbon stock changes

In the land-use change module, only biomass removal linked to the switch from the original land-use to a new one are computed. The tool estimates the difference between the biomass carbon stock under the initial land-use and a final biomass carbon stock of 0. Default values in t d.m./ha are converted into tC/ha using the carbon fraction of 0.47 tC/t d.m. (see Table 4.3 IPCC [2006]). Biomass gain linked to the land-use changes are computed under the final land-use module, in line with default values provided in Table 5.

For all land-use changes, the tool computes biomass loss after subtracting the dry matter that is considered removed before the land-use change. Specifically, users are requested to input the t d.m. removed from the land prior to the land-use change. This quantity is converted into tonnes of carbon and subtracted from the AGB carbon stock present under the initial land use. It is possible to modify biomass carbon stock values in Tier 2 tab of the tool.

Emissions from biomass burning

Non-CO_2 emissions from burning biomass during land-use change consist of CH_4 and N_2O produced by fire used for land clearing activities during land conversion. CO_2 emissions from biomass burning do not have to be reported, since the carbon released during the combustion process is assumed to be reabsorbed by the vegetation during the next growing season (IPCC, 2019).

Table 5 presents the default values used in the EX-ACT land-use change module, together with the corresponding IPCC table references and in-text table references. It also includes the default data and calculations used to estimate emissions from biomass burning. The associated calculations are triggered when the Fire use option in the land-use change module is set to Yes.

Table 5. Sources of default values for biomass carbon stock changes in land-use change module of the Environmental eXternalities ACcounting Tool

Element	Unit	Details	Table	Table IPCC	Notes
Biomass carbon stock changes (Equation 14) $$\Delta \mathbf{C\ Conversion} = \sum \{(\mathbf{B_{final}} - \mathbf{B_{initial}}) * \Delta\, \mathbf{A_{TO-OTHERS}}\} * \mathbf{CF}$$					
Biomass under (forest) initial land use, for systems before deforestation					
Biomass carbon stock under initial land use (Binitial*CF)	tC/ha	Calculated as the sum of AGB, BGB, LI, and DW. Converted into tC/ha multiplying by the carbon fraction (0.47)	See below	See below	
Above-ground biomass (AGB)	t d.m./ha	For forest systems before deforestation. EX-ACT retains either the value proposed by IPCC directly or the central value when a range of values is available.	A4 – A8	4.12 of IPCC (2006). Values from the latest FAO FRA may also be used to replace the default IPCC values.	Modifiable in Tier 2 in tC/ha
Below-ground biomass (BGB)	Dimensionless	Estimated using a ratio of BGB to AGB, R expressed in t root d.m. per t shoot d.m. For all ecosystem in a boreal climate, when AGB (above-ground biomass) has ≤ 75 t d.m./ha, a default ratio of 0.390 is used, and 0.240 for when AGB has > 75 t d.m./ha. For tropical mountain systems (forest and plantation), a default value of 0.27 is used for all regions.	A9 – 14	4.4 of IPCC (2019). Values from the latest FAO FRA may also be used to replace the default IPCC values.	Modifiable in Tier 2 in tC/ha
Litter (LI) and deadwood (DW)	t d.m./ha	For forest systems before deforestation.	A15	2.2 of IPCC (2019). Values from the latest FAO FRA may also be used to replace the default IPCC values.	Modifiable in Tier 2 in tC/ha
Biomass under (non-forest) initial land use, for systems prior to afforestation/reforestation and other land-use change					
Biomass carbon stock under initial land use (Binitial*CF)	tC/ha	For systems prior to afforestation/reforestation or other land-use changes.	A17, A27 and A28	6.4 of IPCC (2006). 5.1, 5.3 and 5.9 of IPCC (2019).	
N_2O and CH_4 emissions from biomass burning (Equation 17) $$\mathbf{GHG_{fire}} = \mathbf{A} * \mathbf{M_B} * \mathbf{C_f} * \mathbf{G_{ef}} * \mathbf{10^{-3}}$$					
Biomass available for burning (M_B)	t d.m./ha	For all land uses, the default value is equal to biomass or AGB (whenever AGB is specified separately) available at land-use change,	N/A	N/A	Modifiable in Tier 2

Element	Unit	Details	Table	Table IPCC	Notes
		less the biomass removed through user input.			
Combustion factor (C_f)	Dimensionless	For fire used during deforestation	A16	2.6 of IPCC (2019)	Not modifiable
		For fire used prior to afforestation/reforestation	A29		
		For fire used prior to other land-use changes: default value equal to 0.4 for annual fallow, annual crops and irrigate rice and 0.8 for all other land uses.	N/A	N/A	
Emission factor (G_{ef})	g/kg d.m. burnt	For forest systems before deforestation	A16	2.5 of IPCC (2019)	Not modifiable
		For systems prior to afforestation/reforestation and other land-use changes: 2.7 g CH_4/kg d.m. burnt, and 0.07 g N_2O/kg d.m. burnt	A29		

Note: AGB = above-ground biomass; BGB = below-ground biomass; CF = carbon fraction; FRA = Forest Resource Assessment; gCH_4/kg d.m. = grams of methane per kilogram of dry matter; gN_2O/kg d.m. = grams of nitrous oxide per kilogram of dry matter; IPCC = Intergovernmental Panel on Climate Change; R = ratio of below-ground biomass to above-ground biomass; SOC = soil organic carbon stock change; tC/ha = tonnes of carbon per hectare; t d.m./ha = tonnes of dry matter per hectare.

Sources: Authors' own elaboration based on IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Volume 4 – Agriculture, Forestry and Other Land Use. Geneva, Switzerland. https://www.ipcc-nggip.iges.or.jp/public/2019rf/vol4.html; IPCC. 2006. *2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Volume 4 – Agriculture, Forestry and Other Land Use. Kanagawa, Japan. https://www.ipcc-nggip.iges.or.jp/public/2006gl

2.4.2 FOREST MANAGEMENT AND DISTURBANCES

Material used to develop this module can be found in Chapter 2 (Generic methodologies applicable to multiple land-use categories), Chapter 4 (Forest Land) and Chapter 11 (N_2O emissions from managed soils, and CO_2 emissions from lime and urea application) of). The methodological note on forest management Crespi *et al.* (2025) provides further guidance on forest definitions and computations operated in the module.

The forest management module focuses on GHG emissions associated with afforestation and changes in forest conditions. Deforestation emissions are tackled in the land-use change module.

The module covers the following computations:

1. **Biomass carbon gains** linked to natural forest growth and af/reforestation:
   - CO_2 emissions and removals associated with biomass carbon gains are calculated according to the gain-loss methodology illustrated in Section 2.2.
2. **Biomass carbon losses** linked to wood removal activities and disturbances:
   - CO_2 emissions and removals associated with biomass carbon losses are calculated according to the general gain-loss methodology illustrated in Section 2.2.
3. **Emissions from biomass burning,** i.e. the occurrence of fires or deliberate wood burning for energy purposes:
   - CH_4 and N_2O from residue burning are calculated according to the method illustrated in Section 2.3.
4. **Changes in soil organic carbon stocks** linked to land-use changes (i.e. afforestation and deforestation) and changes in management practices:
   - CO_2 emissions and removals associated with soil organic carbon stock changes are calculated according to the methodology illustrated in Section 2.2.

5. **Emissions from mineralization of soil organic matter** associated with loss of soil organic carbon when a change in land use occurs:

– N_2O emissions associated with soil organic carbon stock changes are calculated according to the methodology illustrated in Section 2.2.

Table 6 illustrates the GHGs addressed by the module, with the corresponding level of assessment, i.e. Tier 1 or Tier 2 and includes in-text references to the equations used for the GHG fluxes calculations, as well as the corresponding IPCC equations for easier cross-reference.

Definition of the vegetation

EX-ACT requires specifying the type of vegetation (native or forest plantation) undergoing management or degradation. Different types of vegetation, and corresponding default carbon stocks in the four biomass carbon pools, are associated with varying climatic and geographical information. Table A3 in the annex summarises the vegetation types per climate zone available in EX-ACT. The distinction between forest and plantation is based on the growth rate of trees, which strongly depends on the management regime. Therefore, a distinction is made between intensive forest management (i.e. plantation forestry) and extensive forest management (naturally re-growing stands with reduced or minimum human intervention).

The EX-ACT distinguishes between primary forests, secondary forests and forest plantations, in alignment with IPCC. For AGB, EX-ACT retains either the value proposed by IPCC directly or the central value when a range of values is proposed by IPCC. Values vary according to the continent and the ecological zone. BGB is estimated in alignment with IPCC (2019) using a ratio of below-ground biomass to above-ground biomass, R, expressed in t root d.m. per t shoot d.m.

Biomass carbon stock changes

The biomass carbon stock changes are calculated according to the gain-loss methodology described in Section 2.2, corresponding to an estimate of annual change in the considered carbon pools obtained by subtracting the pools' carbon losses from their carbon gains.

In practical terms, the module assumes a continuous growth of the forest considered in accordance with biomass growth rates and subtracts carbon losses to the pools associated with i) timber harvesting, ii) forest plantation rotations, iii) occurrence of disturbances.

Biomass gains and losses

Biomass gains include total biomass (AGB and BGB) growth are assessed on an annual basis and automatically computed in any given year that the forest under assessment has a biomass level lower than the reference levels. Once the reference level is reached, biomass gains no longer accrue, assuming that gains and losses are balanced.

Biomass losses are estimated annually using different approaches depending on the section of the forest management module applied in the assessment, as follows:

- For calculating biomass carbon gains and losses linked to **rotational planting**, it is assumed that **all of the biomass gained through a rotation period is removed at the end of it,** with the length of the rotation period being specified by the user. This only includes AGB and BGB, as DOM is deemed to remain stable at Tier 1 in line with IPCC (2019). If the biomass is used for energy, non-CO_2 emissions from incomplete combustions are also calculated.
- For calculating biomass carbon gains and losses linked to **timber harvesting**, including logging, thinning and wood collection if relevant, it is **assumed that harvesting practices occur periodically throughout the activity duration**, with their recurrence being specified by the user. Only AGB and BGB are assumed to be removed at each recurrence, in line with Equation 7 reported in Section 2.2.1, DOM is deemed to remain stable at Tier 1 in line with IPCC (2019). If the biomass is used for energy, non-CO_2 emissions from incomplete combustions are also calculated.

- For calculating biomass carbon losses linked to **disturbances** such as fires, pests and diseases, and insects, it is assumed that **disturbances occur periodically throughout the activity duration**, with their recurrence being specified by the user. Only AGB and BGB are assumed to be lost at each recurrence, in line with Equation 12 reported in Section 2.2.1, DOM is deemed to remain stable at Tier 1 in line with IPCC (2019).
- The calculation of biomass carbon losses linked to unspecified, generic causes, i.e. when no disaggregated information on rotations, harvesting, or disturbances affecting biomass levels is available at the activity level, is based on assessing the percentage of total living or dead biomass available on the site (AGB, BGB, deadwood and litter) that is lost due to degradation in each year.

Table 6. Overview of greenhouse gas fluxes estimated in the forest management and disturbances module of Environmental eXternalities ACcounting Tool and corresponding equations

Activity	Tier	GHG	Flux description	Equation	IPCC equation
Forest management and disturbances	Tier 1	CO_2	Biomass carbon stock gains linked to biomass regrowth	8 and 9	2.9 and 2.10 of
			Biomass carbon stock losses linked to wood and fuelwood removal	10 and 11	2.11 – 2.13 of IPCC (2006)
			Biomass carbon stock losses linked to disturbances	10 and 12	2.11 and 2.14 of IPCC (2006)
	Tier 1	CH_4 and N_2O	Non-CO_2 emissions from biomass burning in case of fire or burning fuelwood for energy	17	2.27 of IPCC (2019)
	Tier 2	CO_2	Soil organic carbon stock changes	15	2.25 of IPCC (2019)
Afforestation	Tier 2	CO_2	Soil organic carbon stock changes	15	2.25 of IPCC (2019)
	Tier 1	CH_4 and N_2O	Non-CO_2 emissions from biomass burning in case of fire use during land conversion	17	2.27 of IPCC (2019)
	Tier 1	N_2O	Nitrogen mineralization linked to land-use change	18 and 19	11.2 and 11.8 of IPCC (2019)
	Tier 1	CO_2	Biomass carbon stock gains linked to biomass growth	8 and 9	2.9 and 2.10 of
	Tier 1/2	CO_2	Biomass carbon stock gains following land-use changes	14	2.16 of IPCC (2006)

Note: GHG = greenhouse gas; CO_2 = carbon dioxide; N_2O = nitrous oxide; CH_4 = methane; IPCC = Intergovernmental Panel on Climate Change; non-CO2 = non-carbon dioxide greenhouse gases.

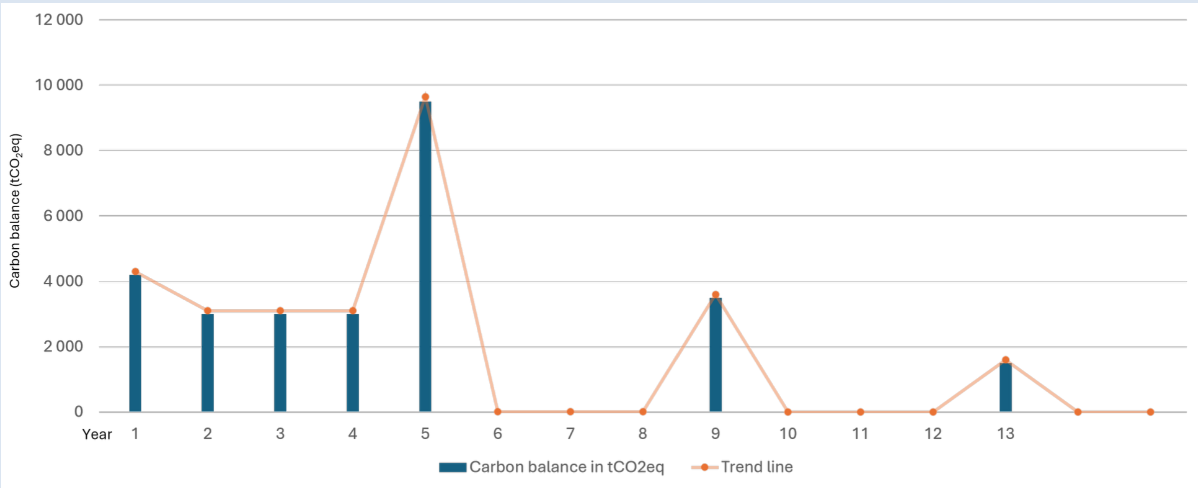
Sources: Authors' own elaboration based on IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Volume 4 – Agriculture, Forestry and Other Land Use. Geneva, Switzerland. https://www.ipcc-nggip.iges.or.jp/public/2019rf/vol4.html; IPCC. 2006. *2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Volume 4 – Agriculture, Forestry and Other Land Use. Kanagawa, Japan. https://www.ipcc-nggip.iges.or.jp/public/2006gl.

In Tier 2, users can specify the start year of rotations, occurrence of harvesting or disturbance. This will incorporate a delay in the initial cutting or biomass loss, process, corresponding to the number of years specified in the section, see Box 1.

Box 1. Examples of greenhouse gas flux dynamics in forest management activities

An example of GHG fluxes linked to timber harvesting is provided in Figure 1. The positive peaks indicate harvesting occurrences. The first occurrence is prolonged over the implementation time in line with the general approach to dynamics of change adopted in the tool. The tool will compute biomass growth in the analyzed forest, unless it is already at its maximum reference biomass level in line with IPCC values. In the harvesting year, the amount of biomass identified by users will be removed from the targeted area.

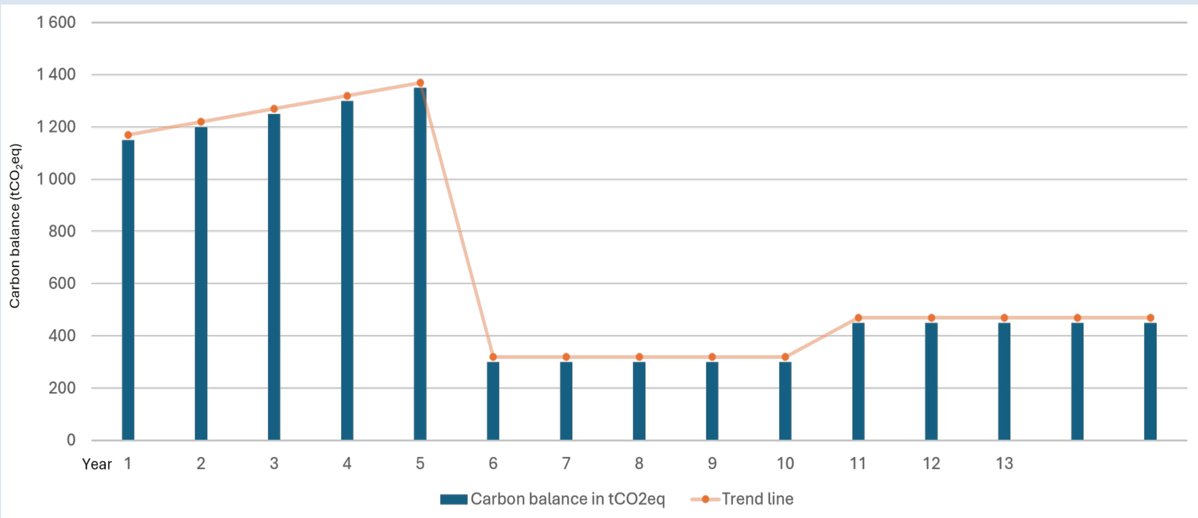
Figure 1. Timber harvesting (four-year recurrence, 50 percent harvest, five years implementation time)



Source: Authors' own elaboration.

An example of GHG fluxes linked to the application of rotations is provided in Figure 2. At the year of the rotation recurrence, the tool will apply rotations to hectares targeted in line with the implementation dynamic. New trees are assumed to be planted in the same year of the cutting. The tool will compute biomass growth in the analyzed plantation, unless it is already at its maximum reference biomass level in line with IPCC values.

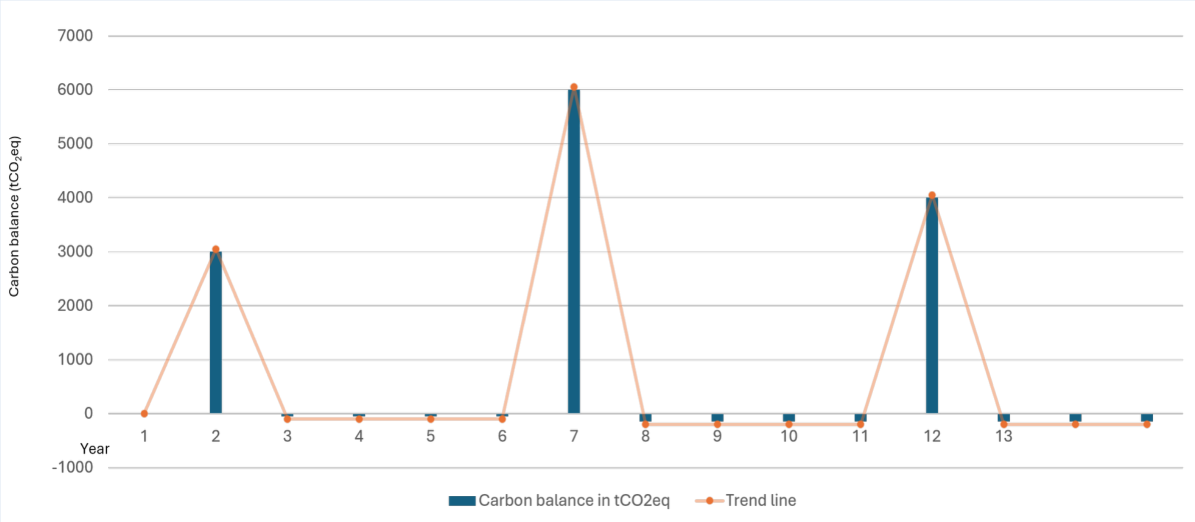
Figure 2. Rotations (five-year rotation length with five years implementation time)



Source: Authors' own elaboration.

An example of GHG fluxes linked to the presence of disturbances is provided in Figure 3. In the disturbance recurrence year, the amount of biomass lost as specified by users will be removed from the targeted area in line with the dynamic of change selected and the duration of the activity. The tool will compute biomass growth in the analyzed forest, unless it is already at its maximum reference biomass level in line with IPCC values.

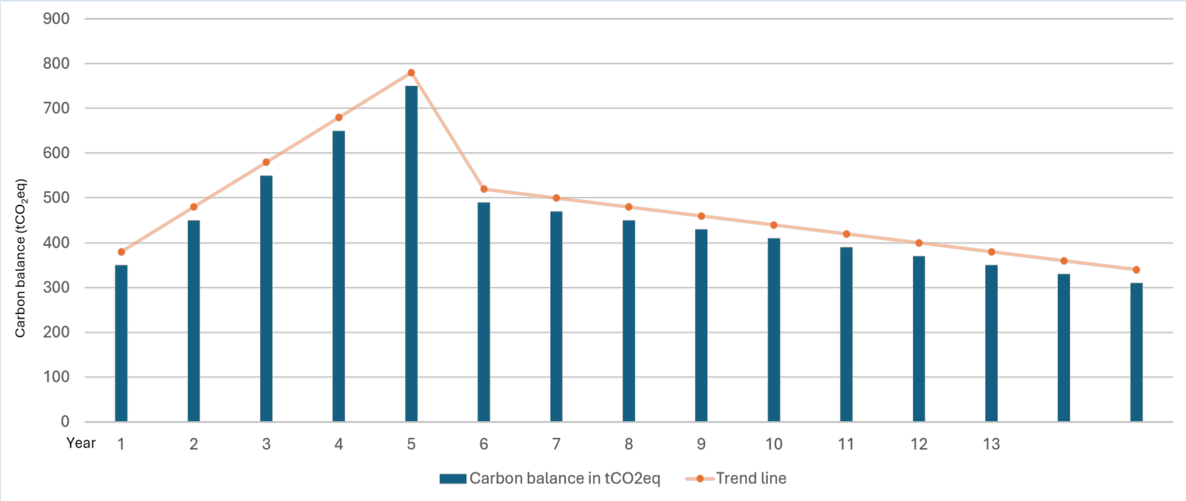
Figure 3. Fire disturbance (five-year recurrence, 50 percent loss, five years implementation time)



Source: Authors' own elaboration.

An example of GHG fluxes linked to the aggregate changes in forest conditions is provided in Figure 4. The amount of biomass lost as specified by users will be removed from the targeted area in line with the dynamic of change selected and the duration of the activity. The tool will compute biomass growth in the analyzed forest, unless it is already at its maximum reference biomass level in line with IPCC values. If no percentage is specified, or the percentage specified is equal to 0, degradation is assumed not to take place, with the result that only biomass growth will be accounted. This is the encouraged approach to use to estimate GHG fluxes from forest regeneration activities.

Figure 4. Aggregate changes in forest conditions: 5 percent yearly degradation, five years implementation time



Source: Authors' own elaboration.

Soil organic carbon stock changes

The methodology used for calculating the annual rates of SOC stock changes in mineral soils is based on the difference of SOC stocks between with and without the project scenario as described in Section 2.2.3. EX-ACT estimates the increase/decrease in SOC_{ref} following land-use changes and associated management practices, if any, in the converted area. In line with IPCC (2019), EX-ACT assumes that under Tier 1 soil organic carbon stock does not change with forest management.

Together with changes and oxidation of SOC storage, land-use management is also accompanied by mineralization of nitrogen (N), which is an additional source of N available for N_2O emissions. Where soil carbon losses occur, the release of N by mineralization is estimated according to the general methodology described in Section 2.3.2, Equations 18 and 19 in this document.

If the forest is subject to fire, and if the wood collected is used for energy purposes, non-CO_2 emissions i.e. CH_4 and N_2O produced by the combustion of biomass are also calculated. EX-ACT estimates the associated non-CO_2 emissions from biomass burning based on the generic methodology summarized in Section 2.3.1. The associated calculations will be triggered if the type of disturbance in the forest management and disturbances module is set to Fire, or if the use of biomass for energy is flagged in the Rotations and harvesting sections. The mass of fuel available for combustion (the term M_B in Equation17) in the forest management and disturbances module includes all above and below-ground biomass available at the time of burning.

Table 7 details the default values adopted in the Forest management module, along with the IPCC table reference and in-text table reference.

Table 7. Sources of default values in forest management and disturbances module of the Environmental eXternalities ACcounting Tool

Element	Unit	Details	Table	Table IPCC	Notes
Biomass carbon stock changes from LUC (Equation 14) $\Delta C\ \mathbf{Conversion} = \sum\{(\mathbf{B_{final}} - \mathbf{B_{initial}}) * \Delta\, \mathbf{A_{TO-OTHERS}}\} * \mathbf{CF}$					
Biomass under forest final land use, for systems after afforestation/reforestation					
Biomass carbon stock under final land use (B_{final}*CF)	tC/ha	Calculated as the sum of AGB, BGB, LI, and DW. Converted into tC/ha multiplying by the carbon fraction (0.47)	See below	See below	-
Above-ground biomass (AGB) growth	t d.m./ha/yr	For forest systems implanted after afforestation/reforestation.	A18 – A26	4.12 of IPCC (2019)	Modifiable in Tier 2 in tC/ha/yr; also, if >20 yrs.
Below-ground biomass (BGB) growth	Dimensionless	Estimated using a ratio of BGB to AGB, R, expressed in t root d.m. per t shoot d.m.	A9 – A14	4.4 of IPCC (2019)	Modifiable in Tier 2 in tC/ha/yr; also, if >20 yrs.
Litter and deadwood	t d.m./ha	For forest systems after afforestation/reforestation.	A15	2.2 of IPCC (2019). Values from the latest FAO FRA can also be used to replace IPCC default values.	Modifiable in Tier 2.

Element	Unit	Details	Table	Table IPCC	Notes
Biomass carbon gains (Equations 8 and 9) $\mathbf{\Delta C_G} = \sum_{i,j}(\mathbf{A_{i,j}} * \mathbf{G_{TOTAL\ i,j}} * \mathbf{CF_{i,j}})$ and $\mathbf{G_{TOTAL}} = \sum\{\mathbf{G_W} * (\mathbf{1} + \mathbf{R})\}$					
Above-ground biomass growth (G_W)	t d.m./ha/ year	Values vary depending on type of forest and the continent and the ecological zone.	A18 – A26	4.12 of IPCC (2019)	Modifiable in Tier 2 in tC/ha/year, with distinction >20 years or <20 years
Ratio of below-ground biomass to above-ground biomass (R)	t root d.m. per t shoot d.m.	N/A	A9 – A14	4.4 of IPCC (2019)	In Tier 2, possibility to report below-ground biomass growth in tC/ha/year, with distinction >20 years or <20 years
Carbon fraction (CF) of AGB	tC/t d.m.	0.47	N/A	4.3 of IPCC (2006)	N/A
Average annual biomass growth (G_{TOTAL})	t d.m./ha/ year	Calculated	N/A	N/A	N/A
Area (A)	ha	Area concerned, User input	N/A	N/A	N/A
Biomass carbon losses (Equations 10 – 12) $\mathbf{\Delta C_L} = \mathbf{L_{wood\ removals}} + \mathbf{L_{fuelwood}} + \mathbf{L_{disturbance}}$ where: $\mathbf{L_{wood\ removals}} + \mathbf{L_{fuelwood}} = \{\mathbf{A_{removal}} * \mathbf{B_W} * (\mathbf{1} + \mathbf{R}) * \mathbf{CF} * \mathbf{fr}\}$ and where: $\mathbf{L_{disturbance}} = \{\mathbf{A_{disturbance}} * \mathbf{B_W} * (\mathbf{1} + \mathbf{R}) * \mathbf{CF} * \mathbf{fd}\}$					
Above-ground biomass (B_W)	t d.m./ha	Natural forest Values vary according to the continent and the ecological zone	A4–A8	4.12 of IPCC (2019)	EX-ACT retains either the value proposed by IPCC directly or the central value when a range of possible values is proposed. Modifiable in Tier 2 in tC/ha
Ratio of below-ground biomass to above-ground biomass (R)	t root d.m. per t shoot d.m.	Used to estimate BGB, given R and B_W.	A9 – A14	4.4 of IPCC (2019)	In Tier 2, possibility to report below-ground biomass in tC/ha
Carbon fraction (CF) of AGB	tC/t d.m.	0.47	N/A	4.3 of IPCC (2006)	N/A
Share of total biomass removed (fr)	Dimensionless	User input (as percentage)	N/A	N/A	N/A
Share of total biomass destroyed by the disturbance (fd)	Dimensionless	User input (as percentage)	N/A	N/A	N/A
Area ($A_{removal}$)	ha	Area concerned by the removal, User input	N/A	N/A	N/A
Area ($A_{disturbance}$)	ha	Area affected by the disturbance, User input	N/A	N/A	N/A

Element	Unit	Details	Table	Table IPCC	Notes
Soil organic carbon stock change (Equation 15) - only Tier 2 assessment $$SOC = \sum (SOC_{REF} * F_{LU} * F_{MG} * F_I * A)$$					
Reference soil organic carbon stock (SOC_{REF})	tC/ha	References for SOC stocks for mineral soils to a depth of 30 cm	A2	2.3 of IPCC (2019)	Modifiable in Tier 2
SOC stock change factor (combined for F_{LU}, F_{MG}, F_I)	Dimensionless	By default, set to 1	N/A	N/A	Modifiable in Tier 2
N_2O and CH_4 emissions from biomass burning (Equation 17) $$GHG_{fire} = A * M_B * C_f * G_{ef} * 10^{-3}$$					
Area (A)	ha	Area concerned by fire. User input.	N/A	N/A	N/A
Mass of fuel available for combustion (M_B)	t d.m./ha	Biomass of forest type concerned.	A4 – A8	4.12 of IPCC (2019)	Modifiable in Tier 2
Fuel biomass consumption values ($M_B * C_f$)	t d.m./ha	Estimated values of the product of M_B and Cf.	N/A	2.4 of IPCC (2019)	N/A
Combustion factor (C_f)	percentage	For fire during forest management and disturbances.	A16	2.6 of IPCC (2019)	N/A
Emission factor (G_{ef})	g/kg d.m. burnt	For fire during forest management and disturbances.	A16	2.5 of IPCC (2019)	N/A

Note: AGB = above-ground biomass; BGB = below-ground biomass; CF = carbon fraction; EX-ACT = Environmental eXternalities ACcounting Tool; FRA = Forest Resource Assessment; gCH_4/kg d.m. burnt = grams of methane per kilogram of dry matter burnt; gN_2O/kg d.m. = grams of nitrous oxide per kilogram of dry matter; IPCC = Intergovernmental Panel on Climate Change; N/A = not applicable; R = ratio of below-ground biomass to above-ground biomass; SOC = soil organic carbon stock change; tC/ha = tonnes of carbon per hectare; t d.m./ha = tonnes of dry matter per hectare.

Sources: Authors' own elaboration based on IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Volume 4 – Agriculture, Forestry and Other Land Use. Geneva, Switzerland. https://www.ipcc-nggip.iges.or.jp/public/2019rf/vol4.html; IPCC. 2006. *2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Volume 4 – Agriculture, Forestry and Other Land Use. Kanagawa, Japan. https://www.ipcc-nggip.iges.or.jp/public/2006gl.

2.4.3 ANNUAL CROPLAND MANAGEMENT

Material used to develop this module can be found in Chapter 2 (Generic methodology applicable to multiple land-use categories), Chapter 5 (Cropland) and Chapter 11 (N_2O emissions from managed soils, and CO_2 emissions from lime and urea application) of IPCC (2019).

The annual cropland module concerns land that is used for growing annual and pluriannual crops, including upland rice, cotton and sugarcane. It considers annual systems remaining annual systems, but which are subject to changes in management. EX-ACT considers the following crops: grains, beans and pulses, tubers, barley, maize, oats, potatoes, soybeans, wheat, millet, sorghum, rye, peanuts, dry beans, root crops. A default option is also available.

The module covers the following computations (see Table 8):

1. **Changes in soil organic carbon stocks** linked to land-use changes and changes in management practices (i.e. tillage and nutrient management):
   - CO_2 emissions and removals associated with soil organic carbon stock changes are calculated according to the methodology illustrated in Section 2.2.3.
2. **Emissions from nitrogen mineralization** associated with loss of soil organic matter:

- N_2O emissions associated with soil organic carbon stock changes are calculated according to the methodology illustrated in Section 2.3.2.

3. **Changes in biomass carbon stocks** linked to land-use changes:
   - CO_2 fluxes associated with biomass gains following a land-use change are calculated according to the methodology illustrated in Section 2.2.1, using the stock-difference method.
4. **Emissions from residue management practices**, including burning and retention on fields:
   - CH_4 and N_2O from **residue burning** are calculated according to the method illustrated in Section 2.3.1.
   - N_2O from **residue retention** are calculated according to the method illustrated in Section 2.3.3.

Table 8. Overview of greenhouse gas fluxes estimated in the annual cropland module of the Environmental eXternalities ACcounting Tool and corresponding equations

Activity		GHG	Flux description	Equation	IPCC equation
Annual cropland management	Tier 1	CO_2	Soil organic carbon stock changes linked to change in practices (tillage and nutrient management)	15	2.25 of IPCC (2019)
	Tier 1	CH_4 and N_2O	Non-CO_2 emissions from residue burning	17 and 23	2.27 of IPCC (2019)
	Tier 1/2	CO_2	Biomass carbon stock gains following land-use changes	14	2.16 of IPCC (2006)
	Tier 1	N_2O	N emissions from residue retention	20 – 23	See Table 11.2 , Volume 4 of IPCC (2019)
	Tier 1	N_2O	N mineralization linked to change in management practice	18 and 19	11.2 and 11.8 of IPCC (2019)

Note: GHG = greenhouse gas; CO_2 = carbon dioxide; N_2O = nitrous oxide; CH_4 = methane; IPCC = Intergovernmental Panel on Climate Change; Non-CO_2 = non-carbon dioxide greenhouse gas.

Source: Authors' own elaboration based on IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Volume 4 – Agriculture, Forestry and Other Land Use. Geneva, Switzerland. https://www.ipcc-nggip.iges.or.jp/public/2019rf/vol4.html; IPCC. 2006. *2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Volume 4 – Agriculture, Forestry and Other Land Use. Kanagawa, Japan. https://www.ipcc-nggip.iges.or.jp/public/2006gl.

Table 8 illustrates the GHGs addressed by the module, with the corresponding level of assessment, i.e. Tier 1 or Tier 2 and includes in-text references to the equations used for the GHG fluxes calculations, as well as the corresponding IPCC equations for easier cross-reference.

Changes in soil organic carbon stocks

In Tier 1, EX-ACT assumes zero CO_2 emissions and removals from biomass. Since annual crops are planted and harvested within a single year, there is no net gain or loss of biomass. Annual croplands also contain little or no deadwood or litter; therefore, EX-ACT does not account for carbon stock changes in DOM for annual systems.

Management of annual cropping systems can significantly affect the amount of carbon stock in soil. The carbon storage potential depends on the type of crop and soil, prevalent climate and soil management practices. EX-ACT calculates related CO_2 emissions and removals, distinguishing between type of crop, its residue management, tillage practices, fertilizer management (both mineral fertilizers and organic amendments), irrigation management and intensity of cropping management (i.e. continuous cropping versus cropping rotations with periods of bare fallow or mixed systems with cropping and pasture or hay in rotating sequences).

The methodology used for calculating the annual rates of SOC stock changes in mineral soils is based on the difference of SOC stocks between with and without the project scenario as described in Section 2.1. EX-ACT estimates the increase/decrease in SOC_{ref} following changes in management practices on a given land and following a land-use change and associated management practices, if any, on the converted area.

The SOC stock for the with and without project scenario estimated though Equation 15 are calculated multiplying the SOC_{ref} by the default soil stock change factors for land use, management and inputs (F_{LU}, F_{MG}, F_I). These factors quantify the influence of land-use and management activities on soil carbon to a depth of 30 cm and are defined as follows:

- **Long-term cultivated (F_{LU})**: Represents area that has been converted from native conditions and continuously managed for predominantly annual crops over 50 years.
- **Tillage management (F_{MG})**: Represents the different tillage practices in cropland (IPCC, 2019):
  - Full: Substantial soil disturbance with full inversion and/or frequent (within year) tillage operations. At planting time, little (i.e. <30 percent) of the surface is covered by residues.
  - Reduced: Primary and/or secondary tillage but with reduced soil disturbance (usually shallow and without full soil inversion). Normally leaves surface with >30 percent coverage by residues at planting.
  - No-till: Direct seeding without primary tillage, with only minimal soil disturbance in the seeding zone. Herbicides are typically used for weed control.
- **Input of organic material (F_I)**: Represents the different levels of C input to soil (IPCC, 2019):
  - Low: Low residue return occurs when there is removal of residues (via collection or burning), frequent bare-fallowing, production of crops yielding low residues (i.e. vegetables, tobacco and cotton), no mineral fertilization or N-fixing crops.
  - Medium: Representative for annual cropping with cereals where all crop residues are returned to the field. If residues are removed, then supplemental organic matter (i.e. manure) is added. Also requires mineral fertilization or N-fixing crop in rotation.
  - High without manure: Represents significantly greater crop residue inputs over medium carbon input cropping systems due to additional practices, such as production of high residue yielding crops, use of green manures, cover crops, improved vegetated fallows, irrigation, frequent use of perennial grasses in annual crop rotations, but without manure applied.
  - High with manure: Represents significantly higher carbon input over medium carbon input cropping systems due to an additional practice of regular addition of animal manure.

Soil management practices of a minor cropping season should be weighted and represented through the main season section if they are applied for more than 1/3 of the yearly cropping period and should be excluded in all other situations (IPCC, 2019). Table 9 details the sources of the default values for the parameters in Equation 15 embedded in EX-ACT, along with the IPCC table reference and in-text table reference.

Emissions from residue management practices

Non-CO_2 emissions from burning crop residues consist of CH_4 and N_2O gases produced by the combustion of (a percentage of) crop residues burnt on-site. CO_2 emissions from burning agricultural residues do not have to be reported, since the carbon released during the combustion process is assumed to be reabsorbed by the vegetation during the next growing season (IPCC, 2019).

The EX-ACT estimates the associated non-CO_2 emissions from crop residue burning based on the generic methodology summarized in Section 2.3.1. The associated calculations will be triggered if the residue management option in the annual cropland module is set to Burned.

The mass of fuel available for combustion (the term $\mathrm{M_B}$ in Equation 17) in the Annual cropland module is defined as the above-ground residue dry matter (AGDM), of a specific crop (in kg d.m./ha). AGDM is estimated using Equation 23 from the updated Table 11.2 of IPCC (2019) and crop yield data (t d.m./ha) for the year of 2020 obtained from FAOSTAT (FAO, 2022a).

The combustion factor (the term $\mathrm{C_f}$ in Equation 17) is a measure of the proportion of the biomass actually combusted. The emission factor (the term $\mathrm{G_{ef}}$ in Equation 17) gives the amount of a particular GHG emitted per unit of crop residue dry matter combusted (g/kg d.m. burnt).

The methodology used to quantify the effect of crop residue retention on N_2O emissions follows the generic methodology summarized in Section 2.3.3 (Table 11.2 of IPCC [2019]), or Equations 20 –23 in this document. Residue management in EX-ACT can be assessed also for a minor cropping season.

Together with changes and oxidation of SOC storage, land-use management is also accompanied by mineralization of N, which is an additional source of N available for N_2O emissions. Where soil carbon losses occur, the release of N by mineralization is estimated according to the general methodology described in Section 2.3.2 and Equations 18 and 19 in this document.

Table 9 details the default values adopted in the Annual cropland module, along with the IPCC table reference and in-text table reference.

Table 9. Sources of default values in annual cropland module of the Environmental eXternalities ACcounting Tool

Element	Unit	Details	Table	Table IPCC	Notes
Biomass carbon stock changes from LUC (Equation 14) $\Delta \mathbf{C\ Conversion} = \sum \{(\mathbf{B_{final}} - \mathbf{B_{initial}}) * \Delta\, \mathbf{A_{TO-OTHERS}}\} * \mathbf{CF}$					
Biomass under annual cropland final land use, for systems after deforestation and other land-use change					
Biomass carbon stock under final land use (B_{final})	t d.m./ha	Values for annual crops correspond to the default biomass carbon stocks in the year following land conversion.	A17	Table 6.4 of IPCC (2006), Table 5.2, Table 5.3 and 5.9 of IPCC (2019)	-
Biomass carbon stock under initial land use ($B_{Initial}$)	t d.m./ha	See section of initial land-use type	N/A	-	-
Carbon fraction (CF)	Dimensionless	Default value: 0.47	N/A	-	-
Soil organic carbon stock changes (Equation 15) $\mathbf{SOC} = \sum (\mathbf{SOC_{REF}} * \mathbf{F_{LU}} * \mathbf{F_{MG}} * \mathbf{F_I} * \mathbf{A})$					
Soil organic carbon (SOC_{REF})	tC/ha	References for SOC stocks for mineral soils to a depth of 30 cm	A2	2.3 of IPCC (2019)	Modifiable in Tier 2
Land-use factor (F_{LU})	Dimensionless	Type of crop and land use	A31	5.5 of IPCC (2006) and 5.5 of IPCC (2019)	Modifiable in Tier 2
Management factor (F_{MG})	Dimensionless	Type of tillage and disturbance level	A31	5.5 of IPCC (2006) and 5.5 IPCC (2019)	Modifiable in Tier 2
Input factor (F_I)	Dimensionless	Type and level of carbon inputs	A31	5.5 of IPCC (2006) and 5.5 IPCC (2019)	Modifiable in Tier 2
N_2O and CH_4 emissions from residue burning (Equations 17 and 23) $\mathbf{GHG_{fire}} = \mathbf{A} * \mathbf{M_B} * \mathbf{C_f} * \mathbf{G_{ef}} * \mathbf{10^{-3}}$ with; $\mathbf{M_B} = \mathbf{AG_{DM(T)}} = \mathbf{Crop_{(T)}} \times \mathbf{Slope_{(T)}} + \mathbf{Intercept_{(T)}}$					
Biomass available for burning (M_B)	t d.m./ha	Above-ground residue dry matter (AG_{DM}) of a specific crop.	N/A	AG_{DM}: 11.2 of IPCC (2019)	-

Element	Unit	Details	Table	Table IPCC	Notes
Crop yield ($Crop_{(T)}$)	t/ha	For application rate of organic amendment (ROA) calculation (AG_{DM}).	N/A	FAOSTAT values are continuously updated.	-
$Slope_{(T)}$	Dimensionless	For ROA calculation. Default values per $Crop_{(T)}$ given by IPCC.	N/A	11.2 of IPCC (2019)	-
$Intercept_{(T)}$	Dimensionless	For ROA calculation Default values per $Crop_{(T)}$ given by IPCC.	N/A	11.2 of IPCC (2019)	-
Combustion factor (C_F)	Percentage	Measure of the proportion of the biomass actually combusted (0.90 for wheat residues, 0.80 for maize residues and 0.85 for other agricultural residues).	A29	2.6 of IPCC (2019)	-
Emission factor (G_{ef})	g/kg d.m. burnt	2.7 gCH_4/kg d.m. burnt, and 0.07 gN_2O/kg d.m. burnt.	A29	2.5 of IPCC (2019)	-
N_2O emissions from residue retention (Equations 20-23) $\mathbf{N_2O - N_{CR} = F_{CR} \times EF_1}$ where: $\mathbf{F_{CR} = AGR_{(T)} \times N_{AG(T)} + BGR_{(T)} \times N_{BG(T)}}$ and where $\mathbf{AGR_{(T)} = AG_{DM(T)} \times Area_{(T)} \mid BGR_{(T)} = \left(Crop_{(T)} + AG_{DM(T)}\right) \times RS_{(T)} \times Area_{(T)} \times Frac_{renew(T)}}$ and where $\mathbf{AG_{DM(T)} = Crop_{(T)} \times Slope_{(T)} + Intercept_{(T)}}$					
Above-ground residue dry matter for crop T ($AG_{DM(T)}$)	kg d.m./ha	AG_{DM}(T) is estimated using crop yield data combined with the parameters and equation in Table 11.2.	N/A	11.1A and 11.2 of IPCC (2019)	Modifiable in Tier 2.
Harvested annual dry matter yield ($Crop_{(T)}$)	kg d.m./ha	Based on crop yield defaults from FAOSTAT or user input.	N/A	FAOSTAT values are continuously updated.	
$N_{AG(T)}$ = N content of above-ground residues	kg N/kg d.m.	-	A32	11.1A of IPCC (2019)	-
Residue availability (below-ground) ($BGR_{(T)}$)	kg d.m./yr	Annual total amount of below-ground crop residue for crop T.	N/A	11.1A of IPCC (2019)	Modifiable in Tier 2.
$N_{BG(T)}$ = N content of below-ground residues	kg N/kg d.m.	-	A32	11.1A of IPCC (2019)	-
Ratio of below-ground root biomass to above-ground shoot biomass for crop T ($RS_{(T)}$)	kg d.m./ha/(kg d.m./ha)	-	A32	11.1A of IPCC (2019)	-

Element	Unit	Details	Table	Table IPCC	Notes
Emission factor for N_2O from N addition through crop residues (EF_1)	kg N_2O–N/kg N	0.006 kg N_2O–N/kg N for all climates wet and moist, 0.005 kg N_2O–N/kg N for dry climates and 0.010 kg N_2O–N/kg N for an aggregated climate.	N/A	11.1 of IPCC (2019)	-
FracRenew(T)	Dimensionless	Fraction of total area under crop T that is renewed annually. Default value for annual crops: 1	N/A	N/A	-
$Slope_{(T)}$	Dimensionless	Default values per $Crop_{(T)}$ given by IPCC.	A32	11.2 of IPCC (2019)	-
$Intercept_{(T)}$	Dimensionless	Default values per $Crop_{(T)}$ given by IPCC.	A32	11.2 of IPCC (2019)	-
N_2O emissions from N mineralization (Equations 18 and 19) (if LUC or change in management practice) $\mathbf{N_2O - N_{SOM} = F_{SOM} \times EF}$ where: $\mathbf{F_{SOM}} = \sum(\frac{\mathbf{\Delta C_{mineral}}}{\mathbf{R}} \times \mathbf{1000})$					
C:N ratio of the soil organic matter (R)	Dimensionless	Default values: 10 for cropland remaining cropland 15 for land-use changes from Forest land or Grassland to Cropland (annual and perennial)	N/A	11.1 of IPCC (2019)	-
Average annual loss of soil carbon ($\Delta C_{mineral}$)	tC/yr	-	N/A	-	-
Net annual amount of N mineralized in mineral soils (F_{SOM})	kg N/yr	As a result of loss of soil carbon through change in land-use or management	N/A	-	-
Emission factor for N_2O–N from N mineralised (EF)	kg N_2O–N/kg N	0.006 kg N_2O–N/kg N for all climates wet and moist 0.005 kg N_2O–N/kg N for dry climates 0.010 kg N_2O–N/kg N for an aggregated climate	N/A	11.1 of IPCC (2019)	-

Note: AGB = above-ground biomass; BGB = below-ground biomass; CF = carbon fraction; EX-ACT = Environmental eXternalities ACcounting Tool; FLU = land-use factor; FMG = management factor; FI = input factor; FRA = Forest Resource Assessment; gCH_4/kg d.m. burnt = grams of methane per kilogram of dry matter burnt; gN_2O/kg d.m. burnt = grams of nitrous oxide per kilogram of dry matter burnt; IPCC = Intergovernmental Panel on Climate Change; N/A = not applicable; non-CO_2 = non-carbon dioxide greenhouse gases; ROA = application rate of organic amendment; R = ratio of below-ground biomass to above-ground biomass; SOC = soil organic carbon stock change; SOCREF = reference soil organic carbon stock; tC/ha = tonnes of carbon per hectare; t d.m./ha = tonnes of dry matter per hectare.

Sources: Authors' own elaboration based on IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Volume 4 – Agriculture, Forestry and Other Land Use. Geneva, Switzerland. https://www.ipcc-nggip.iges.or.jp/public/2019rf/vol4.html; IPCC. 2006. *2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Volume 4 – Agriculture, Forestry and Other Land Use. Kanagawa, Japan. https://www.ipcc-nggip.iges.or.jp/public/2006gl.

2.4.4 PERENNIAL CROPLAND MANAGEMENT

Material used to develop this module can be found in Chapter 2 (Generic methodology applicable to multiple land-use categories), Chapter 5 (Cropland) and Chapter 11 (N_2O emissions from managed soils, and CO_2 emissions from lime and urea application) of Volume 4 of IPCC (2019).

The perennial cropland module concerns plants and other woody species that persist for many growing seasons. Generally, the top portion of the plant dies back each winter and regrows the following spring from the same root system. Perennial cropland systems can help mitigate climate change by storing carbon in biomass and removing CO_2 from the atmosphere. Perennial woody crops accumulate C for an amount of time equal to a nominal harvest maturity cycle (IPCC, 2019). This assumption implies that perennial woody crops accumulate biomass for a finite period until they are removed through harvest or reach a steady state where there is no net accumulation of carbon in biomass because growth rates have slowed and incremental gains from growth are offset by losses from natural mortality pruning or other losses (Tier 1 assumption).

The perennial cropland module includes trees and shrubs, in combination with herbaceous crops (e.g. agroforestry) or as orchards, vineyards and plantations, except where these lands meet the criteria for categorisation as forest land. The module can also be used to represent mixed systems in which annual and perennial woody crops coexist.

The module covers the following computations:

1. **Changes in biomass carbon stocks** linked to changes in agroforestry system types:
   - CO_2 emissions and removals associated with biomass carbon stock changes are calculated according to the methodology illustrated in Section 2.2.1, using a gain-loss method.
2. **Changes in soil organic carbon stocks** linked to land-use changes and changes in management practices (i.e. tillage and nutrient management):
   - CO_2 emissions and removals associated with soil organic carbon stock changes are calculated according to the methodology illustrated in Section 2.2.3.
3. **Emissions from residue burning practices**:
   - CH_4 and N_2O from **residue/biomass burning** linked to management practices like pruning are calculated according to the method illustrated in Section 2.3.1.
4. **Emissions from nitrogen mineralization** associated with loss of soil organic matter:
   - N_2O emissions associated with soil organic carbon stock changes are calculated according to the methodology illustrated in Section 2.3.2.

Table 10 illustrates the GHGs addressed by the module, with the corresponding level of assessment, i.e. Tier 1 or Tier 2 and includes in-text references to the equations used for the GHG fluxes calculations, as well as the corresponding IPCC equations for easier cross-reference.

Table 10. Overview of greenhouse gas fluxes estimated in the perennial cropland module of the Environmental eXternalities ACcounting Tool and corresponding equations

Activity		GHG	Flux description	Equation	IPCC equation
Perennial cropland management	Tier 1	CO_2	Biomass carbon stock changes	7	2.4 of IPCC (2019)
	Tier 1	CO_2	Soil organic carbon stock changes linked to change in practices (tillage and nutrient management)	15	2.25 of IPCC (2019)
	Tier 1	CH_4 and N_2O	Non-CO_2 emissions linked to biomass burning	17	2.27 of IPCC (2019)
	Tier 1	N_2O	N mineralization linked to change in management practices	18 and 19	11.2 and 11.8 of IPCC (2019)
	Tier 1/2	CO_2	Biomass carbon stock gains following land-use changes	14	2.16 of IPCC (2006)

Note: GHG = greenhouse gas; CO_2 = carbon dioxide; N_2O = nitrous oxide; CH_4 = methane; IPCC = Intergovernmental Panel on Climate Change; non-CO_2 = non-carbon dioxide greenhouse gases; N mineralization linked to change in management practices = nitrogen released from soil organic matter due to changes in land management.

Sources: Authors' own elaboration based on IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Volume 4 – Agriculture, Forestry and Other Land Use. Geneva, Switzerland. https://www.ipcc-nggip.iges.or.jp/public/2019rf/vol4.html; IPCC. 2006. *2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Volume 4 – Agriculture, Forestry and Other Land Use. Kanagawa, Japan. https://www.ipcc-nggip.iges.or.jp/public/2006gl.

Carbon stock changes in biomass and soil organic carbon

The Environmental eXternalities ACcounting Tool estimates carbon stock changes in the above-ground and below-ground biomass of agroforestry systems. Users can define the maturity state of the agroforestry system in Tier 2. If the system is mature, EX-ACT takes into account the maximum above-ground biomass stock at harvest provided in Table 5.1 of IPCC (2019). If not, the tool takes into account the average above-ground biomass stock at harvest available in the same table (see Tables A27, A28, A33, A40, A48 – A49 in the annex).

The methodology used for calculating the annual rates of SOC stock changes in mineral soils is based on the difference of SOC stocks between with and without the project scenario as described in Section 2.2.3**.** EX-ACT estimates the increase/decrease in SOC_{ref} following land-use changes and associated management practices, if any, on the converted area. Refer to Section 2.4.3 on annual cropland to see definitions and parameters for management practices.

Emissions from residue burning

Non-CO_2 emissions from burning pruning materials consist of CH_4 and N_2O produced by the combustion of pruning. CO_2 emissions from burning pruning materials do not have to be reported, since the carbon released during the combustion process is assumed to be reabsorbed by the vegetation during the next growing season (IPCC, 2019). EX-ACT estimates the associated non-CO_2emissions from crop residue burning based on the generic methodology summarized in Section 2.3.1**.** The associated calculations will be triggered if the residue management option in the perennial cropland module is set to Burned. The mass of fuel available for combustion (the term $M_B$in in Equation 17) in the perennial cropland module is set by default to be equal to 50 percent of the AGB yearly growth rate, but users can enter Tier 2 values and specify the pruning biomass available for burning.

Emissions from nitrogen mineralization

Together with changes and oxidation of SOC storage, land-use management is also accompanied by mineralization of N, which is an additional source of N available for N_2O emissions. Where soil carbon losses occur, the release of N by mineralization is estimated according to the general methodology described in Section 2.3.2, and Equations 18 and 19 this document.

Table 11 details the default values adopted in the perennial cropland module, along with the IPCC table reference and in-text table reference.

Table 11. Sources of default values in perennial cropland module of the Environmental eXternalities ACcounting Tool

Element	Unit	Details	Table	Table IPCC	Notes
Biomass carbon stock changes under perennial cropland land use					
From LUC, for perennial systems after deforestation and other land-use change (Equation 14) $\Delta \mathbf{C\ Conversion} = \sum\{(\mathbf{B}_{\mathbf{final}} - \mathbf{B}_{\mathbf{initial}}) * \Delta\, \mathbf{A}_{\mathbf{TO-OTHERS}}\} * \mathbf{CF}$ From annual accumulation in non-mature perennial systems (Equation 7) $\Delta \mathbf{C\ Biomass} = \Delta \mathbf{C}_{\mathbf{G}} - \Delta \mathbf{C}_{\mathbf{L}}$ where; $\Delta \mathbf{C} = (\mathbf{AGB(T)} + \mathbf{G}_{\mathbf{AGB}} + \mathbf{BGB(T)} + \mathbf{G}_{\mathbf{BGB}}) * \mathbf{CF}$					
Biomass carbon stock under final land use (B_{final})	t d.m./ha	Calculated as the sum of AGB and BGB after one year	See below	See below	Modifiable in Tier 2
Carbon Fraction (CF)	Dimensionless	Used if biomass is given in t d.m. instead of TC Default value = 0.47	N/A	-	-
Above-ground biomass carbon stock (maximum level) (AGB(T))	tC/ha	ABG of a mature perennial cropland system	A48 and A49	5.1 of IPCC (2019)	Modifiable in Tier 2
Above-ground biomass carbon stock (mean level) (AGB(T))	t d.m./ha	This is the EX-ACT default value for ABG in an existing, non-mature perennial cropland system	A27 and A28	5.1 of IPCC (2019)	Modifiable in Tier 2 (as initial AGB)
Above-ground biomass growth rate (G_{AGB})	tC/ha/yr	For non-mature perennial cropland systems	A33 – A40	5.2 and 5.3 of IPCC (2019)	Modifiable in Tier 2
Below-ground biomass growth rate (G_{BGB})	tC/ha/yr	For the year after LUC or non-mature perennial cropland systems	A41 – A47	5.2 and 5.3 of IPCC (2019)	Modifiable in Tier 2
Soil organic carbon stock changes Equation 15) (also if LUC) $\mathbf{SOC} = \sum(\mathbf{SOC}_{\mathbf{REF}} * \mathbf{F}_{\mathbf{LU}} * \mathbf{F}_{\mathbf{MG}} * \mathbf{F}_{\mathbf{I}} * \mathbf{A})$					
Soil organic carbon (SOC_{REF})	tC/ha	References for SOC stocks for mineral soils to a depth of 30 cm	A2	2.3 of IPCC (2019)	Modifiable in Tier 2
Land-use factor (F_{LU})	Dimensionless	Type of crop and land use	A31	5.5 of IPCC (2019)	Modifiable in Tier 2
Management factor (F_{MG})	Dimensionless	Type of tillage and disturbance level	A31	5.5 of IPCC (2019)	Modifiable in Tier 2
Input factor (F_I)	Dimensionless	Type and level of carbon inputs	A31	5.5 of IPCC (2019)	Modifiable in Tier 2
N_2O and CH_4 emissions from residue/biomass burning (Equation 17) $\mathbf{GHG}_{\mathbf{fire}} = \mathbf{A} * \mathbf{M}_{\mathbf{B}} * \mathbf{C}_{\mathbf{f}} * \mathbf{G}_{\mathbf{ef}} * \mathbf{10}^{-3}$					
Biomass available for burning (M_B)	kg d.m./ha	Default value for residues: 50 % of the AGB yearly growth rate	N/A	N/A	Modifiable in Tier 2
Combustion factor (C_F)	Dimensionless	Default value: 0.85	A29	2.6 of IPCC (2019)	Not modifiable

Element	Unit	Details	Table	Table IPCC	Notes
Emission factor (G_{ef})	g/kg d.m. burnt	Determined by crop type	A29	2.5 of IPCC (2019)	Not modifiable
N_2O emissions from N mineralization (Equations 18 and 19) (if LUC or if change in management practices) $\mathbf{N_2O - N_{SOM}} = \mathbf{F_{SOM}} \times \mathbf{EF}$ where: $\mathbf{F_{SOM}} = \sum(\frac{\Delta C_{mineral}}{R} \times \mathbf{1000})$					
Net annual amount of N mineralized in mineral soils (F_{SOM})	kg N/yr	As a result of loss of soil carbon through change in land use or management	N/A	-	-
Average annual loss of soil carbon ($\Delta C_{mineral}$)	tC/yr	-	N/A	-	-
C:N ratio of the soil organic matter (R)	Dimensionless	Default values: 10 for cropland remaining cropland. 15 for land-use changes from Forest land or Grassland to Cropland (annual and perennial).	N/A	11.1 of IPCC (2019)	-
Emission factor for N_2O–N from N mineralised (EF)	kg N_2O–N/kg N	0.006 kg N_2O–N/kg N for all climates wet and moist 0.005 kg N_2O–N/kg N for dry climates 0.010 kg N_2O–N/kg N for an aggregated climate	N/A	11.1 of IPCC (2019)	-

Note: AGB = above-ground biomass; BGB = below-ground biomass; EX-ACT = Environmental eXternalities ACcounting Tool; gCH_4/kg d.m. burnt = grams of methane per kilogram of dry matter burnt; gN_2O/kg d.m. burnt = grams of nitrous oxide per kilogram of dry matter burnt; IPCC = Intergovernmental Panel on Climate Change; LUC = land-use change; N/A = not applicable; non-CO_2 = non-carbon dioxide greenhouse gases; SOC = soil organic carbon stock change; tC/ha = tonnes of carbon per hectare; tC/ha/yr = tonnes of carbon per hectare per year; t d.m./ha = tonnes of dry matter per hectare; t d.m./ha/yr = tonnes of dry matter per hectare per year.

Source: IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Volume 4 – Agriculture, Forestry and Other Land Use. Geneva, Switzerland. https://www.ipcc-nggip.iges.or.jp/public/2019rf/vol4.html.

2.4.5 FLOODED RICE MANAGEMENT

Material used to develop this module can be found in Chapter 2 (Generic methodology applicable to multiple land-use categories) for the soil and GHG emissions from residues burning, Chapter 5 (Cropland) for methane emission from rice cultivation and Chapter 11 (N_2O emissions from managed soils, and CO_2 emissions from lime and urea application) of IPCC (2019) for the quantity of residues.

The flooded rice module only concerns rice fields that are permanently (or during part of the year) flooded. Upland rice is not a source of CH_4, since it is grown in aerated soils that never become flooded for a significant period of time. Thus, upland rice is considered an annual crop in EX-ACT and the underlying methodology is described in the annual cropland module (Section 2.4.3).

The module covers the following computations:

1. **Changes in soil organic carbon stocks** linked to land-use changes and changes in management practices:
   - CO_2 emissions and removals associated with soil organic carbon stock changes are calculated adapting the methodology illustrated in Section 2.2.3.
2. **Emissions from nitrogen mineralization** associated with loss of soil organic matter:
   - N_2O emissions associated with soil organic carbon stock changes are calculated according to the methodology illustrated in Section 2.3.2.
3. **Changes in biomass carbon stocks** linked to land-use changes

- CO_2 fluxes associated with biomass gains following a land-use change are calculated according to the methodology illustrated in Section 2.2.1, using the stock-difference method.

4. **Emissions produced from anaerobic decomposition of organic matter**:
   - CH_4 emissions from rice cultivation are calculated according to the methodology illustrated in Equation 27.
5. **Emissions from residue management practices**, including burning and retention on fields:
   - CH_4 and N_2O from residue burning are calculated according to the method illustrated in Section 2.3.1.
   - N_2O from residue retention is calculated according to the method illustrated in Section 2.3.3.

Table 12 illustrates the GHGs addressed by the module, with the corresponding level of assessment, i.e. Tier 1 or Tier 2 and includes in-text references to the equations used for the GHG fluxes calculations, as well as the corresponding IPCC equations for easier cross-reference.

Table 12. Overview of greenhouse gas fluxes estimated in the flooded rice module of the Environmental eXternalities ACcounting Tool and corresponding equations

Activity		GHG	Flux description	Equation	IPCC equation
Flooded rice management	Tier 1	CO_2	Soil organic carbon stock changes linked to change in practices (tillage and nutrient management)	15 and 26	2.25 of IPCC (2019)
	Tier 1/2	CO_2	Biomass carbon stock gains following land-use changes	14	2.16 of IPCC (2006)
	Tier 1	CH_4	Methane emissions linked to anaerobic decomposition of organic matter	27 – 29	5.1, 5.2 and of IPCC (2019)
	Tier 1	CH_4 and N_2O	Non-CO_2 emissions linked to residue burning	17	2.27 of IPCC (2019)
	Tier 1	N_2O	Non-CO_2 emissions linked to residue retention	20 – 23	11.2 and 11.6 of IPCC (2019)
	Tier 1	N_2O	N mineralization linked to change in management practices	18 and 19	11.2 and 11.6 of IPCC (2019)
	Tier 1/2	CO_2	Biomass carbon stock gains following land-use changes	14	2.16 of IPCC (2006)

Note: GHG = greenhouse gas; CO_2 = carbon dioxide; N_2O = nitrous oxide; CH_4 = methane; IPCC = Intergovernmental Panel on Climate Change; non-CO_2 = non-carbon dioxide greenhouse gases.

Sources: Authors' own elaboration based on IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Volume 4 – Agriculture, Forestry and Other Land Use. Geneva, Switzerland. https://www.ipcc-nggip.iges.or.jp/public/2019rf/vol4.html; IPCC. 2006. *2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Volume 4 – Agriculture, Forestry and Other Land Use. Kanagawa, Japan. https://www.ipcc-nggip.iges.or.jp/public/2006gl.

Soil organic carbon stock changes

Emissions from biomass pools are set to zero assuming that there are no net gains or losses of living biomass in flooded rice systems.

The total change in soil organic carbon stock for flooded rice systems is estimated using Equation 2.25 of IPCC (2019). The soil organic carbon stock changes for mineral soils of flooded rice systems are computed in EX-ACT from the default reference

and default stock change factors (F_{LU}) following Equation 15 as described in Section 2.2.3. Users can change the default soil reference and stock change factors using the corresponding Tier 2 section. For paddy rice, tillage and input factors are not used. Therefore, Equation 2.25 of the IPCC (2019) is simplified accordingly as in Equation 26:

$$SOC_{mineral} = SOC_{REF} \times F_{LU} \times A$$

(Equation 26)

where:

- $SOC_{mineral}$ = total mineral SOC at a defined time, in tC;
- SOC_{REF} = SOC in the reference condition, in tC/ha;
- F_{LU} = stock change factor for mineral SOC land-use system or sub-systems, dimensionless; and
- A = land area under the different management, in ha.

Emissions from N mineralization

Together with changes and oxidation of SOC storage, land-use management is also accompanied by mineralization of N, which is an additional source of N available for N_2O emissions. Where soil carbon losses occur, the release of N by mineralization is estimated according to the general methodology described in Section 2.3.2, Equations 18 and 19 in this document.

Methane emissions from rice cultivation

The methodology used for estimating CH_4 emissions from rice cultivation follows Equation 5.1 of IPCC (2019) as described in (Equation 27):

$$CH_{4\,Rice} = \sum_{i,j,k} (EF_{i,j,k} \times t_{i,j,k} \times A_{i,j,k} \times 10^{-6})$$

(Equation 27)[24]

where:

- $CH_{4\,Rice}$ = annual CH_4 emissions from rice cultivation, in $GgCH_4/yr$ (subsequently converted to tCO_2eq/yr);
- $EF_{i,j,k}$ = a daily emission factor for i, j, and k conditions, in $kgCH_4/ha/day$;
- $t_{i,j,k}$ = cultivation period of rice for i, j, and k conditions, in days;
- $A_{i,j,k}$ = harvested area of rice for i, j, and k conditions, in ha/yr;
- i, j, and k represent different ecosystems, water regimes, type and amount of organic amendments, and other conditions under which CH_4 emissions from rice may vary.

Under Tier 1, the adjusted daily emission factor for a particular harvested area (EF_i) is estimated as in Equation 28, where scaling factors are used to adjust the baseline emission factor:

$$EF_i = EF_c \times SF_w \times SF_p \times SF_o$$

(Equation 28)[25]

where:

- EF_i= the adjusted daily emission factor for a particular harvested area, $kgCH_4/ha/day$;
- EF_c = the baseline emission factor for continuously flooded fields without organic amendments, $kgCH_4/ha/day$;
- SF_w= the scaling factor to account for the differences in water regime during the cultivation period, dimensionless;
- SF_p= the scaling factor to account for the differences in water regime in the pre-season before the cultivation period, dimensionless;
- SF_o= the scaling factor that is varied by type and amount of organic amendment applied in the field.

The scaling factor for the water regime during the cultivation period (SF_w) considers the three following situations:

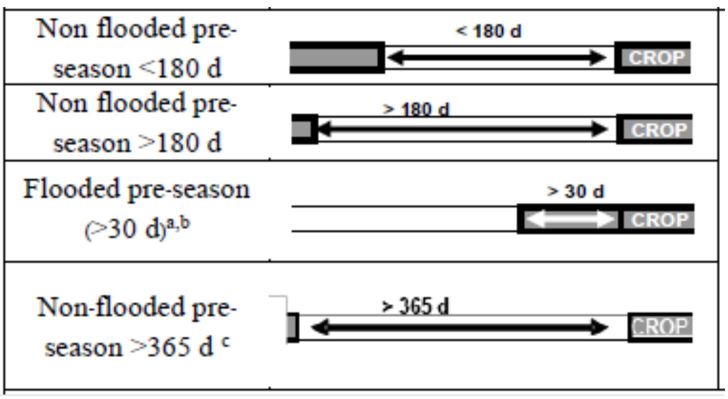
[24] Corresponding to Equation 5.1 in Chapter 5 of IPCC (2019).
[25] Corresponding to Equation 5.2 in Chapter 5 of IPCC (2019).

- **Irrigated - continuously flooded:** fields have standing water throughout the rice growing season and may only dry out for harvest (end-season drainage);
- **Irrigated - intermittently flooded:** fields have at least one aeration period of more than three days during the cropping period. EX-ACT distinguishes between single or multiple aeration;
  - Single aeration/drainage**:** one mid-season drainage and an end-season drainage are adopted over the entire rice growing season.
  - Multiple aeration/drainage: multiple drainage refers to the management water regime, also called 'intermittent irrigation', in which the number of drainage events was not clear, but there are more than one events during the growing season.
- **Rainfed and deep water:** fields are flooded for a significant period of time and water regime depends solely on precipitation. It includes the following subcases:
  - Rainfed, wet season/regular rainfed: rice cultivation that relies on rainfall for water, in this case the field is flood prone during the rice-growing season (the water level may rise up to 50 cm during the cropping season).
  - Rainfed, dry season/drought prone**:** rice cultivation that relies on rainfall for water, in this case the field is drought prone during the rice-growing season (drought periods occur during every cropping season).
  - Rainfed, deep water**:** rice grown in flooded conditions with water depth more than 50 cm deep (floodwater rises to more than 50 cm for a significant period of time during the cropping season).

The scaling factor for the water regime before the cultivation period (SF_p) considers the following situations (see Figure 5):

- Non-flooded pre-season <180 days**:** Rice is planted more than once a year, but there is more than one month of fallow time between the two seasons, 'non-flooded pre-season' e.g. which often occurs under double cropping of rice.
- Non-flooded pre-season >180 days**:** If rice is planted once a year and the field is not flooded in the non-rice growing season e.g. single rice crop following a dry fallow period.
- Flooded pre-season >30 days: Flooded pre-season in which the minimum flooding interval is set to 30 days, i.e. shorter flooding periods (usually done to prepare the soil for ploughing) will not be included in this category.

Figure 5. Description of the different water regime before the cultivation period



Source: Table 5.13 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Volume 4 – Agriculture, Forestry and Other Land Use. Geneva, Switzerland. https://www.ipcc-nggip.iges.or.jp/public/2019rf/vol4.html.

The scaling factor for organic amendments (SF_o, i.e. compost, farmyard manure, green manure and rice straw) is estimated using Equation 29:

$$SF_o = (1 + \sum_i ROA_i \times CFOA_i)^{0.59}$$

(Equation 29)[26]

where:

- SF_o = the scaling factor for both type and organic amendment applied;
- ROA_i = the application rate of organic amendment (i) in dry weight for straw(t d.m./ha); in fresh weight for others (t/ha).[27]
- $CFOA_i$ = the conversion factor for organic amendment (i).

The amount of rice straw (ROA_i), in t d.m./ha, is based on the Equation 11.6 of IPCC (2019) and crop yield data obtained from FAOSTAT (FAO, 2022a) and estimated following the methodology provided in Table 11.2 of IPCC (2019) to estimate the above-ground residue dry matter, in t d.m./ha, illustrated in Equation 22 in Section 2.3.3.

Five default variables are proposed for the organic amendments (taken from Table 5.A.2.1 from IPCC (2019))[28]:

- **Straw incorporated shortly (<30 days before cultivation):** Straw applied just before rice transplanting as on season; straw that is left on the soil surface in the fallow season and incorporated into the soil before the next rice transplanting.
- **Straw incorporated long (>30 days before cultivation):** Straw incorporated into soils in the previous season (upland crop or fallow).
- **Compost:** Organic matter that has been decomposed and recycled as a fertilizer and soil amendment.
- **Farmyard manure:** A mixture of animal dung and urine with straw and litter used as bedding material, decomposed to varying degrees.
- **Green manure:** N-fixing biomass incorporated into the soil, such as Azolla, which contributes organic carbon and biologically fixed nitrogen.

Straw application refers to the straw that is incorporated into the soil; it does not include the case in which straw is placed on soil surface (as this option is not considered in IPCC [2019] nor IPCC [2006]).

Emissions from residue burning

Non-CO_2 emissions from residue burning based on the generic methodology summarized in Section 2.3.1. The associated calculations will be triggered if the residue management option in the Flooded rice module is set to Burned.

The mass of fuel available for combustion (the term M_B in Equation 17) in the flooded rice module by default corresponds to residue amounts calculated according to the above methodology. The combustion factor (the term C_f in Equation 17) corresponds to the proportion of pre-fire fuel biomass consumed by fire. The emission factor (the term G_{ef} in Equation 17) gives the amount of a particular GHG emitted per unit of crop residue dry matter combusted (g/kg d.m. burnt). N_2O and CH_4 emissions are converted into CO_2eq using the respective GWP (see Section 2.1.2).

Table 13 details the default values adopted in the flooded rice module, along with the IPCC table reference and in-text table reference.

[26] Corresponding to Equation 5.3 in Chapter 5 of IPCC (2019).
[27] By default, at Tier 1 EX-ACT uses rice straw amounts as proxy for all organic amendment types, due to the lack of data on average organic amendment application rates. In the online tool, Tier 2 tab allows users to refine these values.
[28] Given the lack of data for compost, farmyard manure and green manure, the tool currently uses the same ROA value as rice straw, with a default value CFOA from IPCC (2019) Table 5.14

Table 13. Sources of default values in flooded rice module of the Environmental eXternalities ACcounting Tool

Element	Unit	Details	Table	Table IPCC	Notes
Biomass carbon stock changes from LUC (Equation 14) $\Delta \mathbf{C\ Conversion} = \sum \{(\mathbf{B}_{\mathbf{final}} - \mathbf{B}_{\mathbf{initial}}) * \Delta\, \mathbf{A}_{\mathbf{TO-OTHERS}}\} * \mathbf{CF}$					
Biomass under flooded rice final land use, for systems after deforestation and other land-use change					
Biomass carbon stock under final land use (B_{final})	t d.m./ha	For systems implanted after deforestation and other land-use change. Values in the year following the conversion.	A17	5.9 of IPCC (2019)	-
Biomass carbon stock under initial land use ($B_{Initial}$)	t d.m./ha	See section of initial land-use type	N/A	-	-
Carbon Fraction (CF)	Dimensionless	Default value: 0.47	N/A	-	-
Soil organic carbon stock changes (Equation 26) (also if LUC) $\mathbf{SOC}_{\mathbf{mineral}} = \mathbf{SOC}_{\mathbf{REF}} \times \mathbf{F}_{\mathbf{LU}} \times \mathbf{A}$					
Soil organic carbon (SOC_{REF})	tC/ha	References for SOC stocks for mineral soils to a depth of 30 cm, differentiated by climate and soil type	A2	2.3 of IPCC (2019)	Modifiable in Tier 2
Land-use factor (F_{LU})	Dimensionless	Default value for paddy rice: 1.35	A30	5.5 of IPCC (2019)	Modifiable in Tier 2
CH_4 emissions from rice cultivation (Equations 23, 27 –29) $\mathbf{CH}_{\mathbf{4\ Rice}} = \sum_{\mathbf{i,j,k}} (\mathbf{EF}_{\mathbf{i,j,k}} \times \mathbf{t}_{\mathbf{i,j,k}} \times \mathbf{A}_{\mathbf{i,j,k}} \times \mathbf{10}^{-6})$ where $\mathbf{EF}_{\mathbf{i}} = \mathbf{EF}_{\mathbf{c}} \times \mathbf{SF}_{\mathbf{w}} \times \mathbf{SF}_{\mathbf{p}} \times \mathbf{SF}_{\mathbf{o}}$ and where $\mathbf{SF}_{\mathbf{o}} = (\mathbf{1} + \sum_{\mathbf{i}} \mathbf{ROA}_{\mathbf{i}} \times \mathbf{CFOA}_{\mathbf{i}})^{\mathbf{0.59}}$ and where ROAi is geven by AGDM(T): $\mathbf{AG}_{\mathbf{DM(T)}} = \mathbf{Crop}_{\mathbf{(T)}} \times \mathbf{Slope}_{\mathbf{(T)}} + \mathbf{Intercept}_{\mathbf{(T)}}$					
Cultivation period (t)	Number of days	Disaggregated by region.	A50	5.11A of IPCC (2019)	Modifiable in Tier 2
Emission factor (EF_i)	kg CH_4/ha/day	Adjusted daily emission factor.	A51 – A53	5.11 – 5.14 of IPCC (2019)	Modifiable in Tier 2
Baseline emission factor (EF_c)	kg CH_4/ha/day	Baseline emission factor for continuously flooded fields without organic amendments; according to regions.	A51	5.11 of IPCC (2019)	Modifiable in Tier 2
Scaling factor for Water regime during the cultivation period (SFw)	Dimensionless	N/A	A52	5.12 of IPCC (2019)	Modifiable in Tier 2
Scaling factor for Water regime	Dimensionless	N/A	A53	5.13 of IPCC (2019)	Modifiable in Tier 2

Element	Unit	Details	Table	Table IPCC	Notes
before the cultivation period (SFp)					
Scaling factor for Organic amendment (SF_o)	Dimensionless	Calculated in application using Equation 29	N/A	N/A	Modifiable in Tier 2
Application rate of organic amendment (ROA_i)	t d.m./ha or t/ha	For rice straw: dry weight. Available straw calculated according to $AG_{DM(T)}$ in the residue retention section of this table. Other organic amendments: fresh weight (t/ha). In Tier 1, rice straw is used as proxy. Users can specify Tier 2 depending on data availability	N/A	11.2 of IPCC (2019)	-
Conversion factor for organic amendment (CFOA)	Dimensionless	Relative effect of organic amendments respective to straw incorporated shortly before cultivation	A54	5.14 of IPCC (2019)	-
Harvested annual dry matter yield for crop T ($Crop_{(T)}$)	kg d.m./ha	Default yield values from FAOSTAT or optional user input	N/A	FAOSTAT values are continuously updated.	-
$Slope_{(T)}$	Dimensionless	Default value for rice: 0.95	N/A	11.2 of IPCC (2019)	Used if no rice straw availability value is given in Tier 2.
$Intercept_{(T)}$	Dimensionless	Default value for rice: 2.46	N/A	11.2 of IPCC (2019)	Used if no rice straw availability value is given in Tier 2.
N_2O and CH_4 emissions from residue burning (Equation 17) $\mathbf{GHG_{fire} = A * M_B * C_f * G_{ef} * 10^{-3}}$					
Biomass available for burning (M_B)	kg d.m./ha	Available rice straw residues correspond to $AG_{DM(T)}$ computed in the CH_4 emissions table section above	N/A	11.2 of IPCC (2019)	Modifiable in Tier 2
Combustion factor (C_F)	Dimensionless	Default value for rice residues: 0.8	N/A	2.6 of IPCC (2019)	-
Emission factor (G_{ef})	kg/t d.m. burnt	set to 0.07 kgN_2O/t d.m. for N_2O and 2.7 kg CH_4/t d.m. for CH_4	N/A	2.5 of IPCC (2019)	-
N_2O emissions from N mineralization (Equations 18 and 19) (if LUC or if change in management practices)					

Element	Unit	Details	Table	Table IPCC	Notes
$N_2O - N_{SOM} = F_{SOM} \times EF_1$ where $F_{SOM} = \sum(\frac{\Delta C_{mineral}}{R} \times 1000)$					
C:N ratio of the soil organic matter (R)	Dimensionless	Default values: 10 for cropland remaining cropland. 15 for land-use changes from Forest land or Grassland to Cropland (annual and perennial).	N/A	11.1 of IPCC (2019)	-
Average annual loss of soil carbon ($\Delta C_{mineral}$)	tC/yr	-	N/A	-	-
Net annual amount of N mineralized in mineral soils (F_{SOM})	kg N/yr	Calculated in the tool as a result of loss of soil carbon through change in land use or management	N/A	-	-
Emission factor for N_2O–N from N mineralised (EF_1)	kg N_2O–N/kg N	0.006 kg N_2O–N/kg N for all climates wet and moist. 0.005 kg N_2O–N/kg N for dry climates. 0.010 kg N_2O–N/kg N for an aggregated climate	N/A	11.1 of IPCC (2019)	-

Note: GHG = greenhouse gas; CO_2 = carbon dioxide; CH_4 = methane; N_2O = nitrous oxide; N_2O–N = nitrous oxide expressed as nitrogen; SOC = soil organic carbon; SOC_REF = reference soil organic carbon stock; LUC = land-use change; EF = emission factor; EFc = baseline emission factor; EF_1 = emission factor for N_2O–N from N mineralization; ROA = rate of organic amendment; CF = carbon fraction; IPCC = Intergovernmental Panel on Climate Change; N/A = not applicable.

Source: Authors' own elaboration based on IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Volume 4 – Agriculture, Forestry and Other Land Use. Geneva, Switzerland. https://www.ipcc-nggip.iges.or.jp/public/2019rf/vol4.html

2.4.6 GRASSLAND MANAGEMENT

Material used to develop this sub-module can be found in Chapter 2 (Generic methodology applicable to multiple land-use categories) and Chapter 6 (Grassland) in IPCC (2019).

The grassland module addresses changes in grassland management activities that are known to influence carbon input to the soil and thus SOC stocks. The management activities on grasslands that improve SOC stocks include irrigation, fertilization, liming, organic amendments, more productive grass varieties (labelled as improved), medium or high inputs. Prolonged heavy grazing or biomass removal will lead to degradation of grassland and hence to SOC stock losses. The module covers the following computations:

1. **Changes in soil organic carbon stocks** linked to land-use changes and changes in management practices:
    - CO_2 emissions and removals associated with soil organic carbon stock changes are calculated according to the methodology illustrated in Section 2.2.3.
2. **Changes in biomass carbon stocks** linked to land-use changes
    - CO_2 fluxes associated with biomass gains following a land-use change are calculated according to the methodology illustrated in Section 2.2.1, using the stock-difference method.
3. **Emissions from fire on grasslands**:

- CH_4 and N_2O from periodic grassland burning are calculated according to the method illustrated in Section 2.3.1.

4. **Emissions from nitrogen mineralization** associated with loss of soil organic matter:
   - N_2O emissions associated with soil organic carbon stock changes are calculated according to the methodology illustrated in Section 2.3.2.

Table 14 illustrates the GHGs addressed by the module, with the corresponding level of assessment, i.e. Tier 1 or Tier 2 and includes in-text references to the equations used for the GHG fluxes calculations, as well as the corresponding IPCC equations for easier cross-reference.

Table 14. Overview of greenhouse gas fluxes estimated in the grassland module of the Environmental eXternalities ACcounting Tool and corresponding equations

Activity		GHG	Flux description	Equation	IPCC equation
Grassland management	Tier 1	CO_2	Soil organic carbon stock changes linked to change in practices (tillage and nutrient management)	15	2.25 of IPCC (2019)
	Tier 1/2	CO_2	Biomass carbon stock gains following land-use changes	14	2.16 of IPCC (2006)
	Tier 1	CH_4 and N_2O	Biomass (grassland) burning as management practice	17	2.27 of IPCC (2019)
	Tier 1	N_2O	N mineralization linked to change in management practice	18 and 19	11.2 and 11.8 of IPCC (2019)

Note: GHG = greenhouse gas; CO_2 = carbon dioxide; N = nitrogen; N_2O = nitrous oxide; IPCC = Intergovernmental Panel on Climate Change.

Source: Authors' own elaboration based on IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Volume 4 – Agriculture, Forestry and Other Land Use. Geneva, Switzerland. https://www.ipcc-nggip.iges.or.jp/public/2019rf/vol4.html; IPCC. 2006. *2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Volume 4 – Agriculture, Forestry and Other Land Use. Kanagawa, Japan. https://www.ipcc-nggip.iges.or.jp/public/2006gl

Carbon stock changes in biomass and soil organic carbon

Biomass carbon stocks in grassland are influenced by human activities including harvesting, grazing, fires, soil and vegetation rehabilitation. EX-ACT aligns with the IPCC default assumption that biomass is at steady state in grasslands remaining grasslands (IPCC, 2019). At Tier 1, EX-ACT only accounts for changes in biomass carbon stock linked to land-use change.

The methodology used for calculating the annual rates of SOC stock changes in mineral soils is based on the difference of SOC stocks between with and without the project scenario as described in Section 2.2.3. EX-ACT estimates the increase/decrease in SOC_{ref} following changes in management practices on a given land or following a land-use change and associated management practices, if any, on the converted area. The SOC difference between the with project scenario (SOC_{With}) and without project scenario ($SOC_{W/out}$) indicates changes due to grassland management in the topsoil (0–30 cm, see Table A2) There are five grassland management states provided by the IPCC, each corresponding to a specific level of SOC (from Table 6.2 in IPCC [2019]):

- **Nominally managed (non-degraded)**: Represents low or medium intensity grazing regimes, in addition to periodic cutting and removal of above-ground vegetation, without significant management improvements.
- **High intensity grazing**: Represents high intensity grazing systems (or cutting and removal of vegetation) with shifts in vegetation composition and possibly productivity but is not severely degraded.
- **Severely degraded**: Implies major long-term loss of productivity and vegetation cover, due to severe mechanical damage to the vegetation and/or severe soil erosion.

- **Improved, with medium inputs**: Applies to improved grassland where no additional management inputs have been used.
- **Improved, with high inputs**: Applies to improved grassland where one or more additional management inputs/improvements have been used (beyond that required to be classified as improved grassland).

Emissions from periodic biomass burning

Grasslands may be subject to fire as a result of management practices. As for cropland, CO_2 emissions from biomass burning in grassland remaining grassland are not reported since they are largely balanced by the CO_2 reincorporated back into the biomass via photosynthetic activity within the year. Non-CO_2 emissions should however be reported. EX-ACT estimates non-CO_2 emissions from crop residue burning based on the generic methodology summarized in Section 2.3.1. The mass of fuel available for combustion (the term M_B in Equation 17) in the grassland module is equal to the AGB of the grassland in the specific geographical context of analysis. Table 15 details the default values for the parameters in Equation 17 embedded in EX-ACT, along with the IPCC table reference and in-text table reference.

Emissions from nitrogen mineralization

Together with changes and oxidation of SOC storage, land-use management is also accompanied by mineralization of N, which is an additional source of N available for N_2O emissions. Where soil carbon losses occur, the release of N by mineralization is estimated according to the general methodology described in Section 2.3.2, Equations 18 and 19 in this document.

Table 15 details the default values adopted in the grassland module, along with the IPCC table reference and in-text table reference.

Table 15. Sources of default values in grassland module of the Environmental eXternalities ACcounting Tool

Element	Unit	Details	Table	Table IPCC	Notes
Biomass carbon stock changes (Equation 14) $\Delta \mathbf{C\ Conversion} = \sum \{(\mathbf{B_{final}} - \mathbf{B_{initial}}) * \Delta\, \mathbf{A_{TO-OTHERS}}\} * \mathbf{CF}$					
Biomass under (non-forest) final land use, for systems after deforestation and other land-use change					
Biomass carbon stock under final land use (B_{final})	t d.m./ha	For systems implanted after deforestation and other land-use changes. Values in the year following the conversion.	A17	6.4 of IPCC (2006)	-
Biomass carbon stock under initial land use ($B_{initial}$)	t d.m./ha	See section of initial land-use type	N/A	-	-
Carbon Fraction (CF)	Dimensionless	Default value: 0.47	N/A	-	-
Soil organic carbon stock changes (Equation 15) $\mathbf{SOC} = \sum (\mathbf{SOC_{REF}} * \mathbf{F_{LU}} * \mathbf{F_{MG}} * \mathbf{F_I} * \mathbf{A})$					
Soil organic carbon (SOC_{REF})	tC/ha	References for SOC stocks for mineral soils to a depth of 30 cm over a period of 20 years.	A2	2.3 of IPCC (2019)	Modifiable in Tier 2
Land-use factor (F_{LU})	Dimensionless	For all grassland states; by climate regime.	A56	6.2 of IPCC (2019)	Modifiable in Tier 2

Element	Unit	Details	Table	Table IPCC	Notes
Management factor (F_{MG})	Dimensionless	For non-degraded, high intensity grazing, severely degraded, and Improved grassland states; by climate regime	A56	6.2 of IPCC (2019)	Modifiable in Tier 2
Input factor (F_I)	Dimensionless	For high and medium inputs, by climate regime	A56	6.2 of IPCC (2019)	Modifiable in Tier 2
N_2O and CH_4 emissions from biomass burning (Equation 17) $\mathbf{GHG_{fire} = A * M_B * C_f * G_{ef} * 10^{-3}}$					
Biomass available for burning (M_B)	kg d.m./ha	Corresponds to above-ground biomass for grassland.	A56	6.4 of IPCC (2006)	Modifiable in Tier 2
Combustion factor (C_F)	Dimensionless	Proportion of pre-fire fuel biomass consumed Default value: 0.77	A29	2.6 of IPCC (2019)	Modifiable in Tier 2
Emission factor (G_{ef})	g/kg d.m. burnt	Default values for grassland and savannah, for CH_4 and N_2O; respectively 2.3 and 0.21 g/kg d.m. burnt.	A29	2.5 of IPCC (2019)	-
N_2O emissions from N mineralization (Equations 18 and 19) (if LUC or change in management practice) $\mathbf{N_2O - N_{SOM} = F_{SOM} \times EF}$ where $\mathbf{F_{SOM}} = \sum\left(\frac{\mathbf{\Delta C_{mineral}}}{\boldsymbol{R}} \times \mathbf{1000}\right)$					
C:N ratio of the soil organic matter (R)	Dimensionless	10 for cropland remaining cropland 15 for land-use changes from Forest land or Grassland to Cropland (annual and perennial)	N/A	11.1 of IPCC (2019)	-
Average annual loss of soil carbon ($\Delta C_{mineral}$)	tC/yr	-	N/A	-	-
Net annual amount of N mineralized in mineral soils (F_{SOM})	kg N/yr	As a result of loss of soil carbon through change in land use or management	N/A	-	-
Emission factor for N_2O–N from N mineralised (EF)	kg N_2O–N/kg N	0.006 kg N_2O–N/kg N for all climates wet and moist 0.005 kg N_2O–N/kg N for dry climates 0.010 kg N_2O–N/kg N for an aggregated climate	N/A	11.1 of IPCC (2019)	-

Note: AGB = above-ground biomass; BGB = below-ground biomass; gCH_4/kg d.m. burnt = grams of methane per kilogram of dry matter burnt; gN_2O/kg d.m. burnt = grams of nitrous oxide per kilogram of dry matter burnt; IPCC = Intergovernmental Panel on Climate Change; LUC = land-use change; N/A = not applicable; non-CO_2 = non-carbon dioxide greenhouse gases; SOC = soil organic carbon stock change; tC/ha = tonnes of carbon per hectare; tC/ha/yr = tonnes of carbon per hectare per year; t d.m./ha = tonnes of dry matter per hectare; t d.m./ha/yr = tonnes of dry matter per hectare per year.

Sources: Authors' own elaboration based on IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Volume 4 – Agriculture, Forestry and Other Land Use. Geneva, Switzerland. https://www.ipcc-nggip.iges.or.jp/public/2019rf/vol4.html; IPCC. 2006. *2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Volume 4 – Agriculture, Forestry and Other Land Use. Kanagawa, Japan. https://www.ipcc-nggip.iges.or.jp/public/2006gl

2.4.7 SET ASIDE LAND

Material used to develop this sub-module can be found in Chapter 5 (Cropland) Chapter 11 (N_2O emissions from managed soils, and CO_2 emissions from lime and urea application) and Chapter 2 (Generic methodology applicable to multiple land-use categories) in IPCC (2019).

The set aside module represents temporary set-aside of annual cropland (e.g. conservation reserves) or other idle cropland that has been revegetated with perennial grasses. The module covers the following computations:

1. **Changes in biomass carbon stocks** linked to land-use changes:
   - CO_2 fluxes associated with biomass gains following a land-use change are calculated according to the methodology illustrated in Section 2.2, using the stock-difference method.
2. **Changes in soil organic carbon stocks** linked to land-use changes and changes in management practices (the latter through Tier 2 only):
   - CO_2 emissions and removals associated with soil organic carbon stock changes are calculated according to the methodology illustrated in Section 2.2.3.
3. **Emissions from nitrogen mineralization** associated with loss of soil organic matter:
   - N_2O emissions associated with soil organic carbon stock changes are calculated according to the methodology illustrated in Section 2.3.2.

Table 16 illustrates the GHGs addressed by the module, with the corresponding level of assessment, i.e. Tier 1 or Tier 2 and includes in-text references to the equations used for the GHG fluxes calculations, as well as the corresponding IPCC equations for easier cross-reference.

Table 16. Overview of greenhouse gas fluxes estimated in the set aside module of the Environmental eXternalities ACcounting Tool and corresponding equations

Activity		GHG	Flux description	Equation	IPCC equation
Set aside land management	Tier 1, Tier 2	CO_2	Soil organic carbon stock changes linked to land-use changes and/or changes in practices	15	2.25 of IPCC (2019)
	Tier 1/2	CO_2	Biomass carbon stock gains following land-use changes	14	2.16 of IPCC (2006)
	Tier 1	N_2O	Nitrogen mineralization linked to change in management practice	18 and 19	11.2 and 11.of IPCC (2019)

Note: GHG = greenhouse gas; CO_2 = carbon dioxide; N_2O = nitrous oxide; CH_4 = methane; IPCC = Intergovernmental Panel on Climate Change.

Sources: Authors' own elaboration based on IPCC. 2019. 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories. Volume 4 – Agriculture, Forestry and Other Land Use. Geneva, Switzerland. *https://www.ipcc-nggip.iges.or.jp/public/2019rf/vol4.html*; IPCC. 2006. 2006 IPCC Guidelines for National Greenhouse Gas Inventories. Volume 4 – Agriculture, Forestry and Other Land Use. Kanagawa, Japan. *https://www.ipcc-nggip.iges.or.jp/public/2006gl*

Carbon stock changes in biomass and soil organic carbon

EX-ACT adopts the default assumption that biomass is at steady state in set aside land remaining under the same land use. At Tier 1, EX-ACT only accounts for changes in biomass carbon stock linked to land-use change.

The methodology used for calculating the annual rates of SOC stock changes in mineral soils is based on the difference of SOC stocks between with and without the project scenario as described in Section 2.2.3. EX-ACT estimates the increase/decrease in SOC_{ref} following changes in management practices on a given land and following a land-use change and associated management practices, if any, on the converted area.

Emissions from nitrogen mineralization

In case of carbon losses linked to a change in land-use or management practices, the release of N by mineralization is estimated according to the general methodology described in Section 2.3.2, and Equations 18 and 19 in this document.

Table 17 details the default values adopted in the set aside module, along with the IPCC table reference and in-text table reference.

Table 17. Sources of default values in set aside module of the Environmental eXternalities ACcounting Tool

Element	Unit	Details	Table	Table IPCC	Notes
Biomass carbon stock changes (Equation 14) $\Delta C\ Conversion = \sum\{(B_{final} - B_{initial}) * \Delta A_{TO-OTHERS}\} * CF$					
Biomass under (non-forest) final land use, for systems after deforestation and other land-use change					
Biomass carbon stock under final land use (B_{final})	t d.m./ha	For systems implanted after deforestation and other land-use changes. Values in the year following the conversion.	A17	5.9 of IPCC (2019)	-
Biomass carbon stock under initial land use ($B_{Initial}$)	t d.m./ha	See section of initial land-use type	N/A	-	-
Carbon Fraction (CF)	Dimensionless	Default value: 0.47	N/A	-	-
Soil organic carbon stock changes (Equation 15) $SOC = \sum(SOC_{REF} * F_{LU} * F_{MG} * F_I * A)$					
Soil organic carbon (SOC_{REF})	tC/ha	References for SOC stocks for mineral soils to a depth of 30 cm	A2	2.3 of IPCC (2019)	Modifiable in Tier 2
Land-use factor (F_{LU})	Dimensionless	Default value for a tropical montane climate: 0.88 Default value for other climate types: Dry; 0.93, and moist/wet: 0.82.	A30	5.5 of IPCC (2019)	Modifiable in Tier 2
Management factor (F_{MG})	Dimensionless	Set to 1 by default	N/A	-	Modifiable in Tier 2
Input factor (F_I)	Dimensionless	Set to 1 by default	N/A	-	Modifiable in Tier 2
N_2O emissions from N mineralization (Equations 18 and 19) if LUC or change in management practice $N_2O - N_{SOM} = F_{SOM} \times EF$ where $F_{SOM} = \sum(\frac{\Delta C_{mineral}}{R} \times 1000)$					
C:N ratio of the soil organic matter (R)	Dimensionless	Default values: 10 for cropland remaining cropland. 15 for land-use changes from forest land or Grassland to Cropland (annual or perennial).	N/A	11.1of IPCC (2019)	-

Element	Unit	Details	Table	Table IPCC	Notes
Average annual loss of soil carbon ($\Delta C_{mineral}$)	tC/yr	-	N/A	-	-
Net annual amount of N mineralized in mineral soils (F_{SOM})	kg N/yr	As a result of loss of soil carbon through change in land use or management	N/A	-	-
Emission factor for N_2O–N from N mineralised (EF)	kg N_2O–N/kg N	0.006 kg N_2O–N/kg N for all climates wet and moist 0.005 kg N_2O–N/kg N for dry climates 0.010 kg N_2O–N/kg N for an aggregated climate	N/A	11.1of IPCC (2019)	-

Note: AGB = above-ground biomass; BGB = below-ground biomass; IPCC = Intergovernmental Panel on Climate Change; LUC = land-use change; N/A = not applicable; SOC = soil organic carbon stock change; tC/yr = tonnes of carbon per year.

Source: Authors' own elaboration based on IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Volume 4 – Agriculture, Forestry and Other Land Use. Geneva, Switzerland. https://www.ipcc-nggip.iges.or.jp/public/2019rf/vol4.html

2.4.8 OTHER LAND

Material used to develop this sub-module can be found in Chapter 9 (Other land) Chapter 11 (N_2O emissions from managed soils, and CO_2 emissions from lime and urea application) and Chapter 2 (Generic methodology applicable to multiple land-use categories) in IPCC (2006, 2019).

The other land module represents includes bare soil, rock, ice, and all land areas that do not fall into any of the other five IPCC categories. The module covers the following computations:

1. **Changes in biomass carbon stocks** linked to land-use changes
   - CO_2 fluxes associated with biomass gains following a land-use change are calculated according to the methodology illustrated in Section 2.2.2, using the stock-difference method.
2. **Changes in soil organic carbon stocks** linked to land-use changes and changes in management practices (the latter through Tier 2 only):
   - CO_2 emissions and removals associated with soil organic carbon stock changes are calculated according to the methodology illustrated in Section 2.2.3**.**
3. **Emissions from nitrogen mineralization** associated with loss of soil organic matter:
   - N_2O emissions associated with soil organic carbon stock changes are calculated according to the methodology illustrated in Section 2.3.2.

Table 18 illustrates the GHGs addressed by the module, with the corresponding level of assessment, i.e. Tier 1 or Tier 2 and includes in-text references to the equations used for the GHG fluxes calculations, as well as the corresponding IPCC equations for easier cross-reference.

Table 18. Overview of greenhouse gas fluxes estimated in the other land module of the Environmental eXternalities ACcounting Tool and corresponding equations

Activity		GHG	Flux description	Equation	IPCC equation
Other land management	Tier 1, Tier 2	CO_2	Soil organic carbon stock changes linked to land-use changes and/or changes in practices	15	2.25 of IPCC (2019)
	Tier 1/2	CO_2	Biomass carbon stock gains following land-use changes	14	2.16 of IPCC (2006)
	Tier 1	N_2O	N mineralization linked to change in management practice	18 and 19	11.2 and 11.8 of IPCC (2019)

Note: GHG = greenhouse gas; CO_2 = carbon dioxide; N_2O = nitrous oxide; IPCC = Intergovernmental Panel on Climate Change.

Sources: Authors' own elaboration based on IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Volume 4 – Agriculture, Forestry and Other Land Use. Geneva, Switzerland. https://www.ipcc-nggip.iges.or.jp/public/2019rf/vol4.html; IPCC. 2006. *2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Volume 4 – Agriculture, Forestry and Other Land Use. Kanagawa, Japan. https://www.ipcc-nggip.iges.or.jp/public/2006gl

Carbon stock changes in biomass and soil organic carbon

The Environmental eXternalities ACcounting Tool adopts the default assumption that biomass is at steady state in other land remaining under the same land use. At Tier 1, EX-ACT only accounts for changes in biomass carbon stock linked to land-use change.

The methodology used for calculating the annual rates of SOC stock changes in mineral soils is based on the difference of SOC stocks between with and without the project scenario as described in Section 2.2.3. EX-ACT estimates the increase/decrease in SOC_{ref} following changes in management practices on a given land and following a land-use change and associated management practices, if any, on the converted area.

Emissions from nitrogen mineralization

In case of carbon losses linked to a change in land-use or management practices, the release of N by mineralization is estimated according to the general methodology described in Section 2.3.2 and Equations 18 and 19 in this document.

Table 19 details the default values adopted in the other land module, along with the IPCC table reference and in-text table reference.

Table 19. Sources of default values in other land module of the Environmental eXternalities ACcounting Tool

Element	Unit	Details	Table	Table IPCC	Notes
Biomass carbon stock changes (Equation 14) $$\Delta C\ Conversion = \sum\{(B_{final} - B_{initial}) * \Delta A_{TO-OTHERS}\} * CF$$					
Biomass under (non-forest) final land use, for systems after deforestation and other land-use change					
Biomass carbon stock under final land use (B_{final})	t d.m./ha	For systems implanted after deforestation. Values in the year following the conversion. Default value: 0. This category should be used where no vegetation is present (chapter 9.3.1, IPCC (2019), Or for land that does not fall under any of the other IPCC land-use categories in chapter 3.2 of IPCC (2019).	A17	-	-
Biomass carbon stock under initial land use ($B_{Initial}$)	t d.m./ha	See section of initial land-use type	N/A	-	-
Carbon Fraction (CF)	Dimensionless	Default value: 0.47	N/A	-	-
Soil organic carbon stock changes (Equation 15) $$SOC = \sum(SOC_{REF} * F_{LU} * F_{MG} * F_I * A)$$					
Soil organic carbon (SOC_{REF})	tC/ha	References for SOC stocks for mineral soils to a depth of 30 cm	A2	2.3 of IPCC (2019)	Modifiable in Tier 2
Land-use factor (F_{LU})	Dimensionless	Set to 1 by default	A30	5.5 of IPCC (2019)	Modifiable in Tier 2
Management factor (F_{MG})	Dimensionless	Set to 1 by default	N/A	-	Modifiable in Tier 2
Input factor (F_I)	Dimensionless	Set to 1 by default	N/A	-	Modifiable in Tier 2
N2O emissions from N mineralization (Equation 18)(Equation 19) (if LUC or change in management practice) $$N_2O - N_{SOM} = F_{SOM} \times EF \text{ where}$$ $$F_{SOM} = \sum\left(\frac{\Delta C_{mineral}}{R} \times 1000\right)$$					
C:N ratio of the soil organic matter (R)	Dimensionless	Default values: 10 for cropland remaining cropland. 15 for land-use changes from forest land or Grassland to Cropland (annual or perennial).	N/A	Chapter 11 IPCC (2019)	-
Average annual loss of soil carbon ($\Delta C_{mineral}$)	tC/yr	-	N/A	-	-

Element	Unit	Details	Table	Table IPCC	Notes
Net annual amount of N mineralized in mineral soils (F_{SOM})	kg N/yr	As a result of loss of soil carbon through change in land use or management	N/A	-	-
Emission factor for N_2O–N from N mineralised (EF)	kg N_2O–N/kg N	0.006 kg N_2O–N/kg N for all climates wet and moist 0.005 kg N_2O–N/kg N for dry climates 0.010 kg N_2O–N/kg N for an aggregated climate	N/A	11.1 of IPCC (2019)	-

Note: AGB = above-ground biomass; BGB = below-ground biomass; IPCC = Intergovernmental Panel on Climate Change; LUC = land-use change; N/A = not applicable; SOC = soil organic carbon stock change; tC/yr = tonnes of carbon per year.

Source: Authors' own elaboration based on IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Volume 4 – Agriculture, Forestry and Other Land Use. Geneva, Switzerland. https://www.ipcc-nggip.iges.or.jp/public/2019rf/vol4.html

2.4.9 SETTLEMENTS AND INFRASTRUCTURE

Material used to develop this sub-module can be found in Chapter 8 (Settlements), Chapter 11 (N_2O emissions from managed soils, and CO_2 emissions from lime and urea application) and Chapter 2 (Generic methodology applicable to multiple land-use categories) in IPCC (2006, 2019).

The settlements and infrastructure module includes all developed land, including transportation infrastructure and human settlements of any size, unless they are already included under other categories. The module covers the following computations:

1. **Changes in biomass carbon stocks** linked to land-use changes
   - CO_2 fluxes associated with biomass gains following a land-use change are calculated according to the methodology illustrated in Section 2.2.2, using the stock-difference method.
2. **Changes in soil organic carbon stocks** linked to land-use changes and changes in management practices;
   - CO_2 emissions and removals associated with soil organic carbon stock changes are calculated according to the methodology illustrated in Section 2.2.3.
3. **Emissions from nitrogen mineralization** associated with loss of soil organic matter:
   - N_2O emissions associated with soil organic carbon stock changes are calculated according to the methodology illustrated in Section 2.3.2.
4. **Emissions from infrastructure building**, linked to construction and use of materials:
   - CO_2eq emissions associated with the construction of new infrastructure for agricultural or post-harvest use, calculated according to the methodology illustrated in Section 2.1.

Table 20 illustrates the GHGs addressed by the module, with the corresponding level of assessment, i.e. Tier 1 or Tier 2 and includes in-text references to the equations used for the GHG fluxes calculations, as well as the corresponding IPCC equations for easier cross-reference.

Table 20. Overview of greenhouse gas fluxes estimated in the settlements and infrastructure module of the Environmental eXternalities ACcounting Tool and corresponding equations

Activity		GHG	Flux description	Equation	IPCC equation
Settlements and infrastructure management	Tier 1, Tier 2	CO_2	Soil organic carbon stock changes linked to land-use changes and/or changes in settlement type	15	2.25 of IPCC (2019)
	Tier 1/2	CO_2	Biomass carbon stock gains following land-use changes	14	2.16 of IPCC (2006)
	Tier 1	N_2O	N mineralization linked to land-use changes and/or changes in settlement type	18 and 19	11.2 and 11.8 of IPCC (2019)
	Tier 1	CO_2eq	Emissions from buildings and roads construction	16	2.6 of IPCC (2019)

Note: GHG = greenhouse gas; CO_2 = carbon dioxide; CO_2eq = carbon dioxide equivalent; N_2O = nitrous oxide; CH_4 = methane; IPCC = Intergovernmental Panel on Climate Change.

Sources: Authors' own elaboration based on IPCC. 2019. 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories. Volume 4 – Agriculture, Forestry and Other Land Use. Geneva, Switzerland. *https://www.ipcc-nggip.iges.or.jp/public/2019rf/vol4.html*; IPCC. 2006. 2006 IPCC Guidelines for National Greenhouse Gas Inventories. Volume 4 – Agriculture, Forestry and Other Land Use. Kanagawa, Japan. *https://www.ipcc-nggip.iges.or.jp/public/2006gl*

Carbon stock changes in biomass and soil organic carbon

EX-ACT adopts the default assumption that biomass is at steady state in settlements remaining under the same land use. At Tier 1, EX-ACT only accounts for changes in biomass carbon stock linked to land-use change.

The methodology used for calculating the annual rates of SOC stock changes in mineral soils is based on the difference of SOC stocks between with and without the project scenario as described in Section 2.2.3. EX-ACT estimates the increase/decrease in SOC_{ref} following changes in management practices on a given land and following a land-use change and associated management practices, if any, on the converted area.

Emissions from nitrogen mineralization

In case of carbon losses linked to a change in land use or management practices, the release of N by mineralization is estimated according to the general methodology described in Section 2.3.2 and Equations 18 and 19 in this document.

Emissions from infrastructure building

Agrifood commodities often require the construction of infrastructure such as storage facilities, processing units, and market centers to support production and commercialization. EX-ACT accounts for greenhouse gas emissions associated with the **construction of such infrastructure**. Default emission factors for infrastructure building are sourced from ADEME (2020) and provided in Table A148 in annex.

Table 21 details the default values adopted in the Settlements and infrastructure module, along with the IPCC table reference and in-text table reference.

Table 21. Sources of default values in settlements and infrastructure module of the Environmental eXternalities ACcounting Tool

Element	Unit	Details	Table	Table IPCC	Notes
Biomass carbon stock changes (Equation 14) $\Delta C\ Conversion = \sum\{(B_{final} - B_{initial}) * \Delta A_{TO-OTHERS}\} * CF$					
Biomass under (non-forest) final land use, for systems after deforestation and other land-use change					
Biomass carbon stock under final land use (B_{final}*CF)	t d.m./ha	Values in the year following the conversion. Default value for settlements: 0	N/A	-	-
Biomass carbon stock under initial land use ($B_{Initial}$)	t d.m./ha	See section of initial land-use type	N/A	-	-
Carbon Fraction (CF)	Dimensionless	Default value: 0.47	N/A	-	-
Soil organic carbon stock changes (Equation 15) $SOC = \sum(SOC_{REF} * F_{LU} * F_{MG} * F_I * A)$					
Soil organic carbon (SOC_{REF})	tC/ha	References for SOC stocks for mineral soils to a depth of 30 cm.	A2	2.3 of IPCC (2019)	Modifiable in Tier 2
Land-use factor (F_{LU}) **Management factor (F_{MG})** **Input factor (F_I)**	Dimensionless	Depends on settlement system.	A151	5.5 and 6.2 of IPCC (2019)	Modifiable in Tier 2
N_2O emissions from N mineralization (Equations 18 and 19) (if LUC or change in management practice) $N_2O - N_{SOM} = F_{SOM} \times EF$ where $F_{SOM} = \sum(\frac{\Delta C_{mineral}}{R} \times 1000)$					
C:N ratio of the soil organic matter (R)	Dimensionless	Default values: 10 for cropland remaining cropland. 15 for land-use changes from forest land or Grassland to Cropland (annual or perennial).	N/A	Chapter 11 IPCC (2019)	-
Average annual loss of soil carbon ($\Delta C_{mineral}$)	tC/yr	-	N/A	-	-
Net annual amount of N mineralized in mineral soils (F_{SOM})	kg N/yr	As a result of loss of soil carbon through change in land use or management	N/A	-	-

Element	Unit	Details	Table	Table IPCC	Notes
Emission factor for $N_2O–N$ from N mineralised (EF)	kg $N_2O–N$/kg N	0.006 kg $N_2O–N$/kg N for all climates wet and moist 0.005 kg $N_2O–N$/kg N for dry climates 0.010 kg $N_2O–N$/kg N for an aggregated climate	N/A	11.1 of IPCC (2019)	-
Emissions from buildings and roads construction (Equation 16) **Emissions = A * EF**					
Surface area of the construction (A)	m^2	User input.	N/A	-	-
Emissions from the construction and material use (EF)	$kgCO^2eq/m^2$	Depends on the selected building type and construction material.	A148	-	Modifiable in Tier 2

Note: AGB = above-ground biomass; BGB = below-ground biomass; IPCC = Intergovernmental Panel on Climate Change; LUC = land-use change; N/A = not applicable; SOC = soil organic carbon stock change; tC/yr = tonnes of carbon per year; m^2 = meter square; $kgCO^2eq/m^2$ = kilogram of carbon dioxide equivalent per meter square.

Source: Authors' own elaboration based on IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Volume 4 – Agriculture, Forestry and Other Land Use. Geneva, Switzerland. https://www.ipcc-nggip.iges.or.jp/public/2019rf/vol4.html

2.4.10 LIVESTOCK MANAGEMENT

Material used to develop this module can be found in Chapter 10 (Emissions from livestock and manure management) of IPCC (2019). Most of the methodology provided in the livestock chapter of IPCC (2019) is based on the FAO's Global Livestock Environmental Assessment Model (GLEAM) (FAO, 2022b), which embeds country specific emission factors. These emission factors have been aggregated by IPCC into regions, which slightly differ from the grouping of countries used in the other EX-ACT modules. The full regional distribution of countries in the livestock module is provided in Table A1 to inform the users and help refine the emission factors at Tier 2 as may be desired. GHGs covered by the livestock module include:

1. Enteric fermentation:
   - CH_4 emissions linked to enteric fermentation processes according to livestock categories
2. Manure management:
   - CH_4 emissions associated with manure management practices.
   - Direct and indirect N_2O emissions associated with manure management practices.

The CH_4 and N_2O emissions associated with enteric fermentation and manure management are calculated according to the general methodology detailed in the sections below.

Table 22 illustrates the GHGs addressed by the module, with the corresponding level of assessment, i.e. Tier 1 or Tier 2 and includes in-text references to the equations used for the GHG fluxes calculations, as well as the corresponding IPCC equations for easier cross-reference.

Table 22. Overview of greenhouse gas fluxes estimated in the livestock module of the Environmental eXternalities ACcounting Tool and corresponding equations

Activity		GHG	Flux description	Equation	IPCC equation
Livestock management	Tier 1	CH_4	Enteric fermentation	30	10.19 of IPCC (2019)
	Tier 1	CH_4	Manure management	31 and 32	10.22 and 10.22A of IPCC (2019)
	Tier 1	N_2O	Direct N_2O emissions from manure management	33 and 34	10.25 and 10.30 of IPCC (2019)
	Tier 1	N_2O	Indirect N_2O emissions from N volatilisation, linked with manure management	34 - 36	10.26, 10.28 and 10.30 of IPCC (2019)
	Tier 1	N_2O	Indirect N_2O emissions from N leaching and runoff, linked with manure management	34, 37, 38	10.27, 10.29 and 10.30 of IPCC (2019)

Note: GHG = greenhouse gas; N_2O = nitrous oxide; CH_4 = methane; IPCC = Intergovernmental Panel on Climate Change.

Sources: Authors' own elaboration based on IPCC. 2019. 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories. Volume 4 – Agriculture, Forestry and Other Land Use. Geneva, Switzerland. https://www.ipcc-nggip.iges.or.jp/public/2019rf/vol4.html; IPCC. 2006. 2006 IPCC Guidelines for National Greenhouse Gas Inventories. Volume 4 – Agriculture, Forestry and Other Land Use. Kanagawa, Japan. https://www.ipcc-nggip.iges.or.jp/public/2006gl

Livestock emissions depend on the livestock category (T), productivity system (P), and – for manure management-related emissions, the manure management system (S):

- The main livestock categories are based on Table 10.9 of IPCC (2019) with a level of disaggregation determined by emissions factors for manure management (see next sections): cattle (dairy/other), buffalo (dairy/other), sheep (dairy/other), swine (growing/breeding), goats, camels, horses, mules and asses, deer, ostrich, poultry (chicken layer/chicken broiler), llamas and alpacas.
- The productivity systems available in the tool are: high-productivity, low-productivity and default (IPCC, 2019). Productivity levels affect both enteric fermentation and manure management emissions. A description of productivity systems available in the tool is available in Table A60 in the annex.
- The following manure management systems are available in the tool (for their definitions consult Table A61 in the annex):
  - pasture/range/paddock
  - daily spread
  - solid storage
  - dry lot
  - liquid/slurry
  - lagoons (uncovered anaerobic)
  - pit storage below animal confinements
  - anaerobic digester
  - burned for fuel
  - poultry manure
  - composting

Methane emissions from enteric fermentation

The EX-ACT uses the IPCC Tier 1 method so that only animal population data (in addition to information already provided in the description section) is needed to estimate the emissions, according to Equation 30:

$$E_T = \sum_{(P)} EF_{(T,P)} \times \left(\frac{N_{(T,P)}}{10^6}\right)$$

(Equation 30)[29]

where:

- E_T = methane emissions from enteric fermentation, in Gg CH_4/yr;
- $EF_{(T,P)}$ = emission factor for the defined livestock population T and productivity system P, in kg CH4/head/yr;
- $N_{(T,P)}$= number of livestock heads within category T and productivity system P.

For poultry (chicken layer and chicken broiler) the IPCC does not provide emission factor for enteric fermentation, due to the lack of data. Yet poultry is one of the most important meat sources in many countries and there are large poultry flocks in many territories. Therefore, EX-ACT uses the value provided by Wang and Huang (2005) for all productivity levels of poultry animal categories.

Methane emissions from manure management

The calculations of CH_4 emissions from manure management draw on Section10.4 in Chapter 10 of IPCC (2019). These emissions correspond to CH_4 produced during the storage and treatment of manure (including both dung and urine) and from manure deposited on pastures (Pasture-Range-Paddock, PRP). The default is calculated as the average of the different manure management systems available in the specific geographical region of interest. EX-ACT uses the advanced Tier 1a method based on the total amount of volatile solid (VS) excreted (for all livestock and categories except deer and ostrich) in each type of manure management system multiplied by an emission factor for that type of livestock category within selected productivity system, the specified climate zone and manure management system (see Equations 31 and 32):

$$CH_{4(mm)} = \sum [N_{(T,P)} \times VS_{(T,P)} \times AWMS_{(T,S,P)} \times EF_{(T,S,P)}]/1000$$

(Equation 31)[30]

where:

- $CH_{4(mm)}$ = methane emissions from manure management, in kg CH_4/yr;
- $N_{(T,P)}$ = number of head of livestock species/category T in the project area/country, for productivity system P;
- $VS_{(T,P)}$ = annual average volatile solid (VS) excretion per head of species/category T, for productivity system P, in kg VS/animal/yr;
- $AWMS_{(T,S,P)}$ = fraction of total annual VS excretion for each livestock species/category T that is managed in manure management system S, dimensionless;
- $EF_{(T,S,P)}$ = emission factor for direct CH_4 emissions from manure management system S, by animal species/category T, in manure management system S, for productivity system P, when applicable, in g CH_4/kg VS.

The annual average VS excretion per head is calculated as per Equation 32:

$$VS_{(T,P)} = \left(VS_{rate(T,P)} \times \frac{TAM_{(T,P)}}{1000}\right) \times 365$$

(Equation 32)[31]

where:

- $VS_{rate(T,P)}$= default VS excretion rate, for productivity system P for livestock species/category T in kg VS/animal/yr;
- $TAM_{(T,P)}$= typical animal mass for livestock categry T, for productivity system P, in kg/animal.

[29] Corresponding to Equation 10.19 of Chapter 10 of IPCC (2019).
[30] Corresponding to Equation 10.22 of Chapter 10 of IPCC (2019).
[31] Corresponding to Equation 10.22A of Chapter 10 of IPCC (2019).

Nitrous oxide emissions from manure management

These emissions correspond to N_2O produced, directly or indirectly, during the storage and treatment of manure (solids and liquids). They include volatisation and leaching of N excreted by livestock. Calculations are based on Tier 1 approach that consists in multiplying the total amount of N excretion of species category by a default emission factor (see Section10.5 of IPCC [2019] for further details). **Direct emissions** occur via combined nitrification and denitrification of nitrogen contained in the manure. Emissions of N_2O from manure during storage and treatments depend on the nitrogen and carbon content of manure as well as on the duration of the storage and type of treatment. **Indirect emissions** result from volatile nitrogen losses that occur primarily in the forms of NH_3 (ammonia) and NO_x (nitrogen oxide). Nitrogen losses begin at the point of excretion in animal production areas and continue through on-site management in storage and treatment systems: i.e. manure management systems. Nitrogen is also lost through runoff and leaching into soils from the solid storage of manure at outdoors areas, in feedlots, pastures, ranges or paddocks. For lamas and alpacas, the IPCC does not provide any emissions factor to account for nitrogen excretion rates. Therefore, direct and indirect N2O emissions for this category are not accounted for in EX-ACT at Tier 1.

Direct nitrous oxide emissions from manure management are calculated according to Equation 33, and depend on the total amount of N excretion (from all livestock species/categories) in each type of manure management system, related to the emission factor for that type of manure management system:

$$N_2O_{\ D(mm)} = \left[\sum_{S}\left[\sum_{T,P}\left[\left(N_{(T,P)} \times Nex_{(T,P)}\right) \times AWMS_{(T,S,P)}\right]\right] \times EF_{3(S)}\right] \times \frac{44}{28}$$

(Equation 33)[32]

where:

- $N_2O_{\ D(mm)}$ = nitrous oxide emissions from manure management, in kg N_2O/yr;
- $N_{(T,P)}$ = number of head of livestock species/category T in the project area/country, for productivity system P;
- $Nex_{(T,P)}$ = annual average N excretion per head of species/category T and productivity system P, in kg N/Animal/yr;
- $AWMS_{(T,S,P)}$ = fraction of total annual nitrogen excretion for each livestock species/category T that is managed in manure management system S, dimensionless; to consider productivity class P, if using a Tier 1a approach;
- $EF_{3(S)}$ = emission factor for direct N_2O emissions, in manure management system S, when applicable, in kg N_2O/kg N;
- $\frac{44}{28}$ = conversion of N2O-N(mm) emissions to N2O(mm) emissions.

The annual average N excretion per head (in kg N/Animal/yr) is calculated by Equation 34:

$$Nex_{(T,P)} = \left(N_{rate(T,P)} \times \frac{TAM_{(T,P)}}{1000}\right) \times 365$$

(Equation 34)[33]

where:

- $N_{rate(T,P)}$ = default N excretion rate, for productivity system P for livestock species/category T, in kg N/1000 kg animal mass)/ day;
- $TAM_{(T,P)}$ = typical animal mass for livestock category T, for productivity system P, in kg/animal.

Indirect nitrous oxide emissions due to volatilisation of N in the form of NH_3 and NO_x from manure management systems are calculated according to Equation 35, and based on the multiplication of the amount of N excreted

[32] Based on Equation 10.25 of Chapter 10 of IPCC (2019).
[33] Corresponding to Equation 10.30 of Chapter 10 of IPCC (2019).

and managed in each manure management system by a fraction of volatilised N. Nitrogen losses are then summed overall manure management systems, as described below:

$$N_2O_{\,G(mm)} = N_{vol(MMS)} \times EF_4 \times \frac{44}{28}$$

(Equation 35)[34]

where:

- $N_2O_{\,D(mm)}$ = indirect nitrous oxide emissions due to volatilisation of N from manure management, in kg N_2O/yr;
- $N_{vol(MMS)}$= amount of N from manure that is lost due to volatilisation of NH_3 and NO_x, in kg N/yr;
- EF_4= emission factor for N2O emissions from atmospheric deposition of nitrogen on soils and water surfaces, in kg N2O-N/(kg NH3-N + NOx-N volatilised);
- $\frac{44}{28}$ = conversion of N2O-N(mm) emissions to N2O(mm) emissions.

The amount of N from manure that is lost due to volatilisation of NH_3 and NO_x, in kg N/yr, is calculated by Equation 36:

$$N_{vol(MMS)} = \sum_{S}\left[\sum_{T,P}\left[\left(\left(N_{(T,P)} \times Nex_{(T,P)}\right) \times AWMS_{(T,S,P)}\right) * Frac_{gasMS(T,S)}\right]\right]$$

(Equation 36)[35]

where:

- $N_{(T,P)}$ = number of head of livestock species/category T in the project area/country, for productivity system P;
- $Nex_{(T,P)}$ = annual average N excretion per head of species/category T and productivity system P, in kg N/Animal/yr; calculated by Equation 34;
- $AWMS_{(T,S,P)}$ = fraction of total annual nitrogen excretion for each livestock species/category T that is managed in manure management system S, dimensionless; to consider productivity class P, if using a Tier 1a approach;

- $Frac_{gasMS(T,S)}$ = fraction of managed manure N for livestock category T that volatilises as NH_3 and NO_x in the manure management system S.

[34] Corresponding to Equation 10.28 of Chapter 10 of IPCC (2019).
[35] Corresponding to Equation 10.26 of Chapter 10 of IPCC (2019).

Indirect nitrous oxide emissions due to leaching of N from manure management are calculated according to Equation 37, and based on the multiplication of the amount of N excreted and managed in each manure management system by a fraction of leached N. N losses are then summed overall manure management systems, as described below:

$$N_2O_{\,L(mm)} = N_{leaching(MMS)} \times EF_5 \times \frac{44}{28}$$

(Equation 37)[36]

where:

- $N_2O_{\,L(mm)}$ = indirect nitrous oxide emissions due to volatilisation of N from manure management, in kg N_2O/yr;
- $N_{leaching(MMS)}$ = amount of N from manure that is lost due to leaching and runoff of NH_3 and NO_x, in kg N/yr;
- EF_5 = emission factor for N2O emissions from atmospheric deposition of nitrogen on soils and water surfaces, in kg N2O-N/(kg NH3-N + NOx-N volatilised);
- $\frac{44}{28}$ = conversion of N2O-N(mm) emissions to N2O(mm) emissions.

The amount of N from manure that is lost due to leaching and runoff of NH_3 and NO_x, in kg N/yr, is calculated by Equation 38:

$$N_{leaching(MMS)} = \sum_{S}\left[\sum_{T,P}\left[\left(N_{(T,P)} \times Nex_{(T,P)} \times AWMS_{(T,S,P)}\right) \times Frac_{leachMS(T,S)}\right]\right]$$

(Equation 38)[37]

where:

- $N_{(T,P)}$ = number of head of livestock species/category T in the project area/country, for productivity system P;
- $Nex_{(T,P)}$ = annual average N excretion per head of species/category T and productivity system P, in kg N/Animal/yr; calculated by Equation 34;
- $AWMS_{(T,S,P)}$ = fraction of total annual nitrogen excretion for each livestock species/category T that is managed in manure management system S, dimensionless; to consider productivity class P, if using a Tier 1a approach;
- $Frac_{gasMS(T,S)}$ = fraction of managed manure N for livestock category T that is leached from the manure management system S.

Table 23 details the default values adopted in the livestock management module, along with the IPCC table reference and in-text table reference.

[36] Corresponding to Equation 10.29 of Chapter 10 of IPCC (2019).
[37] Corresponding to Equation 10.27 of Chapter 10 of IPCC (2019).

Table 23. Sources of default values in livestock module of the Environmental eXternalities ACcounting Tool

Element	Unit	Details	Table	Table IPCC	Notes
Methane emissions from enteric fermentation (Equations 16 and 30) **Emissions = AD * EF** where $\mathbf{E_T} = \sum_{(P)} \mathbf{EF_{(T,P)}} \times \left(\frac{\mathbf{N_{(T,P)}}}{\mathbf{10^6}}\right)$					
Methane EF for enteric fermentation ($EF_{(T,P)}$)	kg CH_4/head/yr	By livestock category and production system.	A57 and A59	10.10 and 10.11of IPCC (2019)	Modifiable in Tier 2.
Methane emissions from manure management (Equations 31 and 32) $\mathbf{CH_{4(mm)}} = \sum[\mathbf{N_{(T,P)}} \times \mathbf{VS_{(T,P)}} \times \mathbf{AWMS_{(T,S,P)}} \times \mathbf{EF_{(T,S,P)}}]/\mathbf{1000}$ where $\mathbf{VS_{(T,P)}} = \left(\mathbf{VS_{rate(T,P)}} \times \frac{\mathbf{TAM_{(T,P)}}}{\mathbf{1000}}\right) \times \mathbf{365}$					
Methane EF for manure mgmt. ($EF_{(T,S,P)}$)	g CH_4/ kg VS	By livestock category, manure management system, productivity system.	A85 and A102	10.14 and 10.15 of IPCC (2019)	The combined EF, expressed in GHG per head per year, is modifiable in Tier 2.
Fraction of total annual VS excretion managed in manure management system S (AWMS $_{(T, S, P)}$)	Dimensionless	By livestock **category, manure management system, and productivity system.** Default values for regional averages, or specification of system and fraction as user input.	A68 and A84	10A.6 - 10A.9 of IPCC (2019)	-
Excretion rate (VS $_{rate}$)	kg VS/ (1000 kg animal mass)/day	By livestock category and productivity system.	A65 – A67	10.13A and 10.15 of IPCC (2019)	-
Typical animal mass (TAM)	kg/animal	By livestock category and productivity system.	A62 – A64	A.5 of IPCC (2019)	-
Direct nitrous oxide emissions from manure management (Equations 33 and 34) $\mathbf{N_2O}\ _{\mathbf{D(mm)}} = \left[\sum_S\left[\sum_{T,P}\left[\left(\mathbf{N_{(T,P)}} \times \mathbf{Nex_{(T,P)}}\right) \times \mathbf{AWMS_{(T,S,P)}}\right]\ \right] \times \mathbf{EF_{3(S)}}\right] \times \frac{44}{28}$ where $\mathbf{Nex_{(T,P)}} = \left(\mathbf{N_{rate(T,P)}} \times \frac{\mathbf{TAM_{(T,P)}}}{\mathbf{1000}}\right) \times \mathbf{365}$					
Direct nitrous oxide EF for manure mgmt. ($EF_{3(S)}$)	kg N_2O-N/kg N	By manure management system.	A106 and A107	10.21of IPCC (2019)	The combined EF, expressed in GHG per head per year, is modifiable in Tier 2.
Fraction of total annual nitrogen excretion managed in manure management system S (AWMS $_{(T, S, P)}$)	Dimensionless	By animal **category, manure management system, and productivity system.** Default values for regional averages, or specification of	A68 – A84	10A.6 to 10A.9 of IPCC (2019)	-

Element	Unit	Details	Table	Table IPCC	Notes
		system and fraction as user input.			
Nitrogen excretion rate ($N_{rate (T, P)}$)	kg N/ (1000 kg animal mass)/day	By livestock category and productivity system.	A103 – A105	10.19 of IPCC (2019)	-
Typical animal mass (TAM)	kg/animal	By livestock category and productivity system.	A62 – A64	A.5 of IPCC (2019)	-
Indirect nitrous oxide emissions due to N volatilisation from manure management (Equations 34 – 36)**(Equation 37** $\mathbf{N_2O_{G(mm)}} = \mathbf{N_{vol(MMS)}} \times \mathbf{EF_4} \times \frac{44}{28}$ where $\mathbf{N_{vol(MMS)}} = \sum_S [\sum_{T,P} [(\mathbf{N_{(T,P)}} \times \mathbf{Nex_{(T,P)}}) \times \mathbf{AWMS_{(T,S,P)}}]\ ] \times \mathbf{Frac_{gasMS(T,S)}}$ and where $\mathbf{Nex_{(T,P)}} = \left(\mathbf{N_{rate(T,P)}} \times \frac{\mathbf{TAM_{(T,P)}}}{\mathbf{1000}}\right) \times \mathbf{365}$					
Fraction of volatilized N from managed manure ($\mathbf{Frac_{gasMS(T,S)}}$)	Dimensionless	By animal category and manure management system.	A108 – A112	10.22 of IPCC (2019)	-
EF for atmospheric N deposition (EF_4)	kg N_2O-N/ (kg NH_3-N + NO_x-N volatilised)	-	N/A	11.3 of IPCC (2019)	-
Fraction of total annual nitrogen excretion managed in manure management system S ($AWMS_{(T, S, P)}$)	Dimensionless	By animal **category, manure management system, and productivity system.** Default values for regional averages, or specification of system and fraction as user input.	A68 – A84	10A.6 - 10A.9 of IPCC (2019)	-
Nitrogen excretion rate ($N_{rate (T, P)}$)	kg N/(1000 kg animal mass)/day	By livestock category and productivity system.	A103 – A105	10.19 of IPCC (2019)	-
Typical animal mass (TAM)	kg/animal	By livestock category and productivity system.	A62 – A64	A.5 of IPCC (2019)	-
Indirect nitrous oxide emissions due to N leaching from manure management (Equations 34, 37 and 38) $\mathbf{N_2O_{L(mm)}} = \mathbf{N_{leaching(MMS)}} \times \mathbf{EF_5} \times \frac{44}{28}$ where $\mathbf{N_{leaching(MMS)}} = \sum_S [\sum_{T,P} [(\mathbf{N_{(T,P)}} \times \mathbf{Nex_{(T,P)}}) \times \mathbf{AWMS_{(T,S,P)}}]\ ] \times \mathbf{Frac_{leachMS(T,S)}}$ and where $\mathbf{Nex_{(T,P)}} = \left(\mathbf{N_{rate(T,P)}} \times \frac{\mathbf{TAM_{(T,P)}}}{\mathbf{1000}}\right) \times \mathbf{365}$					
Fraction of leached N from managed manure ($\mathbf{Frac_{leachMS(T,S)}}$)	Dimensionless	By animal category and manure management system.	A113	10.22 of IPCC (2019)	-
EF for N_2O emissions from N leaching and runoff (EF_5)	kg N_2O-N/kg N leached and runoff	-	N/A	11.3 of IPCC (2019)	-
Fraction of total annual nitrogen excretion managed in manure management system S ($AWMS_{(T, S, P)}$)	Dimensionless	By animal **category, manure management system, and productivity system.** Default values for regional averages, or specification of system and fraction as user input.	A68 – A84	10A.6 - 10A.9 of IPCC (2019)	-

Element	Unit	Details	Table	Table IPCC	Notes
Nitrogen excretion rate ($N_{rate (T, P)}$)	kg N/ (1000 kg animal mass)/day	By livestock category and productivity system.	A103 – A105	10.19 of IPCC (2019)	-
Typical animal mass (TAM)	kg/animal	By livestock category and productivity system.	A62 – A64	A.5 of IPCC (2019)	-

Note: AD = activity data; AWMS = fraction of total annual excretion managed in a manure management system; CH_4 = methane; EF = emission factor; EF_3 = emission factor for direct N_2O emissions from manure management; EF_4 = emission factor for atmospheric nitrogen deposition; EF_5 = emission factor for N_2O emissions from nitrogen leaching and runoff; GHG = greenhouse gases; IPCC = Intergovernmental Panel on Climate Change; kg CH_4/head/yr = kilograms of methane per head per year; kg N/(1000 kg animal mass)/day = kilograms of nitrogen per 1000 kilograms of animal mass per day; kg N_2O–N/kg N = kilograms of nitrous oxide nitrogen per kilogram of nitrogen; kg VS/(1000 kg animal mass)/day = kilograms of volatile solids per 1000 kilograms of animal mass per day; MMS = manure management systems; N = nitrogen; N_2O = nitrous oxide; N_2O–N = nitrous oxide expressed as nitrogen; TAM = typical animal mass; VS = volatile solids; VS rate = excretion rate of volatile solids.

Source: Authors' own elaboration based on IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Volume 4 – Agriculture, Forestry and Other Land Use. Geneva, Switzerland. https://www.ipcc-nggip.iges.or.jp/public/2019rf/vol4.html

2.4.11 ORGANIC SOILS

Material used to develop this module can be found in Chapter 7 (Wetlands) of IPCC (2006) and Chapters 2 (Drained inland organic soils) and Chapter 3 (Rewetted organic soils) of IPCC (2014).

The organic soils module addendum calculates GHG fluxes associated with the management of inland wetlands organic soils. Organic soils, or histosols, are defined as ([FAO, 2015] cited by IPCC [2019])[38]:

- Starting at the soil surface and having a thickness of ≥ 10 cm and directly overlying ice, continuous rock or technic hard material or coarse fragments, the interstices of which are filed with organic material; or
- Starting ≤ 40 cm from the soil surface and having within ≤ 100 cm of the soil surface a combined thickness of either ≥ 60 cm, if ≥ 75 percent (by volume) of the materials consist of moss fibres, or ≥ 40 cm in other materials.

The module covers the following computations:

1. Emissions from changes in water table management (i.e. drainage and rewetting):
   - On-site emissions: (i) CO_2 emissions removals from organic soils due to organic matter decomposition, (ii) CH_4 emissions and removals, and (iii) N_2O emissions.
   - Off-site emissions: loss of DOC and CH_4 in the drainage waters.
2. Peat extraction:
   - CO_2eq emissions from the extraction and off-site use of peat.
3. Emissions from soil fire and related management practices:
   - CO_2, CO and CH_4 from prescribed or wildfire burning of organic soils.

The module addendum complements other land-use modules by targeting emissions linked to organic soil management. All other emissions are targeted in the respective land-use modules according to methodologies reported in the respective sections of this handbook.

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[38] Additionally, according to FAO (1998) Soils are classified as "organic soils" when they also satisfy the following statements (1 and 2, or 1 and 3):
1) Thickness of organic horizon greater than or equal to 10 cm. A horizon of less than 20 cm must have 12 percent or more organic carbon when mixed to a depth of 20 cm.
2) Soils that are never saturated with water for more than a few days must contain more than 20 percent organic carbon by weight (i.e. about 35 percent organic matter).
3) Soils are subject to water saturation episodes and have either i) at least 12 percent organic carbon by weight (i.e. about 20 percent organic matter) if the soil has no clay; or ii) at least 18 percent organic carbon by weight (i.e. about 30 percent organic matter) if the soil has 60 percent or more clay; or iii) an intermediate, proportional amount of organic carbon for intermediate amounts of clay.

Table 24 illustrates the GHGs addressed by the module, with the corresponding level of assessment, i.e. Tier 1 or Tier 2 and includes in-text references to the equations used for the GHG fluxes calculations, as well as the corresponding IPCC equations for easier cross-reference.

Table 24. Overview of greenhouse gas fluxes estimated in the organic soil module addendum of the Environmental eXternalities ACcounting Tool and corresponding equations

Activity		GHG	Flux description	Equation	IPCC equation
Organic soil drainage	Tier 1	CO_2	On-site emissions linked to water table management	See Table 25	2.3 of IPCC (2014)
	Tier 1	CO_2	Off-site DOC emissions linked to water table management		2.4 and 2.5 of IPCC (2014)
	Tier 1	CH_4	On-site and off-site emissions linked to water table management		2.6 of IPCC (2014)
	Tier 1	N_2O	On-site emissions linked to water table management		2.7 of IPCC (2014)
Organic soil rewetting	Tier 1	CO_2	On-site emissions linked to water table management		3.4 of IPCC (2014)
	Tier 1	CO_2	Off-site DOC emissions linked to water table management		3.5 and 3.6 of IPCC (2014)
	Tier 1	CH_4	On-site and off-site emissions linked to water table management		3.8 of IPCC (2014)
	Tier 1	N_2O	On-site emissions linked to water table management		3.9 of IPCC (2014)
Organic soil extraction	Tier 1	CO_2	Emissions from soils in peatland being drained for peat extraction		7.5 of IPCC (2006)
Fire on organic soil	Tier 1	CO_2, CO, CH_4	Emissions from fire on soils	See Table 25 and Equation 17	2.8 of IPCC (2014), and 2.27 of IPCC (2019)

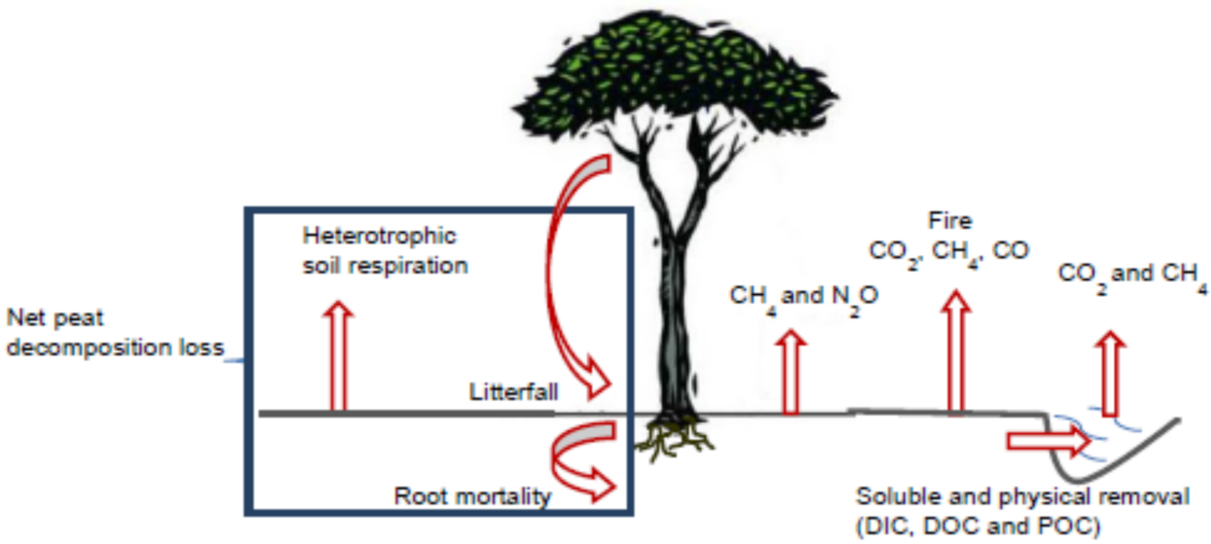
Note: GHG = greenhouse gas; CO_2 = carbon dioxide; N_2O = nitrous oxide; CH_4 = methane; IPCC = Intergovernmental Panel on Climate Change.

Sources: Authors' own elaboration based on IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Volume 4 – Agriculture, Forestry and Other Land Use. Geneva, Switzerland. https://www.ipcc-nggip.iges.or.jp/public/2019rf/vol4.html; IPCC. 2006. *2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Volume 4 – Agriculture, Forestry and Other Land Use. Kanagawa, Japan. https://www.ipcc-nggip.iges.or.jp/public/2006gl; IPCC. 2014. *2013 supplement to the 2006 IPCC guidelines for national greenhouse gas inventories: wetlands : methodological guidance on lands with wet and drained soils, and constructed wetlands for wastewater treatment*. Hayama, Japan, IPCC, Intergovernmental Panel on Climate Change.

Emissions from drainage

Drainage is a practice used in agriculture and forestry to improve site conditions for plant growth. Carbon stored in organic soils will readily decompose when conditions become aerobic following soil drainage, with an increase in CO_2 and N_2O emissions due to increased oxidation of soil organic material, and a loss of dissolved organic carbon (DOC) leading to off-site CO_2 emissions, particulate organic carbon (POC) and waterborne transport of dissolved inorganic carbon (DIC) (see Figure 6)(IPCC, 2014). Currently, there is no methodology to assess emissions from POC and DIC; as a result, these are not taken into account in EX-ACT.

Figure 6. Summary of fluxes from drained organic soils



Source: IPCC. 2014. *2013 supplement to the 2006 IPCC guidelines for national greenhouse gas inventories: wetlands : methodological guidance on lands with wet and drained soils, and constructed wetlands for wastewater treatment*. Hayama, Japan, IPCC, Intergovernmental Panel on Climate Change.

The basic methodology to assess drainage GHG fluxes is to multiply targeted areas by climate and land-use specific yearly emission factors that are stratified by global climate region. EX-ACT uses emission factors in Tables 2.1 to 2.5 of IPCC (2014) and provided in Table A114 – A121 in the annex. EX-ACT allows for calculating the impact of drainage of organic soils according to the following types of land uses defined in IPCC (2014): forest, plantations, annual crops, perennial crops, flooded rice, grasslands. EX-ACT land use definitions are associated to each of these categories to provide specific emission factors to users. If one or more of the selected land uses is not linked to the above categories, EX-ACT will require users to insert a Tier 2 to perform computations. When water table management activities target land remaining under the same land use, the emission factors are calculated based on the specified land use. When management activity includes a land-use conversion, EX-ACT associates the initial and final (without and with project) emission factors according to the respective portions of land under each land use.

- **On-site CO_2 emissions/removal from drained inland organic soil:** The emission factor is relative to the climate domain, nutrient class and drainage class. At Tier 1, EX-ACT assumes that the organic soil is deep-drained, as this is the most suitable practice for a wide range of management intensities. The emission factors are by default set for nutrient-poor soil.
- **Off-site CO_2 emissions via waterborne carbon losses from drained inland organic soils:** Waterborne carbon comprises of different fractions of DOC, POC, DIC, CH_4 and particulate inorganic carbon (PIC). As DOC forms the largest component of waterborne carbon export, and there is no current methodology for DIC and POC, off-site CO_2 emissions are estimated from the loss of carbon from conversion of DOC.
- **CH_4 emissions and removal from drained inland organic soils:** CH_4 emissions can occur from the drained land surface and also from the ditch network during drainage.
- **N_2O emission from drained inland organic soils:** When drained, organic soils also emit N_2O from the mineralization of the organic matter.

Emissions from rewetting

Rewetting refers to the deliberate raising of the water table on previously drained soils to restore water-saturated conditions (i.e. by blocking drainage ditches, disabling pumping facilities or breaching obstructions) (IPCC, 2014). Rewetting can have several objectives, including wetland restoration or allowing other management practices on saturated organic soils such as paludiculture. Rewetting re-establishes the carbon sink function if covered with vegetation, resulting in a balance between CO_2 and CH_4 emissions and removals. N_2O emissions from rewetting activities are assumed to be negligible at Tier 1 (IPCC, 2014).

Emission factors are dependent on climate and nutrient status of the rewetted soil. Default emission factors in EX-ACT are based on those in Tables 3.1 to 3.3 in Chapter 3 of IPCC (2014). Emission factors for nutrient-poor soil are provided by default in EX-ACT, but users can change the nutrient class of the soil in the Tier 2 section.

EX-ACT allows for calculating the impact of rewetting of organic soils according to six types of land uses defined in IPCC (2014): forest, plantations, annual crops, perennial crops, flooded rice and grasslands. The EX-ACT land-use definitions are associated to each of these categories to provide with specific emission factors. If one or more of the selected land uses is not linked to the above categories, EX-ACT will require users to insert a Tier 2 to perform computations.

- **On-site and off-site CO_2 emissions**: They include carbon emissions/removals from soil and non-tree vegetation, off-site emissions from DOC (exported from rewetted soil) and emissions from burning of rewetted soils.
- CH_4 **emissions/removals from rewetted organic soils.** CH_4 emissions and removals from rewetted soils result from (i) the balance between CH_4 production and oxidation, and (ii) emission of CH_4 produced by the combustion of soil organic matter due to fire.

Greenhouse gas emissions from fire on organic soils

Fires on organic soils (uncontrolled wildfires and managed, prescribed fires) consume surface vegetation (above-ground biomass, litter and duff) and ground fires burn into and below the surface, consuming the soil organic matter and deadwood biomass, both of which generate CO_2, CO and CH_4 emissions. Emissions from organic soils are estimated according to the general methodology described in Section 2.1. Emission factors are available in Table A128 in the annex.

Greenhouse gas emissions from peatlands undergoing active peat extraction

Peat is widely used for energy, horticulture, landscapes, industrial wastewater treatment and other purposes. Peat extraction starts with drainage (i.e. construction of draining ditches, lowering of the water table and vegetation clearing). In the second phase, peat is extracted and stockpiled. GHG emissions associated with drainage of peatlands are calculated using the same methodology outlined above, with associated emission factors reported in Table A114 - A121 in the annex. Additionally, once the peat extracted, it can be used for different purposes that may lead to additional emissions (off-site CO_2 emissions) depending on the quantity of peat extracted. Emissions are calculated by multiplying a conversion factor with the quantity extracted per year and converted in tCO_2eq.

Table 25 details the default values for the parameters embedded in the Organic soil module addendum, along with the IPCC table reference and in-text table reference.

Table 25. Sources of default values in organic soil module addendum of the Environmental eXternalities ACcounting Tool

Element	Unit	Details	Table	Table IPCC	Notes
On-site CO_2 emissions linked to drainage (Equation 2.3 of IPCC [2014]) $CO_2 - C_{on-site} = \sum_{c,n,d}(A * EF_{c,n,d})$					
On-site emissions or removals from drained organic soils (CO_2-$C_{on-site}$)	tC/yr	For all land use categories, including peat extraction.	N/A	-	-
Area of drained organic soils (A)	ha	User input. Location determines land-use category, climate	N/A	-	-

Element	Unit	Details	Table	Table IPCC	Notes
		domain c, nutrient status n, and drainage class d.			
emission factors for drained organic soils (EFc,n,d)	tC/ha/yr	For land-use category, climate domain c, nutrient status n, and drainage class d.	A114 and A115	2.1 of IPCC (2014)	Modifiable in Tier 2
Off-site CO_2 emissions from DOC linked to drainage (Equations 2.4 and 2.5 of IPCC [2014]) $\mathbf{CO_2 - C_{DOC}} = \sum_{c,n}(\mathbf{A} * \mathbf{EF_{DOC}})$ where $\mathbf{EF_{DOC}} = \mathbf{DOC_{FLUX+NATURAL}} * (\mathbf{1} + \mathbf{\Delta DOC_{DRAINAGE}}) * \mathbf{FRAC_{DOC-CO_2}})$					
Off-site emissions due to DOC loss from drained soils (CO_2-C_{DOC})	tC/yr	For land-use category (including peat extraction), climate domain c, nutrient status n, and drainage class d.	N/A		-
Area of drained organic soils ($A_{c,\,n}$)	Ha	User input. Location determines land-use category in climate domain c and nutrient status n.	N/A		-
Emission factor for CO_2 emissions from DOC losses ($EF_{DOCc,n}$)	tC/ha/yr	For land-use category in climate domain c and nutrient status n.	A116	2.2 of IPCC (2014)	Modifiable in Tier 2.
Flux of DOC from natural (undrained) organic soil ($DOC_{FLUX\_NATURAL}$)	tC/ha/yr	-	N/A	2.2 of IPCC (2014)	-
Proportional increase in DOC flux from drained sites relative to undrained sites ($\Delta DOC_{DRAINAGE}$)	Dimensionless	Default value: 0.60	N/A	2.2 of IPCC (2014)	-
Conversion factor for proportion of DOC converted to CO_2 following export from site ($Frac_{DOC\text{-}CO2}$)	Dimensionless	Default value: 0.9	N/A	2.2 of IPCC (2014)	-
On- and off-site CH_4 emissions linked to drainage (Equation 2.6 of IPCC [2014]) $\mathbf{CH_{4_{organic}}} = \sum_{c,n,p}\left(\mathbf{A_{c,n,p}} * \left((\mathbf{1} - \mathbf{Frac_{ditch}}) * \mathbf{EF_{CH_{4_{land(c,n)}}}} + \mathbf{Frac_{ditch}} * \mathbf{EF_{CH_{4_{ditch(c,p)}}}}\right)\right)$					
Annual CH_4 loss from drained organic soils ($CH_{4\_organic}$)	kg CH_4/yr	For land use-category (including peat extraction), climate zone c, nutrient status n, and soil type p.	N/A	-	-
Area of drained organic soil ($A_{c,n,p}$)	ha	User input Location determines land-use category, climate zone c, nutrient status n, and soil type p.	N/A	-	-

Element	Unit	Details	Table	Table IPCC	Notes
Emission factors for direct CH_4 emissions from drained organic soils ($EF_{CH4\_landc,n}$)	kg CH_4/ha/yr	For climate zone c and nutrient status n.	A117 and A118	2.3 of IPCC (2014)	Modifiable in Tier 2.
Emission factors for CH_4 emissions from drainage ditches ($EF_{CH4\_ditchc,p}$)	kg CH_4/ha/yr	For climate zone c and nutrient status n.	A119	2.4 of IPCC (2014)	Modifiable in Tier 2.
Fraction of the total area of drained organic soil which is occupied by ditches ($Frac_{ditch}$)	Dimensionless	Calculated from user input of the area of the ditches.	N/A	-	-
On- site N_2O emissions linked to drainage (Equation 2.7of IPCC [2014]) $$N_2O - N_{OS} = \sum_{CG,F,Temp,Trop,NR,NP} (F_{OS} * EF_2)$$					
Direct N_2O–N emissions from managed/drained organic soils (N_2O–N_{OS})	kg N_2O–N/yr	Per land-use category (including peat extraction).	N/A	-	-
Area of managed/drained organic soils (F_{OS})	Ha	User input.	N/A	-	-
Emission factor for N_2O emissions from drained/managed organic soils (EF_2)	kg N_2O–N/ha/yr	Depends on CG, F, Temp, Trop, NR, NP.	A120 and A121	2.5 of IPCC (2014)	-
CG, F, Temp, Trop, NR, NP	-	Correspond to respectively: Cropland and Grassland, Forest Land, Temperate, Tropical, Nutrient-Rich, Nutrient-Poor	N/A	-	-
On-site CO_2 emissions linked to rewetting (Equation 3.4 of IPCC [2014]) $$CO_2 - C_{composite} = \sum_{c,n} (A * EF_{CO2})$$					
CO_2-C emissions from the soil and non-tree vegetation (CO_2-$C_{composite}$)	tC/yr	Emissions from the soil and non-tree vegetation	N/A	-	-
Area (Ac,n)	Ha	Based on user input. Location determines climate zone c and nutrient status	N/A	-	-
Emission factor for rewetted organic soils ($EF_{CO2\ c,n}$)	tC/ha/yr	For climate zone c and nutrient status n.	A122 and A123	3.1 of IPCC (2014)	Modifiable in Tier 2

Element	Unit	Details	Table	Table IPCC	Notes
Off-site CO_2 (DOC) emissions linked to rewetting (Equations 3.5 and 3.6 of IPCC [2014]) $\mathbf{CO_2 - C_{DOC} = \sum_c (A * EF_{DOC\_REWETTED})}$ where $\mathbf{EF_{DOC\_REWETTED} = DOC_{FLUX} * Frac_{DOC-CO2}}$					
Off-site CO_2-C emissions from dissolved organic carbon exported from rewetted organic soils (CO_2-C_{DOC})	tC/yr	-	N/A	-	-
Area of rewetted organic soil (Ac)	ha	Based on user input. Location determines climate zone c	N/A	-	-
Emission factor for DOC from rewetted organic soils ($EF_{DOC\_rewetted, c}$)	tC/ha/yr	For climate zone c	A124	3.2 of IPCC (2014)	Modifiable in Tier 2
Net flux of DOC from natural (undrained) and rewetted organic soils (DOC_{FLUX})	tC/ha/yr	-	N/A	3.2 of IPCC (2014)	-
Conversion factor for proportion of DOC converted to CO2 following export from site ($Frac_{DOC\_CO2}$)	Dimensionless	Default value: 0.9	N/A	-	-
On-site CH_4 emissions linked to rewetting (Equation 3.8 of IPCC [2014]) $\mathbf{CH_4 - C_{soil} = \frac{\sum_{c,n}(A * EF_{CH4\ soil})}{1000}}$					
Emissions from rewetted organic soils (CH_4-C_{soil})	tC/yr	-	N/A	-	-
Area of rewetted organic soils ($A_{c,n}$)	ha	Based on user input. Location determines climate zone c and nutrient status n.	N/A	-	-
Emission factor from rewetted organic soils ($EF_{CH4\ soil}$)	tC/ha/yr	For climate zone c and nutrient status n.	A125 and A126	3.3 of IPCC (2014)	Modifiable in Tier 2
On-site N_2O emissions linked to rewetting (Equation 3.9 and modified Equation 3.8 of IPCC [2014]) $\mathbf{N_2O_{rewetting\ org\ soil} - N = N_2O_{soil} - N + L_{fire}\text{-}N_2O - N}$ where $\mathbf{N_2O_{soil} - N = \frac{\sum_{c,n}(A * EF_{N2O\ soil})}{1000}}$ **(Tier 2)**					
N_2O-N emissions from rewetted organic soils ($N2O_{rewetting\ org\ soil}$-N)	tN_2O-N/yr	Considered negligible Tier 1, default value: 0.	N/A	-	-
N_2O-N emissions from the soil pool of rewetted	tN_2O-N/yr		N/A	-	-

Element	Unit	Details	Table	Table IPCC	Notes
organic soils (N_2O_{soil}-N)					
N_2O-N emissions from burning of rewetted organic soils (L_{fire}-N_2O-N)	tN_2O-N/yr		N/A	-	-
Emission factor for N_2O-N emissions on organic soil ($EF_{N2O\ soil}$)	tN_2O-N/ha/yr		N/A	-	Modifiable in Tier 2
Area (A)	ha	Based on user input.	N/A	-	-
Emissions from fire on organic soil (Equation 17 in this document and Equation 2.8 of IPCC [2014]) $$\mathbf{L_{fire} = A * M_B * C_f * G_{ef} * 10^{-3}}$$					
Number of emissions from fire (L_{fire})	t GHG	CO_2, CO, or CH_4.	N/A	-	-
Area of fire on organic soil land (A)	ha	User input through percentage of soil fire impact.	N/A	-	-
Mass of fuel available for combustion (M_B)	t/ha	**Mass of dry organic soil fuel.** Estimates of product M_B and C_f are available in IPCC (2014).	A127	2.6 of IPCC (2014)	Modifiable in Tier 2
Combustion factor (C_f)	Dimensionless	Estimates of product M_B and C_f are available in IPCC (2014).		2.6 of IPCC (2014)	-
Emission factor (G_{ef})	g/kg d.m. burnt	Per GHG.	A128	2.7 of IPCC (2014)	Modifiable in Tier 2
Emissions from peat extraction, due to drainage (CO_2 & DOC, CH_4, N_2O). Drainage-emissions for peat extraction are identical to the drainage emissions on organic soils detailed above, with dedicated emission factors.					
Off-site emissions from extracted peat (Equation 7.5 of IPCC [2006]) $$\mathbf{CO_2\text{-}C_{WW\_peat\_off-site} = Vol_{dry\_peat} * Cfraction_{vol\_peat}}$$					
Off-site CO2–C emissions from removed peat (CO2-$C_{WW\_peat\_off-site}$)	tC/yr	Considers only the carbon balance from the extracted peat in the year of extraction.	N/A	-	-
Volume of air-dry extracted peat (Wt_{dry_peat})	m3/yr	Default: Computed based on target area and extraction height.[39]	N/A	-	Modifiable in Tier 2 (Volume of air-dried peat extracted per hectare)

[39] A Tier 1 conversion factor to compute the air-dry volume of peat from the extracted volume is not available in IPCC guidelines. Therefore, EX-ACT equates the extracted (bulk) volume to the air-dry volume. Users are invited to insert project-specific data on the air-dry volume of peat extracted per hectare with the Tier 2 panel.

Element	Unit	Details	Table	Table IPCC	Notes
Carbon fraction of air-dry peat by volume ($Cfraction_{vol\_peat}$)	tC/m^3 air-dry peat	Depends on climate conditions. Boreal and temperate: Nutrient-Poor: 0.07, Nutrient-Rich: 0.24 Tropical: 0.26	N/A	7.5 of IPCC (2006)	Modifiable in Tier 2

Note: CF = carbon fraction; CH_4 = methane; CO_2 = carbon dioxide; DOC = dissolved organic carbon; EF = emission factor; EF_2 = emission factor for N_2O emissions from drained/managed organic soils; GHG = greenhouse gases; IPCC = Intergovernmental Panel on Climate Change; kg CH_4/ha/yr = kilograms of methane per hectare per year; kg CH_4/yr = kilograms of methane per year; kg N_2O–N/ha/yr = kilograms of nitrous oxide nitrogen per hectare per year; kg N_2O–N/yr = kilograms of nitrous oxide nitrogen per year; MB = mass of fuel available for combustion; N_2O = nitrous oxide; N_2O–N = nitrous oxide expressed as nitrogen; tC/ha/yr = tonnes of carbon per hectare per year; tC/yr = tonnes of carbon per year; t GHG = tonnes of greenhouse gases; tN_2O–N/ha/yr = tonnes of nitrous oxide nitrogen per hectare per year; tN_2O–N/yr = tonnes of nitrous oxide nitrogen per year.

Sources: Authors' own elaboration based on IPCC. 2006. *2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Volume 4 – Agriculture, Forestry and Other Land Use. Kanagawa, Japan. https://www.ipcc-nggip.iges.or.jp/public/2006gl; IPCC. 2014. *2013 supplement to the 2006 IPCC guidelines for national greenhouse gas inventories: wetlands : methodological guidance on lands with wet and drained soils, and constructed wetlands for wastewater treatment*. Hayama, Japan, IPCC, Intergovernmental Panel on Climate Change.

2.4.12 COASTAL WETLANDS

Materials used to develop this module can be found in Chapter 4 (Coastal wetlands) of the IPCC (2014).

The coastal wetlands module allows to estimate CO_2 emission/removal from management practices in coastal wetlands, such as (i) extraction/excavation, (ii) drainage and (iii) rewetting, and CH_4 emissions associated to rewetting of drained land with brackish waters.

A coastal wetland is defined as a wetland covered by vascular plants at or near to the coast that is influenced by brackish/saline water and/or tides and which may occur on both organic and mineral soils (IPCC, 2014). Wetlands covered in this chapter include mangrove, seagrass and tidal marsh.

The module covers the following computations:

1. **Emissions from loss of SOC and biomass** associated with drainage and excavation/extraction of soils:
   - Biomass CO_2 emissions are calculated according to the methodology illustrated in Section 2.2.
   - Soil CO_2 emissions are calculated assuming all the soil carbon is oxidized within the year of extraction up to 1m depth.
2. **Carbon sequestration in soil and biomass and methane emissions** associated to restoration activities in coastal wetlands, following restoration of the hydrology and/or revegetation:
   - Biomass CO_2 fluxes are calculated according to the methodology illustrated in Section 2.2.
   - Soil CO_2 emissions are calculated assuming rewetting allows for the instantaneous reinstatement of all the soil carbon at an equivalent level to that in natural settings.
   - CH_4 emissions from the microbial decomposition of organic matter during the shift from an aerobic to an anaerobic environment

Table 266 illustrates the GHGs addressed by the module, with the corresponding level of assessment i.e. Tier 1 or Tier 2 and includes in-text references to the equations used for the GHG fluxes calculations, as well as the corresponding IPCC equations for easier cross-reference.

Table 26. Overview of greenhouse gas fluxes estimated in the coastal wetlands module of the Environmental eXternalities ACcounting Tool and corresponding equations

Activity		GHG	Flux description	Equation	IPCC equation
Coastal wetland drainage and excavation	Tier 1	CO_2	Soil organic carbon stock changes linked to soil extraction and excavation	See equations in the Table 27	4.6 of IPCC (2014) and 2.8A of IPCC (2006)
	Tier 1	CO_2	Biomass carbon stock changes linked to soil extraction and excavation		4.3 and 4.4 of IPCC (2014), and Equation 2,3 and 2.8a of IPCC (2006)
	Tier 1	CO_2	Emissions linked to wetland drainage		4.8 of IPCC (2014)
Coastal wetland rewetting and restoration	Tier 1	CO_2 and CH_4	Soil emissions linked to rewetting		4.7 and 4.9 of IPCC (2014)
	Tier 1	CO_2	Biomass carbon stock changes linked to wetland revegetation		2,3 and 2.8A of IPCC (2006)

Note: GHG = greenhouse gas; CO_2 = carbon dioxide; CH_4 = methane; IPCC = Intergovernmental Panel on Climate Change.

Sources: Authors' own elaboration based on IPCC. 2006. *2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Volume 4 – Agriculture, Forestry and Other Land Use. Kanagawa, Japan. https://www.ipcc-nggip.iges.or.jp/public/2006gl; IPCC. 2014. *2013 supplement to the 2006 IPCC guidelines for national greenhouse gas inventories: wetlands : methodological guidance on lands with wet and drained soils, and constructed wetlands for wastewater treatment*. Hayama, Japan, IPCC, Intergovernmental Panel on Climate Change.

Drainage

Drainage activities in coastal wetlands refer to the diking and removal of water to convert wetland areas into pastures, cropland, or settlements. At Tier 1, EX-ACT assumes complete removal of biomass and dead organic matter in the first year, and full drainage - defined as lowering the water table to 1 metre below the soil surface in both organic and mineral soils.

EX-ACT uses default data for estimating carbon stock changes in coastal wetland ecosystems provided in Chapter 4 of IPCC (2014), specifically Table A132 – A138. Soil carbon emissions are calculated under the assumption that (i) emissions persist as long as the soil remains drained or until the original soil carbon stock is fully oxidized, and (ii) drainage affects the top 1 metre of soil. For drained mangroves and tidal marshes, EX-ACT applies a default emission factor of 7.9 tC/ha/year (see Table 4.13 in Chapter 4 of IPCC [2014]). Emissions are projected until the soil carbon stock is depleted, as indicated in Table 4.11 of the same chapter, regardless of whether a land-use change occurs. If extraction activities target the drained area, emissions linked to soil carbon stocks are calculated according to the extraction methodology below, to avoid double counting.

Extraction/excavation

Extraction activities refer to excavation, with associated dredging, used to provide soil for raising the elevation of land, or excavation for construction of harbour, marina and aquaculture/salt ponds. In EX-ACT, excavation implies the removal of the above and below-ground biomass, litter and deadwood, and soil extraction over 1 m depth. At Tier 1 the IPCC (2014) methodology assumes that the biomass, litter, deadwood and soil are all removed and exposed under aerobic conditions, in which all the carbon is converted into CO2 within the first year of the extraction.

For biomass, since tidal marshes and seagrass do not have reference biomass levels in IPCC, EX-ACT does not compute emissions at Tier 1, but users can include information on biomass levels in the Tier 2 section. For mangroves EX-ACT retains AGB default values from Table 4.3 of IPCC (2014), i.e. 92 tonne of dry matter per hectare (t d.m./ha) for tropical dry climate, 192 t d.m./ha for tropical moist and wet climate, and 75 t d.m./ha for warm temperate climate. Values are converted in tonne of carbon per hectare (tC/ha) using the carbon fraction of above-ground biomass of 0.451 tC/t d.m. (see Table 4.2 of IPCC [2014]). For mangrove BGB, the tools considers a specific R ratio of below-ground to above-ground biomass according to climate zone, i.e. 0.49 for tropical wet climate, 0.29 for tropical dry climate and 0.91 for warm temperate climate, (IPCC, 2014). The litter in mangrove is 0.7tC/ha, and the deadwood carbon stock is 10.7 tC/ha from Table 4.7 (IPCC, 2014).

For biomass carbon stock changes, EX-ACT uses default data provided in Chapter 4 of IPCC (2014) and reported in Table A132 – A135. SOC stocks for organic and mineral soils represent the top 1 metre depth of soil undergoing extraction/excavation (see Table A136 – A138). It is assumed that 96 percent of the SOC stock in 1 m depth is oxidized immediately after extraction, and only 4 percent of the SOC stock is considered refractory. EX-ACT includes data from Atwood *et al.* (2017) for mangrove soil carbon stocks, as reported in Table A139. These values can be selected in Tier 2. By default EX-ACT uses the IPCC values (see Table 4.11 in Chapter 4 of IPCC [2014]).

Rewetting and revegetation

Rewetting refers to the restoration of hydrological conditions in coastal wetlands, serving as a natural precursor to the establishment of mangrove and tidal marsh vegetation, or to environmental changes that enable seagrass growth in already waterlogged soils. At Tier 1, EX-ACT assesses changes in biomass and dead organic matter for mangroves using default carbon stock values previously described (see Table A132 – A135). Due to the lack of mangrove biomass growth rates in IPCC, the tool adopts a conservative approach to biomass growth estimations, assuming that biomass reference levels will be reached after 20 years from the start of revegetation on the targeted area.

For soils, rewetting and revegetation of previously drained areas allow carbon accumulation at rates equivalent to those in natural settings, with default values provided in Table A140. In brackish water environments, rewetting shifts soil conditions from aerobic to anaerobic, triggering CH_4 emissions due to microbial decomposition of organic matter. EX-ACT applies a default emission factor of 193.7 kg CH_4/ha/year for mangroves, seagrass, and tidal marshes (see Section 4.3.1 of IPCC [2014]).

Table 27 details the default values for the parameters embedded in the coastal wetland module, along with the IPCC table reference and in-text table reference.

Table 27. Sources of default values in coastal wetlands module of the Environmental eXternalities ACcounting Tool

Element	Unit	Details	Table	Table IPCC	Notes
Soil organic carbon stock changes linked to soil extraction and excavation (based on Equation 4.6 of IPCC [2014] and 2.8a of IPCC [2006]) $\Delta C_{SO-EX} = SOC_{1m} * C_{lost} * A_{EX}$					
Carbon loss due to excavation and extraction (ΔC_{SO-EX})	tC	-	N/A	-	-
Carbon content of the top-metre of soil (SOC_{1m})	tC/ha	In line with IPCC (Chapter 4 of IPCC [2014]) specifications of full drainage and excavation. EX-ACT can use default values from the IPCC or from Atwood *et al.* (2017) (selectable with Tier 2).	A136 – A139	4.11 of IPCC (2014)	Modifiable in Tier 2.
Fraction of carbon lost after excavation (C_{lost})	Dimensionless	Default value: 0.96.	N/A	4A.4 of IPCC (2014)	Modifiable in Tier 2
Area excavated (A_{EX})	ha	Excavated area is a subset of the drained area. User input.	N/A	-	-

Element	Unit	Details	Table	Table IPCC	Notes
Biomass carbon stock changes linked to soil drainage or extraction and excavation (based on Equations 4.3 and 4.4 of IPCC [2014], and Equations 2.3 and 2.8a of IPCC [2006]) $\Delta C_{B-EX} = (\Delta C_{AGB} + \Delta C_{BGB} + \Delta C_{DW} + \Delta C_{LI}) * A$ where $\Delta C_{AGB} = AGB * CF$ $\Delta C_{BGB} = \Delta C_{AGB} * R$					
Carbon loss from biomass following drainage and/or excavation (ΔC_{B-EX})	tC	Assuming a complete loss of biomass after drainage and/or excavation is complete.	N/A	-	-
Carbon stock change from above-ground biomass (ΔC_{AGB})	tC/ha	Stock change as the absolute value of carbon losses.	N/A	-	Modifiable in Tier 2
Carbon stock change from below-ground biomass (ΔC_{BGB})	tC/ha	Stock change as the absolute value of carbon losses.	N/A	-	Modifiable in Tier 2
Carbon stock change from deadwood (ΔC_{DW})	tC/ha	Stock change as the absolute value of carbon losses. Default value for mangroves: 10.7	A134	4.7 of IPCC (2014)	Modifiable in Tier 2
Carbon stock-change from litter (ΔC_{LI})	tC/ha	Stock change as the absolute value of carbon losses. Default value for mangroves: 0.7	A134	4.7 of IPCC (2014)	Modifiable in Tier 2
Area drained or excavated (A)	ha	Excavated area, when present, is a subset of the drained area. User input.	N/A	-	-
Above ground biomass (AGB)	t d.m./ha	Tier 1 **data only for mangroves.** For tidal marshes and seagrass meadows, only in **Tier 2 calculations (user input necessary).** By climate and vegetation type.	A132	4.3 of IPCC (2014)	Modifiable in Tier 2
Carbon fraction (CF)	Dimensionless	Carbon fraction per t d.m. Default value for mangroves: 45.1%	N/A	4.2 of IPCC (2014)	-
Root to shoot ratio (R)	t d.m./t d.m.	Ratio of BGB to AGB. By climate and vegetation type.	A133	4.5, 4.9 and 4.10 of IPCC (2014)	-
Soil emissions linked to soil drainage (based on eq. 4.8 in IPCC [2014]) $CO_{2-SO-DR} = (A_{DR} * EF_{DR})$					
CO_2 emissions from aggregated organic and mineral soil carbon ($CO_{2\text{-}SO\text{-}DR}$)	tC/yr	Carbon emissions last for the project duration or until the top-metre of soil (SOC_{1m}) is depleted, given by SOC_{1m}/EF_{DR}.	N/A	-	-

Element	Unit	Details	Table	Table IPCC	Notes
Land area under drainage (A_{DR})	ha	User input.	N/A	-	-
Emission factor for carbon from organic or mineral soil associated with drainage (EF_{DR})	tC/ha/yr	Default value: 7.9 tC/ha/yr	N/A	4.13 of IPCC (2014)	Modifiable in Tier 2.
Soil emissions linked to coastal wetland rewetting and revegetation (Equations 4.7 and 4.9 of IPCC [2014]). $$CO_{2SO-RE} = \sum_{V,S,C} (A_{RE} * EF_{RE-CO2})_{V,S,C}$$ $$CH_{4-SO-RE} = \sum_{V} (A_{RE} * EF_{RE-CH4})_{V}$$					
CO_2 emissions are associated with rewetting, revegetation and creation activities ($CO_{2-SO-RE}$)	tC/yr	By vegetation type (v), soil type (s) and climate (c).	N/A	-	-
Area of soil that has been influenced by rewetting, revegetation and creation activities (A_{RE})	ha	Area rewetted is a subset of the drained area at the start of the project. User input.	N/A	-	-
Emission factor for CO2 from rewetted or revegetated aggregated mineral and organic soils. (EF_{RE-CO2})	tC/ha/yr	By vegetation type (v), soil type (s) and climate (c).	A140	4.12 of IPCC (2014)	Modifiable in Tier 2
CH_4 emissions associated with rewetting ($CH_{4-SO-RE}$)	kg CH_4/yr	CH**4 emissions occur only in fresh- and brackish water, where the salinity < 18 ppt.** By vegetation type (v).	A131	A4.1of IPCC (2014)	Salinity modifiable in Tier 2.
Emission factor for CH_4 from rewetted aggregated mineral and organic soils. (EF_{RE-CH4})	kg CH_4/ha/yr	By vegetation type (v) and water salinity. Default values: Fresh or brackish water; 193.7. Saline water; 0.	N/A	4.14 of IPCC (2014)	Modifiable in Tier 2.
Biomass carbon stock changes linked to coastal wetland revegetation (Equations 2.3, 2.8A of IPCC [2006]) $$C_{B-REV} = \frac{(\Delta C_{AGB} + \Delta C_{BGB} + \Delta C_{DW} + \Delta C_{LI})}{20} * A_{REV} \text{ where}$$ $$\Delta C_{AGB} = AGB * CF$$ $$\Delta C_{BGB} = \Delta C_{AGB} * R$$					
Sequestrated carbon because of	tCO_2/yr	Assumed full restoration of biomass in 20 years.	N/A	-	-

Element	Unit	Details	Table	Table IPCC	Notes
biomass gains during revegetation ($CO_{2\text{-}B\text{-}REV}$)					
Revegetated area (A_{REV})	ha	Area must be a subset of the rewetted area. User input.	N/A	-	-
Carbon stock change from above-ground biomass (ΔC_{AGB})	tC/ha	Stock change as a result of biomass accumulation.	N/A	-	Modifiable in Tier 2
Carbon stock change from below-ground biomass (ΔC_{BGB})	tC/ha	Stock change as a result of biomass accumulation	N/A	-	Modifiable in Tier 2
Carbon stock change from deadwood (ΔC_{DW})	tC/ha	Stock change as a result of biomass accumulation	A135	4.7 of IPCC (2014)	Modifiable in Tier 2
Carbon stock-change from litter (ΔC_{LI})	tC/ha	Stock change as a result of biomass accumulation	A134	4.7 of IPCC (2014)	Modifiable in Tier 2
Above ground biomass (AGB)	t d.m./ha	Tier 1 **data only for mangroves.** For tidal marshes and seagrass meadows, only in Tier 2 calculations (user input necessary).	A132	4.3 of IPCC (2014)	-
Carbon fraction (CF)	Dimensionless	Carbon fraction per **t d.m.** Default value for mangroves: 45.1%	N/A	4.2 of IPCC (2014)	-
Root to shoot ratio (R)	t d.m./t d.m.	Ratio of BGB to AGB.	Table A133	4.5, 4.9 and 4.10 of IPCC (2014)	-

Note: AGB = above-ground biomass; ADR = land area under drainage; ARE = area rewetted, revegetated or created; AREV = revegetated area; BGB = below-ground biomass; C = climate; CF = carbon fraction; CH_4 = methane; CO_2 = carbon dioxide; DW = deadwood; EF = emission factor; EFDR = emission factor for carbon emissions from drained soils; EFRE–CH_4 = emission factor for methane emissions from rewetted soils; EFRE–CO_2 = emission factor for carbon dioxide emissions from rewetted soils; IPCC = Intergovernmental Panel on Climate Change; LI = litter; R = root-to-shoot ratio; SOC = soil organic carbon; SOC1m = soil organic carbon content in the top one metre; tC = tonnes of carbon; tC/ha = tonnes of carbon per hectare; tC/ha/yr = tonnes of carbon per hectare per year; tC/yr = tonnes of carbon per year; tCO_2/yr = tonnes of carbon dioxide per year.

Sources: Authors' own elaboration based on IPCC. 2006. *2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Volume 4 – Agriculture, Forestry and Other Land Use. Kanagawa, Japan. https://www.ipcc-nggip.iges.or.jp/public/2006gl; IPCC. 2014. *2013 supplement to the 2006 IPCC guidelines for national greenhouse gas inventories: wetlands : methodological guidance on lands with wet and drained soils, and constructed wetlands for wastewater treatment*. Hayama, Japan, IPCC, Intergovernmental Panel on Climate Change.

2.4.13 WATERBODIES

Materials used to develop this module are based on methodology, guidance, and default emission factors from Chapter 7 (Wetlands) of IPCC (2019).

The inland waterbodies module addresses flooded lands, defined as water bodies where human activities have altered the surface area covered by water typically through water level regulation. This includes: (i) natural waterbodies whose hydrology has been modified, affecting water residence times and sedimentation rates and thereby altering natural greenhouse gas fluxes; and (ii) artificial waterbodies created by excavation, such as canals, ditches, and ponds.

The module covers the following computations:

- Methane emissions linked to the trophic state of the waterbody, linked to management practices and nutrient input.

Table 28 illustrates the GHGs addressed by the module, the corresponding tier level of assessment, and includes references to the equations used for GHG flux calculations, along with the relevant IPCC equations for easier cross-reference.

Table 28. Overview of greenhouse gas fluxes estimated in the waterbodies module of the Environmental eXternalities ACcounting Tool and corresponding equations

Activity		GHG	Flux description	Equation	IPCC equation
Waterbodies management	Tier 1	CH_4	Methane emissions linked to changes in the trophic state of the waterbody	39 and 40	7.11 and 7.12 of IPCC (2019)

Note: GHG = greenhouse gas; CH_4 = methane; IPCC = Intergovernmental Panel on Climate Change.

Source: Authors' own elaboration based on IPCC. 2019. 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories. Volume 4. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Methane emissions

In waterbodies management, EX-ACT only addresses CH_4 emissions. The trophic class of the waterbody (default, oligotrophic, mesotrophic, eutrophic and hypertrophic) defines the total yearly methane emissions in union with the emission factor linked to the specific waterbody targeted (freshwater ponds, canal and ditches, saline and brackish ponds). Annual CH_4 emission from waterbodies are calculated according to Equation 39 (see Section 7.3.1 and Equation 7.12 in Chapter 7 of IPCC [2019] for further details):

$$F_{CH4} = A \times EF_{CH4} \times \alpha$$

(Equation 39)[40]

where:

- F_{CH4} = total annual of CH_4 from ponds and ditches, in kg CH_4/yr;
- A = area of waterbody, in ha;
- EF_{CH4} = emission factor for other waterbody, in kg CH_4/ha/yr; and
- α = emission factor adjustment for trophic state of other waterbody, dimensionless.

EX-ACT uses EF_{CH4} for waterbodies following Table 7.12 in Chapter 7 of IPCC (2019) as reported in Table A129 in the annex. EX-ACT uses default values for the emission factor adjustment for trophic state of other waterbody estimated according to the Equation 40 and based on Equation 7.11 in Chapter 7 of IPCC (2019). The default value for α is 1 (IPCC, 2019):

$$\alpha_i = 0.26 \times [Chlo.a]_i$$

(Equation 40)[41]

where:

- α = emission factor adjustment for trophic state i of other waterbody, dimensionless;

[40] Based on Equation 7.12 of Chapter 7 of IPCC (2019).
[41] Corresponding to Equation 7.11 of Chapter 7 of .

- Chlo.a = mean annual chlorophyll-a concentration in reservoir i, in µg/L.

Table 0 details the default values for the parameters embedded in the Waterbodies module, along with the IPCC 29 table reference and in-text table reference.

Table 29. Sources of default values in the waterbodies module of the Environmental eXternalities ACcounting Tool

<table>
<tr><th>Element</th><th>Unit</th><th>Details</th><th>Table</th><th>Table IPCC</th><th>Notes</th></tr>
<tr><td colspan="6">Annual CH4 emissions from other constructed waterbodies (Equation 39)(Equation 40)
$\mathbf{F_{CH4}} = \mathbf{A} * \mathbf{EF_{CH4}} * \boldsymbol{\alpha}$ where
$\boldsymbol{\alpha} = \mathbf{0.26} * \mathbf{Chlo.a_i}$</td></tr>
<tr><td>Total CH_4 flux from waterbodies (F_{CH4})</td><td>kg CH_4/yr</td><td>-</td><td>N/A</td><td>-</td><td>-</td></tr>
<tr><td>Area of the waterbody (A)</td><td>ha</td><td>User input</td><td>N/A</td><td>-</td><td>-</td></tr>
<tr><td>Emission factor for CH_4 (EF_{CH4})</td><td>kg CH_4/ha/yr</td><td>Based on climate zone and type of constructed waterbody.</td><td>A129</td><td>7.9, 7.12, and 7.15 of IPCC (2019)</td><td>Modifiable in Tier 2.</td></tr>
<tr><td>EF adjustment factor for waterbody trophic state (α)</td><td>Dimensionless</td><td>Default value: 1.0
Other waterbody classes: Default value based on trophic class.</td><td>A130</td><td>7.11 of IPCC (2019)</td><td>Modifiable in Tier 2.</td></tr>
<tr><td>Chlorophil-a concentration in waterbody ($Chlo.a_i$)</td><td>µg/L</td><td>Mean annual concentration in waterbody i.</td><td>A131</td><td>-</td><td>Modifiable in Tier 2.</td></tr>
</table>

Note: GHG = greenhouse gas; CH_4 = methane; IPCC = Intergovernmental Panel on Climate Change; ha = hectares; kg CH_4/ha/yr = kilograms of methane per hectare per year; kg CH_4/yr = kilograms of methane per year.

Source: Authors' own elaboration based on Volume 4 of IPCC. 2019. 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html and Volume 4 of IPCC. 2006. 2006 IPCC Guidelines for National Greenhouse Gas Inventories. Kanagawa, Japan. www.ipcc-nggip.iges.or.jp/public/2006gl

2.4.14 SMALL SCALE FISHERIES

Materials used to develop this module can be found in Maestripieri *et al.* (2025).

Small-scale fisheries can be defined as those fisheries characterised by a limited amount of production, taking place at a family or community level (FAO, 2022c). In line with the data sources used to develop the methodology of this module (FAO, 2023), small-scale fisheries can be further defined by their limited (or lack of) fresh storage, close shoreline or short travel distance, smaller vessels, less-formally engaged economic activities, fewer crew, use of passive gears, among other characteristics.

The module covers the following computations:

1. **Fuel related emissions** linked to the fuel use intensity (FUI), which is the litres of fuel used to land one metric tonne (t) of catch
   - CO_2, CH_4, N_2O emissions linked to fossil fuel use, calculated according to the methodology illustrated in Section 2.1.
2. **Refrigeration emissions** linked to leakage of HCFC refrigerants or the on-shore production of ice

- HCFC emissions linked to the leakage of the refrigerant R22, calculated according to the methodology illustrated in Section 2.1.
- $CO2eq$ emissions linked to electricity use for the production of ice, calculated according to the methodology illustrated in Section 2.1.

Emissions are computed for each appraisal scenario in line with Equation 41:

$$E_{scenario} = \left(FUI * Q * EF_{fuel}\right) + \left(Pct_{ice} * Q * \ EF_{ice}\right) + ((Pct_R * Q) * L * EF_R)$$

(Equation 41)

where:

- $E_{scenario}$= the total emissions for each appraisal scenario;
- Q = quantity of catch in tonnes;
- FUI = the fuel use intensity;
- EF_{fuel} = the emission factor for fuel;
- Pct_{ice} = the percentage of catch frozen with onshore produced ice;
- EF_{ice} = the emission factor for ice;
- Pct_R = the percentage of catch refrigerated on board;
- L = the leakage of the refrigerant; and
- EF_R = the emission factor for the refrigerant used.

Table 30 illustrates the GHGs addressed by the module, with the corresponding level of assessment, i.e. Tier 1 or Tier 2 and includes in-text references to the equations used for the GHG fluxes calculations, as well as the corresponding IPCC equations for easier cross-reference.

Table 30. Overview of greenhouse gas fluxes estimated in the small-scale fisheries module of the Environmental eXternalities ACcounting Tool and corresponding equations

Activity		GHG	Flux description	Equation	IPCC equation
Fuel use in fishing operations	Tier 1	CO_2, CH_4, N_2O	Emissions from fuel required to land one tonne of catch	41	3.5.1 of IPCC (2006)
Refrigeration in fishing operations	Tier 1	HCFC	Emissions from refrigerant leakage	41	7.13 of IPCC (2006)
	Tier 1	CO_2eq	Emissions from electricity required to produce ice per tonne of catch	41	2.1 of IPCC (2006)

Note: GHG = greenhouse gas; CO_2 = carbon dioxide; N_2O = nitrous oxide; CH_4=methane; IPCC = Intergovernmental Panel on Climate Change; HCFC = hydrochlorofluorocarbons.

Source: Authors' own elaboration based on IPCC. 2006. *2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Volume 4 – Agriculture, Forestry and Other Land Use. Kanagawa, Japan. https://www.ipcc-nggip.iges.or.jp/public/2006gl

Fuel-related emissions

EX-ACT estimates CO_2, CH_4, N_2O emissions from the catch, i.e. multiplying the quantity of fish caught by a standardized fuel use intensity (FUI). Unmotorized vessels are included for user reference and activity tracking, but they are assumed to not consume fuel. Default FUI data provided by the tool is provided in Table A141.

On-board refrigerant leakage

Fisheries are known to use storage to preserve fresh catch. Especially among large-scale fisheries, refrigeration is used, and leakage of compressed refrigerants presents a non-zero impact on global warming. EX-ACT accounts for R-22 leakage in line with Maestripieri *et al.(*2025). If users have special knowledge of other refrigeration systems in use, they can input alternative refrigerant leakage estimates in the Tier 2 portion of the module.

Greenhouse gas emissions from ice produced on land

The onshore production of ice for fish preservation is characteristic of many traditional fishing settings. In EX-ACT, ice production relies on electricity consumption generating GHG emissions. The emission factors and the calculation of electricity emissions is in line with the electricity section of the energy module. EX-ACT considers that the average ratio ice/fish is about 2.8 (kg of ice per kg of fish) and electricity consumption is by default 60 kWh for the production of 1 tonne of ice (Ghosh *et al.*, 2014; Sciortino, 2010).

Table 31 details the default values for the parameters embedded in the small scale and large-scale fisheries modules, along with the IPCC table reference and in-text table reference.

Table 31. Sources of default values in the small- and large fisheries modules of the Environmental eXternalities ACcounting Tool

Emissions from fisheries from fuel and refrigeration (Equation 41)

$$\mathbf{C_{balance}} = \sum\left(\frac{\mathbf{FUI*Q*EF_{fuel}*GWP}}{\mathbf{1000}}\right) + \frac{\mathbf{Pct_R*Q*L*GWP_{Refrigerant}}}{\mathbf{1000}} + (\mathbf{Pct_{ice} * Q * EF_{ice}}) \text{ where}$$

$$\mathbf{EF_{ice}} = \mathbf{Q_{ice} * E_{ice} * \frac{EF_{el}}{1000}} \text{ and}$$

$$\mathbf{EF_{fuel}} = \frac{\boldsymbol{\rho}}{\mathbf{10^9}} * \mathbf{NCV * EF_{GHG}}$$

Element	Unit	Details	Table	Table IPCC	Notes
Carbon balance ($C_{balance}$)	tCO_2eq/yr	-	N/A	-	-
Quantity of catch (Q)	t/yr	User input.	N/A	-	-
Fuel use intensity (FUI)	L/t catch	By fishery type and gear category.	A141 and A142		
Emission factor for fuel (EF_{fuel})	$tGHG/m^3$	Emission factor per fuel type and GHG species. Default values: based on average of off-road diesel types in IPCC table 3.3.1.	A146	3.3.1 of IPCC (2006)	Modifiable in Tier 2.
Percentage of catch frozen with onshore produced ice (Pct_{ice})	Dimensionless	User input	N/A	-	-
Emission factor for ice-preserved catch (EF_{ice})	tCO2eq/t ice-preserved catch	On-shore produced ice. By quantity of ice, ice production electricity use, and operation margin of the local electricity.	N/A	-	Modifiable in Tier 2.
Percentage of catch refrigerated on board (Pct_R)	Dimensionless	User input	N/A	-	-
Refrigerant leakage (L)	kg/t catch	Default value: 0.08	N/A	-	Modifiable in Tier 2.
Quantity of ice (Q_{ice})	t ice/t catch	Default value: 2.80 t ice/t catch.	N/A	-	Modifiable in Tier 2.
Ice production electricity use (E_{ice})	kWh/t ice	Default value: 60.00 kWh/t ice	N/A	-	Modifiable in Tier 2.
Emission factor of electricity (EF_{el})	tCO_2eq/MWh	EF for the operating margin by country where the activity takes place.	A145	-	-

Element	Unit	Details	Table	Table IPCC	Notes
Fuel density (ρ)	kg/m³	Per fuel type.	A146	-	-
Net Calorific Value (NCV)	TJ/Gg	Per fuel type.	A146	1.2 of IPCC (2006)	-
Emission GHG intensity of fuel (EF_{GHG})	kg/TJ	Per fuel type and GHG species. Denoted as emission factor in the annex and IPCC tables.	A146	3.1.1 of IPCC (2006)	-

Note: GHG = greenhouse gas; CO_2 = carbon dioxide; N_2O = nitrous oxide; CH_4 = methane; eq. = equation; IPCC = Intergovernmental Panel on Climate Change.

Source: Authors' own elaboration based on IPCC. 2006. 2006 IPCC Guidelines for National Greenhouse Gas Inventories. Volume 2. Kanagawa, Japan. www.ipcc-nggip.iges.or.jp/public/2006gl

2.4.15 LARGE SCALE FISHERIES

Materials used to develop this module can be found in Maestripieri *et al.* (2025).

Large scale fisheries can be generally identified as those fisheries working at an industrial level (FAO, 2010) with vessels 12 metres in length or greater (FAO, 2022b). The module covers the following computations:

1. **Fuel related emissions** linked to the fuel use intensity (FUI), which is the litres of fuel used to land one metric tonne (t) of catch
   - CO_2, CH_4, N_2O emissions linked to fossil fuel use.
2. **Refrigeration emissions** linked to leakage of HCFC refrigerants or the onshore production of ice
   - HCFC emissions linked to the leakage of the refrigerant R22.
   - CO_2eq emissions linked to electricity use for the production of ice.

The methodology linked to calculation emissions is the same of the small-scale fisheries module. Emission factors for large-scale fisheries FUIs are reported in Table A142.

2.4.16 AQUACULTURE MANAGEMENT

Materials used to develop this module can be found in Chapter 4 of IPCC (2014) and in Chapter 7 of IPCC (2019). Aquaculture is the farming of aquatic organisms, including fish, molluscs, crustaceans and aquatic plants. The module covers the following computations:

- **Nitrous oxide emissions** linked to the production of fish.

GHG emissions associated with feed can be significant, stemming from the manufacturing of feed and transportation (Avadí *et al.*, 2014; Fréon *et al.*, 2017; Winther *et al.*, 2009). As a result, EX-ACT also proposes to include them in the calculations under the Inputs module of the tool.

Table 3 illustrates the GHGs addressed by the module, with the corresponding level of assessment, i.e. Tier 1 or Tier 2 and includes in-text references to the equations used for the GHG fluxes calculations, as well as the corresponding IPCC equations for easier cross-reference.

Table 32. Overview of greenhouse gas fluxes estimated in the aquaculture module of the Environmental eXternalities ACcounting Tool and corresponding equations

Activity		GHG	Flux description	Equation	IPCC equation
Fish production and management	Tier 1	N_2O	Nitrogen emissions linked to management of fisherlings	See Table 33	4.10 of IPCC (2014)

Note: GHG = greenhouse gas; N_2O = nitrous oxide; IPCC = Intergovernmental Panel on Climate Change.

Source: Authors' own elaboration based on IPCC. 2014. *2013 supplement to the 2006 IPCC guidelines for national greenhouse gas inventories: wetlands : methodological guidance on lands with wet and drained soils, and constructed wetlands for wastewater treatment*. Hayama, Japan, IPCC, Intergovernmental Panel on Climate Change.

Fish management emissions

Nitrous oxide a by-product of fish rearing, resulting from the nitrification and denitrification of ammonia excreted by the fish. Thus, it is possible to quantify N_2O emission from the fish production data, when aquaculture systems are in the stocked phase. EX-ACT uses the global emission factor, 0.00169 tonnes N_2O-N per tonne fish produced, provided by IPCC (see Section4.3.2 of IPCC [2014] for further details).

Table 33 details the default values for the parameters embedded in the aquaculture module, along with the IPCC table reference and in-text table reference.

Table 33. Sources of default values in the aquaculture module of the Environmental eXternalities ACcounting Tool

Element	Unit	Details	Table	Table IPCC	Notes
N_2O emissions from aquaculture (Equation 4.10, IPCC [2014]) $\mathbf{N_2O - N_{AQ} = F_F * EF_F}$					
Direct N_2O emissions from aquaculture (N_2O-N_{AQ})	tN_2O-N/yr	-	N/A	-	-
Annual fish production (F_F)	t/yr	User input.	N/A	-	-
Emission factor for N_2O-N emissions from fish production (EF_F)	tN_2O-N/t production	Default value: 0.00169	N/A	4.15 of IPCC (2014)	Modifiable in Tier 2.

Note: GHG = greenhouse gas; CO_2 = carbon dioxide; N_2O = nitrous oxide; CH_4 = methane; eq. = equation; IPCC = Intergovernmental Panel on Climate Change; N/A = not applicable; N_2O-N = nitrous oxide–nitrogen; t = tonne; tN_2O-N = tonnes of N_2O–N; yr = year.

Source: Authors' own elaboration based on IPCC. 2014. 2013 supplement to the 2006 IPCC guidelines for national greenhouse gas inventories: wetlands : methodological guidance on lands with wet and drained soils, and constructed wetlands for wastewater treatment. Hayama, Japan, IPCC, Intergovernmental Panel on Climate Change.

2.4.17 INPUTS

Materials used to develop this module can be found in Chapter 11 (N_2O emissions from managed soils, and CO_2 emissions from lime and urea application) of IPCC (2019) and Lal (2004).

The Inputs module covers the use of agricultural inputs including lime, urea and other synthetic and organic fertilizers, as well as feed. The module covers the following computations:

1. Direct emissions from fertilizer use:
   - CO_2 emissions from lime and urea application, calculated according to the methodology illustrated in Section 2.1.
   - N_2O emissions from synthetic and organic nitrogen-based fertilizers (except manure management, treated in livestock module), calculated according to the methodology illustrated in Section 2.

2. Indirect upstream emissions (in tCO_2eq) from fertilizers and pesticides, calculated according to the

methodology illustrated in Section 2.1.

3. Emissions (in tCO_2eq) from feed use, calculated according to the methodology illustrated in Section 2.1.

Table 34 illustrates the GHGs addressed by the module, with the corresponding level of assessment, i.e. Tier 1 or Tier 2 and includes in-text references to the equations used for the GHG fluxes calculations, as well as the corresponding IPCC equations for easier cross-reference.

Table 34. Overview of greenhouse gas fluxes estimated in the inputs module of the Environmental eXternalities ACcounting Tool

Activity		GHG	Flux description	Equation	IPCC equation
Application of agricultural inputs	Tier 1	CO_2	Emissions from lime and urea application	See Table 35	11.12 and 11.13 of IPCC (2019)
	Tier 1	N_2O	Direct emissions from the application of nitrogen-based fertilizers	See Table 35	11.1 of IPCC (2019)
	Tier 1	CO_2eq	Indirect upstream emissions linked to inputs production	16	2.6 of IPCC (2019)
Use of animal feed	Tier 1	CO_2eq	Emissions linked to the life cycle of feeds	16	2.6 of IPCC (2019)

Note: GHG = greenhouse gas; CO_2 = carbon dioxide; N_2O = nitrous oxide; CH_4 = methane; eq. = equation; IPCC = Intergovernmental Panel on Climate Change.

Source: Authors' own elaboration based on IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Volume 4 – Agriculture, Forestry and Other Land Use. Geneva, Switzerland. https://www.ipcc-nggip.iges.or.jp/public/2019rf/vol4.html

CO_2 emissions from agricultural inputs application

Liming is the addition of carbonates to soils in the form of calcic limestone or dolomite, which lead to CO_2 emissions from carbonate dissolution. CO_2 emissions are estimated from the quantity of lime applied on the soil associated with a default emission factors, i.e. 0.12 tC/t for limestone and 0.13 tC/t for dolomite (IPCC, 2019).

The application of urea ($CO(NH_2)_2$) to agricultural soils results in direct CO_2 emissions arising from the breakdown of urea into ammonium and carbon dioxide. These emissions are estimated in proportion to the quantity of urea applied, using a default emission factor of 0.2 t C per tonne of urea, which reflects the carbon content of urea on a molecular-weight basis. This coefficient represents the fraction of carbon in urea that is released as CO_2 once the compound decomposes in soil conditions, following the standardized approach outlined in the IPCC (2019).

CO_2 emission from animal feed. EX-ACT allows users to specify an emission factor (through Tier 2) to be used in the calculation of feed emissions whenever livestock feed is created and manufactured outside the perimeter of the project. If feed is instead produced through cultivation or sourced from grasslands within the project's perimeter, users should include the related information in the corresponding sections of the tool.

N_2O emissions from nitrogen application on managed soils

This section covers direct N_2O emissions from: (i) synthetic N fertilizers, (ii) N fertilizer in irrigated rice systems, and (iii) organic fertilizers. It excludes N_2O emissions from the manure left on pasture as it is already covered in the Livestock management module. Emissions are calculated based on amount of N-fertilizer applied and an emission factor associated with the type of input (see Table A143 in the annex).

Emissions from production, transportation, storage and transfer of agricultural inputs

Embodied GHG emissions associated with the production, transportation storage and transfer of agricultural chemicals are automatically accounted for based on the amount of inputs specified and emission factors from Lal (2004), reported in Table A144 in the annex.

Table 35 details the default values adopted in the inputs module, along with the IPCC table reference and in-text table reference.

Table 35. Sources of default values in the Inputs module of the Environmental eXternalities ACcounting Tool

Element	Unit	Details	Table	Table IPCC	Notes
CO_2 emissions from lime application (Equation 11.12 of IPCC [2006]) $\mathbf{CO_2 - C = (M_{Limestone} * EF_{Limestone}) + (M_{Dolomite} * EF_{Dolomite})}$					
Annual C emissions from lime applications (CO_2-C)	tC/yr	-	N/A	-	-
Annual application of limestone or dolomite (M)	t/yr	Limestone: $CaCO_3$ Dolomite: $CaMg(CO_3)_2$	N/A	-	-
Emission factor (EF)	tC/t limestone or dolomite	Default values: Limestone: 0.12 Dolomite: 0.13	N/A	-	Modifiable in Tier 2.
CO_2 emissions from urea application (Equation 11.12 of IPCC [2006]) $\mathbf{CO_2 - C = M * EF}$					
Annual C emissions from urea applications (CO_2-C)	tC/yr	-	N/A	-	
Annual application of urea (M)	t/yr	-	N/A	-	
Emission factor (EF)	tC/t urea	Default value: 0.20 Based on the carbon content of urea ($CO(NH_2)_2$)	N/A	-	
N_2O emissions from nitrogen-based fertilizers application (Equation 11.1of IPCC [2019]) $\mathbf{N_2O - N_{Inputs} = (F_{SN} + F_{ON}) * EF_1 + (F_{SN} + F_{ON}) * EF_{1FR}}$					
(N_2O-N_{inputs})	kg N_2O-N/yr	-	N/A	-	-
Amount of synthetic fertilizer N applied to soils (F_{SN})	kg N/yr	-	N/A	-	-
Amount of organic N additions applied to soils (F_{ON})	kg N/yr	Including animal manure, compost, sewage sludge and other organic N additions	N/A	-	-
Emission factor for N inputs (EF_1)	kg N_2O-N/kg N input	-	A143	11.1 of IPCC (2019)	Modifiable in Tier 2.
Emission factor for N inputs to flooded rice (EF_{1FR})	kg N_2O-N/kg N input	-	N/A	11.1 of IPCC (2019)	Modifiable in Tier 2.
Indirect CO_2eq upstream emissions linked to input production (Equation 16) **$CO_2eq = F_i * EF_i$**					
CO_2eq emissions (CO_2eq)	tCO_2eq/yr	-	N/A	-	-
Annual application of input(i) (F_i)	t/yr	Input of input (i)	N/A	-	-

Element	Unit	Details	Table	Table IPCC	Notes
Emission factor (EF_i)	tCO_2eq/t input(i)	EF for carbon emissions corresponding to input (i).	A144	-	Modifiable in Tier 2.
CO_2eq emissions linked to the production of animal feeds (Equation 16) **$CO_2eq = F * EF$**					
CO_2eq emissions (CO_2eq)	tCO_2eq/yr	-	N/A	-	-
Annual consumption of animal feed (F)	t/yr	Only if not accounted for with any other EX-ACT module.	N/A	-	-
Emission factor (EF)	tCO_2eq/t feed	Tier 1 values not available.	N/A	-	Only with Tier 2.

Note: GHG = greenhouse gas; CO_2 = carbon dioxide; CO_2eq = carbon dioxide equivalent; IPCC = Intergovernmental Panel on Climate Change; N/A = not applicable; t/yr = tonne per year; tCO_2eq/yr = tonne of carbon dioxide equivalent per year.

Sources: Authors' own elaboration based on IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Volume 4 – Agriculture, Forestry and Other Land Use. Geneva, Switzerland. https://www.ipcc-nggip.iges.or.jp/public/2019rf/vol4.html; IPCC. 2006. *2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Volume 4 – Agriculture, Forestry and Other Land Use. Kanagawa, Japan. https://www.ipcc-nggip.iges.or.jp/public/2006gl

2.4.18 ENERGY

Material used to develop this module can be found in Volume 2 of IPCC (2006) on energy (fuel and solid biofuels) and in UNFCCC's list of methodologies for estimating emissions from energy consumption assembled by the International Financial Institutions (IFI) Technical Working Group (UNFCCC, 2021).

The energy module covers the use of electricity and fuel, including liquid, gaseous and solid biofuels. The module covers the following computations:

1. GHG emissions from electricity consumption:
   - CO_2 emissions from electricity use based on grid emission factors, calculated according to the methodology illustrated in Section 2.1.
2. GHG emissions from fuel consumption:
   - CO_2, CH_4 and N_2O emissions from liquid, gaseous and solid biofuels, calculated according to the methodology illustrated in Section 2.1.

The Environmental eXternalities ACcounting Tool estimates GHG emissions from (i) electricity consumption and (ii) fuel consumption using the generic methodology for non-CO_2 emissions of IPCC (2019) (see Equation 2.6) with default emission factors provided in Tables A145 and A146 in the annex.

Table 36 illustrates the GHGs addressed by the module, with the corresponding level of assessment, i.e. Tier 1 or Tier 2 and includes in-text references to the equations used for the GHG fluxes calculations, as well as the corresponding IPCC equations for easier cross-reference.

Table 36. Overview of greenhouse gas fluxes estimated in the energy module of the Environmental eXternalities ACcounting Tool and corresponding equations

Activity		GHG	Flux description	Equation	IPCC equation
Electricity consumption	Tier 1	CO_2eq	Emissions linked to grid electricity use	16	2.6 of IPCC (2019)
Energy consumption	Tier 1	CO_2, CH_4, N_2O	Emissions linked to the use of fossil fuels	16	2.1, 2.2, 3.2.1 and 3.2.3 of IPCC (2006)

Note: GHG = greenhouse gas; CO_2 = carbon dioxide; N_2O = nitrous oxide; CH_4 = methane; CO_2eq = carbon dioxide equivalent; IPCC = Intergovernmental Panel on Climate Change.

Sources: Authors' own elaboration based on IPCC. 2019. 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories. Volume 4. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html; IPCC. 2006. 2006 IPCC Guidelines for National Greenhouse Gas Inventories. Volume 4. Kanagawa, Japan. www.ipcc-nggip.iges.or.jp/public/2006gl; IPCC. 2006. 2006 IPCC Guidelines for National Greenhouse Gas Inventories. Kanagawa, Japan. www.ipcc-nggip.iges.or.jp/public/2006gl

CO_2eq emissions from electricity consumption

For electricity consumption, EX-ACT proposes two sets of emission factors related to the energy grid following the Harmonized IFI Default Grid Factors (UNFCCC, 2021). The default is the Operating Margin (OM) emission factor, representing electricity generation from existing power plants with the highest variable operating costs to dispatch the electricity across the system. Users can also choose to adopt a Combined Margin (CM) emission factor, which additionally accounts for the annual emission intensities of new electricity generation projected over the next 8 years under the Stated Policy Scenario (STEPS) of the most recent IEA World Economic Outlook. As a result, CM emission factors thus take stock of projected changes in emissions of the electricity generation systems. Default emission factors are provided in Table A145 in the annex.

Greenhouse gas emissions from fuel consumption

The Environmental eXternalities ACcounting Tool computes CO_2, CH_4 and N_2O emissions linked to the use of fossil fuels, with default emission factors described in Table A146 in the annex. The tool relies on IPCC (2019) default net calorific values, which indicate how many terajoules are generated per unit mass or volume of a given fuel, and on corresponding emission factors for CO_2, CH_4 and N_2O. For solid fuels, such as wood, peat or charcoal, the user-provided mass of fuel is sufficient, and the tool disregards the CO_2 sink generated by biomass-related carbon removal during tree growth. For liquid and gaseous fuels, which are usually reported in volume units, EX-ACT uses externally sourced fuel-density values (available in Table A146 in the annex) to convert liters or cubic meters into mass, enabling the same energy-based calculation approach.

Table 37 details the default values adopted in the energy module, along with the IPCC table reference and in-text table reference.

Table 37. Sources of default values in the energy module of the Environmental eXternalities ACcounting Tool

Element	Unit	Details	Table	Table IPCC	Notes
CO_2eq emissions linked to grid electricity use (Equation 16) $$\mathbf{E = F * EF_C * (1 + T)}$$					
Annual emissions from electricity use (CO_2)	tCO_2eq/yr	-	N/A	-	-
Annual electricity use (F)	MWh/yr	User input.	N/A	-	-
Emission factor (EF_C)	tCO_2eq/MWh	Tier 1: **EF for the operating margin in project country (C).** Default value for renewable energy: 0.	A145	-	Value, country of origin, and margin type modifiable in Tier 2.
Fraction of transport losses (T)	-	Default value: 0.1	N/A		Modifiable in Tier 2.
Emissions linked to the use of solid and liquid fuels (Equation 16) $$\mathbf{E = \sum F * EF_{GHG}}$$					
Annual GHG emissions from fuel consumption (E)	tGHG/yr	Emissions corresponding to fuel type and GHG species.	N/A	-	-
Annual fuel consumption (F)	m3/yr t d.m./yr	User input. Unit differs per fuel type: Liquid fuel: m3/yr Solid fuel: t d.m./yr	N/A	-	-
Emission factor for burnt fuel (EF)	tCO**2/unit fuel** tCH**4/unit fuel** tN_2O/unit fuel	Specific EFs for each GHG species (CO_2, CH_4, and N_2O) from fuel combustion.	A146	-	Modifiable in Tier 2.

Note: GHG = greenhouse gas; CO_2 = carbon dioxide; N_2O = nitrous oxide; CH_4=methane; eq. = equation; IPCC = Intergovernmental Panel on Climate Change; N/A = not applicable; d.m. = dry matter; EFC = emission factor for electricity (carbon intensity of grid electricity); MWh/yr = megawatt hour per year.

Sources: Authors' own elaboration based on IPCC. 2019. 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories. Volume 4. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html; IPCC. 2006. 2006 IPCC Guidelines for National Greenhouse Gas Inventories. Volume 4. Kanagawa, Japan. www.ipcc-nggip.iges.or.jp/public/2006gl; IPCC. 2006. 2006 IPCC Guidelines for National Greenhouse Gas Inventories. Kanagawa, Japan. www.ipcc-nggip.iges.or.jp/public/2006gl

2.4.19 IRRIGATION

Materials used to develop this module come from Lal (2004) and UNFCCC (2021).

The irrigation module estimates GHG emissions from the installation of new irrigation systems, as well as emissions linked to energy use from the operational phase of irrigation systems. The adopted definitions have been developed following expert consultations based on Lal (2004) irrigation categories.

The module covers the following computations:

1. Emissions from the installation of irrigation systems:
   - GHG emissions associated with the installation of different types of irrigation systems calculated according to the methodology illustrated in Section 2.1.
2. Emissions from the energy use related to the operational phase of irrigation:
   - GHG emissions from fuel consumption: CO_2, CH_4 and N_2O emissions from liquid or gaseous fuels.
   - GHG emissions from electricity consumption: CO_2 emissions from electricity use based on grid emission factors.

Table 38 illustrates the GHGs addressed by the module, with the corresponding level of assessment, i.e. Tier 1 or Tier 2 and includes in-text references to the equations used for the GHG fluxes calculations, as well as the corresponding IPCC equations for easier cross-reference.

Table 38. Overview of greenhouse gas fluxes estimated in the irrigation module of the Environmental eXternalities ACcounting Tool and corresponding equations

Activity		GHG	Flux description	Equation	IPCC equation
Cropland irrigation	Tier 1	CO_2eq	Installation emissions from new irrigation systems	16	2.6 of IPCC (2019)
	Tier 1	CO_2eq	Operational emissions from existing irrigation systems	42 and 43	-

Note: GHG = greenhouse gas; CO_2eq = carbon dioxide equivalent; IPCC = Intergovernmental Panel on Climate Change.

Source: Authors' own elaboration based on IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Volume 4. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html; IPCC. 2006. *2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Volume 4. Kanagawa, Japan. www.ipcc-nggip.iges.or.jp/public/2006gl; IPCC. 2006. 2006 IPCC Guidelines for National Greenhouse Gas Inventories. Kanagawa, Japan. www.ipcc-nggip.iges.or.jp/public/2006gl

Greenhouse gas emissions from installation of irrigation systems. GHG emissions from the installation of new irrigation systems are calculated using the generic methodology for non-CO_2 emissions described in SECTION2.1.2. Default emission factors from Lal (2004) are reported in Table A147 in the annex.

Greenhouse gas emissions from the operational phase of irrigation systems. The operational phase of groundwater pumping systems encompasses the processes involved in the abstraction, conveyance and application of water, considering these are the main components of energy use associated with irrigated agriculture (Rothausen and Conway, 2011). The total energy required (ER) for an irrigation system is estimated in Equation 42:

$$ER = \frac{ERh \times GIWR \times (PR + D)}{Efficiency}$$

(Equation 42)[42]

where:

- ER = energy required to pump water from groundwater well, in MWh per total growing season per hectare;
- ERh = energy required to lift 1 m^3 of water over 1 m height, in MWh or m^3 per metre;
- GIWR = gross irrigation water requirement, in m3/ha/growing season;
- PR = pressure requirement of the irrigation system including energy losses, in metres head;
- D = depth of the well, in metres;
- Efficiency = efficiency of the pumping system, in percentage.

[42] Equation from Rothausen and Conway (2011)

Once the energy required is calculated, it is possible to estimate the GHG emissions with Equation 43:

$$GHG = ER \times EF \times S$$

(Equation 43)

where:

- GHG = the GHG emissions from groundwater pumping, in tCO_2eq over a growing period;
- ER = energy requirement, in MWh per growing season per ha;
- EF = emission Factor of energy or fuel used, in tCO_2eq/MWh or tCO_2eq/m^3, respectively;
- S = crop's area under groundwater irrigation, in ha.

Fossil fuels emission factors are based on IPCC (2006), while the emission factors for the electricity grid are country-specific and have been reported in the energy section.

Table 39 details the default values adopted in the *Irrigation* module, along with the IPCC table reference and in-text table reference.

Table 39. Sources of default values in the irrigation module of the Environmental eXternalities ACcounting Tool

Element	Unit	Details	Table	Table IPCC	Notes
Greenhouse gas emissions from installation of new irrigation systems (Equation 16) $\mathbf{E = \frac{S * EF}{1000}}$					
Emissions from irrigation system installation (E)	tCO_2eq	-	N/A	-	-
Area under groundwater irrigation (S)	ha	User input.	N/A	-	-
Emission factor (EF)	kg CO_2eq/ha	Per irrigation system type. Includes	A147	-	Modifiable in Tier 2.
Greenhouse gas emissions from the operational phase of irrigation systems (Equation 42 and 43) $\mathbf{GHG = ER \times EF \times S}$ **where** $\mathbf{ER = \frac{ERh \times GIWR \times (PR+D)}{\eta} * (1 + T)}$ **where;** **PR = Average pressure(i) * 10.19, and** $\mathbf{ERh_{fuel} = \frac{9.81}{NCV/\rho} * 10^{-3}}$					
GHG emissions from irrigation system operation (GHG)	tGHG/yr	Emissions corresponding to fuel or energy type and GHG species.	N/A	-	-
Total energy required (ER)	MWh/ha/yr	Energy required to pump water from groundwater well.	N/A	-	-
Emission factor of energy or fuel used (EF)	tCO_2eq/MWh $tCO_2eq/m3$	Energy, Tier 1: EF for the operating margin in project country.	A145 and A146	-	Value and country of origin modifiable in Tier 2.
Area under groundwater irrigation (S)	ha	User input.	N/A	-	-
Energy required to lift 1 m^3 of water over 1 m height (ERh)	MWh/m m^3/m	Default value: $ERh_{electricity}$ = $2.725*10^{-6}$ MWh/m ERh_{fuel} per fuel type.	N/A	-	-

Element	Unit	Details	Table	Table IPCC	Notes
Gross irrigation water requirement (GIWR)	m^3/ha/yr	User input in mm/yr	N/A	-	-
Pressure requirement of the irrigation system (PR)	m head	Including energy losses	N/A	-	Modifiable in Tier 2.
Average pressure(i)	bar	Default value per irrigation system(i)	-	-	Modifiable in Tier 2.
Well depth (D)	m	User input	N/A	-	
Efficiency of the pumping system (η)	Dimensionless	Efficiency as a percentage. Default value: 45%	N/A	-	Modifiable in Tier 2.
Fraction of transport losses (T)	Dimensionless	Applied to ER only when electricity is used. Default value for electricity losses: 0.1 Default value when using fuel: 0	N/A	-	Modifiable in Tier 2.
9.81	MJ	Energy required to lift 1 m^3 of water 1 m in height.	N/A	-	-
Fuel net calorific value (NCV)	kg/m^3	Per fuel type.	A146	1.2 of IPCC (2006)	-
Fuel density (ρ)	TJ/Gg	Per fuel type.	A146	-	-
10.19	Dimensionless	Bar to m head conversion factor.	N/A	-	-

Note: GHG = greenhouse gas; IPCC = Intergovernmental Panel on Climate Change; N/A = not applicable; tCO_2eq = tonnes of carbon dioxide equivalent; ha = hectare; kg CO_2eq/ha = kilograms of carbon dioxide equivalent per hectare; tGHG/yr = tonnes of greenhouse gases per year; MWh/ha/yr = megawatt-hours per hectare per year; tCO_2eq/MWh = tonnes of carbon dioxide equivalent per megawatt-hour; tCO_2eq/m^3 = tonnes of carbon dioxide equivalent per cubic metre; MWh/m = megawatt-hours per metre; m^3/m = cubic metres per metre; m^3/ha/yr = cubic metres per hectare per year; m head = metres of head; bar = unit of pressure in bars; m = metre; dimensionless = unit without physical dimension; MJ = megajoule; kg/m^3 = kilograms per cubic metre; TJ/Gg = terajoules per gigagram.

Source: Authors' own elaboration based on IPCC. 2006. *2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Volume 4. Kanagawa, Japan. www.ipcc-nggip.iges.or.jp/public/2006gl; IPCC. 2006. 2006 IPCC Guidelines for National Greenhouse Gas Inventories. Kanagawa, Japan. www.ipcc-nggip.iges.or.jp/public/2006gl

2.4.20 POST-HARVEST EMISSIONS (DOWNSTREAM)

Materials used to develop this section of the tool come from IPCC (, 2007, 2013, 2019, 2021), UNFCCC (2021), and Berneers-Lee and Hoolohan (2012).

The downstream section of the tool, represented by the packaging, processing, storage and transportation modules, estimates GHG emissions from energy use, refrigeration leakage and material production linked to agrifood commodities.

The module covers the following computations:

1. Emissions from the energy use:
   - GHG emissions from electricity consumption: CO_2 emissions from electricity use based on grid emission factors, calculated according to the methodology illustrated in Section 2.4.18.
   - GHG emissions from fuel consumption: CO_2, CH_4 and N_2O emissions from liquid, gaseous and solid biofuels, Section 2.4.18.
2. Emissions from refrigerant leakage:
   - Hydrofluorocarbon (HFC), hydrofluoro-olefins (HFO) and hydrochlorofluoro-olefins (HCFO),

emissions linked to leakage of refrigerants in cold storage, calculated according to the methodology illustrated in Section 2.3.4.

3. Emissions from material production:
   - GHG emissions from the production of materials used in packaging processes linked to product preparation, calculated according to the methodology illustrated in Section 2.4.18.

Table 40 illustrates the GHGs addressed by the module, with the corresponding level of assessment i.e. Tier 1 or Tier 2 and includes in-text references to the equations used for the GHG fluxes calculations, as well as the corresponding IPCC equations for easier cross-reference.

Table 40. Overview of greenhouse gas fluxes estimated in the downstream section of the Environmental eXternalities ACcounting Tool and corresponding equations

Activity		GHG	Flux description	Equation	IPCC equation
Energy use in processing, packaging, storage and transportation	Tier 1	CO_2eq	Emissions linked to grid electricity use	16	2.6 of IPCC (2019)
	Tier 1	CO_2, CH_4, N_2O	Emissions linked to the use of fossil fuels	16	2.1 and 2.2, 3.2.1 and 3.2.3 of IPCC (2006)
Refrigerant leakage in storage	Tier 1	HFC, HFO, HCFO	Emissions from refrigerant leakage	24 and 25	7.13 of IPCC (2006) 2.6 of IPCC (2019)
Material production in packaging	Tier 1	CO_2eq	Emissions linked to production of packaging materials	16	2.6 of IPCC (2019)

Note: GHG = greenhouse gas; IPCC = Intergovernmental Panel on Climate Change; N/A = not applicable; CO_2eq = carbon dioxide equivalent; CO_2 = carbon dioxide; CH_4 = methane; N_2O = nitrous oxide; HFC = hydrofluorocarbons; HFO = hydrofluoroolefins; HCFO = hydrochlorofluoroolefins.

Source: Authors' own elaboration based on IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Volume 4. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html; IPCC. 2006. *2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Volume 4. Kanagawa, Japan. www.ipcc-nggip.iges.or.jp/public/2006gl; IPCC. 2006. 2006 IPCC Guidelines for National Greenhouse Gas Inventories. Kanagawa, Japan. www.ipcc-nggip.iges.or.jp/public/2006gl

Emissions from the energy use

Energy consumption emissions are calculated for processing, packaging, storage and transportation. EX-ACT computes CO_2, CH_4 and N_2O emissions linked to the use of fossil fuels, with default emission factors described in Table A146 in the annex. For electricity use, EX-ACT proposes two sets of emission factors related to the energy grid following the Harmonized IFI Default Grid Factors (UNFCCC, 2021), provided in Table A145 in the annex. More information can be found in the Section 2.4.18.

Refrigerant leakage

Agrifood commodities may require refrigerated storage to preserve quality and avoid food losses prior to commercialization. EX-ACT accounts for refrigerant leakage depending on the type of refrigerant used (HFC, HFO, HCFO). Default GWP of refrigerants are provided in Table A149 in the annex. If users have special knowledge of other refrigeration systems in use, they can input alternative refrigerant leakage estimates in the Tier 2 portion of the module.

Material production

Agrifood commodities often require packaging and infrastructure materials that contribute to greenhouse gas emissions during their production. EX-ACT accounts for emissions linked to the production of aluminum, paper, plastic (LDPE or mixed), and wood, which are commonly used across agrifood supply chains. These materials are considered due to their relevance in storage, transport, and marketing stages (Berneers-Lee and Hoolohan, 2012)

Default emission factors for each material type are provided in Table A150 in the annex. If users have specific knowledge of alternative materials or production processes in use, they can input customized emission estimates in the Tier 2 portion of the module.

Table 41 details the default values adopted in the storage, transportation, processing and packaging (i.e. downstream*)* modules, along with the IPCC table reference and in-text table reference.

Table 41. Sources of default values in the downstream modules of Environmental eXternalities ACcounting Tool

Element	Unit	Details	Table	Table IPCC	Notes
CO_2eq emissions linked to grid electricity use in downstream modules (Equation 16) $\mathbf{E} = \sum \mathbf{F} * \mathbf{EF_C} * (\mathbf{1} + \mathbf{T})$					
Annual energy emissions from electricity use (E)	tCO_2eq/yr	Sum of all energy usages in the module.	N/A	-	-
Annual electricity use (F)	MWh/yr	User input. Default value for renewable energy: 0.	N/A	-	-
Emission factor (EF_C)	tCO_2eq/MWh	Tier 1: EF for the operating margin in project country (C).	A145	-	Value, country of origin, and margin type modifiable in Tier 2.
Fraction of transport losses (T)	-	Default value: 0.1	N/A		Modifiable in Tier 2.
Emissions linked to the use of solid and liquid fuels in downstream modules (Equation 16) $\mathbf{E} = \sum \mathbf{F} * \mathbf{EF_{GHG}}$					
Annual GHG emissions from fuel consumption (E)	tGHG/yr	Sum of all energy usages in the module. Emissions corresponding to fuel type and GHG species.	N/A	-	-
Annual fuel consumption (F)	m^3/yr t d.m./yr	User input. Unit differs per fuel type: Liquid fuel: m3/yr Solid fuel: t d.m./yr	N/A	-	-
Emission factor for burnt fuel (EF)	tCO_2/unit fuel tCH_4/unit fuel tN_2O/unit fuel	Specific EFs for each GHG species (CO2, CH4, and N2O) from fuel combustion.	Table A146	-	Modifiable in Tier 2.
Emissions from refrigerant leakage in storage (Equation 16 and Equation 7.13 in Volume 3 of IPCC (2006) $\mathbf{E} = \mathbf{R} \times \mathbf{EF}$ where $\mathbf{EF} = \frac{\mathbf{GWP}}{\mathbf{1000}}$					
Annual refrigerant emissions from leakage during operation (E)	tCO_2eq q/yr	-	N/A	-	-
Annual leakage of refrigerant (R)	kg refrigerant/yr	User input.	N/A	-	-

Element	Unit	Details	Table	Table IPCC	Notes
Emission factor (EF)	tCO_2eq/kg refrigerant	-	N/A	-	-
Refrigerant global warming potential (GWP)	Dimensionless	The set of GWP values corresponds to the IPCC Assessment Report specified in the project description.	A149	-	-
Emissions from material use in packaging (Equation 16) $$\mathbf{E} = \sum \mathbf{M} * \mathbf{EF}$$					
Emissions from packaging use (E)	tCO_2eq/yr	Sum of all packaging materials used in the project.	N/A	-	-
Weight of annually used packaging material (M)	t/yr	User input.	N/A	-	-
Emission factor (EF)	tCO_2eq/t material	Based on Berneers-Lee and Hoolohan (2012)	A150	-	Modifiable in Tier 2.

Note: GHG = greenhouse gas; IPCC = Intergovernmental Panel on Climate Change; N/A = not applicable; t/yr = tonne per year; tCO_2eq = tonnes of carbon dioxide equivalent; tGHG = tonnes of greenhouse gases; CO_2 = carbon dioxide; CH_4 = methane; N_2O = nitrous oxide; HFC = hydrofluorocarbons; HFO = hydrofluoroolefins; HCFO = hydrochlorofluoroolefins; tCO_2/unit fuel = tonnes of carbon dioxide per unit of fuel; tCH_4/unit fuel = tonnes of methane per unit of fuel; tN_2O/unit fuel = tonnes of nitrous oxide per unit of fuel.

Source: Authors' own elaboration based on IPCC. 2006. 2006 IPCC Guidelines for National Greenhouse Gas Inventories. Kanagawa, Japan. http://www.ipcc-nggip.iges.or.jp/public/2006g. UNFCCC. 2021. *Methodological approach for the common fault grid emissions factors dataset. Technical Working Group on Greenhouse Gas Accounting*. IFI TWG- AHG-001. https://unfccc.int/climate-action/sectoral-engagement/ifis-harmonization-of-standards-for-ghg-accounting/ifi-twg-list-of-methodologies; Berneers-Lee, M. & Hoolohan, C. 2012. *The Greenhouse Gas Footprint of Booths*. Preston, United Kingdom. https://www.booths.co.uk/wp-content/themes/booths/images/Booths%20GHG%20Report%202012%20Final.pdf

2.5 Results in the Environmental eXternalities ACcounting Tool

Having detailed the methodology applied across all modules, this section explains how EX-ACT aggregates and communicates results to users. The following subsections detail how the carbon balance is computed (Section 2.5.1), how the dynamics of change influence results over time (Section 2.5.2), and how GHG fluxes are monetized through the shadow price of carbon (Section 2.5.3).

The Environmental eXternalities ACcounting Tool produces results through three complementary outputs: an interactive dashboard, a narrative report, and an Excel report. Together, these outputs provide a comprehensive view of GHG impacts, carbon balances, and key indicators for decision-making.

The dashboard offers an at-a-glance summary of the project's performance. It displays the overall carbon balance, GHG fluxes with and without the project, and the yearly carbon balance disaggregated by impacted GHGs. It also includes an economic efficiency indicator, expressed as the cost of mitigation in terms of tonnes of CO_2eq abated per dollar spent.

The narrative report provides detailed contextual information. It explains the methodology behind the tool and presents the carbon balance both at the project level and for each activity. This report is designed to support transparency and facilitate communication with stakeholders by combining technical rigor with clear explanations.

The excel report delivers granular data for in-depth analysis. It includes the carbon balance per year at both project and activity levels, disaggregated by GHG source in alignment with the UNFCCC Enhanced Transparency Framework (ETF) (UNFCCC, 2025). The report also contains metadata on user choices made within the tool, including comments and notes, and a dedicated section on the shadow price of carbon (see Section 2.5.3 for more details).

Both the narrative and Excel reports feature a land-use change matrix consistent with Chapter 3 of the IPCC (2019) Guidelines, tracking conversions between initial and final land-use categories and reporting the total area of unchanged land by category.

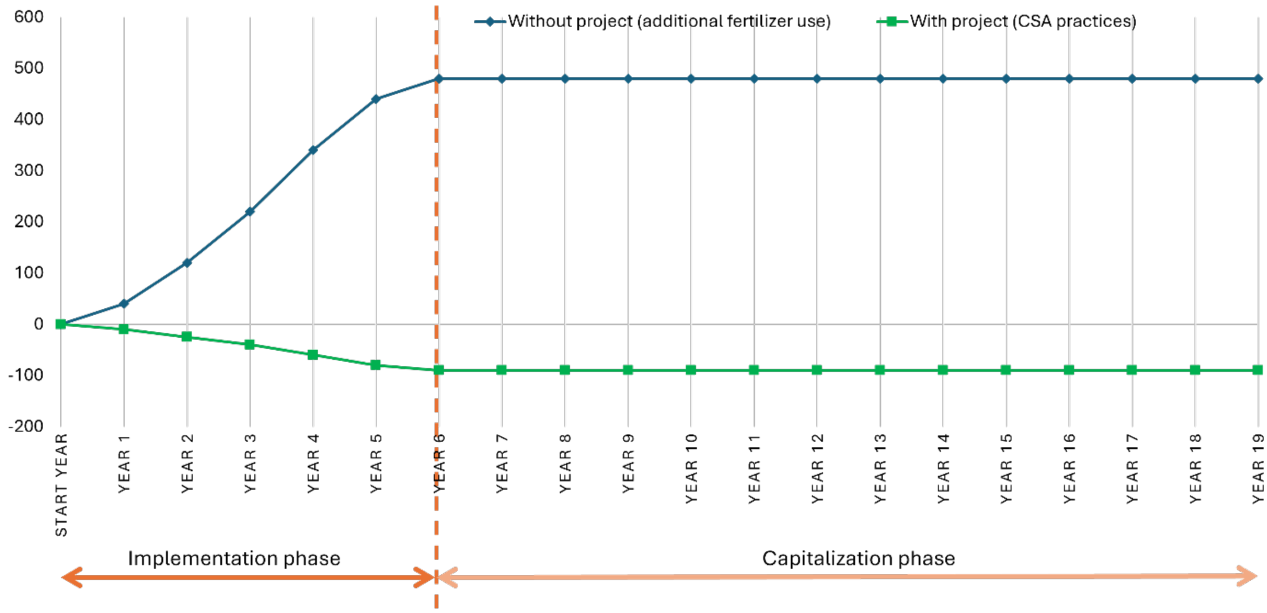
2.5.1 THE CARBON BALANCE

This section aims to provide an overview of how the carbon balance is computed in EX-ACT, clarifying yearly GHG fluxes to enhance user understanding on results obtained by applying the tool.

Figures 7 and 8 present a simplified example of how the carbon balance is calculated in EX-ACT. Figure 7 represents the different activities included in the GHG assessment, namely the distribution of additional fertilizer without the project and the introduction of climate smart agriculture (CSA) practices with the project. The activities are gradually introduced during the implementation period and remain stable throughout the capitalization period, up to the last year of accounting considered.

Figure 8 represents explicitly what is computed as the carbon balance. The carbon balance represents the difference between GHG fluxes generated with the project and the reference scenario (i.e. without project). It is represented on the figure by the total area marked in yellow between the two lines.

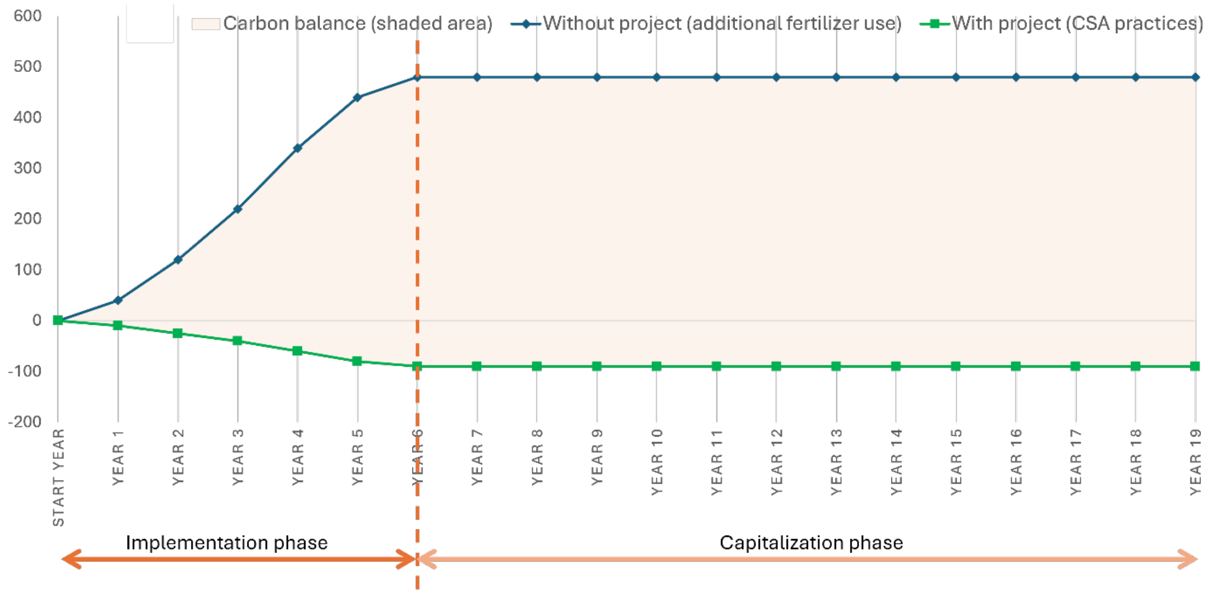
Figure 7. Calculating carbon balance in the Environmental eXternalities ACcounting Tool



Note: CSA= climate-smart agriculture.

Source: Authors' own elaboration.

Figure 8. Illustration of the carbon balance in the Environmental eXternalities ACcounting Tool



Note: CSA= climate-smart agriculture.

Source: Authors' own elaboration.

It is important to note that the tool always assumes that, once the implementation period is over, GHG fluxes generated by implemented activities will continue until the end of the total accounting period. In order to avoid gross overestimations of carbon stock accumulation, however, EX-ACT includes specific rules for soil carbon dynamics and biomass growth:

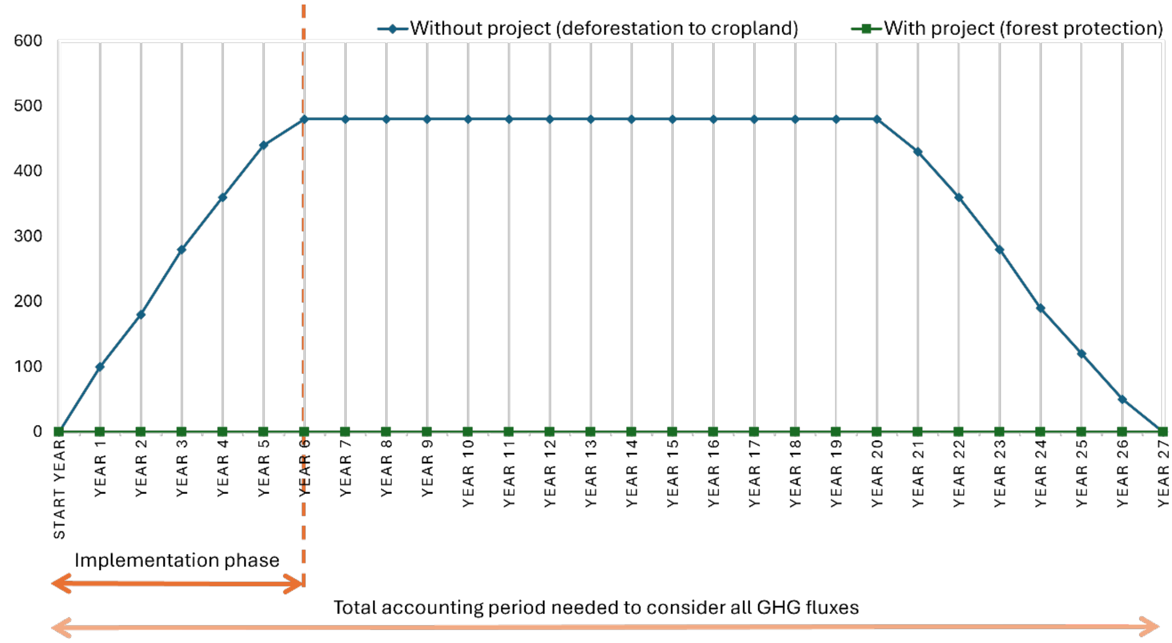
Soil organic carbon (SOC). EX-ACT applies a fixed 20-year time dependence for SOC stock changes, aligned with IPCC guidance that soil systems typically transition between equilibrium states over this duration (IPCC, 2019). Once the soil of a targeted area reaches the final SOC level associated with the final activities, the tool stops calculating further SOC change even if the overall accounting period is longer.

Biomass growth in forests and perennial crops. Biomass accumulation follows growth rates based on IPCC Tier 2 coefficients (IPCC 2006, 2019). Growth proceeds until it reaches a defined reference level from IPCC, after which EX-ACT stops accounting for additional changes, assuming the yearly gains and losses from the biomass will compensate one another. This prevents unrealistic projections of infinite biomass accumulation.

The total accounting period is the crucial aspect defining the total carbon balance. A shorter accounting period may fail to include all GHG fluxes generated by the activities implemented; a longer accounting period may increase the carbon balance indefinitely depending on the activities included in the appraisal. The following aspects should be kept in mind when defining the total accounting period:

- In order to fully account for SOC stock changes, users should take into consideration the dynamics of change (see Section 2.5.2) and accommodating the total accounting period accordingly (an example is provided in Figure 9).

Figure 9. Total accounting period needed for soil organic carbon stock changes

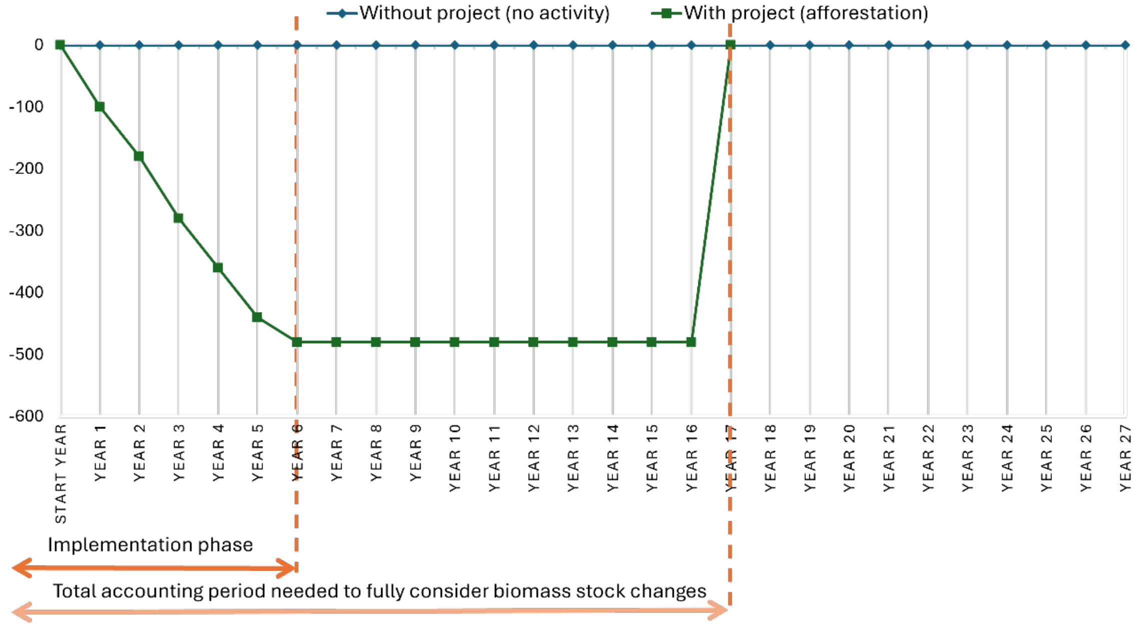


Note: CSA= climate-smart agriculture.

Source: Authors' own elaboration.

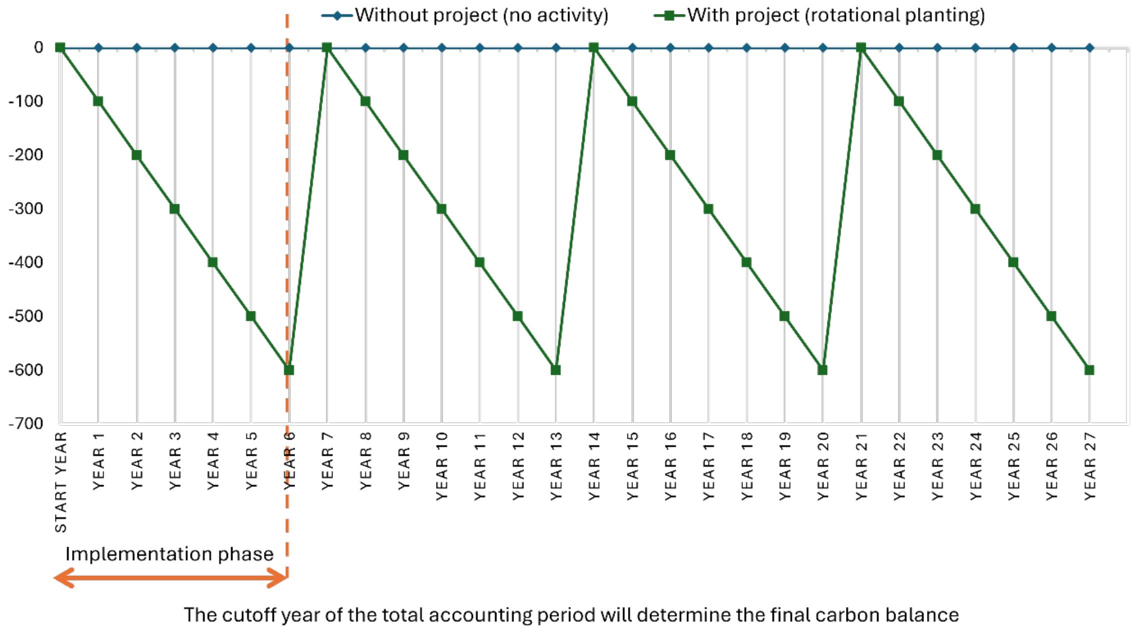
- Biomass carbon stock changes require different timing depending on the type of forest vegetation/agroforestry system selected. In order to fully account for biomass stock changes, the total accounting period may consider all GHG fluxes from biomass until biomass levels reach an equilibrium. This is shown in Figure 10. The above considerations also apply to systems with frequent changes in biomass levels (e.g. rotational planting); however, in these cases, activity data and the total accounting period are the most relevant determinants of biomass gain and loss. An example of this is shown in Figure 11.

Figure 10. Total accounting period and biomass stock changes in afforestation systems



Source: Authors' own elaboration.

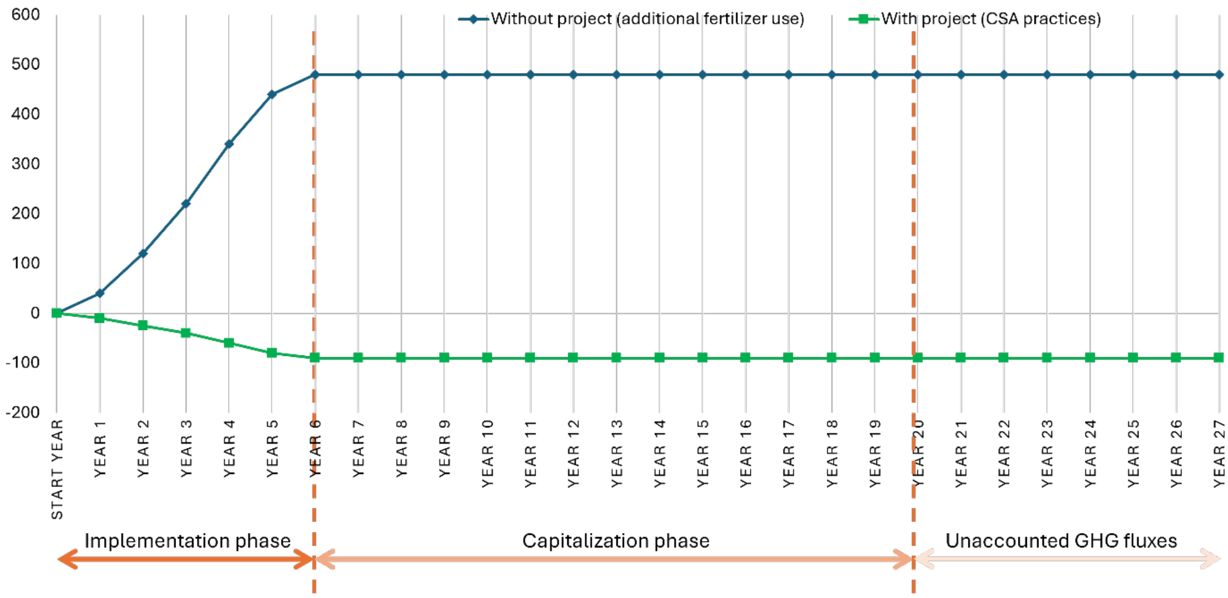
Figure 11. Total accounting period and biomass stock changes in rotational planting



Source: Authors' own elaboration.

- All other activities will emit/mitigate emissions in an incremental fashion through the implementation phase and will maintain stable at the reached level of emissions or mitigation throughout the capitalization phase. The total accounting period will thus determine the final carbon balance, as shown in Figure 12.

Figure 12. Total accounting period and carbon balance implications



Note: CSA= climate-smart agriculture; GHG = greenhouse gas.

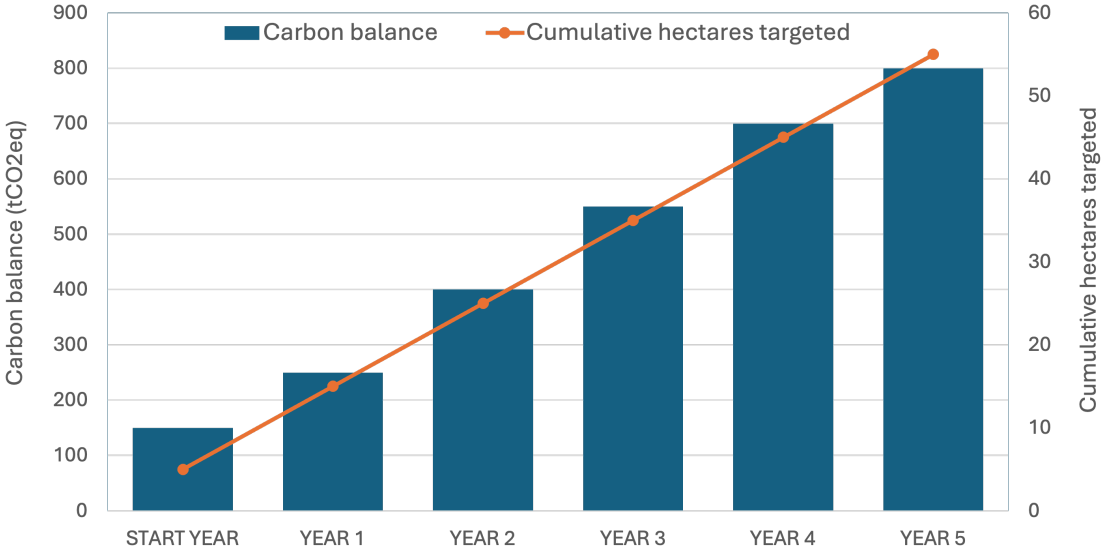
Source: Authors' own elaboration.

2.5.2 DYNAMICS OF CHANGE

The carbon balance is also shaped by the pace at which activities are performed during the implementation phase. EX-ACT allows users to select from three pre-defined dynamics of change (linear, immediate, and exponential) each reflecting a different pace of implementation of an activity over time, as illustrated in Figures 13 to 15.

For example, afforesting a patch of land over 5 years may imply that tree planting is performed entirely in the first year, that trees are gradually planted every year, as well as other options. The default dynamic of change adopted by the tool is linear, which means that the tool assumes the activity is realized gradually and stably across the implementation phase. Users have the possibility to change the dynamic type either to immediate or exponential [43] (see Figures 13 to 15).

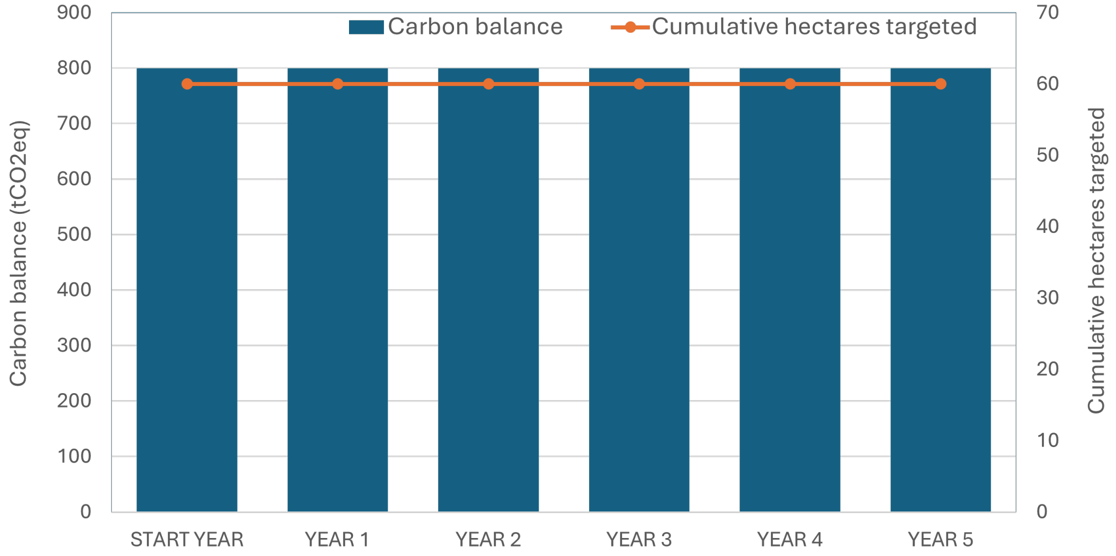
Figure 13. Schematic representation of the linear dynamic in the implementation phase



Source: Authors' own elaboration based on FAO. 2022. *EX-Ante Carbon balance Tool | EX-ACT – Guidelines*. Second edition – Tool version 9. Rome.

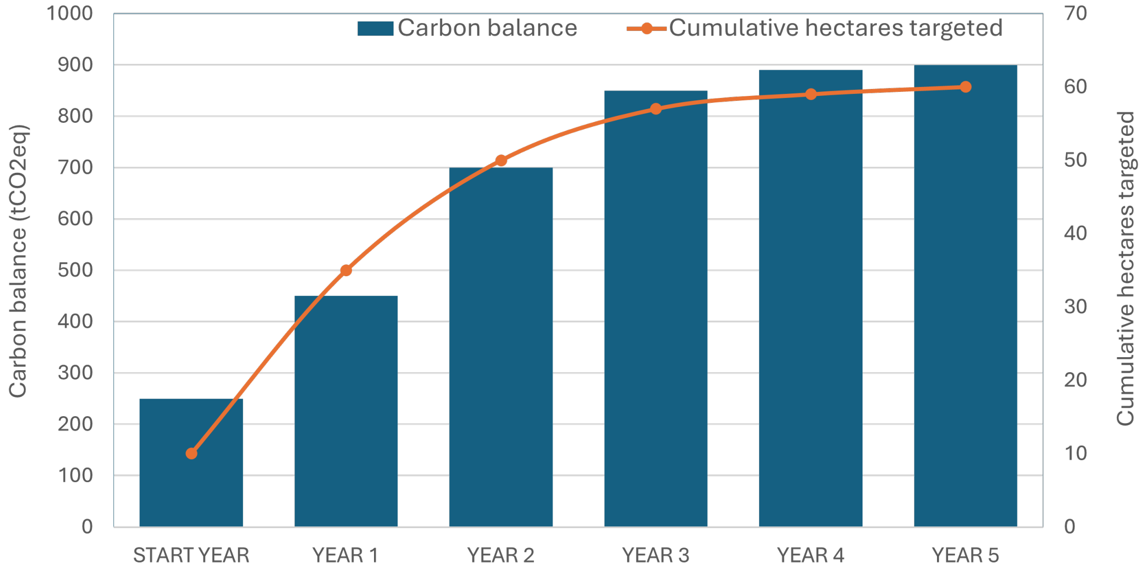
[43] Exponential refers to exponential decay, increasing form.

Figure 14. Schematic representation of the immediate dynamic in the implementation phase



Source: Authors' own elaboration based on FAO. 2022. EX-Ante Carbon balance Tool | EX-ACT – Guidelines. Second edition – Tool version 9. Rome.

Figure 15. Schematic representation of the exponential dynamic in the implementation phase



Source: Authors' own elaboration based on FAO. 2022. EX-Ante Carbon balance Tool | EX-ACT – Guidelines. Second edition – Tool version 9. Rome.

Under the linear dynamics (Figure 13), the change introduced in the activity is constant; for example, when a farmer progressively increases the surface of crops to which he applies fertilisers. The immediate dynamics (Figure14) considers an abrupt change of level when the activity considered has an immediate effect on the carbon balance. For example, a farmer may have one hundred cows at a start and due to a project, he will immediately add an additional ten cows to his herd. As the effect on carbon balance will be immediate, the user should use the immediate dynamics of change in the calculations. The exponential dynamics of change (Figure 15) would be preferred in situations when the rate of change is faster at the beginning of the project and slower at the end of implementation phase of the project.

2.5.3 SHADOW PRICE OF CARBON

Beyond physical GHG accounting, EX-ACT integrates the shadow price of carbon (SPC) to monetize the GHG fluxes associated with agrifood system interventions, based on the World Bank's guidance note on the subject (World Bank, 2024). This valuation enables the estimation of the economic impact of changes in GHG emissions resulting from project activities, expressed in terms of avoided or incurred costs.

The SPC represents the marginal cost of carbon emissions consistent with achieving global climate stabilization targets, particularly those outlined in the Paris Agreement. In line with the World Bank's 2024 guidance, EX-ACT applies a dual-trajectory approach, using both low and high SPC estimates. These trajectories are derived from the High-Level Commission on Carbon Prices and are expressed in constant 2017 USD. The low and high SPC values start at USD 40 and USD 80 per tCO_2eq in 2020, respectively, and increase annually by 2.25 percent, reaching USD 78 and USD 156 per tCO_2eq by 2050.

In EX-ACT, the economic value of the carbon balance is computed by multiplying the annual GHG flux (in tCO_2eq) in each project scenario by the corresponding SPC for each year of the accounting period. The tool reports the resulting economic values under both the low and high SPC scenarios. This allows users to assess the sensitivity of the economic valuation to different carbon pricing assumptions.

The calculation is performed as follows (Equation 44):

$$\text{Economic Value}_t = \text{Net GHG Flux}_t \times \text{SPC}_t$$

(Equation 44)

where:

- Net GHG Flux_t is the annual change in GHG emissions (positive for emissions, negative for mitigation) in each project scenario in year (t),
- SPC_t is the shadow price of carbon applicable in year (t), based on the selected trajectory.

The resulting time series of economic values is aggregated to provide a cumulative estimate of the economic benefit (or cost) associated with the intervention's carbon balance. This valuation supports the integration of climate externalities into broader economic assessments and facilitates alignment with international climate finance and policy frameworks.

2.5.4 INVENTORY

The EX-ACT application results also provide an Inventory overview that displays carbon stocks and GHG emissions associated with the START situation of the project. Carbon stocks in biomass and soils are reported as removals, while GHG emissions from all other sources are reported as emissions. Unlike the standard EX-ACT results, which focus on the changes in emissions and removals induced by project activities, the Inventory function provides a snapshot of the initial state of the system before project implementation. This allows users to relate estimated GHG fluxes to the carbon stocks and emission sources already present in the project area. The Inventory function can support the interpretation of results by:

- providing a reference point against which estimated mitigation outcomes can be assessed;
- helping users understand the relative magnitude of project-induced changes compared with existing carbon stocks and emission levels;
- identifying the main carbon pools and emission sources characterizing the project area at the outset;
- supporting communication and reporting by placing projected impacts within the context of the broader carbon and GHG profile of the intervention area;
- facilitating comparisons across projects or scenarios that may achieve similar GHG balances despite differing initial conditions.

By complementing the analysis of GHG fluxes with information on the initial stock and emission levels, the Inventory function provides a more complete picture of the carbon and GHG characteristics of the system being assessed.

2.6 Building a robust analysis

The Environmental eXternalities ACcounting Tool is an app providing default coefficients and operating on a set of pre-defined computations. It is not, per se, a modelling application; rather, it is a calculator, and the reliability of its results depends on the quality of the input data. Building a robust analysis thus means taking into account a series of factors affecting data input quality prior to the use of the tool and ensuring that implications and caveats of the results are duly considered. The following section will introduce the main elements to take into account for ensuring that the EX-ACT appraisal constitutes a robust analysis of climate outcomes of the targeted intervention:

- Boundary-setting and leakage
- Baseline scenario and additionality
- Permanence
- Uncertainty
- Transparency

A more in-depth appraisal of selected elements is available in Dobrovich *et al.* (2026) and Guzmán Estrada *et al.* (2026).

2.6.1 BOUNDARIES OF THE PROJECT

Setting the boundary of an analysis implies delimiting the scope of the activities and consequences included in the estimate of GHG fluxes. Depending on the need of users, EX-ACT can take into account field emissions only, cover the entire agrifood value chain, and include or exclude the risk of emission leakage, defined as the increase in GHG fluxes outside of the area of intervention that is a direct consequence of the implementation of activities within the area of intervention. Leakage estimates are not provided through an automated system but remain dependent on user analysis and input.

The EX-ACT can also cover scope-related analyses, as it aims to cover upstream and downstream GHG fluxes estimates alike; however, it does not automatically build a scope-specific overview of GHG fluxes, because the categorization of activities within a certain scope depends on the definition of what emissions are directly owned or controlled by the analyzed stakeholder.

2.6.2 BASELINE SCENARIO AND ADDITIONALITY

The Environmental eXternalities ACcounting Tool primarily targets the dynamic carbon balance of project results vis-à-vis what would happen in the absence of the project. A fundamental challenge in this accounting is establishing additionality, i.e. ensuring that reported emission reductions or removals are attributable to a given intervention, and would not have occurred in the absence of that intervention (WRI and WBCSD, 2020).

While representing the activities planned or implemented by the project is eased by the availability of official documentation and reporting, building a scenario representing the absence of such activities relies on assumptions and control methodologies. There is no standard, one-size-fits-all methodology to build the baseline scenario. Yet a few key criteria should always be followed:

a. The project boundaries should be maintained stable to ensure a fair comparison of activities, for example in terms of total area considered with and without the project.
b. The baseline scenario should represent the most plausible evolution of the situation, including the most credible options of land-use, possible land-use changes and main management practices that could have occurred on the land within the project boundary.
c. All underlying assumptions should draw on quality data and information sources, be well documented and easily verifiable.

Great attention has been devoted to the credibility of baseline scenarios in recent years, especially in the realm of carbon crediting schemes (e.g. West *et al.*, 2023). This is linked to the fact that building a credible baseline scenario is instrumental to establishing the additionality of activities analyzed, which is a key requirement for funding in international settings and in the context of carbon crediting. Dedicated research from Dobrovich *et al.* (2026) illustrates in further detail the concept and application of additionality considerations across different financing institutions at the global level.

2.6.3 PERMANENCE

Permanence is a concept related to the duration of climate mitigation through time. When assessing an investment or project, the amount of GHGs reduced or sequestered is linked, among other aspects, to the duration of accounting and the assumptions made on how the activities analyzed will influence the present and the future.

Dedicated research from Guzmán Estrada *et al.* (2026) is targeting the concept and application of permanence considerations across different financing institutions at the global level. From the practical point of view, EX-ACT does not include automated permanence (or non-permanence) estimates. As such, permanence considerations are dependent on user input linked to the duration of the individual activities and the total accounting period of the project.

2.6.4 UNCERTAINTY

Uncertainty arises from the lack of precise knowledge of the true value of a variable and is commonly described through a probability distribution reflecting the range and likelihood of possible outcomes. In a GHG appraisal, uncertainty determines both the degree of confidence in the estimated mitigation benefits and the robustness of the variables feeding the analysis. EX-ACT does not currently include an automated calculation of uncertainty for the variables used in its assessments.

Because EX-ACT relies primarily on Tier 1 coefficients from the IPCC, generally based on broad regional aggregates, default parameters often carry substantial uncertainty. Replacing such coefficients with location-specific Tier 2 values or, when available, direct measurements from the field is therefore one of the most effective ways to reduce uncertainty in the GHG analysis. Using values that more accurately reflect the biophysical reality of the project area improves both the precision and the accuracy of the appraisal, provided these values come from reliable and context-appropriate sources.

Another dimension of uncertainty relates to how project activities will actually be implemented on the ground. This type of uncertainty is less about biophysical parameters and more about the credibility and durability of the mitigation outcome. It is therefore better captured through considerations on additionality and permanence.

2.6.5 TRANSPARENCY OF THE CARBON APPRAISAL

The transparency of a carbon appraisal is dependent on the free, continuous and consistent availability of clear information linked to the methodological basis and the data used.

The Environmental eXternalities ACcounting Tool is based on the IPCC methodology, complemented with other sound scientific references, and provides a complete overview of the methodological underpinnings of the tool through the present document, allowing for a complete screening of references, sources and values of the default parameters used in the calculations.

Another essential dimension of transparency concerns the quality of data inputs and the replicability of the assessment. For an appraisal to be verifiable and reproducible, all assumptions, data sources, and methodological choices must be clearly documented. This includes detailing any modifications to default parameters, the use of local or project specific data, and the rationale underpinning these choices.

The Environmental eXternalities ACcounting Tool provides dedicated spaces within the tool specifically designed to support this level of documentation. It is strongly recommended to use these sections systematically to record all data used in the analysis, along with explicit references to their sources. This should cover bibliographic citations, links to datasets, descriptions of measurement methods, and any expert judgments used when no direct data is available.

Comprehensive documentation not only enhances the credibility and transparency of the GHG appraisal but also enables others to independently revisit, update, or replicate the analysis. This is particularly important when using location specific values or project generated data, where clear traceability is essential to ensure that the assessment remains consistent with good practice standards and can be robustly audited.

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Annex

The annex tables present default values used in EX-ACT, organized by module, geographic location, climate, and other relevant characteristics. They are provided to support users in understanding the underlying parameters applied in computations performed in the tool.

Table A1. Geographical distribution of countries within the Environmental eXternalities ACcounting Tool

Geographical division	Affiliated countries
Eastern Africa	British Indian Ocean Territory; Burundi; Comoros; Djibouti; Eritrea; Ethiopia; French Southern Territories; Kenya; Madagascar; Malawi; Mauritius; Mayotte; Mozambique; Réunion; Rwanda; Seychelles; Somalia; South Sudan; Uganda; United Republic of Tanzania; Zambia; Zimbabwe.
Central Africa	Angola; Cameroon; Central African Republic; Chad; Congo; Democratic Republic of the Congo; Equatorial Guinea; Gabon; Sao Tome and Principe.
Northern Africa	Algeria; Egypt; Libya; Morocco; Sudan; Tunisia; Western Sahara.
Southern Africa	Botswana; Eswatini; Lesotho; Namibia; South Africa.
Western Africa	Benin; Burkina Faso; Cabo Verde; Côte d'Ivoire; Gambia; Ghana; Guinea; Guinea-Bissau; Liberia; Mali; Mauritania; Niger; Nigeria; Saint Helena; Senegal; Sierra Leone; Togo.
Central Asia	Kazakhstan; Kyrgyzstan; Tajikistan; Turkmenistan; Uzbekistan.
Eastern Asia	China; China, Hong Kong SAR; China, Macao SAR; Democratic People's Republic of Korea; Japan; Mongolia; Republic of Korea.
South-Eastern Asia	Brunei Darussalam; Cambodia; Indonesia; Lao People's Democratic Republic; Malaysia; Myanmar; Philippines; Singapore; Thailand; Timor-Leste; Viet Nam.
Southern Asia	Afghanistan; Bangladesh; Bhutan; India; Iran (Islamic Republic of); Maldives; Nepal; Pakistan; Sri Lanka.
Western Asia	Armenia; Azerbaijan; Bahrain; Cyprus; Georgia; Iraq; Israel; Jordan; Kuwait; Lebanon; Oman; Qatar; Palestine: Saudi Arabia; Syrian Arab Republic; Türkiye; United Arab Emirates; Yemen.
Europe	Åland Islands; Albania; Andorra; Austria; Belarus; Belgium; Bosnia and Herzegovina; Bulgaria; Croatia; Czechia; Denmark; Estonia; Faroe Islands; Finland; France; Germany; Gibraltar; Greece; Guernsey; Holy See; Hungary; Iceland; Ireland; Isle of Man; Italy; Jersey; Latvia; Liechtenstein; Lithuania; Luxembourg; Malta; Monaco; Montenegro; Netherlands (Kingdom of the); North Macedonia; Norway; Russian Federation; Serbia; Slovakia; Slovenia; Spain; Svalbard and Jan Mayen Islands; Sweden; Switzerland; Ukraine; United Kingdom of Great Britain and Northern Ireland.
Oceania	American Samoa; Australia; Christmas Island; Cocos (Keeling) Islands; Cook Islands; Fiji; French Polynesia; Guam; Heard Island and McDonald Islands; Kiribati; Marshall Islands; Micronesia (Federated States of); Naoero; New Caledonia; New Zealand; Niue; Norfolk Island; Northern Mariana Islands; Palau; Papua New Guinea; Pitcairn; Samoa; Solomon Islands; Tokelau; Tonga; Tuvalu; United States Minor Outlying Islands; Vanuatu; Wallis and Futuna Islands.
North America	Bermuda; Canada; Greenland; Saint Pierre and Miquelon; United States of America.
South America	Argentina; Bolivia (Plurinational State of); Bouvet Island; Brazil; Chile; Colombia; Ecuador; Falkland Islands (Malvinas); French Guiana; Guyana; Paraguay; Peru; South Georgia and the South Sandwich Islands; Suriname; Uruguay; Venezuela (Bolivarian Republic of).
Central America	Belize; Costa Rica; El Salvador; Guatemala; Honduras; Mexico; Nicaragua; Panama.
Caribbean	Anguilla; Antigua and Barbuda; Aruba; Bahamas; Barbados; Bonaire, Sint Eustatius and Saba; British Virgin Islands; Cayman Islands; Cuba; Curaçao; Dominica; Dominican Republic; Grenada; Guadeloupe; Haiti; Jamaica; Martinique; Montserrat; Puerto Rico; Saint Barthélemy; Saint Kitts and Nevis; Saint Lucia; Saint Martin (French part); Saint Vincent and the Grenadines; Sint Maarten (Dutch part); Trinidad and Tobago; Turks and Caicos Islands; United States Virgin Islands.

Source: Author's own elaboration based on United Nations Statistics Division. 2022. *Methodology – Standard country or area codes for statistical use (M49)*. Cited 25 March 2022. https://unstats.un.org/unsd/methodology/m49

Table A2. Default carbon stocks for mineral soils to a depth of 30 cm, in tC/ha

IPCC climate zones	High activity clay soils	Low activity clay soils	Sandy soils	Spodic soils	Volcanic soils	Wetland soils
Boreal moist/dry	63	N/A	10	117	20	116
Cool temperate dry	43	33	13	No	20	87
Cool temperate moist	81	76	51	128	136	128
Warm temperate dry	24	19	10	N/A	84	74
Warm temperate moist	64	55	36	143	138	135
Tropical dry	21	19	9	N/A	50	22
Tropical moist	40	38	27	N/A	70	68
Tropical wet	60	52	46	N/A	77	49
Tropical montane	51	44	52	N/A	96	82

Notes: IPCC = Intergovernmental Panel on Climate Change. N/A denotes that the soil category may occur in a climate zone, but no data was available. No denotes that the soil type does not normally occur within a climate zone.

Source: Authors' own elaboration based on Table 2.3 IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Volume 4 – Agriculture, Forestry and Other Land Use. Geneva, Switzerland. https://www.ipcc-nggip.iges.or.jp/public/2019rf/vol4.html

Table A3. Vegetation types (natural or plantation) per climate zone available in the Environmental eXternalities ACcounting Tool

Tropical	Warm temperate	Cool temperate	Boreal	Tropical montane
Tropical rainforest	Subtropical humid forest	Temperate oceanic forest	Boreal coniferous forest	Tropical mountain systems
Tropical moist deciduous forest	Subtropical dry forest	Temperate continental forest	Boreal tundra woodland	Planted tropical mountain systems
Tropical dry forest	Subtropical steppe	Temperate mountain systems	Boreal mountain systems	
Tropical shrubland	Subtropical mountain systems	Temperate desert	Planted boreal coniferous forest	
Planted tropical rainforest	Planted subtropical humid forest	Temperate steppe	Planted boreal tundra woodland	
Planted tropical moist deciduous forest	Planted subtropical dry forest	Planted temperate oceanic forest	Planted boreal mountain systems	
Planted tropical dry forest	Planted subtropical steppe	Planted temperate continental forest		
Planted tropical shrubland	Planted subtropical mountain systems	Planted temperate mountain systems		

Source: Authors' own elaboration based on Table 4.4 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A4. Above-ground biomass, in tonne of dry matter per hectare, for tropical climate

Geographical regions	Tropical rainforest	Tropical moist deciduous forest	Tropical dry forest	Tropical shrubland	Planted tropical rainforest	Planted tropical moist deciduous forest	Planted tropical dry forest	Planted tropical shrubland
Africa	404	237	70	48	250	302	127	20
Asia	413	68	185	38	281	177	67	147
Europe	**300**	**180**	**130**	**70**	**150**	**120**	**60**	**30**
Oceania	**300**	**180**	**130**	**70**	**150**	**120**	**60**	**30**
The Americas and the Caribbean	307	187	128	72	225	166	80	45

Note: Values for plantations are calculated as the average of a given range.

Sources: Values in bold are taken from Table 4.12 of IPCC. 2006. *2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Kanagawa, Japan. www.ipcc-nggip.iges.or.jp/public/2006gl; other values are taken from Table 4.12 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A5. Above-ground biomass, in tonne of dry matter per hectare, for warm temperate climate

Geographical regions	Subtropical humid forest	Subtropical dry forest	Subtropical steppe	Subtropical mountain systems	Planted subtropical humid forest	Planted subtropical dry forest	Planted subtropical steppe	Planted subtropical mountain systems
Africa	54	65	51	35	**140**	65	18	90
Asia	323	71	42	250	140	111	50	**90**
Europe	**220**	**130**	**70**	**140**	**140**	**60**	**30**	**90**
Oceania	**220**	**130**	**70**	**140**	**140**	**60**	**30**	**90**
The Americas and the Caribbean	85	116	44	75	141	85	45	98

Note: Values for plantations are calculated as the average of a given range.

Sources: Values in bold are taken from Table 4.12 of IPCC. 2006. *2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Kanagawa, Japan. www.ipcc-nggip.iges.or.jp/public/2006gl; other values are taken from Table 4.12 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A6. Above-ground biomass, in tonne of dry matter per hectare, for temperate climate

Geographical regions	Temperate oceanic forest	Temperate continental forest	Temperate mountain systems	Temperate desert	Temperate steppe	Planted temperate oceanic forest	Planted temperate continental forest	Planted temperate mountain systems	Planted temperate desert	Planted temperate steppe
Africa	**180**	**120**	**100**	**44**	**119**	**160**	**100**	100	44	44
Asia	290	**120**	**100**	**44**	**119**	175	93	100	44	44
Europe	126	332	301	**44**	**119**	175	**100**	100	44	44
Oceania	353	**120**	**100**	**44**	**119**	**160**	**100**	100	44	44
The Americas and the Caribbean	**180**	**120**	**100**	**44**	**119**	104	59	59	44	44

Note: Values for plantations are calculated as the average of a given range.

Sources: Values in bold are taken from Table 4.12 of IPCC. 2006. *2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Kanagawa, Japan. www.ipcc-nggip.iges.or.jp/public/2006gl; other values are taken from Table 4.12 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A7. Above-ground biomass, in tonne of dry matter per hectare, for boreal climate

Geographical regions	Boreal coniferous forest	Boreal tundra woodland	Boreal mountain systems	Planted boreal coniferous forest	Planted boreal tundra woodland	Planted boreal mountain systems
Africa	**50**	**15**	**30**	**40**	**15**	**30**
Asia	63	**15**	**30**	45	25	45
Europe	63	**15**	**30**	45	25	45
Oceania	**50**	**15**	**30**	**40**	**15**	**30**
North America	63	**15**	**30**	45	25	45
South and Central America and the Caribbean	**50**	**15**	**30**	**40**	**15**	**30**

Note: Values for plantations are calculated as the average of a given range.

Sources: Values in bold are taken from Table 4.12 of IPCC. 2006. *2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Kanagawa, Japan. www.ipcc-nggip.iges.or.jp/public/2006gl; other values are taken from Table 4.12 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A8. Above-ground biomass, in tonne of dry matter per hectare, for tropical mountain climate

Geographical regions	Tropical mountain systems	Planted tropical mountain systems
Africa	190	90
Asia	434	88
Europe	**140**	**90**
Oceania	**140**	**90**
The Americas and the Caribbean	195	100

Note: Values for plantations are calculated as the average of a given range.

Sources: Values in bold are taken from Table 4.12 of IPCC. 2006. *2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Kanagawa, Japan. www.ipcc-nggip.iges.or.jp/public/2006gl; other values are taken from Table 4.12 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A9. Ratio of below-ground biomass to above-ground biomass (R) in t root d.m./t shoot d.m., for above-ground biomass ≤ 125 t d.m./ha in tropical climate

Geographical regions	Tropical rainforest	Tropical moist deciduous forest	Tropical dry forest	Tropical shrubland	Planted tropical rainforest	Planted tropical moist deciduous forest	Planted tropical dry forest	Planted tropical shrubland
Africa	0.825	0.232	0.332	**0.400**	**0.370**	**0.200**	**0.560**	**0.400**
Asia	0.207	0.323	0.381	0.322	0.325	**0.200**	**0.560**	**0.400**
Europe	**0.370**	**0.200**	**0.560**	**0.400**	**0.370**	**0.200**	**0.560**	**0.400**
Oceania	**0.370**	**0.200**	**0.560**	**0.400**	**0.370**	**0.200**	**0.560**	**0.400**
The Americas and the Caribbean	0.221	0.284	0.334	0.348	0.170	**0.200**	**0.560**	2.158

Sources: Values in bold are taken from Table 4.4 of IPCC. 2006. *2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Kanagawa, Japan. www.ipcc-nggip.iges.or.jp/public/2006gl; other values are taken from Table 4.4 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A10. Ratio of below-ground biomass to above-ground biomass (R) in t root d.m./t shoot d.m., for above-ground biomass ≤ 125 t d.m./ha in warm temperate climate

Geographical regions	Subtropical humid forest	Subtropical dry forest	Subtropical steppe	Subtropical mountain systems	Planted subtropical humid forest	Planted subtropical dry forest	Planted subtropical steppe	Planted subtropical mountain forest
Africa	0.232	**0.560**	**0.320**	0.270	0.200	0.560	**0.320**	**0.270**
Asia	0.230	0.440	**0.320**	0.270	0.200	0.560	2.158	**0.270**
Europe	**0.200**	**0.560**	**0.320**	0.270	0.200	0.560	**0.320**	**0.270**
Oceania	**0.200**	**0.560**	**0.320**	0.270	0.200	0.560	**0.320**	**0.270**
The Americas and the Caribbean	0.175	0.336	1.338	0.270	0.200	0.560	**0.320**	**0.270**

Sources: Values in bold are taken from Table 4.4 of IPCC. 2006. *2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Kanagawa, Japan. www.ipcc-nggip.iges.or.jp/public/2006gl; other values are taken from Table 4.4 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories.* Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A11. Ratio of below-ground biomass to above-ground biomass (R) in t root d.m./t shoot d.m., for above-ground biomass ≤ 125 t d.m./ha in temperate climate

Geographical regions	Temperate oceanic forest	Temperate continental forest	Temperate mountain systems	Temperate desert	Temperate steppe	Planted temperate oceanic forest	Planted temperate continental forest	Planted temperate mountain systems	Planted temperate desert	Planted temperate steppe
Africa	**0.440**	**0.440**	**0.440**	**0.440**	**0.440**	**0.440**	**0.440**	**0.440**	**0.440**	**0.440**
Asia	*0.267*	*0.243*	*0.323*	**0.440**	**0.440**	0.307	*0.305*	*0.254*	**0.440**	**0.440**
Europe	*0.276*	**0.440**	1.155	**0.440**	**0.440**	*0.880*	0.340	**0.440**	**0.440**	**0.440**
Oceania	*0.339*	**0.440**	0.145	**0.440**	**0.440**	*0.513*	**0.440**	0.293	**0.440**	**0.440**
The Americas and the Caribbean	*0.402*	0.481	*0.301*	**0.440**	**0.440**	**0.440**	0.237	0.302	**0.440**	**0.440**

Note: Values in italic are the average of a given range including conifers.

Sources: Values in bold are taken from Table 4.4 of IPCC. 2006. *2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Kanagawa, Japan. www.ipcc-nggip.iges.or.jp/public/2006gl; other values are taken from Table 4.4 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories.* Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html relative

Table A12. Ratio of below-ground biomass to above-ground biomass (R) in t root d.m./t shoot d.m., for above-ground biomass >125 t d.m./ha in tropical climate

Geographical regions	Tropical rainforest	Tropical moist deciduous forest	Tropical dry forest	Tropical shrubland	Planted tropical rainforest	Planted tropical moist deciduous forest	Planted tropical dry forest	Planted tropical shrubland
Africa	0.532	0.232	0.379	**0.400**	**0.370**	**0.240**	**0.280**	**0.400**
Asia	0.212	0.246	0.379	0.345	**0.370**	**0.240**	**0.280**	**0.400**
Europe	**0.370**	**0.240**	**0.280**	**0.400**	**0.370**	**0.240**	**0.280**	**0.400**
Oceania	**0.370**	**0.240**	**0.280**	**0.400**	**0.370**	**0.240**	**0.280**	**0.400**
The Americas and the Caribbean	0.221	0.284	0.379	0.283	0.170	**0.240**	**0.280**	**0.400**

Sources: Values in bold are taken from Table 4.4 of IPCC. 2006. *2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Kanagawa, Japan. www.ipcc-nggip.iges.or.jp/public/2006gl; other values are taken from Table 4.4 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A13. Ratio of below-ground biomass to above-ground biomass (R) in t root d.m./t shoot d.m., for above-ground biomass > 125 t d.m./ha in warm temperate climate

Geographical regions	Subtropical humid forest	Subtropical dry forest	Subtropical steppe	Subtropical mountain systems	Planted subtropical humid forest	Planted Subtropical dry forest	Planted Subtropical steppe	Planted subtropical mountain systems
Africa	0.232	**0.280**	**0.320**	**0.270**	**0.240**	**0.280**	**0.320**	**0.270**
Asia	0.246	0.440	1.338	**0.270**	**0.240**	**0.280**	**0.320**	**0.270**
Europe	**0.240**	**0.280**	**0.320**	**0.270**	**0.240**	**0.280**	**0.320**	**0.270**
Oceania	**0.240**	**0.280**	**0.320**	**0.270**	**0.240**	**0.280**	**0.320**	**0.270**
The Americas and the Caribbean	0.284	0.352	**0.320**	**0.270**	**0.240**	**0.280**	**0.320**	**0.270**

Sources: Values in bold are taken from Table 4.4 of IPCC. 2006. *2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Kanagawa, Japan. www.ipcc-nggip.iges.or.jp/public/2006gl; other values are taken from Table 4.4 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A14. Ratio of below-ground biomass to above-ground biomass (R) in t root d.m./t shoot d.m., for above-ground biomass >125 t d.m./ha in temperate climate

Geographical regions	Temperate oceanic forest	Temperate continental forest	Temperate mountain systems	Temperate desert	Temperate steppe	Planted temperate oceanic forest	Planted temperate continental forest	Planted temperate mountain systems	Planted temperate desert	Planted temperate steppe
Africa	**0.220**	**0.220**	**0.220**	**0.220**	**0.220**	**0.220**	**0.220**	**0.220**	**0.220**	**0.220**
Asia	0.229	*0.288*	*0.265*	**0.220**	**0.220**	*0.240*	*0.234*	*0.233*	**0.220**	**0.220**
Europe	0.172	0.477	0.394	**0.220**	**0.220**	*0.283*	**0.220**	**0.220**	**0.220**	**0.220**
Oceania	*0.253*	**0.220**	0.302	**0.220**	**0.220**	*0.242*	**0.220**	0.201	**0.220**	**0.220**
The Americas and the Caribbean	*0.264*	0.277	*0.231*	**0.220**	**0.220**	0.203	**0.220**	**0.220**	**0.220**	**0.220**

Note: Values in italic are the average of a given range including conifers.

Sources: Values in bold are taken from Table 4.4 of IPCC. 2006. *2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Kanagawa, Japan. www.ipcc-nggip.iges.or.jp/public/2006gl; other values are taken from Table 4.4 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A15. Default litter and deadwood carbon stocks

Forest type stratified by climate	Litter carbon stocks (tC/ha)	Deadwood carbon stocks (tC/ha)
Tropical rainforest	4.8	14.8
Tropical moist deciduous forest	5.9	8.0
Tropical dry forest	2.4	9.0
Tropical shrubland	0	0
Mangrove Forest	0.7	10.7
Subtropical humid forest	8.7	13.2
Subtropical dry forest	0	64.4
Subtropical steppe	0	6.8
Subtropical mountain systems	0	11.8
Temperate oceanic forest	2.9	36.8
Temperate continental forest	47.8	23
Temperate mountain systems	3.7	37.6
Temperate desert	0	10.5
Temperate steppe	28.7	21.7
Boreal coniferous forest	31.4	19.7
Boreal tundra woodland	49.5	3.1
Boreal mountain systems	0	0
Tropical mountain systems	0	3.3
Planted tropical rainforest	4.8	14.8
Planted tropical moist deciduous forest	5.9	8.0
Planted tropical dry forest	2.4	9.0
Planted tropical shrubland	0	0

Forest type stratified by climate	Litter carbon stocks (tC/ha)	Deadwood carbon stocks (tC/ha)
Planted subtropical humid forest	8.7	13.2
Planted subtropical dry forest	0	64.4
Planted subtropical steppe	0	6.8
Planted subtropical mountain systems	0	11.8
Planted temperate oceanic forest	2.9	36.8
Planted temperate continental forest	47.8	23
Planted temperate mountain systems	3.7	37.6
Planted temperate desert	0	10.5
Planted temperate steppe	28.7	21.7
Planted boreal coniferous forest	31.4	19.7
Planted boreal tundra woodland	49.5	3.1
Planted boreal mountain systems	0	0
Planted tropical mountain systems	0	3.3

Note: tC/ha = tonnes of carbon per hectare.

Source: Authors' own elaboration based on Table 2.2 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A16. Combustion factor, dimensionless and emission factor, in gram per kilogram of dry matter burnt, used for fire used during deforestation

Forest types stratified by IPCC climate	Combustion factor	Emission factor (CH_4)	Emission factor (N_2O)
Tropical rainforest	0.32	6.8	0.2
Tropical moist deciduous forest	0.36	6.8	0.2
Tropical dry forest	0.36	6.8	0.2
Tropical shrubland	0.72	6.8	0.2
Mangrove forest	0.32	6.8	0.2
Planted tropical rainforest	0.32	6.8	0.2
Planted tropical moist deciduous forest	0.36	6.8	0.2
Planted tropical dry forest	0.36	6.8	0.2
Planted tropical shrubland	0.72	6.8	0.2
Subtropical humid forest	0.36	4.7	0.26
Subtropical dry forest	0.36	4.7	0.26
Subtropical steppe	0.74	4.7	0.26
Subtropical mountain systems	0.36	4.7	0.26
Planted subtropical humid forest	0.36	4.7	0.26
Planted subtropical dry forest	0.36	4.7	0.26
Planted subtropical steppe	0.74	4.7	0.26
Planted subtropical mountain systems	0.36	4.7	0.26
Temperate oceanic forest	0.45	4.7	0.26
Temperate continental forest	0.45	4.7	0.26
Temperate mountain systems	0.45	4.7	0.26
Temperate desert	0.45	4.7	0.26
Temperate steppe	0.45	4.7	0.26
Planted temperate oceanic forest	0.45	4.7	0.26

Forest types stratified by IPCC climate	Combustion factor	Emission factor (CH_4)	Emission factor (N_2O)
Planted temperate continental forest	0.45	4.7	0.26
Planted temperate mountain systems	0.45	4.7	0.26
Planted temperate desert	0.45	4.7	0.26
Planted temperate steppe	0.45	4.7	0.26
Boreal coniferous forest	0.34	4.7	0.26
Boreal tundra woodland	0.34	4.7	0.26
Boreal mountain systems	0.34	4.7	0.26
Planted boreal coniferous forest	0.34	4.7	0.26
Planted boreal tundra woodland	0.34	4.7	0.26
Planted boreal mountain systems	0.34	4.7	0.26
Tropical mountain systems	0.36	6.8	0.2
Planted tropical mountain systems	0.36	6.8	0.2

Note: N_2O = nitrous oxide; CH_4 = methane; IPCC = Intergovernmental Panel on Climate Change.

Source: Authors' own elaboration based on Table 2.5 and Table 2.6 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A17. Default biomass carbon stock, in tC/ha, for system prior to afforestation, implanted after deforestation, or other land-use change for all regions

IPCC climate zones	Annual cropland	Annual fallow	Paddy rice	Grassland	Other land
Boreal	4.7	4.7	4.7	**4.00**	0
Cool temperate dry	4.7	4.7	4.7	**3.06**	0
Cool temperate moist	4.7	4.7	4.7	**6.39**	0
Warm temperate dry	4.7	4.7	4.7	**2.87**	0
Warm temperate moist	4.7	4.7	4.7	**6.35**	0
Tropical montane and tropical dry	4.7	4.7	4.7	**4.09**	0
Tropical moist and tropical wet	**4.7**	**4.7**	**4.7**	7.57	0

Notes: tC/ha = tonnes of carbon per hectare. Bold values are from IPCC (2006). The value for paddy rice is assumed the same as for annual crop.

Source: Authors' own elaboration based on Table 5.9 of IPCC. 2006. *2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Kanagawa, Japan. www.ipcc-nggip.iges.or.jp/public/2006gl and of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html and Table 6.4 of IPCC. 2006. *2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Kanagawa, Japan. www.ipcc-nggip.iges.or.jp/public/2006gl

Table A18. Above-ground net biomass growth in natural forests in tropical climate, in tonne d.m./ha/yr

Geographical regions	Tropical rainforest		Tropical moist deciduous forest		Tropical dry forest		Tropical shrubland	
	AGB > 20 yrs	AGB ≤ 20 yrs	AGB > 20 yrs	AGB ≤ 20 yrs	AGB > 20 yrs	AGB ≤ 20 yrs	AGB > 20 yrs	AGB ≤ 20 yrs
Africa	3.5	7.6	0.9	2.9	1.6	3.9	0.9	*0.5*
Asia	2.7	3.4	0.9	2.4	1.6	3.9	*1.2*	*3.5*
Europe	**7.0**	**7.0**	**5.0**	**5.0**	**2.4**	**2.4**	**1.0**	**1.0**
Oceania	**7.0**	**7.0**	**5.0**	**5.0**	**2.4**	**2.4**	**1.0**	**1.0**
The Americas and the Caribbean	2.3	5.9	2.7	5.2	1.6	3.9	1.0	4.0

Notes: Values in bold are from IPCC (2006). Values in italic are the average of a given range.

Source: Authors' own elaboration based on Table 4.12 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories.* Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A19. Above-ground net biomass growth in plantations forests in tropical climate, in tonne d.m./ha/yr

Geographical regions	Tropical rainforest		Tropical moist deciduous forest		Tropical dry forest		Tropical shrubland	
	AGB > 20 yrs	AGB ≤ 20 yrs	AGB > 20 yrs	AGB ≤ 20 yrs	AGB > 20 yrs	AGB ≤ 20 yrs	AGB > 20 yrs	AGB ≤ 20 yrs
Africa	**15.0**	*6.5*	**10.0**	*9.0*	*9.5*	*12.0*	*8.3*	*5.0*
Asia	*5.0*	*5.0*	8.0	8.0	*13.5*	*13.5*	*6.5*	*6.5*
Europe	**15.0**	**15.0**	**10.0**	**10.0**	**8.0**	**8.0**	**5.0**	**5.0**
Oceania	**15.0**	**15.0**	**10.0**	**10.0**	**8.0**	**8.0**	**5.0**	**5.0**
The Americas and the Caribbean	*22.5*	*22.5*	*12.0*	*12.0*	*17.0*	*17.0*	*12.5*	*12.5*

Notes: AGB = above-ground biomass. Values in bold are from IPCC (2006). Values in italic are the average of a given range.

Source: Authors' own elaboration based on Table 4.12 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories.* Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A20. Above-ground net biomass growth in natural forests in warm temperate climate, in tonne d.m./ha/yr

Geographical regions	Subtropical humid forest		Subtropical dry forest		Subtropical steppe		Subtropical mountain systems	
	AGB > 20 yrs	AGB ≤ 20 yrs	AGB > 20 yrs	AGB ≤ 20 yrs	AGB > 20 yrs	AGB ≤ 20 yrs	AGB > 20 yrs	AGB ≤ 20 yrs
Africa	1.0	2.5	1.8	2.4	0.9	1.2	0.5	2.5
Asia	1.0	2.5	*1.8*	6.5	*1.15*	*3.5*	0.5	*2.5*
Europe	**5.0**	**5.0**	**2.4**	**2.4**	**1.0**	**1.0**	**1.0**	**1.0**
Oceania	**5.0**	**5.0**	**2.4**	**2.4**	**1.0**	**1.0**	**1.0**	**1.0**
The Americas and the Caribbean	1.0	2.5	1.0	4.0	1.0	4.0	0.5	2.5

Notes: AGB = above-ground biomass. Values in bold are from IPCC (2006). Values in italic are the average of a given range.

Source: Authors' own elaboration based on Table 4.12 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories.* Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A21. Above-ground net biomass growth in plantations forests in warm temperate climate, in tonne d.m./ha/yr

Geographical regions	Subtropical humid forest		Subtropical dry forest		Subtropical steppe		Subtropical mountain systems	
	AGB > 20 yrs	AGB ≤ 20 yrs	AGB > 20 yrs	AGB ≤ 20 yrs	AGB > 20 yrs	AGB ≤ 20 yrs	AGB > 20 yrs	AGB ≤ 20 yrs
Africa	**10.0**	**10.0**	8.0	*12.0*	*8.3*	*7.8*	10.0	10.0
Asia	8.0	8.0	*13.5*	*13.5*	*6.5*	*6.5*	*6.5*	*6.5*
Europe	**10.0**	**10.0**	**8.0**	**8.0**	**5.0**	**5.0**	**5.0**	**5.0**
Oceania	**10.0**	**10.0**	**8.0**	**8.0**	**5.0**	**5.0**	**5.0**	**5.0**
The Americas and the Caribbean	*17.5*	*17.5*	*16.5*	*16.5*	*12.5*	*12.5*	*10.0*	*10.0*

Notes: AGB = above-ground biomass. Values in bold are from IPCC (2006). Values in italic are the average of a given range.

Source: Authors' own elaboration based on Table 4.12 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A22. Above-ground net biomass growth in natural forests in temperate climate, in tonne d.m./ha/yr

Geographical regions	Temperate oceanic forest		Temperate continental forest		Temperate mountain systems		Temperate desert		Temperate steppe	
	AGB > 20 yrs	AGB ≤ 20 yrs	AGB > 20 yrs	AGB ≤ 20 yrs	AGB > 20 yrs	AGB ≤ 20 yrs	AGB > 20 yrs	AGB ≤ 20 yrs	AGB > 20 yrs	AGB ≤ 20 yrs
Africa		**4.4**	**4.0**	**4.0**	**3.0**	**3.0**	**0.6**	**0.5**	**3.5**	**2.3**
Asia	**4.4**	**4.4**	**4.0**	**4.0**	**3.0**	**3.0**	**0.6**	**0.5**	**3.5**	**2.3**
Europe	2.3	2.3	**4.0**	**4.0**	**3.0**	**3.0**	**0.6**	**0.5**	**3.5**	**2.3**
Oceania	2.1	3.1	**4.0**	**4.0**	**3.0**	**3.0**	**0.6**	**0.5**	**3.5**	**2.3**
The Americas and the Caribbean	*9.1*	*6.3*	*3.6*	*3.3*	*4.4*	*3.1*	0.6	0.5	3.5	2.3

Notes: AGB = above-ground biomass. Values in bold are from IPCC (2006). Values in italic are the average of a given range.

Source: Authors' own elaboration based on Table 4.12 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A23. Above-ground net biomass growth in plantations forests in temperate climate, in tonne d.m./ha/yr

Geographical regions	Temperate oceanic forest		Temperate continental forest		Temperate mountain systems		Temperate desert		Temperate steppe	
	AGB > 20 yrs	AGB ≤ 20 yrs	AGB > 20 yrs	AGB ≤ 20 yrs	AGB > 20 yrs	AGB ≤ 20 yrs	AGB > 20 yrs	AGB ≤ 20 yrs	AGB > 20 yrs	AGB ≤ 20 yrs
Africa	**4.4**	**4.4**	**4.0**	**4.0**	**3.0**	**3.0**	0	0	11.0	4.0
Asia	4.4	4.4	4.0	4.0	3.0	3.0	0	0	11.0	4.0
Europe	4.4	4.4	4.0	4.0	3.0	3.0	0	0	11.0	4.0
Oceania	4.4	4.4	**4.0**	**4.0**	**3.0**	**3.0**	0	0	11.0	4.0
The Americas and the Caribbean	*10.0*	*6.0*	*4.0*	*5.0*	*9.0*	*10.0*	*0*	*0*	*11.0*	*4.0*

Notes: AGB = above-ground biomass. Values in bold are from IPCC (2006). Values in italic are the average of a given range.

Source: Authors' own elaboration based on Table 4.12 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A24. Above-ground net biomass growth in natural forests in boreal climate, in tonne d.m./ha/yr

Geographical regions	Boreal coniferous forest		Boreal tundra woodland		Boreal mountain systems	
	AGB > 20 yrs	AGB ≤ 20 yrs	AGB > 20 yrs	AGB ≤ 20 yrs	AGB > 20 yrs	AGB ≤ 20 yrs
Africa	0	0	0	0	0	0
Asia	*1.15*	*1.2*	0.4	0.4	*1.3*	*1.05*
Europe	*1.15*	*1.2*	0.4	0.4	*1.3*	*1.05*
Oceania	*1.15*	*1.2*	0.4	0.4	*1.3*	*1.05*
The Americas and the Caribbean	*1.15*	*1.2*	0.4	0.4	*1.3*	*1.05*

Notes: AGB = above-ground biomass. Values in bold are from IPCC (2006). Values in italic are the average of a given range.

Source: Authors' own elaboration based on Table 4.12 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A25. Above-ground net biomass growth in plantations forests in boreal climate, in tonne d.m./ha/yr

Geographical regions	Boreal coniferous forest		Boreal tundra woodland		Boreal mountain systems	
	AGB > 20 yrs	AGB ≤ 20 yrs	AGB > 20 yrs	AGB ≤ 20 yrs	AGB > 20 yrs	AGB ≤ 20 yrs
Africa	1.0	1.0	0.4	0.4	1.0	1.0
Asia	1.0	1.0	0.4	0.4	1.0	1.0
Europe	1.0	1.0	0.4	0.4	1.0	1.0
Oceania	1.0	1.0	0.4	0.4	1.0	1.0
The Americas and the Caribbean	1.0	1.0	0.4	0.4	1.0	1.0

Note: AGB = above-ground biomass.

Source: Authors' own elaboration based on Table 4.12 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A26. Above-ground net biomass growth in natural forests and plantations forests in tropical mountain climate, in tonne d.m./ha/yr

Geographical regions	Forest tropical mountain systems		Planted tropical mountains systems	
	AGB > 20 yrs	AGB ≤ 20 yrs	AGB > 20 yrs	AGB ≤ 20 yrs
Africa	1.8	5.5	10.0	10.0
Asia	1.1	2.9	*5.5*	*5.5*
Europe	**1.0**	**1.0**	**5.0**	**5.0**
Oceania	**1.0**	**1.0**	**5.0**	**5.0**
The Americas and the Caribbean	1.8	4.4	*13.0*	*13.0*

Notes: AGB = above-ground biomass. Values in bold are from IPCC (2006). Values in italic are the average of a given range.

Source: Authors' own elaboration based on Table 4.12 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A27. Default biomass carbon stock for various agroforestry systems, in tC/ha, prior afforestation/reforestation or non-forest land-use change according to the climate characteristics

IPCC climate zones	Agroforestry - Default	Fallow	Hedgerow	Alley cropping	Multistrata	Parkland	Shaded perennial	Silvoarable	Silvopasture
Temperate (all) and boreal (all)	20.60	0	13.1	0	0	0	0	13.7	**35.0**
Tropical (all)	**20.89**	**11.1**	**4.7**	**23.7**	**32.5**	**5.9**	**24.0**	**36.1**	**29.1**

Notes: Temperate default values are used as proxy for boreal climate. Agroforestry – default is computed as the average of the other agroforestry systems. Values in bold are from IPCC (2006). Values in italic are the average of a given range.

Source: Authors' own elaboration based on Table 5.1 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A28. Default biomass carbon stock, in tC/ha, for monoculture agroforestry system implanted prior afforestation/reforestation or non-forest land-use change according to the climate characteristics

IPCC climate zones	Olive	Orchard	Vine	Short rotation coppice	Oil palm	Rubber	Tea
Temperate (all) and boreal (all)	6.9	6.4	2.8	6.4	0	0	**18.3**
Tropical (all)	**0**	**0**	**0**	**0**	**30.0**	**40.1**	**18.3**

Note: Temperate default values are used as proxy for boreal climate. Values in bold are from IPCC (2006).

Source: Authors' own elaboration based on Table 5.3 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A29. Combustion factor, dimensionless, and emission factor (G_{ef}), in g per kg of dry matter burnt, used for fire used prior to afforestation/reforestation

Non-forestry land use	Combustion factor	G_{ef} CO_2	G_{ef} CH_4	G_{ef} N_2O
Annual cropland	0.4	1 515	2.7	0.07
Agroforestry	0.8	1 613	2.3	0.21
Annual fallow	0.4	1 515	2.7	0.07
Degraded land	0.8	1 613	2.3	0.21
Other land	0	0	0	0
Flooded rice	0.4	1 515	2.7	0.07
Grassland	0.8	1 613	2.3	0.21
Alley cropping	0.8	1 613	2.3	0.21
Perennial fallow	0.8	1 613	2.3	0.21
Hedgerow	0.8	1 613	2.3	0.21
Multi-strata	0.8	1 613	2.3	0.21
Parkland	0.8	1 613	2.3	0.21
Shaded perennial	0.8	1 613	2.3	0.21
Silvoarable	0.8	1 613	2.3	0.21
Silvopasture	0.8	1 613	2.3	0.21
Degraded land	0.8	1 613	2.3	0.21

Non-forestry land use	Combustion factor	G_{ef} CO_2	G_{ef} CH_4	G_{ef} N_2O
Olive	0.8	1 613	2.3	0.21
Orchard	0.8	1 613	2.3	0.21
Short rotation coppice	0.8	1 613	2.3	0.21
Tea	0.8	1 613	2.3	0.21
Vine	0.8	1 613	2.3	0.21
Oil palm	0.8	1 613	2.3	0.21
Rubber	0.8	1 613	2.3	0.21

Note: G_{ef} = emission factor per unit of dry matter burned, G_{ef} CO_2 = grams of CO_2 emitted per kg of biomass burned, G_{ef} CH_4 = grams of methane emitted per kg burned, G_{ef} N_2O = grams of nitrous oxide emitted per kg burned.

Source: Authors' own elaboration based on Tables 2.5 and 2.6 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A30. Relative stock change factor for land-use (Flu) dimensionless

IPCC climate zones	Annual cropland	Set-aside land	Flooded rice	Grassland	Other land	Agroforestry
Boreal dry	0.77	0.93	1.35	1.0	**1.0**	0.72
Boreal moist	0.70	0.82	1.35	1.0	**1.0**	0.72
Cool temperate dry	0.77	0.93	1.35	1.0	**1.0**	0.72
Cool temperate moist	0.70	0.82	1.35	1.0	**1.0**	0.72
Warm temperate dry	0.76	0.93	1.35	1.0	**1.0**	0.72
Warm temperate moist	0.69	0.82	1.35	1.0	**1.0**	0.72
Tropical dry	0.92	0.93	1.35	1.0	**1.0**	1.01
Tropical moist	0.83	0.82	1.35	1.0	**1.0**	1.01
Tropical wet	0.83	0.82	1.35	1.0	**1.0**	1.01
Tropical montane	**0.64**	0.88	1.35	1.0	**1.0**	1.00

Note: Values in bold are from IPCC (2006).

Source: Authors' own elaboration based on Table 5.5 and 6.2 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A31. Relative stock change factors (FLu, FMG, FI) over 20 years for management practices on cropland

IPCC climate zones	FL_u	F_{MG} full tillage	F_{MG} reduced tillage	F_{MG} no tillage	F_I low carbon input	F_I medium carbon input	F_I high carbon input, no manure	F_I high carbon input, with manure
Boreal dry	0.77	1.00	0.98	1.03	0.92	1.00	1.11	1.44
Boreal moist	0.70	1.00	1.04	1.09	0.95	1.00	1.04	1.37
Cool temperate dry	0.77	1.00	0.98	1.03	0.95	1.00	1.04	1.37
Cool temperate moist	0.70	1.00	1.04	1.09	0.92	1.00	1.11	1.44
Warm temperate dry	0.76	1.00	0.99	1.04	0.95	1.00	1.04	1.37
Warm temperate moist	0.69	1.00	1.05	1.10	0.92	1.00	1.11	1.44
Tropical dry	0.92	1.00	0.99	1.04	0.95	1.00	1.04	1.37
Tropical moist	0.83	1.00	1.04	1.10	0.92	1.00	1.11	1.44
Tropical wet	0.83	1.00	1.04	1.10	0.92	1.00	1.11	1.44
Tropical montane	**0.64**	1.00	**1.09**	**1.16**	0.94	1.00	1.08	1.41

Notes: FLU = land-use factor, FMG = management factor, including full tillage, reduced tillage, and no-tillage practices, FI = input factor, including low, medium, and high carbon input levels, with or without manure application. Values in bold are from IPCC (2006).

Sources: Authors' own elaboration based on Table 5.5 of IPCC. 2006. *2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Kanagawa, Japan. www.ipcc-nggip.iges.or.jp/public/2006gl and of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A32. Default values for the slope and intercept to estimate above-ground residue, and nitrogen content of above-ground residues, below-ground residues and ratio of below-ground to above-ground biomass

Crops	Slope	Intercept	N AG(T)	N BG(T)	RS(T)
Generic value	None	None	0.008	0.009	0.22
Generic grains	1.09	0.88	0.006	0.009	0.22
Beans & pulses	1.13	0.85	0.008	0.008	0.19
Tubers	0.1	1.06	0.019	0.014	0.2
Root crops	1.07	1.54	0.019	0.014	0.2
N-fixing forages	0.3	0	0.027	0.022	0.4
Non-N-fixing forages	0.3	0	0.015	0.012	0.54
Maize	1.03	0.61	0.006	0.007	0.22
Wheat	1.51	0.52	0.006	0.009	0.24
Winter wheat	1.61	0.4	0.006	0.009	0.23
Spring wheat	1.29	0.75	0.006	0.009	0.28
Rice	0.95	2.46	0.007	0.009	0.16
Barley	0.98	0.59	0.007	0.014	0.22
Oats	0.91	0.89	0.007	0.008	0.25
Millet	1.43	0.14	0.007	0.009	0.22
Sorghum	0.88	1.33	0.007	0.006	0.22

Crops	Slope	Intercept	N AG(T)	N BG(T)	RS(T)
Rye	1.09	0.88	0.005	0.011	0.22
Soybeans	0.93	1.35	0.008	0.008	0.19
Dry bean	0.36	0.68	0.01	0.01	0.22
Potatoes	0.1	1.06	0.019	0.014	0.2
Peanuts	1.07	1.54	0.016	0.009	0.22
Non-legume hay	0.18	0	0.015	0.012	0.54
Perennial grasses	0.3	0	0.015	0.012	0.8
Grass-clover mixtures	0.3	0	0.025	0.016	0.8

Note: N = nitrogen, N AG(T) = nitrogen content of above-ground residues, N BG(T) = nitrogen content of below-ground residues, RS (T) = ratio of below-ground to above-ground biomass.

Source: Authors' own elaboration based on Table 11.1A and Table 11.2 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A33. Default above-ground biomass growth rate, in tC/ha/yr, for monoculture agroforestry system according to the climate characteristics

IPCC climate zones	Olive	Orchard	Vine	Short rotation coppice	Oil palm	Rubber	Tea
Temperate (all) and boreal (all)	0.46	0.43	0.28	3.2	0	0	**0.7**
Tropical (all)	0	0	0	0	2.4	3.0	**0.7**
All	**0**	**0**	**0**	**0**	**0**	**0**	**0.7**

Note: Values in bold are from IPCC (2006).

Source: Authors' own elaboration based on Table 5.3 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A34. Above-ground biomass growth rate for various agroforestry systems, in tC/ha/yr, for cool temperate dry and moist climate

Geographical regions	Agroforestry	Alley cropping	Perennial fallow	Hedgerow	Multistrata	Parkland	Shaded perennial	Silvoarable	Silvopasture
Africa	1.27			0.87				1.12	1.81
Asia	1.88			0.87				2.97	1.81
Europe	1.39			0.87				1.12	2.17
Oceania	1.27			0.87				1.12	1.81
North America	0.81			0.87				0.59	0.97
South and Central America and the Caribbean	1.06			0.87				1.12	1.18

Notes: Values for agroforestry systems are the average of a given regional range. Values in red indicate generic values for all regions under the same climate.

Source: Authors' own elaboration based on Table 5.2 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A35. Above-ground biomass growth rate for various agroforestry systems, in tC/ha/yr for boreal climate

Geographical regions	Agroforestry	Alley cropping	Perennial fallow	Hedgerow	Multistrata	Parkland	Shaded perennial	Silvoarable	Silvopasture
Africa	1.37			0.87				0.91	2.33
Asia	1.37			0.87				0.91	2.33
Europe	1.37			0.87				0.91	2.33
Oceania	1.37			0.87				0.91	2.33
North America	1.37			0.87				0.91	2.33
South and Central America and the Caribbean	1.37			0.87				0.91	2.33

Notes: Values for agroforestry systems are the average of a given regional range. Values in red indicate generic values for all regions under the same climate. For Boreal climate, the default values of temperate were used.

Source: Authors' own elaboration based on Table 5.2 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A36. Above-ground biomass growth rate for various agroforestry systems, in tC/ha/yr, for warm temperate climate

Geographical regions	Agroforestry	Alley cropping	Perennial fallow	Hedgerow	Multistrata	Parkland	Shaded perennial	Silvoarable	Silvopasture
Africa	1.37			0.87				0.91	2.33
Asia	1.37			0.87				0.91	2.33
Europe	1.37			0.87				0.52	3.11
Oceania	1.37			0.87				0.91	2.33
The Americas and the Caribbean	1.37			0.87				0.91	2.33

Notes: Values for agroforestry systems are the average of a given regional range. Values in red indicate generic values for all regions under the same climate.

Source: Authors' own elaboration based on Table 5.2 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A37. Above-ground biomass growth rate for various agroforestry systems, in tC/ha/yr, for tropical dry climate

Geographical regions	Agroforestry	Alley cropping	Perennial fallow	Hedgerow	Multistrata	Parkland	Shaded perennial	Silvoarable	Silvopasture
Africa	2.74	1.88	5.61	0.48	1.63	0.59	2.40	6.24	3.07
Asia	2.85	2.79	5.61	0.48	1.63	0.59	2.40	6.24	3.07
Europe	2.79	2.27	5.61	0.48	1.63	0.59	2.40	6.24	3.07
Oceania	2.79	2.27	5.61	0.48	1.63	0.59	2.40	6.24	3.07
The Americas and the Caribbean	2.79	2.27	5.61	0.48	1.63	0.59	2.40	6.24	3.07

Notes: Values for agroforestry systems are the average of a given regional range. Values in red indicate generic values for all regions under the same climate

Source: Authors' own elaboration based on Table 5.2 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A38. Above-ground biomass growth rate for various agroforestry systems, in tC/ha/yr, for tropical moist climate

Geographical regions	Agroforestry	Alley cropping	Perennial fallow	Hedgerow	Multistrata	Parkland	Shaded perennial	Silvoarable	Silvopasture
Africa	2.74	2.75	5.30	0.47	2.98	0.59	1.82	5.09	2.91
Asia	2.31	2.59	5.30	0.47	3.03	0.59	2.07	1.50	2.91
Europe	2.49	2.59	5.30	0.47	3.02	0.59	2.43	2.63	2.91
Oceania	2.49	2.59	5.30	0.47	3.02	0.59	2.43	2.63	2.91
North America	2.49	2.59	5.30	0.47	3.02	0.59	2.43	2.63	2.91
South America	2.57	2.59	5.30	0.47	3.02	0.59	3.06	2.63	2.91
Central America	2.45	2.28	5.30	0.47	3.02	0.59	2.43	2.63	2.91
Caribbean	2.49	2.59	5.30	0.47	3.02	0.59	2.43	2.63	2.91

Notes: Values for agroforestry systems are the average of a given regional range. Values in red indicate generic values for all regions under the same climate

Source: Authors' own elaboration based on Table 5.2 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A39. Above-ground biomass growth rate for various agroforestry systems, in tC/ha/yr, for tropical wet climate

Geographical regions	Agroforestry	Alley cropping	Perennial fallow	Hedgerow	Multistrata	Parkland	Shaded perennial	Silvoarable	Silvopasture
Africa	2.35	1.88	6.21	0.43	2.89	0.59	3.16	3.61	0.06
Asia	1.90	1.88	2.00	0.43	4.83	0.59	1.79	3.61	0.06
Europe	2.10	1.88	4.59	0.43	3.25	0.59	2.36	3.61	0.06
Oceania	2.10	1.88	4.59	0.43	3.25	0.59	2.36	3.61	0.06
North America	2.10	1.88	4.59	0.43	3.25	0.59	2.36	3.61	0.06
South America	2.09	1.88	4.59	0.43	3.25	0.59	2.28	3.61	0.06
Central America	2.11	1.88	4.76	0.43	2.60	0.59	2.96	3.61	0.06
Caribbean	2.10	1.88	4.59	0.43	3.25	0.59	2.36	3.61	0.06

Notes: Values for agroforestry systems are the average of a given regional range. Values in red indicate generic values for all regions under the same climate.

Source: Authors' own elaboration based on Table 5.2 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A40. Above-ground biomass growth rate for various agroforestry systems, in tC/ha/yr, for tropical montane climate

Geographical regions	Agroforestry	Alley cropping	Perennial fallow	Hedgerow	Multistrata	Parkland	Shaded perennial	Silvoarable	Silvopasture
Africa	2.16	2.37	3.12	0.67	3.25	0.59	2.40	2.26	2.62
Asia	2.32	2.37	4.42	0.67	3.25	0.59	2.40	2.26	2.62
Europe	2.32	2.37	4.42	0.67	3.25	0.59	2.40	2.26	2.62
Oceania	2.32	2.37	4.42	0.67	3.25	0.59	2.40	2.26	2.62
The Americas and the Caribbean	2.32	2.37	4.42	0.67	3.25	0.59	2.40	2.26	2.62

Notes: Values for agroforestry systems are the average of a given regional range. Values in red indicate generic values for all regions under the same climate.

Source: Authors' own elaboration based on Table 5.2 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A41. Below-ground biomass growth rate for various agroforestry systems, in tC/ha/yr, for cool temperate dry and moist climate

Geographical regions	Agroforestry	Alley cropping	Perennial fallow	Hedgerow	Multistrata	Parkland	Shaded perennial	Silvoarable	Silvopasture
Africa	0.33			0.23				0.28	0.48
Asia	0.49			0.23				0.77	0.48
Europe	0.36			0.23				0.28	0.56
Oceania	0.33			0.23				0.28	0.48
North America	0.16			0.23				0.14	0.11
The Americas and the Caribbean	0.34			0.23				0.28	0.52

Notes: Values for agroforestry systems are the average of a given regional range. Values in red indicate generic values for all regions under the same climate

Source: Authors' own elaboration based on Table 5.2 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A42. Below-ground biomass growth rate for various agroforestry systems, in tC/ha/yr, for boreal climate

Geographical regions	Agroforestry	Alley cropping	Perennial fallow	Hedgerow	Multistrata	Parkland	Shaded perennial	Silvoarable	Silvopasture
Africa	0.39			0.23				0.23	0.70
Asia	0.39			0.23				0.23	0.70
Europe	0.39			0.23				0.23	0.70
Oceania	0.39			0.23				0.23	0.70
North America	0.39			0.23				0.23	0.70
The Americas and the Caribbean	0.39			0.23				0.23	0.70

Notes: Values for agroforestry systems are the average of a given regional range. Values in red indicate generic values for all regions under the same climate. For boreal climate, the default values of Temperate were used.

Source: Authors' own elaboration based on Table 5.2 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A43. Below-ground biomass growth rate for various agroforestry systems, in tC/ha/yr, for warm temperate climate

Geographical regions	Agroforestry	Alley cropping	Perennial fallow	Hedgerow	Multistrata	Parkland	Shaded perennial	Silvoarable	Silvopasture
Africa	0.39			0.23				0.23	0.70
Asia	0.39			0.23				0.23	0.70
Europe	0.47			0.23				0.14	1.03
Oceania	0.39			0.23				0.23	0.70
The Americas and the Caribbean	0.39			0.23				0.23	0.70

Notes: Values for agroforestry systems are the average of a given regional range. Values in red indicate generic values for all regions under the same climate.

Source: Authors' own elaboration based on Table 5.2 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A44. Below-ground biomass growth rate for various agroforestry systems, in tC/ha/yr, for tropical dry climate

Geographical regions	Agroforestry	Alley cropping	Perennial fallow	Hedgerow	Multistrata	Parkland	Shaded perennial	Silvoarable	Silvopasture
Africa	0.85	0.45	2.54	0.12	0.46	0.21	0.55	1.62	0.84
Asia	0.63	0.67	0.53	0.12	0.46	0.21	0.55	1.62	0.84
Europe	0.79	0.54	1.95	0.12	0.46	0.21	0.55	1.62	0.84
Oceania	0.79	0.54	1.95	0.12	0.46	0.21	0.55	1.62	0.84
The Americas and the Caribbean	0.79	0.54	1.95	0.12	0.46	0.21	0.55	1.62	0.84

Notes: Values for agroforestry systems are the average of a given regional range. Values in red indicate generic values for all regions under the same climate

Source: Authors' own elaboration based on Table 5.2 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A45. Below-ground biomass growth rate for various agroforestry systems, in tC/ha/yr, for tropical moist climate

Geographical regions	Agroforestry	Alley cropping	Perennial fallow	Hedgerow	Multistrata	Parkland	Shaded perennial	Silvoarable	Silvopasture
Africa	0.67	0.59	1.27	0.11	0.72	0.21	0.44	1.22	0.79
Asia	0.57	0.58	1.27	0.11	0.73	0.21	0.50	0.35	0.79
Europe	0.61	0.58	1.27	0.11	0.73	0.21	0.57	0.62	0.79
Oceania	0.61	0.58	1.27	0.11	0.73	0.21	0.57	0.62	0.79
North America	0.61	0.58	1.27	0.11	0.73	0.21	0.57	0.62	0.79
South America	0.63	0.58	1.27	0.11	0.73	0.21	0.71	0.62	0.79
Central America	0.61	0.55	1.27	0.11	0.73	0.21	0.57	0.62	0.79
Caribbean	0.61	0.58	1.27	0.11	0.73	0.21	0.57	0.62	0.79

Notes: Values for agroforestry systems are the average of a given regional range. Values in red indicate generic values for all regions under the same climate.

Source: Authors' own elaboration based on Table 5.2 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A46. Below-ground biomass growth rate for various agroforestry systems, in tC/ha/yr, for tropical wet climate

Geographical regions	Agroforestry	Alley cropping	Perennial fallow	Hedgerow	Multistrata	Parkland	Shaded perennial	Silvoarable	Silvopasture
Africa	0.57	0.45	1.49	0.10	0.69	0.21	0.71	0.89	0.01
Asia	0.47	0.45	0.48	0.10	1.16	0.21	0.42	0.89	0.01
Europe	0.53	0.45	1.10	0.10	0.91	0.21	0.54	0.89	0.01
Oceania	0.53	0.45	1.10	0.10	0.91	0.21	0.54	0.89	0.01
North America	0.53	0.45	1.10	0.10	0.91	0.21	0.54	0.89	0.01
South America	0.51	0.45	1.10	0.10	0.78	0.21	0.51	0.89	0.01
Central America	0.53	0.45	1.14	0.1	0.70	0.21	0.71	0.89	0.01
Caribbean	0.53	0.45	1.10	0.10	0.91	0.21	0.54	0.89	0.01

Notes: Values for agroforestry systems are the average of a given regional range. Values in red indicate generic values for all regions under the same climate.

Source: Authors' own elaboration based on Table 5.2 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A47. Below-ground biomass growth rate for various agroforestry systems, in tC/ha/yr, for tropical montane climate

Geographical regions	Agroforestry	Alley cropping	Perennial fallow	Hedgerow	Multistrata	Parkland	Shaded perennial	Silvoarable	Silvopasture
Africa	0.59	0.55	1.12	0.17	0.80	0.21	0.55	0.56	0.75
Asia	0.63	0.55	1.49	0.17	0.80	0.21	0.55	0.56	0.75
Europe	0.63	0.55	1.49	0.17	0.80	0.21	0.55	0.56	0.75
Oceania	0.63	0.55	1.49	0.17	0.80	0.21	0.55	0.56	0.75
The Americas and the Caribbean	0.63	0.55	1.49	0.17	0.80	0.21	0.55	0.56	0.75

Notes: Values for agroforestry systems are the average of a given regional range. Values in red indicate generic values for all regions under the same climate.

Source: Authors' own elaboration based on Table 5.2 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A48. Maximum above-ground biomass at harvest for various agroforestry systems, in tC/ha, according to climate

IPCC climate zones	Agroforestry	Alley cropping	Perennial fallow	Hedgerow	Multistrata	Parkland	Shaded perennial	Silvoarable	Silvopasture
Tropical	**41.7**	47.4	22.1	9.4	65	11.8	48	72.2	58.2
Temperate and boreal	**39.7**	**47.4**	**22.1**	26.1	65	11.8	48	27.3	69.9
Tropical montane	**40.7**	**47.4**	**22.1**	**17.75**	**65**	**11.8**	**48**	**49.75**	**64.05**

Notes: IPCC = Intergovernmental Panel on Climate Change.Values in bold are the default values provided by the IPCC for a specific climate but applied to other climate in the absence of data. Tropical montane system values are the average between tropical and temperate.

Source: Authors' own elaboration based on Table 5.1 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A49. Maximum above-ground biomass at harvest for various monoculture agroforestry systems, in tC/ha, according to climate

IPCC climate zones	Olive	Orchard	Vine	Short rotation coppice	Oil palm	Rubber	Tea
Tropical	**9.1**	**8.5**	**5.5**	**12.69**	60	80.2	20.7
Temperate and boreal	9.1	8.5	5.5	12.69	**60**	**80.2**	20.7
Tropical montane	**9.1**	**8.5**	**5.5**	**12.69**	**60**	**80.2**	**20.7**

Notes: IPCC = Intergovernmental Panel on Climate Change. Values in bold are the default values provided by the IPCC for a specific climate but applied to other climate in the absence of data. Tropical montane system values are the average between tropical and temperate.

Source: Authors' own elaboration based on Table 5.3 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A50. Flooded rice: default cultivation period of rice

Geographical regions	Cultivation period, in days	Error range, in days
Africa	113	74–152
East Asia	112	73–147
Southeast Asia	102	78–150
South Asia	112	90–140
Europe	123	111–153
North America	139	110–165
South America	124	110–146
Global	113	74–152

Source: Authors' own elaboration based on Table 5.11A of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A51. Flooded rice: baseline emission factor (EFc) per region, in kg CH_4/ha/day

Geographical regions	Baseline emission factor
Africa	1.19
East Asia	1.32
Southeast Asia	1.22
South Asia	0.85
Europe	1.56
North America	0.65
South America	1.27
Global	1.19

Note: EFc was determined in the following conditions: i.e. (i) no flooding for less than 180 days prior to rice cultivation; and (ii) continuous flooding condition without organic amendments during the rice cultivation period.

Source: Authors' own elaboration based on Table 5.11 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A52. Flooded rice: default methane emission scaling factor for water regime during the cultivation period

Water regimes	Scaling factor (SFw)
Irrigated, continuously flooded	1.00
Irrigated, single drainage period	0.71
Irrigated, multiple drainage periods	0.55
Rainfed, wet season (regular rainfed)	0.54
Rainfed, dry season (drought prone)	0.16
Rainfed, deep water	0.06

Source: Authors' own elaboration based on Table 5.12 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A53. Flooded rice: default methane emission scaling factor for water regimes prior to rice cultivation

Water regimes	Scaling factor (SFp)
Non flooded pre-season <180 days	1.00
Non flooded pre-season >180 days	0.89
Flooded pre-season (>30 days)	2.41

Source: Authors' own elaboration based on Table 5.13 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A54. Flooded rice: Default conversion factor for various types of organic amendments

Organic amendments	Conversion factor (COA)
Straw incorporated shortly (<30 days) before cultivation	1.00
Straw incorporated long (>30 days) before cultivation	0.19
Compost	0.17
Farmyard manure	0.21
Green manure	0.45

Source: Authors' own elaboration based on Table 5.14 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A55. Default above-ground biomass, in t d.m./ha, in grasslands stratified by climate region

IPCC climate zones	Above-ground biomass
Boreal–dry and wet	1.7
Cool temperate–dry	1.7
Cool temperate–moist	2.4
Warm temperate–dry	1.6
Warm temperate–moist	2.7
Tropical–dry	2.3
Tropical–moist	6.2
Tropical–wet	6.2

Note: IPCC = Intergovernmental Panel on Climate Change.

Source: Authors' own elaboration based on Table 6.4 of IPCC. 2006. *2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Kanagawa, Japan. www.ipcc-nggip.iges.or.jp/public/2006gl

Table A56. Relative stock change factors for grassland management

Factor	Level	Climate regime	IPCC default
Land-use, FL_u	All	All	1.0
Management, F_{MG}	Nominally managed (non-degraded)	All	1.0
	High intensity grazing	All	0.90
	Severely degraded	All	0.7
	Improved grassland	Temperate/boreal	1.14
		Tropical	1.17
		Tropical montane	1.16
Input, F_I	Medium	All	1.0
	High	All	1.11

Note: IPCC = Intergovernmental Panel on Climate Change, FLU = land-use factor FMG = management factor, reflecting grazing intensity and grassland condition (e.g. nominally managed, high intensity grazing, degraded, improved), FI = input factor, reflecting the level of carbon inputs (e.g. medium or high).

Source: Authors' own elaboration based on Table 6.2 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A57. Methane emission factors, in kg CH_4/head/year, from enteric fermentation, for high productivity systems

Geographical regions	Dairy cattle	Other cattle	Dairy buffalo	Other buffalo	Dairy sheep	Other sheep	Growing swine	Breeding swine	Goats	Horses	Camels	Mules and asses	Deer	Ostrich	Chicken layer	Chicken broiler	Llamas and alpacas
Sub-Saharan Africa	86	60	81	81	9	9	1.5	1.5	9	18	46	10	20	5	0.00004868	0.00004868	8
East Asia and South-East Asia	96	43	76	76	9	9	1.5	1.5	9	18	46	10	20	5	0.00004868	0.00004868	8
North America	138	64	0	0	9	9	1.5	1.5	9	18	46	10	20	5	0.00004868	0.00004868	8
Latin America and the Caribbean	103	55	68	68	9	9	1.5	1.5	9	18	46	10	20	5	0.00004868	0.00004868	8
Western Europe	126	52	78	78	9	9	1.5	1.5	9	18	46	10	20	5	0.00004868	0.00004868	8
Eastern Europe	93	58	68	68	9	9	1.5	1.5	9	18	46	10	20	5	0.00004868	0.00004868	8
Russian Federation	93	58	68	68	9	9	1.5	1.5	9	18	46	10	20	5	0.00004868	0.00004868	8
Oceania	93	63	0	0	9	9	1.5	1.5	9	18	46	10	20	5	0.00004868	0.00004868	8
Near East and North Africa/Middle East	94	61	67	67	9	9	1.5	1.5	9	18	46	10	20	5	0.00004868	0.00004868	8
South Asia	70	41	85	85	9	9	1.5	1.5	9	18	46	10	20	5	0.00004868	0.00004868	8

Note: Proxy from Eastern Europe have been applied for Russian Federation.

Sources: Authors' own elaboration based on Tables 10.10 and 10.11 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html and on Wang, S-Y., Huang, D-J. 2005. Assessment of Greenhouse Gas Emissions from Poultry Enteric Fermentation. *Asian-Australasian Journal of Animal Sciences*, 18(6): 873-878. doi.org/10.5713/ajas.2005.873 for poultry.

Table A58. Methane emission factors, in kg CH_4/head/year, from enteric fermentation, for low productivity systems

Geographical regions	Dairy cattle	Other cattle	Dairy buffalo	Other buffalo	Dairy sheep	Other sheep	Growing swine	Breeding swine	Goats	Horses	Camels	Mules and asses	Deer	Ostrich	Chicken layer	Chicken broiler	Llamas and alpacas
Sub-Saharan Africa	66	48	81	81	5	5	1	1	5	18	46	10	20	5	0.00004868	0.00004868	8
East Asia and South-East Asia	71	56	76	76	5	5	1	1	5	18	46	10	20	5	0.00004868	0.00004868	8
North America	138	64	0	0	5	5	1	1	5	18	46	10	20	5	0.00004868	0.00004868	8
Latin America and the Caribbean	78	58	68	68	5	5	1	1	5	18	46	10	20	5	0.00004868	0.00004868	8
Western Europe	126	52	78	78	5	5	1	1	5	18	46	10	20	5	0.00004868	0.00004868	8
Eastern Europe	93	58	68	68	5	5	1	1	5	18	46	10	20	5	0.00004868	0.00004868	8
Russian Federation	93	58	68	68	5	5	1	1	5	18	46	10	20	5	0.00004868	0.00004868	8
Oceania	93	63	0	0	5	5	1	1	5	18	46	10	20	5	0.00004868	0.00004868	8
Near East and North Africa/Middle East	62	55	67	67	5	5	1	1	5	18	46	10	20	5	0.00004868	0.00004868	8
South Asia	74	47	85	85	5	5	1	1	5	18	46	10	20	5	0.00004868	0.00004868	8

Note: Proxy from Eastern Europe have been applied for the Russian Federation.

Sources: Authors' own elaboration based on Tables 10.10 and 10.11 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html and on Wang, S-Y., Huang, D-J. 2005. Assessment of Greenhouse Gas Emissions from Poultry Enteric Fermentation. *Asian-Australasian Journal of Animal Sciences*, 18(6): 873-878. doi.org/10.5713/ajas.2005.873 for poultry.

Table A59. Methane emission factors, in kg CH_4/head/year, from enteric fermentation, for default productivity systems

Geographical regions	Dairy cattle	Other cattle	Dairy buffalo	Other buffalo	Dairy sheep	Other sheep	Growing swine	Breeding swine	Goats	Horses	Camels	Mules and asses	Deer	Ostrich	Chicken layer	Chicken broiler	Llamas and alpacas
Sub-Saharan Africa	76	52	81	81	5	5	1	1	5	18	46	10	20	5	0.000 04868	0.000 04868	8
East Asia and South-East Asia	78	54	76	76	5	5	1	1	5	18	46	10	20	5	0.000 04868	0.000 04868	8
North America	138	64	0	0	5	5	1	1	5	18	46	10	20	5	0.000 04868	0.000 04868	8
Latin America and the Caribbean	87	56	68	68	5	5	1	1	5	18	46	10	20	5	0.000 04868	0.000 04868	8
Western Europe	126	52	78	78	5	5	1	1	5	18	46	10	20	5	0.000 04868	0.000 04868	8
Eastern Europe	93	58	68	68	5	5	1	1	5	18	46	10	20	5	0.000 04868	0.000 04868	8
Russian Federation	93	58	68	68	5	5	1	1	5	18	46	10	20	5	0.000 04868	0.000 04868	8
Oceania	93	63	0	0	5	5	1	1	5	18	46	10	20	5	0.000 04868	0.000 04868	8
Near East and North Africa/Middle East	76	60	67	67	5	5	1	1	5	18	46	10	20	5	0.000 04868	0.000 04868	8
South Asia	73	46	85	85	5	5	1	1	5	18	46	10	20	5	0.000 04868	0.000 04868	8

Notes: For sheep, swine and goat, EX-ACT uses the low-productivity proxy. For the Russian Federation, EX-ACT applies the proxy from Eastern Europe.

Sources: Authors' own elaboration based on Tables 10.10 and 10.11 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html and on Wang, S-Y., Huang, D-J. 2005. Assessment of Greenhouse Gas Emissions from Poultry Enteric Fermentation. *Asian-Australasian Journal of Animal Sciences*, 18(6): 873-878. doi.org/10.5713/ajas.2005.873 for poultry.

Table A60. Definition of productivity systems according to livestock categories

Main livestock categories	Productivity systems
Dairy cattle	Mature cows (first lactation and beyond) that are producing milk in commercial quantities for consumption. Dairy cow population should not be confused with multi-purpose cows that may be used for more than one production purpose milk, meat or draft.
Dairy cattle or buffalo in high-productivity systems	High-yielding dairy cows that are concentrated in confinement production systems or grazing on high quality pastures with supplements. The farms are 100-percent market oriented for commercial milk production, for national markets and/or export; Purebred or crossbred cattle are genetically improved through selective breeding for milk production.
Dairy cattle or buffalo in low-productivity systems	Low-yielding dairy cows, grazing non improved pastures, and using locally produced roughage (i.e. crop residues), and agro-industrial by-products. Local breeds or crossbred cows are bred locally, without intensive selection for milk productivity. Milk production is mostly for local market and local consumption.
Other cattle in high-productivity systems	Productivity system based on animal feeding systems using forage (i.e. high-quality grass) and concentrates in confinement production systems or grazing with supplements or on improved pastures, producing high rates of daily weight gain. Animals can be purebred or crossbred and are genetically improved through selective breeding for improved commercial meat production. Growing cattle may be finished young in "intensive grazing with supplements" or feedlot systems, and meat is produced for national markets and/or export.
Other cattle in low-productivity systems	Productivity system based on animal feeding systems where locally produced roughage (i.e. crop residues) or low-quality rangelands represent the major source of feed utilized, producing low rates of daily weight gain. Animals can be represented by local breeds or may be crossbred and can also be used for multiple purposes such as draft, meat and milk for self-consumption and markets.
Other livestock in high-productivity systems	Productivity system 100 percent market oriented with high level of capital input requirements and high level of overall herd (flock) performance. Feed is purchased from local or international market or intensively produced on farm. Animals are improved through breeding practices for commercial production. The high-productivity systems are common in swine, poultry, goats and sheep production.
Other livestock in low-productivity systems	Productivity system mainly driven by local market or by self-consumption, with low capital input requirements and low level of overall herd (fowl) performance typically using large areas for production or backyards. Locally produced feed represents the major source of feed utilized or animals are kept-free range for major part or all of their production cycle, the yield of the activity being linked to the natural fertility of the land and the seasonal production of the pastures. The low-productivity systems are common in swine, poultry, goats and sheep production

Source: Authors' own elaboration based on Chapter 10 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A61. Definition and characteristics of manure management systems

Manure management systems	Definition and characteristics
Pasture/range/paddock	The manure from pasture and range grazing animals is allowed to lie as deposited and is not managed.
Daily spread	Manure is routinely removed from a confinement facility and is applied to cropland or pasture within 24 hours of excretion.
Solid storage	The storage of manure, typically for a period of several months, in unconfined piles or stacks. Manure is able to be stacked because of the presence of a sufficient amount of bedding material or loss of moisture by evaporation.
Dry lot	A paved or unpaved open confinement area without any significant vegetative cover. Dry lots do not require the addition of bedding to control moisture. Manure may be removed periodically and spread on fields.
Liquid/Slurry	Manure is stored as excreted or with some minimal addition of water or bedding material in tanks or ponds outside the animal housing. Manure is removed and spread on fields once or more in a calendar year. Manure is agitated before removal from the tank/ponds to ensure that most of the VS are removed from the tank.
Lagoons (uncovered anaerobic)	A type of liquid storage system designed and operated to combine waste stabilization and storage. Lagoons have a lower depth and a much larger surface compared to liquid slurry stores. Anaerobic lagoons are designed with varying lengths of storage (up to a year or greater), depending on the climate region, the volatile solids loading rate, and other operational factors. The supernatant water from the lagoon may be recycled as flush water or used to irrigate and fertilise fields.
Pit storage below animal confinements	Collection and storage of manure usually with little or no added water typically below a slatted floor in an enclosed animal confinement facility, usually for periods less than one year. Manure may be pumped out of the storage to a secondary storage tank multiple times in one year or stored and applied directly to fields. It is assumed that VS removal rates on tank emptying are >90 percent.
Anaerobic digester	Anaerobic fermentation of slurry and/or solid. Biogas is captured and flared or used as a fuel.
Burned for fuel	The dung and urine are excreted on fields. The sun-dried dung cakes are burned for fuel.
Pit storage below animal confinements	Mixture of excreta and washing water, stored within the livestock building, usually below the confined animals.
Poultry manure	Similar to cattle and swine deep bedding except usually not combined with a dry lot or pasture. Typically used for all poultry breeder flocks, for alternative systems for layers and for the production of meat type chickens (broilers) and other fowl. Litter and manure are left in place with added bedding during the poultry production cycle and cleaned between poultry cycles, typically 5 to 9 weeks in productive systems and greater in lower productivity systems.
Composting	Biological oxidation of a solid waste including manure usually with bedding or another organic carbon source typically at thermophilic temperatures produced by microbial heat production.

Source: Authors' own elaboration based on Chapter 10 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A62. Typical animal mass, in kilogram, for high-productivity system

Geographical regions	Dairy cattle	Other cattle	Dairy buffalo	Other buffalo	Dairy sheep	Other sheep	Growing swine	Breeding swine	Goats	Horses	Camels	Mules and asses	Deer	Ostrich	Chicken layer	Chicken broiler	Llamas and alpacas
Sub-Saharan Africa	250	302	339	339	31	31	54	200	24	238	217	130	120	120	1	0.8	65
East Asia and South-East Asia	485	310	336	336	31	31	56	160	24	238	217	130	120	120	1.4	1	65
North America	650	407	0	0	40	40	61	184	41	377	217	130	120	120	1.4	1.4	65
Latin America and the Caribbean	520	329	315	315	31	31	59	205	24	238	217	130	120	120	1.3	1.2	65
Western Europe	600	405	509	509	40	40	61	190	40	377	217	130	120	120	1.4	1.2	65
Eastern Europe	550	389	467	467	40	40	59	204	36	377	217	130	120	120	1.3	1.1	65
Russian Federation	550	389	467	467	40	40	59	204	36	377	217	130	120	120	1.3	1.1	65
Oceania	488	359	0	0	40	40	41	163	33	377	217	130	120	120	1.3	1.2	65
Near East and North Africa/Middle East	510	362	381	381	31	31	60	157	24	238	217	130	120	120	1.2	1	65
South Asia	350	167	321	321	31	31	55	162	24	238	217	130	120	120	1.2	1	65

Source: Authors' own elaboration based on Tables 10A.5 and 10.10 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A63. Typical animal mass, in kilogram, for low-productivity system

Geographical regions	Dairy cattle	Other cattle	Dairy buffalo	Other buffalo	Dairy sheep	Other sheep	Growing swine	Breeding swine	Goats	Horses	Camels	Mules and asses	Deer	Ostrich	Chicken layer	Chicken broiler	Llamas and alpacas
Sub-Saharan Africa	270	208	339	339	31	31	33	61	24	238	217	130	120	120	0.8	0.7	65
East Asia and South-East Asia	355	296	336	336	31	31	44	102	24	238	217	130	120	120	1	0.7	65
North America	650	407	0	0	40	40	61	184	41	377	217	130	120	120	1.4	1.4	65
Latin America and the Caribbean	500	295	315	315	31	31	47	121	24	238	217	130	120	120	0.9	0.7	65
Western Europe	600	405	509	509	40	40	61	190	40	377	217	130	120	120	1.4	1.2	65
Eastern Europe	550	389	467	467	40	40	59	204	36	377	217	130	120	120	1.3	1.1	65
Russian Federation	550	389	467	467	40	40	59	204	36	377	217	130	120	120	1.3	1.1	65
Oceania	488	359	0	0	40	40	41	163	33	377	217	130	120	120	1.3	1.2	65
Near East and North Africa/Middle East	270	232	381	381	31	31	48	99	24	238	217	130	120	120	0.7	0.5	65
South Asia	265	236	321	321	31	31	48	99	24	238	217	130	120	120	0.8	0.6	65

Source: Authors' own elaboration based on Tables 10A.5 and 10.10 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A64. Typical animal mass, in kilogram, for default productivity system

Geographical regions	Dairy cattle	Other cattle	Dairy buffalo	Other buffalo	Dairy sheep	Other sheep	Growing swine	Breeding swine	Goats	Horses	Camels	Mules and asses	Deer	Ostrich	Chicken layer	Chicken broiler	Llamas and alpacas
Sub-Saharan Africa	260	236	339	339	31	31	33	61	24	238	217	130	120	120	0.9	0.8	65
East Asia and South-East Asia	386	299	336	336	31	31	44	102	24	238	217	130	120	120	1.2	0.8	65
North America	650	407	0	0	40	40	61	184	41	377	217	130	120	120	1.4	1.4	65
Latin America and the Caribbean	508	303	315	315	31	31	47	121	24	238	217	130	120	120	1.1	0.9	65
Western Europe	600	405	509	509	40	40	61	190	40	377	217	130	120	120	1.4	1.2	65
Eastern Europe	550	389	467	467	40	40	59	204	36	377	217	130	120	120	1.3	1.1	65
Russian Federation	550	389	467	467	40	40	59	204	36	377	217	130	120	120	1.3	1.1	65

Geographical regions	Dairy cattle	Other cattle	Dairy buffalo	Other buffalo	Dairy sheep	Other sheep	Growing swine	Breeding swine	Goats	Horses	Camels	Mules and asses	Deer	Ostrich	Chicken layer	Chicken broiler	Llamas and alpacas
Oceania	488	359	0	0	40	40	41	163	33	377	217	130	120	120	1.3	1.2	65
Near East and North Africa/Middle East	349	275	381	381	31	31	48	99	24	238	217	130	120	120	0.9	0.7	65
South Asia	285	226	321	321	31	31	48	99	24	238	217	130	120	120	1	0.8	65

Source: Authors' own elaboration based on Tables 10A.5 and 10.10 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A65. Volatile solid excretion rate, in kgVS/(1 000 kg TAM)/day, for default productivity system

Geographical regions	Dairy cattle	Other cattle	Dairy buffalo	Other buffalo	Dairy sheep	Other sheep	Growing swine	Breeding swine	Goats	Horses	Camels	Mules and asses	Chicken layer	Chicken broiler
Sub-Saharan Africa	18.2	12	12.9	12.9	8.3	8.3	8.2	4.4	10.4	7.2	11.5	7.2	12.6	15.9
East Asia and South-East Asia	9	9.8	13.5	13.5	8.3	8.3	6.8	3.4	10.4	7.2	11.5	7.2	11.2	15.7
North America	9.3	7.6	0	0	8.2	8.2	3.9	1.8	9	5.65	11.5	7.2	14.5	16.8
Latin America and the Caribbean	7.9	8.5	11.2	11.2	8.3	8.3	6.4	2.7	10.4	7.2	11.5	7.2	13.5	15.6
Western Europe	7.5	5.7	7.7	7.7	8.2	8.2	5.3	2.4	9	5.65	11.5	7.2	12.3	16.1
Eastern Europe	6.7	7.6	6.2	6.2	8.2	8.2	4.9	2	9	5.65	11.5	7.2	12.6	16
Russian Federation	6.7	7.6	6.2	6.2	8.2	8.2	4.9	2	9	5.65	11.5	7.2	12.6	16
Oceania	6	8.7	0	0	8.2	8.2	5.6	2.1	9	5.65	11.5	7.2	15.4	18.3
Near East and North Africa/Middle East	10.7	14.1	9.8	9.8	8.3	8.3	4.9	2.5	10.4	7.2	11.5	7.2	14.2	17.7
South Asia	14.1	12.2	0	0	8.3	8.3	8.6	4.6	10.4	7.2	11.5	7.2	14.9	17.7

Source: Authors' own elaboration based on Table 10.13a of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A66. Volatile solid excretion rate, in kgVS/(1 000 kg TAM)/day, for high-productivity system

Geographical regions	Dairy cattle	Other cattle	Dairy buffalo	Other buffalo	Dairy sheep	Other sheep	Growing swine	Breeding swine	Goats	Horses	Camels	Mules and asses	Chicken layer	Chicken broiler
Sub-Saharan Africa	21.7	10.2	12.9	12.9	8.3	8.3	5.3	2.4	10.4	7.2	11.5	7.2	12.3	16
East Asia and South-East Asia	8.1	6.8	13.5	13.5	8.3	8.3	5.1	2.3	10.4	7.2	11.5	7.2	10.6	15.6
North America	9.3	7.6	0	0	8.2	8.2	3.9	1.8	9	5.65	11.5	7.2	14.5	16.8
Latin America and the Caribbean	9	8.1	11.2	11.2	8.3	8.3	4.3	1.7	10.4	7.2	11.5	7.2	13.3	15.5
Western Europe	7.5	5.7	7.7	7.7	8.2	8.2	5.3	2.4	9	5.65	11.5	7.2	12.3	16.1
Eastern Europe	6.7	7.6	6.2	6.2	8.2	8.2	4.9	2	9	5.65	11.5	7.2	12.6	16
Russian Federation	6.7	7.6	6.2	6.2	8.2	8.2	4.9	2	9	5.65	11.5	7.2	12.6	16
Oceania	6	8.7	0	0	8.2	8.2	5.6	2.1	9	5.65	11.5	7.2	15.4	18.3
Near East and North Africa/Middle East	8.4	10.5	9.8	9.8	8.3	8.3	4.4	2.3	10.4	7.2	11.5	7.2	14.1	17.7
South Asia	9.1	13.5	0	0	8.3	8.3	6.5	3	10.4	7.2	11.5	7.2	14.3	17.6

Source: Authors' own elaboration based on Table 10.13a of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A67. Volatile solid excretion rate, in kgVs/(1 000 kg TAM)/day, for low-productivity system

Geographical regions	Dairy cattle	Other cattle	Dairy buffalo	Other buffalo	Dairy sheep	Other sheep	Growing swine	Breeding swine	Goats	Horses	Camels	Mules and asses	Chicken layer	Chicken broiler
Sub-Saharan Africa	15.2	12.7	12.9	12.9	8.3	8.3	9.4	6	10.4	7.2	11.5	7.2	13	15.4
East Asia and South-East Asia	9.2	10.8	13.5	13.5	8.3	8.3	8.1	4.3	10.4	7.2	11.5	7.2	14.3	17.1
North America	9.3	7.6	0	0	8.2	8.2	3.9	1.8	9	5.65	11.5	7.2	14.5	16.8
Latin America and the Caribbean	7.1	8.6	11.2	11.2	8.3	8.3	10	4.8	10.4	7.2	11.5	7.2	15.7	17.8
Western Europe	7.5	5.7	7.7	7.7	8.2	8.2	5.3	2.4	9	5.65	11.5	7.2	12.3	16.1
Eastern Europe	6.7	7.6	6.2	6.2	8.2	8.2	4.9	2	9	5.65	11.5	7.2	12.6	16
Russian Federation	6.7	7.6	6.2	6.2	8.2	8.2	4.9	2	9	5.65	11.5	7.2	12.6	16
Oceania	6	8.7	0	0	8.2	8.2	5.6	2.1	9	5.65	11.5	7.2	15.4	18.3
Near East and North Africa/Middle East	11.8	16.8	9.8	9.8	8.3	8.3	7.8	4.6	10.4	7.2	11.5	7.2	16.5	17.9
South Asia	16.1	12	0	0	8.3	8.3	9.5	5.5	10.4	7.2	11.5	7.2	15.7	18.2

Source: Authors' own elaboration based on Table 10.13a of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A68. Animal waste management system regional average, in percentage, for dairy cattle

Geographical regions	Lagoon	Liquid/slurry	Solid storage	Dry lot	PRP	Daily spread	Digester	Burned for fuel	Pit<1 month	Pit>1 month	Poultry manure	Deep bedding	Composting	Aerobic treatment
Sub-Saharan Africa	0	0	20	29	45	0	0	6	0	0	0	0	0	0
East Asia and South-East Asia	0	1	21	29	38	0	0	11	0	0	0	0	0	0
North America	26	24	24	0	15	11	0	0	0	0	0	0	0	0
Latin America and the Caribbean	0	0	5	38	57	0	0	0	0	0	0	0	0	0
Western Europe	0	43	29	0	26	2	0	0	0	0	0	0	0	0
Eastern Europe	0	5	74	0	20	1	0	0	0	0	0	0	0	0
Russian Federation	0	5	74	0	20	1	0	0	0	0	0	0	0	0
Oceania	5	0	0	0	94	1	0	0	0	0	0	0	0	0
Near East and North Africa/Middle East	0	0	14	35	46	0	0	5	0	0	0	0	0	0
South Asia	0	0	1	49	30	0	0	20	0	0	0	0	0	0

Notes: PRP stands for pasture/range/paddock. Eastern Europe's proxies are used for the Russian Federation.

Source: Authors' own elaboration based on Table 10A.6 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A69. Animal waste management system regional average, in percentage, for other cattle

Geographical regions	Lagoon	Liquid/slurry	Solid storage	Dry lot	PRP	Daily spread	Digester	Burned for fuel	Pit<1 month	Pit>1 month	Poultry manure	Deep bedding	Composting	Aerobic treatment
Sub-Saharan Africa	0	0	15	30	50	0	0	5	0	0	0	0	0	0
East Asia and South-East Asia	0	0	29	28	36	0	0	7	0	0	0	0	0	0
North America	0	1	43	14	42	0	0	0	0	0	0	0	0	0
Latin America and the Caribbean	0	0	3	5	92	0	0	0	0	0	0	0	0	0
Western Europe	0	22	26	0	48	4	0	0	0	0	0	0	0	0
Eastern Europe	0	64	5	0	31	0	0	0	0	0	0	0	0	0
Russian Federation	0	64	5	0	31	0	0	0	0	0	0	0	0	0
Oceania	0	0	0	0	100	0	0	0	0	0	0	0	0	0
Near East and North Africa/Middle East	0	0	5	46	42	0	0	7	0	0	0	0	0	0
South Asia	0	0	1	49	30	0	0	20	0	0	0	0	0	0

Notes: PRP stands for pasture/range/paddock. Eastern Europe's proxies are used for the Russian Federation.

Sources: Authors' own elaboration based on Table 10A.6 of IPCC. 2019. *2019 Refinement to the 2006 IPCC* Source: Authors' own elaboration based on Table 10A.6 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A70. Animal waste management system regional average, in percentage, for dairy buffalo

Geographical regions	Lagoon	Liquid/slurry	Solid storage	Dry lot	PRP	Daily spread	Digester	Burned for fuel	Pit<1 month	Pit>1 month	Poultry manure	Deep bedding	Composting	Aerobic treatment
Sub-Saharan Africa	0	0	0	0	0	0	0	0	0	0	0	0	0	0
East Asia and South-East Asia	0	0	10	58	29	0	0	3	0	0	0	0	0	0
North America	0	43	40	0	17	0	0	0	0	0	0	0	0	0
Latin America and the Caribbean	0	0	2	48	50	0	0	0	0	0	0	0	0	0
Western Europe	0	34	63	0	3	0	0	0	0	0	0	0	0	0
Eastern Europe	0	18	68	0	13	1	0	0	0	0	0	0	0	0
Russian Federation	0	18	68	0	13	1	0	0	0	0	0	0	0	0
Oceania	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Near East and North Africa/Middle East	0	0	18	35	46	0	0	1	0	0	0	0	0	0
South Asia	0	0	1	41	38	0	0	20	0	0	0	0	0	0

Note: PRP stands for pasture/range/paddock. Eastern Europe's proxies are used for the Russian Federation.

Source: Authors' own elaboration based on Table 10A.6 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A71. Animal waste management system regional average, in percentage, for other buffalo

Geographical regions	Lagoon	Liquid/slurry	Solid storage	Dry lot	PRP	Daily spread	Digester	Burned for fuel	Pit<1 month	Pit>1 month	Poultry manure	Deep bedding	Composting	Aerobic treatment
Sub-Saharan Africa	0	0	0	0	0	0	0	0	0	0	0	0	0	0
East Asia and South-East Asia	0	0	6	64	28	0	0	2	0	0	0	0	0	0
North America	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Latin America and the Caribbean	0	0	2	5	93	0	0	0	0	0	0	0	0	0
Western Europe	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Eastern Europe	0	9	64	0	27	0	0	0	0	0	0	0	0	0
Russian Federation	0	9	64	0	27	0	0	0	0	0	0	0	0	0
Oceania	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Near East and North Africa/Middle East	0	0	16	12	57	0	0	15	0	0	0	0	0	0

Geographical regions	Lagoon	Liquid/slurry	Solid storage	Dry lot	PRP	Daily spread	Digester	Burned for fuel	Pit<1 month	Pit>1 month	Poultry manure	Deep bedding	Composting	Aerobic treatment
South Asia	0	0	1	40	39	0	0	20	0	0	0	0	0	0

Note: PRP stands for pasture/range/paddock. Eastern Europe's proxies are used for the Russian Federation.

Source: Authors' own elaboration based on Table 10A.6 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A72. Animal waste management system regional average, in percentage, for breeding swine in high-productivity system

Geographical regions	Lagoon	Liquid/slurry	Solid storage	Dry lot	PRP	Daily spread	Digester	Burned for fuel	Pit<1 month	Pit>1 month	Poultry manure	Deep bedding	Composting	Aerobic treatment
Sub-Saharan Africa	0	7	6	86	0	0	0	0	1	0	0	0	0	0
East Asia and South-East Asia	35	21	0	2	0	0	7	0	35	0	0	0	0	0
North America	28	31	4	3	0	0	0	0	0	34	0	0	0	0
Latin America and the Caribbean	11	34	12	41	0	2	0	0	0	0	0	0	0	0
Western Europe	6	51	15	0	0	1	0	0	2	25	0	0	0	0
Eastern Europe	5	31	55	1	0	0	0	0	4	4	0	0	0	0
Russian Federation	0	24	76	0	0	0	0	0	0	0	0	0	0	0
Oceania	91	0	1	8	0	0	0	0	0	0	0	0	0	0
Near East and North Africa/Middle East	10	29	0	54	0	0	7	0	0	0	0	0	0	0
South Asia	12	23	14	32	0	8	8	0	3	0	0	0	0	0

Note: PRP stands for pasture/range/paddock. Eastern Europe's proxies are used for the Russian Federation.

Source: Authors' own elaboration based on Table 10A.7 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A73. Animal waste management system regional average, in percentage, for breeding swine in low-productivity system

Geographical regions	Lagoon	Liquid/slurry	Solid storage	Dry lot	PRP	Daily spread	Digester	Burned for fuel	Pit<1 month	Pit>1 month	Poultry manure	Deep bedding	Composting	Aerobic treatment
Sub-Saharan Africa	5	30	15	15	5	5	5	0	15	5	0	0	0	0
East Asia and South-East Asia	4	28	22	13	6	6	4	0	13	4	0	0	0	0
North America	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Geographical regions	Lagoon	Liquid/slurry	Solid storage	Dry lot	PRP	Daily spread	Digester	Burned for fuel	Pit<1 month	Pit>1 month	Poultry manure	Deep bedding	Composting	Aerobic treatment
Latin America and the Caribbean	5	30	15	15	5	5	5	0	15	5	0	0	0	0
Western Europe	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Eastern Europe	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Russian Federation	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Oceania	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Near East and North Africa/Middle East	5	30	15	15	5	5	5	0	15	5	0	0	0	0
South Asia	5	30	15	15	5	5	5	0	15	5	0	0	0	0

Notes: PRP stands for pasture/range/paddock. Eastern Europe's proxies are used for the Russian Federation.

Source: Authors' own elaboration based on Table 10A.7 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A74. Animal waste management system regional average, in percentage, for growing swine in high-productivity system

Geographical regions	Lagoon	Liquid/slurry	Solid storage	Dry lot	PRP	Daily spread	Digester	Burned for fuel	Pit<1 month	Pit>1 month	Poultry manure	Deep bedding	Composting	Aerobic treatment
Sub-Saharan Africa	0	7	6	86	0	0	0	0	1	0	0	0	0	0
East Asia and South-East Asia	35	21	0	2	0	0	7	0	35	0	0	0	0	0
North America	28	31	4	3	0	0	0	0	0	34	0	0	0	0
Latin America and the Caribbean	11	34	12	41	0	2	0	0	0	0	0	0	0	0
Western Europe	6	51	14	0	0	1	0	0	2	26	0	0	0	0
Eastern Europe	5	31	55	1	0	0	0	0	4	4	0	0	0	0
Russian Federation	0	24	76	0	0	0	0	0	0	0	0	0	0	0
Oceania	91	0	1	8	0	0	0	0	0	0	0	0	0	0
Near East and North Africa/Middle East	10	29	0	54	0	0	7	0	0	0	0	0	0	0
South Asia	12	23	13	35	0	7	8	2	2	0	0	0	0	0

Notes: PRP stands for pasture/range/paddock. Eastern Europe's proxies are used for the Russian Federation.

Source: Authors' own elaboration based on Table 10A.7 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A75. Animal waste management system regional average, in percentage, for growing swine in low-productivity system

Geographical regions	Lagoon	Liquid/slurry	Solid storage	Dry lot	PRP	Daily spread	Digester	Burned for fuel	Pit<1 month	Pit>1 month	Poultry manure	Deep bedding	Composting	Aerobic treatment
Sub-Saharan Africa	5	30	15	15	5	5	5	0	15	5	0	0	0	0
East Asia and South-East Asia	5	27	18	14	6	6	5	0	14	5	0	0	0	0
North America	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Latin America and the Caribbean	5	30	15	15	5	5	5	0	15	5	0	0	0	0
Western Europe	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Eastern Europe	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Russian Federation	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Oceania	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Near East and North Africa/Middle East	5	30	15	15	5	5	5	0	15	5	0	0	0	0
South Asia	5	30	15	15	5	5	5	0	15	5	0	0	0	0

Notes: PRP stands for pasture/range/paddock. Eastern Europe's proxies are used for the Russian Federation.

Source: Authors' own elaboration based on Table 10A.7 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A76. Animal waste management system regional average, in percentage, for other sheep

Geographical regions	Lagoon	Liquid/slurry	Solid storage	Dry lot	PRP	Daily spread	Digester	Burned for fuel	Pit<1 month	Pit>1 month	Poultry manure	Deep bedding	Composting	Aerobic treatment
Sub-Saharan Africa	0	0	17	3	80	0	0	0	0	0	0	0	0	0
East Asia and South-East Asia	0	0	17	3	80	0	0	0	0	0	0	0	0	0
North America	0	0	54	0	46	0	0	0	0	0	0	0	0	0
Latin America and the Caribbean	0	0	17	3	80	0	0	0	0	0	0	0	0	0
Western Europe	0	0	13	0	87	0	0	0	0	0	0	0	0	0
Eastern Europe	0	0	54	0	46	0	0	0	0	0	0	0	0	0
Russian Federation	0	0	54	0	46	0	0	0	0	0	0	0	0	0
Oceania	0	0	0	0	100	0	0	0	0	0	0	0	0	0
Near East and North Africa/Middle East	0	0	0	50	50	0	0	0	0	0	0	0	0	0
South Asia	0	0	17	3	80	0	0	0	0	0	0	0	0	0

Notes: PRP stands for pasture/range/paddock. Eastern Europe's proxies are used for the Russian Federation.

Source: Authors' own elaboration based on Table 10A.8 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A77. Animal waste management system regional average, in percentage, for dairy sheep

Geographical regions	Lagoon	Liquid/slurry	Solid storage	Dry lot	PRP	Daily spread	Digester	Burned for fuel	Pit<1 month	Pit>1 month	Poultry manure	Deep bedding	Composting	Aerobic treatment
Sub-Saharan Africa	0	0	17	3	80	0	0	0	0	0	0	0	0	0
East Asia and South-East Asia	0	0	17	3	80	0	0	0	0	0	0	0	0	0
North America	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Latin America and the Caribbean	0	0	17	3	80	0	0	0	0	0	0	0	0	0
Western Europe	0	0	21	0	78	0	0	0	0	0	0	0	0	0
Eastern Europe	0	0	42	0	58	0	0	0	0	0	0	0	0	0
Russian Federation	0	0	42	0	58	0	0	0	0	0	0	0	0	0
Oceania	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Near East and North Africa/Middle East	0	0	0	50	50	0	0	0	0	0	0	0	0	0
South Asia	0	0	17	3	80	0	0	0	0	0	0	0	0	0

Notes: PRP stands for pasture/range/paddock. Eastern Europe's proxies are used for the Russian Federation.

Source: Authors' own elaboration based on Table 10A.8 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A78. Animal waste management system regional average, in percentage, for goat

Geographical regions	Lagoon	Liquid/slurry	Solid storage	Dry lot	PRP	Daily spread	Digester	Burned for fuel	Pit<1 month	Pit>1 month	Poultry manure	Deep bedding	Composting	Aerobic treatment
Sub-Saharan Africa	0	0	17	3	80	0	0	0	0	0	0	0	0	0
East Asia and South-East Asia	0	0	50	0	50	0	0	0	0	0	0	0	0	0
North America	0	0	50	0	50	0	0	0	0	0	0	0	0	0
Latin America and the Caribbean	0	0	17	3	80	0	0	0	0	0	0	0	0	0
Western Europe	0	0	28	0	72	0	0	0	0	0	0	0	0	0
Eastern Europe	0	0	9	0	91	0	0	0	0	0	0	0	0	0
Russian Federation	0	0	82	0	18	0	0	0	0	0	0	0	0	0
Oceania	0	0	0	0	100	0	0	0	0	0	0	0	0	0
Near East and North Africa/Middle East	0	0	0	50	50	0	0	0	0	0	0	0	0	0
South Asia	0	0	50	0	50	0	0	0	0	0	0	0	0	0

Notes: PRP stands for pasture/range/paddock. Eastern Europe's proxies are used for the Russian Federation.

Source: Authors' own elaboration based on Table 10A.8 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A79. Animal waste management system regional average, in percentage, for chicken layer

Geographical regions	Lagoon	Liquid/slurry	Solid storage	Dry lot	PRP	Daily spread	Digester	Burned for fuel	Pit<1 month	Pit>1 month	Poultry manure	Deep bedding	Composting	Aerobic treatment
Sub-Saharan Africa	0	0	0	0	0	0	0	0	0	90	10	0	0	0
East Asia and South-East Asia	0	4	0	0	1	1	0	0	0	94	0	0	0	0
North America	1	29	70	0	0	0	0	0	0	0	0	0	0	0
Latin America and the Caribbean	0	58	42	0	0	0	0	0	0	0	0	0	0	0
Western Europe	0	1	20	21	0	1	0	0	0	43	14	0	0	0
Eastern Europe	0	0	0	47	0	0	0	0	0	34	19	0	0	0
Russian Federation	0	0	0	0	0	0	0	0	0	100	0	0	0	0
Oceania	0	0	0	0	23	0	0	0	0	77	0	0	0	0
Near East and North Africa/Middle East	11	7	11	0	0	0	0	0	0	67	4	0	0	0
South Asia	0	0	100	0	0	0	0	0	0	0	0	0	0	0

Notes: PRP stands for pasture/range/paddock. Eastern Europe's proxies are used for the Russian Federation.

Source: Authors' own elaboration based on Table 10A.9 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A80. Animal waste management system regional average, in percentage, for chicken broiler in high-productivity system

Geographical regions	Lagoon	Liquid/slurry	Solid storage	Dry lot	PRP	Daily spread	Digester	Burned for fuel	Pit<1 month	Pit>1 month	Poultry manure	Deep bedding	Composting	Aerobic treatment
Sub-Saharan Africa	0	0	0	0	0	0	0	0	0	0	100	0	0	0
East Asia and South-East Asia	0	0	0	0	0	0	0	0	0	0	100	0	0	0
North America	0	0	0	0	0	0	0	0	0	0	100	0	0	0
Latin America and the Caribbean	0	0	0	0	0	0	0	0	0	0	100	0	0	0
Western Europe	0	0	0	0	0	0	0	0	0	0	100	0	0	0
Eastern Europe	0	0	0	0	0	0	0	0	0	0	100	0	0	0
Russian Federation	0	0	0	0	0	0	0	0	0	0	100	0	0	0
Oceania	0	0	0	0	0	0	0	0	0	0	100	0	0	0
Near East and North Africa/Middle East	0	0	0	0	0	0	0	0	0	0	100	0	0	0
South Asia	0	0	0	0	0	0	0	0	0	0	100	0	0	0

Notes: PRP stands for pasture/range/paddock. Eastern Europe's proxies are used for the Russian Federation.

Source: Authors' own elaboration based on Table 10A.9 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A81. Animal waste management system regional average, in percentage, for chicken broiler in low-productivity system

Geographical regions	Lagoon	Liquid/slurry	Solid storage	Dry lot	PRP	Daily spread	Digester	Burned for fuel	Pit<1 month	Pit>1 month	Poultry manure	Deep bedding	Composting	Aerobic treatment
Sub-Saharan Africa	0	0	0	0	50	50	0	0	0	0	0	0	0	0
East Asia and South-East Asia	0	0	0	0	50	50	0	0	0	0	0	0	0	0
North America	0	0	0	0	50	50	0	0	0	0	0	0	0	0
Latin America and the Caribbean	0	0	0	0	50	50	0	0	0	0	0	0	0	0
Western Europe	0	0	0	0	50	50	0	0	0	0	0	0	0	0
Eastern Europe	0	0	0	0	50	50	0	0	0	0	0	0	0	0
Russian Federation	0	0	0	0	50	50	0	0	0	0	0	0	0	0
Oceania	0	0	0	0	50	50	0	0	0	0	0	0	0	0
Near East and North Africa/Middle East	0	0	0	0	50	50	0	0	0	0	0	0	0	0
South Asia	0	0	0	0	50	50	0	0	0	0	0	0	0	0

Notes: PRP stands for pasture/range/paddock. Eastern Europe's proxies are used for the Russian Federation.

Source: Authors' own elaboration based on Table 10A.9 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A82. Animal waste management system regional average, in percentage, for horse, camels, mules and asses, and other grazing animals

Geographical regions	Lagoon	Liquid/slurry	Solid storage	Dry lot	PRP	Daily spread	Digester	Burned for fuel	Pit<1 month	Pit>1 month	Poultry manure	Deep bedding	Composting	Aerobic treatment
Sub-Saharan Africa	0	0	17	3	80	0	0	0	0	0	0	0	0	0
East Asia and South-East Asia	0	0	50	0	50	0	0	0	0	0	0	0	0	0
North America	0	0	50	0	50	0	0	0	0	0	0	0	0	0
Latin America and the Caribbean	0	0	17	3	80	0	0	0	0	0	0	0	0	0
Western Europe	0	0	28	0	72	0	0	0	0	0	0	0	0	0
Eastern Europe	0	0	9	0	91	0	0	0	0	0	0	0	0	0
Russian Federation	0	0	82	0	18	0	0	0	0	0	0	0	0	0
Oceania	0	0	0	0	100	0	0	0	0	0	0	0	0	0
Near East and North Africa/Middle East	0	0	0	50	50	0	0	0	0	0	0	0	0	0
South Asia	0	0	50	0	50	0	0	0	0	0	0	0	0	0

Notes: PRP stands for pasture/range/paddock. Eastern Europe's proxies are used for the Russian Federation.

Source: Authors' own elaboration based on Table 10A.9 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A83. Animal waste management system regional average, in percentage, for ostrich

Geographical regions	Lagoon	Liquid/slurry	Solid storage	Dry lot	PRP	Daily spread	Digester	Burned for fuel	Pit<1 month	Pit>1 month	Poultry manure	Deep bedding	Composting	Aerobic treatment
Sub-Saharan Africa	0	0	20	0	80	0	0	0	0	0	0	0	0	0
East Asia and South-East Asia	0	0	20	0	80	0	0	0	0	0	0	0	0	0
North America	0	0	20	0	80	0	0	0	0	0	0	0	0	0
Latin America and the Caribbean	0	0	20	0	80	0	0	0	0	0	0	0	0	0
Western Europe	0	0	20	0	80	0	0	0	0	0	0	0	0	0
Eastern Europe	0	0	20	0	80	0	0	0	0	0	0	0	0	0
Russian Federation	0	0	20	0	80	0	0	0	0	0	0	0	0	0
Oceania	0	0	20	0	80	0	0	0	0	0	0	0	0	0
Near East and North Africa/Middle East	0	0	20	0	80	0	0	0	0	0	0	0	0	0
South Asia	0	0	20	0	80	0	0	0	0	0	0	0	0	0

Notes: PRP stands for pasture/range/paddock. Eastern Europe's proxies are used for the Russian Federation.

Source: Authors' own elaboration based on Table 10A.9 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A84. Animal waste management system regional average, in percentage, for deer

Geographical regions	Lagoon	Liquid/slurry	Solid storage	Dry lot	PRP	Daily spread	Digester	Burned for fuel	Pit<1 month	Pit>1 month	Poultry manure	Deep bedding	Composting	Aerobic treatment
Sub-Saharan Africa	0	0	0	0	100	0	0	0	0	0	0	0	0	0
East Asia and South-East Asia	0	0	0	0	100	0	0	0	0	0	0	0	0	0
North America	0	0	0	0	100	0	0	0	0	0	0	0	0	0
Latin America and the Caribbean	0	0	0	0	100	0	0	0	0	0	0	0	0	0
Western Europe	0	0	0	0	100	0	0	0	0	0	0	0	0	0
Eastern Europe	0	0	0	0	100	0	0	0	0	0	0	0	0	0
Russian Federation	0	0	0	0	100	0	0	0	0	0	0	0	0	0
Oceania	0	0	0	0	100	0	0	0	0	0	0	0	0	0
Near East and North Africa/Middle East	0	0	0	0	100	0	0	0	0	0	0	0	0	0
South Asia	0	0	0	0	100	0	0	0	0	0	0	0	0	0

Notes: PRP stands for pasture/range/paddock. Eastern Europe's proxies are used for the Russian Federation.

Source: Authors' own elaboration based on Table 10A.9 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A85. Methane emission factors from manure management, in gCH_4/kg VS, for dairy cattle in high-productivity system

IPCC climate zones	Lagoon	Liquid/slurry	Solid storage	Dry lot	PRP	Daily spread	Digester	Burned for fuel	Pit<1 month	Pit>1 month	Poultry manure	Deep bedding	Composting	Aerobic treatment
Cool temperate moist	96.5	33.8	3.2	1.6	0.6	0.2	3.2	16.1	0	33.8	0	0	0	0
Cool temperate dry	107.1	41.8	3.2	1.6	0.6	0.2	3.2	16.1	0	41.8	0	0	0	0
Boreal moist	80.4	22.5	3.2	1.6	0.6	0.2	3.2	16.1	0	22.5	0	0	0	0
Boreal dry	78.8	22.5	3.2	1.6	0.6	0.2	3.2	16.1	0	22.5	0	0	0	0
Warm temperate moist	117.4	59.5	6.4	2.4	0.6	0.8	3.7	16.1	0	59.5	0	0	0	0
Warm temperate dry	122.2	65.9	6.4	2.4	0.6	0.8	3.7	16.1	0	65.9	0	0	0	0
Tropical montane	122.2	94.9	8.0	3.2	0.6	1.6	3.7	16.1	0	94.9	0	0	0	0
Tropical wet	128.6	122.2	8.0	3.2	0.6	1.6	3.7	16.1	0	122.2	0	0	0	0
Tropical moist	128.6	117.4	8.0	3.2	0.6	1.6	3.7	16.1	0	117.4	0	0	0	0
Tropical dry	128.6	119.0	8.0	3.2	0.6	1.6	3.7	16.1	0	119	0	0	0	0

Note: IPCC = Intergovernmental Panel on Climate Change; PRP stands for pasture/range/paddock.

Source: Authors' own elaboration based on Table 10.14 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A86. Methane emission factors from manure management, in gCH_4/kg VS, for dairy cattle in low-productivity system

IPCC climate zones	Lagoon	Liquid/slurry	Solid storage	Dry lot	PRP	Daily spread	Digester	Burned for fuel	Pit<1 month	Pit>1 month	Poultry manure	Deep bedding	Composting	Aerobic treatment
Cool temperate moist	52.3	18.3	1.7	0.9	0.6	0.1	9.2	8.7	0	18.3	0	0	0	0
Cool temperate dry	58.4	22.6	1.7	0.9	0.6	0.1	9.2	8.7	0	22.6	0	0	0	0
Boreal moist	43.6	12.2	1.7	0.9	0.6	0.1	9.2	8.7	0	12.2	0	0	0	0
Boreal dry	42.7	12.2	1.7	0.9	0.6	0.1	9.2	8.7	0	12.2	0	0	0	0
Warm temperate moist	63.6	32.2	3.5	1.3	0.6	0.4	9.5	8.7	0	32.2	0	0	0	0
Warm temperate dry	66.2	35.7	3.5	1.3	0.6	0.4	9.5	8.7	0	35.7	0	0	0	0
Tropical montane	66.2	51.4	4.4	1.7	0.6	0.9	9.5	8.7	0	51.4	0	0	0	0
Tropical wet	69.7	66.2	4.4	1.7	0.6	0.9	9.5	8.7	0	66.2	0	0	0	0
Tropical moist	69.7	63.6	4.4	1.7	0.6	0.9	9.5	8.7	0	63.6	0	0	0	0
Tropical dry	69.7	64.5	4.4	1.7	0.6	0.9	9.5	8.7	0	64.5	0	0	0	0

Note: IPCC = Intergovernmental Panel on Climate Change; PRP stands for pasture/range/paddock.

Source: Authors' own elaboration based on Table 10.14 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A87. Methane emission factors from manure management, in gCH_4/kg VS, for other cattle in high-productivity system

IPCC climate zones	Lagoon	Liquid/slurry	Solid storage	Dry lot	PRP	Daily spread	Digester	Burned for fuel	Pit<1 month	Pit>1 month	Poultry manure	Deep bedding	Composting	Aerobic treatment
Cool temperate moist	72.4	25.3	2.4	1.2	0.6	0.1	2.4	12.1	0	25.3	0	0	0	0
Cool temperate dry	80.8	31.4	2.4	1.2	0.6	0.1	2.4	12.1	0	31.4	0	0	0	0
Boreal moist	60.3	16.9	2.4	1.2	0.6	0.1	2.4	12.1	0	16.9	0	0	0	0
Boreal dry	59.1	16.9	2.4	1.2	0.6	0.1	2.4	12.1	0	16.9	0	0	0	0
Warm temperate moist	88	44.6	4.8	1.8	0.6	0.6	2.7	12.1	0	44.6	0	0	0	0
Warm temperate dry	91.7	49.4	4.8	1.8	0.6	0.6	2.7	12.1	0	49.4	0	0	0	0
Tropical montane	91.7	71.2	6	2.4	0.6	1.2	2.8	12.1	0	71.2	0	0	0	0
Tropical wet	96.5	91.7	6	2.4	0.6	1.2	2.8	12.1	0	91.7	0	0	0	0
Tropical moist	96.5	88	6	2.4	0.6	1.2	2.8	12.1	0	88	0	0	0	0
Tropical dry	96.5	89.2	6	2.4	0.6	1.2	2.8	12.1	0	89.2	0	0	0	0

Note: IPCC = Intergovernmental Panel on Climate Change; PRP stands for pasture/range/paddock.

Source: Authors' own elaboration based on Table 10.14 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A88. Methane emission factors from manure management, in gCH_4/kg VS, for other cattle in low-productivity system and buffalo (dairy and other)

IPCC climate zones	Lagoon	Liquid/slurry	Solid storage	Dry lot	PRP	Daily spread	Digester	Burned for fuel	Pit<1 month	Pit>1 month	Poultry manure	Deep bedding	Composting	Aerobic treatment
Cool temperate moist	52.3	18.3	1.7	0.9	0.6	0.1	9.2	8.7	0	18.3	0	0	0	0
Cool temperate dry	58.4	22.6	1.7	0.9	0.6	0.1	9.2	8.7	0	22.6	0	0	0	0
Boreal moist	43.6	12.2	1.7	0.9	0.6	0.1	9.2	8.7	0	12.2	0	0	0	0
Boreal dry	42.7	12.2	1.7	0.9	0.6	0.1	9.2	8.7	0	12.2	0	0	0	0
Warm temperate moist	63.6	32.2	3.5	1.3	0.6	0.4	9.5	8.7	0	32.2	0	0	0	0
Warm temperate dry	66.2	35.7	3.5	1.3	0.6	0.4	9.5	8.7	0	35.7	0	0	0	0
Tropical montane	66.2	51.4	4.4	1.7	0.6	0.9	9.5	8.7	0	51.4	0	0	0	0
Tropical wet	69.7	66.2	4.4	1.7	0.6	0.9	9.5	8.7	0	66.2	0	0	0	0
Tropical moist	69.7	63.6	4.4	1.7	0.6	0.9	9.5	8.7	0	63.6	0	0	0	0
Tropical dry	69.7	64.5	4.4	1.7	0.6	0.9	9.5	8.7	0	64.5	0	0	0	0

Note: IPCC = Intergovernmental Panel on Climate Change; PRP stands for pasture/range/paddock.

Source: Authors' own elaboration based on Table 10.14 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A89. Methane emission factors from manure management, in gCH_4/kg VS, for swine (breeding and growing) in high-productivity systems

IPCC climate zones	Lagoon	Liquid/slurry	Solid storage	Dry lot	PRP	Daily spread	Digester	Burned for fuel	Pit<1 month	Pit>1 month	Poultry manure	Deep bedding	Composting	Aerobic treatment
Cool temperate moist	180.9	63.3	6	3	0.6	0.3	6	30.2	18.1	63.3	0	0	0	0
Cool temperate dry	202	78.4	6	3	0.6	0.3	6	30.2	24.1	78.4	0	0	0	0
Boreal moist	150.8	42.2	6	3	0.6	0.3	6	30.2	12.1	42.2	0	0	0	0
Boreal dry	147.7	42.2	6	3	0.6	0.3	6	30.2	12.1	42.2	0	0	0	0
Warm temperate moist	220.1	111.6	12.1	4.5	0.6	1.5	6.8	30.2	39.2	111.6	0	0	0	0
Warm temperate dry	229.1	123.6	12.1	4.5	0.6	1.5	6.8	30.2	45.2	123.6	0	0	0	0
Tropical montane	229.1	177.9	15.1	6	0.6	3	7	30.2	75.4	177.9	0	0	0	0
Tropical wet	241.2	229.1	15.1	6	0.6	3	7	30.2	114.6	229.1	0	0	0	0
Tropical moist	241.2	220.1	15.1	6	0.6	3	7	30.2	108.5	220.1	0	0	0	0
Tropical dry	241.2	223.1	15.1	6	0.6	3	7	30.2	126.6	223.1	0	0	0	0

Note: IPCC = Intergovernmental Panel on Climate Change; PRP stands for pasture/range/paddock.

Source: Authors' own elaboration based on Table 10.14 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A90. Methane emission factors from manure management, in gCH_4/kg VS, for swine (breeding and growing) in low-productivity systems

IPCC climate zones	Lagoon	Liquid/slurry	Solid storage	Dry lot	PRP	Daily spread	Digester	Burned for fuel	Pit<1 month	Pit>1 month	Poultry manure	Deep bedding	Composting	Aerobic treatment
Cool temperate moist	116.6	40.8	3.9	1.9	0.6	0.2	20.6	19.4	11.7	40.8	0	0	0	0
Cool temperate dry	130.2	50.5	3.9	1.9	0.6	0.2	20.6	19.4	15.5	50.5	0	0	0	0
Boreal moist	97.2	27.2	3.9	1.9	0.6	0.2	20.6	19.4	7.8	27.2	0	0	0	0
Boreal dry	95.2	27.2	3.9	1.9	0.6	0.2	20.6	19.4	7.8	27.2	0	0	0	0
Warm temperate moist	141.8	71.9	7.8	2.9	0.6	1	21.1	19.4	25.3	71.9	0	0	0	0
Warm temperate dry	147.7	79.7	7.8	2.9	0.6	1	21.1	19.4	29.1	79.7	0	0	0	0
Tropical montane	147.7	114.6	9.7	3.9	0.6	1.9	21.2	19.4	48.6	114.6	0	0	0	0
Tropical wet	155.4	147.7	9.7	3.9	0.6	1.9	21.2	19.4	73.8	147.7	0	0	0	0
Tropical moist	155.4	141.8	9.7	3.9	0.6	1.9	21.2	19.4	69.9	141.8	0	0	0	0
Tropical dry	155.4	143.8	9.7	3.9	0.6	1.9	21.2	19.4	81.6	143.8	0	0	0	0

Note: IPCC = Intergovernmental Panel on Climate Change; PRP stands for pasture/range/paddock.

Source: Authors' own elaboration based on Table 10.14 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A91. Methane emission factors from manure management, in gCH_4/kg VS, for poultry (chicken layer and chicken broiler) in high-productivity systems

IPCC climate zones	Lagoon	Liquid/slurry	Solid storage	Dry lot	PRP	Daily spread	Digester	Burned for fuel	Pit<1 month	Pit>1 month	Poultry manure	Deep bedding	Composting	Aerobic treatment
Cool temperate moist	156.8	54.9	5.2	2.6	0.6	0	5.2	2.6	0	54.9	0	0	0	0
Cool temperate dry	175.1	67.9	5.2	2.6	0.6	0	5.2	2.6	0	67.9	0	0	0	0
Boreal moist	130.7	36.6	5.2	2.6	0.6	0	5.2	2.6	0	36.6	0	0	0	0
Boreal dry	128	36.6	5.2	2.6	0.6	0	5.2	2.6	0	36.6	0	0	0	0
Warm temperate moist	190.7	96.7	10.5	3.9	0.6	0	10.5	2.6	0	96.7	0	0	0	0
Warm temperate dry	198.6	107.1	10.5	3.9	0.6	0	10.5	2.6	0	107.1	0	0	0	0
Tropical montane	198.6	154.2	13.1	5.2	0.6	0	13.1	2.6	0	154.2	0	0	0	0
Tropical wet	209	198.6	13.1	5.2	0.6	0	13.1	2.6	0	198.6	0	0	0	0
Tropical moist	209	190.7	13.1	5.2	0.6	0	13.1	2.6	0	190.7	0	0	0	0
Tropical dry	209	193.4	13.1	5.2	0.6	0	13.1	2.6	0	193.4	0	0	0	0

Note: IPCC = Intergovernmental Panel on Climate Change; PRP stands for pasture/range/paddock.

Source: Authors' own elaboration based on Table 10.14 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A92. Methane emission factors from manure management, in gCH_4/kg VS, for poultry (chicken layer and chicken broiler) in low-productivity systems

IPCC climate zones	Lagoon	Liquid/slurry	Solid storage	Dry lot	PRP	Daily spread	Digester	Burned for fuel	Pit<1 month	Pit>1 month	Poultry manure	Deep bedding	Composting	Aerobic treatment
Cool temperate moist	2.4	2.4	2.4	2.4	0.6	2.4	2.4	2.4	2.4	2.4	2.4	0	0	0
Cool temperate dry	2.4	2.4	2.4	2.4	0.6	2.4	2.4	2.4	2.4	2.4	2.4	0	0	0
Boreal moist	2.4	2.4	2.4	2.4	0.6	2.4	2.4	2.4	2.4	2.4	2.4	0	0	0
Boreal dry	2.4	2.4	2.4	2.4	0.6	2.4	2.4	2.4	2.4	2.4	2.4	0	0	0
Warm temperate moist	2.4	2.4	2.4	2.4	0.6	2.4	2.4	2.4	2.4	2.4	2.4	0	0	0
Warm temperate dry	2.4	2.4	2.4	2.4	0.6	2.4	2.4	2.4	2.4	2.4	2.4	0	0	0
Tropical montane	2.4	2.4	2.4	2.4	0.6	2.4	2.4	2.4	2.4	2.4	2.4	0	0	0
Tropical wet	2.4	2.4	2.4	2.4	0.6	2.4	2.4	2.4	2.4	2.4	2.4	0	0	0
Tropical moist	2.4	2.4	2.4	2.4	0.6	2.4	2.4	2.4	2.4	2.4	2.4	0	0	0
Tropical dry	2.4	2.4	2.4	2.4	0.6	2.4	2.4	2.4	2.4	2.4	2.4	0	0	0

Note: IPCC = Intergovernmental Panel on Climate Change; PRP stands for pasture/range/paddock.

Source: Authors' own elaboration based on Table 10.14 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A93. Methane emission factors from manure management, in gCH_4/kg VS, for sheep in high-productivity systems

IPCC climate zones	Lagoon	Liquid/slurry	Solid storage	Dry lot	PRP	Daily spread	Digester	Burned for fuel	Pit<1 month	Pit>1 month	Poultry manure	Deep bedding	Composting	Aerobic treatment
Cool temperate moist	0	0	2.5	1.3	0.6	0	0	0	0	0	0	0	0	0
Cool temperate dry	0	0	2.5	1.3	0.6	0	0	0	0	0	0	0	0	0
Boreal moist	0	0	2.5	1.3	0.6	0	0	0	0	0	0	0	0	0
Boreal dry	0	0	2.5	1.3	0.6	0	0	0	0	0	0	0	0	0
Warm temperate moist	0	0	5.1	1.9	0.6	0	0	0	0	0	0	0	0	0
Warm temperate dry	0	0	5.1	1.9	0.6	0	0	0	0	0	0	0	0	0
Tropical montane	0	0	6.4	2.5	0.6	0	0	0	0	0	0	0	0	0
Tropical wet	0	0	6.4	2.5	0.6	0	0	0	0	0	0	0	0	0
Tropical moist	0	0	6.4	2.5	0.6	0	0	0	0	0	0	0	0	0
Tropical dry	0	0	6.4	2.5	0.6	0	0	0	0	0	0	0	0	0

Note: IPCC = Intergovernmental Panel on Climate Change; PRP stands for pasture/range/paddock.

Source: Authors' own elaboration based on Table 10.14 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A94. Methane emission factors from manure management, in gCH_4/kg VS, for sheep in low-productivity systems

IPCC climate zones	Lagoon	Liquid/slurry	Solid storage	Dry lot	PRP	Daily spread	Digester	Burned for fuel	Pit<1 month	Pit>1 month	Poultry manure	Deep bedding	Composting	Aerobic treatment
Cool temperate moist	0	0	1.7	0.9	0.6	0	0	0	0	0	0	0	0	0
Cool temperate dry	0	0	1.7	0.9	0.6	0	0	0	0	0	0	0	0	0
Boreal moist	0	0	1.7	0.9	0.6	0	0	0	0	0	0	0	0	0
Boreal dry	0	0	1.7	0.9	0.6	0	0	0	0	0	0	0	0	0
Warm temperate moist	0	0	3.5	1.3	0.6	0	0	0	0	0	0	0	0	0
Warm temperate dry	0	0	3.5	1.3	0.6	0	0	0	0	0	0	0	0	0
Tropical montane	0	0	4.4	1.7	0.6	0	0	0	0	0	0	0	0	0
Tropical wet	0	0	4.4	1.7	0.6	0	0	0	0	0	0	0	0	0
Tropical moist	0	0	4.4	1.7	0.6	0	0	0	0	0	0	0	0	0
Tropical dry	0	0	4.4	1.7	0.6	0	0	0	0	0	0	0	0	0

Note: IPCC = Intergovernmental Panel on Climate Change; PRP stands for pasture/range/paddock.

Source: Authors' own elaboration based on Table 10.14 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A95. Methane emission factors from manure management, in gCH_4/kg VS, for goat in high-productivity systems

IPCC climate zones	Lagoon	Liquid/slurry	Solid storage	Dry lot	PRP	Daily spread	Digester	Burned for fuel	Pit<1 month	Pit>1 month	Poultry manure	Deep bedding	Composting	Aerobic treatment
Cool temperate moist	0	0	2.4	1.2	0.6	0	0	0	0	0	0	0	0	0
Cool temperate dry	0	0	2.4	1.2	0.6	0	0	0	0	0	0	0	0	0
Boreal moist	0	0	2.4	1.2	0.6	0	0	0	0	0	0	0	0	0
Boreal dry	0	0	2.4	1.2	0.6	0	0	0	0	0	0	0	0	0
Warm temperate moist	0	0	4.8	1.8	0.6	0	0	0	0	0	0	0	0	0
Warm temperate dry	0	0	4.8	1.8	0.6	0	0	0	0	0	0	0	0	0
Tropical montane	0	0	6	2.4	0.6	0	0	0	0	0	0	0	0	0
Tropical wet	0	0	6	2.4	0.6	0	0	0	0	0	0	0	0	0
Tropical moist	0	0	6	2.4	0.6	0	0	0	0	0	0	0	0	0
Tropical dry	0	0	6	2.4	0.6	0	0	0	0	0	0	0	0	0

Note: IPCC = Intergovernmental Panel on Climate Change; PRP stands for pasture/range/paddock.

Source: Authors' own elaboration based on Table 10.14 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A96. Methane emission factors from manure management, in gCH_4/kg VS, for goat in low-productivity systems

IPCC climate zones	Lagoon	Liquid/slurry	Solid storage	Dry lot	PRP	Daily spread	Digester	Burned for fuel	Pit<1 month	Pit>1 month	Poultry manure	Deep bedding	Composting	Aerobic treatment
Cool temperate moist	0	0	1.7	0.9	0.6	0	0	0	0	0	0	0	0	0
Cool temperate dry	0	0	1.7	0.9	0.6	0	0	0	0	0	0	0	0	0
Boreal moist	0	0	1.7	0.9	0.6	0	0	0	0	0	0	0	0	0
Boreal dry	0	0	1.7	0.9	0.6	0	0	0	0	0	0	0	0	0
Warm temperate moist	0	0	3.5	1.3	0.6	0	0	0	0	0	0	0	0	0
Warm temperate dry	0	0	3.5	1.3	0.6	0	0	0	0	0	0	0	0	0
Tropical montane	0	0	4.4	1.7	0.6	0	0	0	0	0	0	0	0	0
Tropical wet	0	0	4.4	1.7	0.6	0	0	0	0	0	0	0	0	0
Tropical moist	0	0	4.4	1.7	0.6	0	0	0	0	0	0	0	0	0
Tropical dry	0	0	4.4	1.7	0.6	0	0	0	0	0	0	0	0	0

Note: IPCC = Intergovernmental Panel on Climate Change; PRP stands for pasture/range/paddock.

Source: Authors' own elaboration based on Table 10.14 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A97. Methane emission factors from manure management, in gCH_4/kg VS, for camels in high-productivity systems

IPCC climate zones	Lagoon	Liquid/slurry	Solid storage	Dry lot	PRP	Daily spread	Digester	Burned for fuel	Pit<1 month	Pit>1 month	Poultry manure	Deep bedding	Composting	Aerobic treatment
Cool temperate moist	0	0	3.5	1.7	0.6	0	0	0	0	0	0	0	0	0
Cool temperate dry	0	0	3.5	1.7	0.6	0	0	0	0	0	0	0	0	0
Boreal moist	0	0	3.5	1.7	0.6	0	0	0	0	0	0	0	0	0
Boreal dry	0	0	3.5	1.7	0.6	0	0	0	0	0	0	0	0	0
Warm temperate moist	0	0	7	2.6	0.6	0	0	0	0	0	0	0	0	0
Warm temperate dry	0	0	7	2.6	0.6	0	0	0	0	0	0	0	0	0
Tropical montane	0	0	8.7	0	0.6	0	0	0	0	0	0	0	0	0
Tropical wet	0	0	8.7	0	0.6	0	0	0	0	0	0	0	0	0
Tropical moist	0	0	8.7	0	0.6	0	0	0	0	0	0	0	0	0
Tropical dry	0	0	8.7	0	0.6	0	0	0	0	0	0	0	0	0

Note: IPCC = Intergovernmental Panel on Climate Change; PRP stands for pasture/range/paddock.

Source: Authors' own elaboration based on Table 10.14 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A98. Methane emission factors from manure management, in gCH_4/kg VS, for camels in low-productivity systems

IPCC climate zones	Lagoon	Liquid/slurry	Solid storage	Dry lot	PRP	Daily spread	Digester	Burned for fuel	Pit<1 month	Pit>1 month	Poultry manure	Deep bedding	Composting	Aerobic treatment
Cool temperate moist	0	0	2.8	1.4	0.6	0	0	0	0	0	0	0	0	0
Cool temperate dry	0	0	2.8	1.4	0.6	0	0	0	0	0	0	0	0	0
Boreal moist	0	0	2.8	1.4	0.6	0	0	0	0	0	0	0	0	0
Boreal dry	0	0	2.8	1.4	0.6	0	0	0	0	0	0	0	0	0
Warm temperate moist	0	0	5.6	2.1	0.6	0	0	0	0	0	0	0	0	0
Warm temperate dry	0	0	5.6	2.1	0.6	0	0	0	0	0	0	0	0	0
Tropical montane	0	0	7	2.8	0.6	0	0	0	0	0	0	0	0	0
Tropical wet	0	0	7	2.8	0.6	0	0	0	0	0	0	0	0	0
Tropical moist	0	0	7	2.8	0.6	0	0	0	0	0	0	0	0	0
Tropical dry	0	0	7	2.8	0.6	0	0	0	0	0	0	0	0	0

Note: IPCC = Intergovernmental Panel on Climate Change; PRP stands for pasture/range/paddock.

Source: Authors' own elaboration based on Table 10.14 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A99. Methane emission factors from manure management, in gCH_4/kg VS, for horses in high-productivity systems

IPCC climate zones	Lagoon	Liquid/slurry	Solid storage	Dry lot	PRP	Daily spread	Digester	Burned for fuel	Pit<1 month	Pit>1 month	Poultry manure	Deep bedding	Composting	Aerobic treatment
Cool temperate moist	0	0	4	2	0.6	0	0	0	0	0	0	0	0	0
Cool temperate dry	0	0	4	2	0.6	0	0	0	0	0	0	0	0	0
Boreal moist	0	0	4	2	0.6	0	0	0	0	0	0	0	0	0
Boreal dry	0	0	4	2	0.6	0	0	0	0	0	0	0	0	0
Warm temperate moist	0	0	8	3	0.6	0	0	0	0	0	0	0	0	0
Warm temperate dry	0	0	8	3	0.6	0	0	0	0	0	0	0	0	0
Tropical montane	0	0	10.1	4	0.6	0	0	0	0	0	0	0	0	0
Tropical wet	0	0	10.1	4	0.6	0	0	0	0	0	0	0	0	0
Tropical moist	0	0	10.1	4	0.6	0	0	0	0	0	0	0	0	0
Tropical dry	0	0	10.1	4	0.6	0	0	0	0	0	0	0	0	0

Note: IPCC = Intergovernmental Panel on Climate Change; PRP stands for pasture/range/paddock.

Source: Authors' own elaboration based on Table 10.14 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A100. Methane emission factors from manure management, in gCH_4/kg VS, for horses in low-productivity systems

IPCC climate zones	Lagoon	Liquid/slurry	Solid storage	Dry lot	PRP	Daily spread	Digester	Burned for fuel	Pit<1 month	Pit>1 month	Poultry manure	Deep bedding	Composting	Aerobic treatment
Cool temperate moist	0	0	3.5	1.7	0.6	0	0	0	0	0	0	0	0	0
Cool temperate dry	0	0	3.5	1.7	0.6	0	0	0	0	0	0	0	0	0
Boreal moist	0	0	3.5	1.7	0.6	0	0	0	0	0	0	0	0	0
Boreal dry	0	0	3.5	1.7	0.6	0	0	0	0	0	0	0	0	0
Warm temperate moist	0	0	7	2.6	0.6	0	0	0	0	0	0	0	0	0
Warm temperate dry	0	0	7	2.6	0.6	0	0	0	0	0	0	0	0	0
Tropical montane	0	0	8.7	3.5	0.6	0	0	0	0	0	0	0	0	0
Tropical wet	0	0	8.7	3.5	0.6	0	0	0	0	0	0	0	0	0
Tropical moist	0	0	8.7	3.5	0.6	0	0	0	0	0	0	0	0	0
Tropical dry	0	0	8.7	3.5	0.6	0	0	0	0	0	0	0	0	0

Note: IPCC = Intergovernmental Panel on Climate Change; PRP stands for pasture/range/paddock.

Source: Authors' own elaboration based on Table 10.14 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A101. Methane emission factors from manure management, in gCH_4/kg VS, for mules and asses in high-productivity systems

IPCC climate zones	Lagoon	Liquid/slurry	Solid storage	Dry lot	PRP	Daily spread	Digester	Burned for fuel	Pit<1 month	Pit>1 month	Poultry manure	Deep bedding	Composting	Aerobic treatment
Cool temperate moist	0	0	4.4	2.2	0.6	0	0	0	0	0	0	0	0	0
Cool temperate dry	0	0	4.4	2.2	0.6	0	0	0	0	0	0	0	0	0
Boreal moist	0	0	4.4	2.2	0.6	0	0	0	0	0	0	0	0	0
Boreal dry	0	0	4.4	2.2	0.6	0	0	0	0	0	0	0	0	0
Warm temperate moist	0	0	8.8	3.3	0.6	0	0	0	0	0	0	0	0	0
Warm temperate dry	0	0	8.8	3.3	0.6	0	0	0	0	0	0	0	0	0
Tropical montane	0	0	11.1	4.4	0.6	0	0	0	0	0	0	0	0	0
Tropical wet	0	0	11.1	4.4	0.6	0	0	0	0	0	0	0	0	0
Tropical moist	0	0	11.1	4.4	0.6	0	0	0	0	0	0	0	0	0
Tropical dry	0	0	11.1	4.4	0.6	0	0	0	0	0	0	0	0	0

Note: IPCC = Intergovernmental Panel on Climate Change; PRP stands for pasture/range/paddock.

Source: Authors' own elaboration based on Table 10.14 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A102. Methane emission factors from manure management, in gCH_4/kg VS, for mules and asses in low-productivity systems

IPCC climate zones	Lagoon	Liquid/slurry	Solid storage	Dry lot	PRP	Daily spread	Digester	Burned for fuel	Pit<1 month	Pit>1 month	Poultry manure	Deep bedding	Composting	Aerobic treatment
Cool temperate moist	0	0	3.5	1.7	0.6	0	0	0	0	0	0	0	0	0
Cool temperate dry	0	0	3.5	1.7	0.6	0	0	0	0	0	0	0	0	0
Boreal moist	0	0	3.5	1.7	0.6	0	0	0	0	0	0	0	0	0
Boreal dry	0	0	3.5	1.7	0.6	0	0	0	0	0	0	0	0	0
Warm temperate moist	0	0	7	2.6	0.6	0	0	0	0	0	0	0	0	0
Warm temperate dry	0	0	7	2.6	0.6	0	0	0	0	0	0	0	0	0
Tropical montane	0	0	8.7	3.5	0.6	0	0	0	0	0	0	0	0	0
Tropical wet	0	0	8.7	3.5	0.6	0	0	0	0	0	0	0	0	0
Tropical moist	0	0	8.7	3.5	0.6	0	0	0	0	0	0	0	0	0
Tropical dry	0	0	8.7	3.5	0.6	0	0	0	0	0	0	0	0	0

Note: IPCC = Intergovernmental Panel on Climate Change; PRP stands for pasture/range/paddock.

Source: Authors' own elaboration based on Table 10.14 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A103. Default values for nitrogen excretion rates, in kgN/1 000kg TAM/day, for all livestock categories

Geographical regions	Dairy cattle	Other cattle	Dairy buffalo	Other buffalo	Dairy sheep	Other sheep	Growing swine	Breeding swine	Goats	Horses	Camels	Mules and asses	Deer	Ostrich	Chicken layer	Chicken broiler
Sub-Saharan Africa	0.44	0.44	0.41	0.41	0.32	0.32	0.49	0.29	0.34	0.46	0.46	0.46	0.67	0.34	1.29	1.4
East Asia and South-East Asia	0.44	0.38	0.44	0.44	0.32	0.32	0.7	0.37	0.34	0.46	0.46	0.46	0.67	0.34	1.1	1.35
North America	0.6	0.4	0	0	0.35	0.35	0.46	0.24	0.46	0.3	0.38	0.3	0.67	0.34	1.45	1.59
Latin America and the Caribbean	0.39	0.31	0.41	0.41	0.32	0.32	0.73	0.35	0.34	0.46	0.46	0.46	0.67	0.34	1.2	1.23
Western Europe	0.5	0.42	0.45	0.45	0.36	0.36	0.76	0.38	0.46	0.26	0.38	0.26	0.67	0.34	0.99	1.14
Eastern Europe	0.42	0.47	0.35	0.35	0.36	0.36	0.77	0.36	0.44	0.3	0.38	0.3	0.67	0.34	0.96	1.12
Russian Federation	0.42	0.47	0.35	0.35	0.36	0.36	0.77	0.36	0.44	0.3	0.38	0.3	0.67	0.34	0.96	1.12
Oceania	0.72	0.46	0	0	0.43	0.43	0.72	0.31	0.42	0.3	0.38	0.3	0.67	0.34	1.42	1.59
Near East and North Africa/Middle East	0.5	0.55	0.39	0.39	0.32	0.32	0.73	0.4	0.34	0.46	0.46	0.46	0.67	0.34	1.29	1.43
South Asia	0.65	0.44	0.57	0.57	0.32	0.32	0.76	0.43	0.34	0.46	0.46	0.46	0.67	0.34	1.62	1.58

Note: EX-ACT uses proxies from Eastern Europe for the Russian Federation.

Source: Authors' own elaboration based on Table 10.19 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A104. Nitrogen excretion rates, in kg N/1 000kg TAM/day, for all livestock categories in low-productivity system

Geographical regions	Dairy cattle	Other cattle	Dairy buffalo	Other buffalo	Dairy sheep	Other sheep	Growing swine	Breeding swine	Goats	Horses	Camels	Mules and asses	Deer	Ostrich	Chicken layer	Chicken broiler
Sub-Saharan Africa	0.45	0.45	0.41	0.41	0.32	0.32	0.54	0.35	0.34	0.46	0.46	0.46	0.67	0.34	1.44	1.58
East Asia and South-East Asia	0.41	0.38	0.44	0.44	0.32	0.32	0.76	0.43	0.34	0.46	0.46	0.46	0.67	0.34	1.62	1.84
North America	0.6	0.4	0	0	0.35	0.35	0.46	0.24	0.46	0.3	0.38	0.3	0.67	0.34	1.45	1.59
Latin America and the Caribbean	0.28	0.29	0.41	0.41	0.32	0.32	0.8	0.43	0.34	0.46	0.46	0.46	0.67	0.34	2.14	2.39
Western Europe	0.5	0.42	0.45	0.45	0.36	0.36	0.76	0.38	0.46	0.26	0.38	0.26	0.67	0.34	0.99	1.14
Eastern Europe	0.42	0.47	0.35	0.35	0.36	0.36	0.77	0.36	0.44	0.3	0.38	0.3	0.67	0.34	0.96	1.12
Russian Federation	0.42	0.47	0.35	0.35	0.36	0.36	0.77	0.36	0.44	0.3	0.38	0.3	0.67	0.34	0.96	1.12
Oceania	0.72	0.46	0	0	0.43	0.43	0.72	0.31	0.42	0.3	0.38	0.3	0.67	0.34	1.42	1.59

Geographical regions	Dairy cattle	Other cattle	Dairy buffalo	Other buffalo	Dairy sheep	Other sheep	Growing swine	Breeding swine	Goats	Horses	Camels	Mules and asses	Deer	Ostrich	Chicken layer	Chicken broiler
Near East and North Africa/Middle East	0.51	0.58	0.39	0.39	0.32	0.32	0.6	0.37	0.34	0.46	0.46	0.46	0.67	0.34	1.79	1.95
South Asia	0.7	0.4	0.57	0.57	0.32	0.32	0.76	0.47	0.34	0.46	0.46	0.46	0.67	0.34	1.83	2.11

Note: EX-ACT uses proxies from Eastern Europe for the Russian Federation.

Source: Authors' own elaboration based on Table 10.19 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A105. Nitrogen excretion rates, in kg N/1 000kg TAM/day, for all livestock categories in high-productivity system

Geographical regions	Dairy cattle	Other cattle	Dairy buffalo	Other buffalo	Dairy sheep	Other sheep	Growing swine	Breeding swine	Goats	Horses	Camels	Mules and asses	Deer	Ostrich	Chicken layer	Chicken broiler
Sub-Saharan Africa	0.41	0.42	0.41	0.41	0.32	0.32	0.39	0.21	0.34	0.46	0.46	0.46	0.67	0.34	1.16	1.34
East Asia and South-East Asia	0.55	0.36	0.44	0.44	0.32	0.32	0.63	0.32	0.34	0.46	0.46	0.46	0.67	0.34	1.0	1.31
North America	0.6	0.4	0	0	0.35	0.35	0.46	0.24	0.46	0.3	0.38	0.3	0.67	0.34	1.45	1.59
Latin America and the Caribbean	0.6	0.36	0.41	0.41	0.32	0.32	0.69	0.32	0.34	0.46	0.46	0.46	0.67	0.34	1.13	1.21
Western Europe	0.5	0.42	0.45	0.45	0.36	0.36	0.76	0.38	0.46	0.26	0.38	0.26	0.67	0.34	0.99	1.14
Eastern Europe	0.42	0.47	0.35	0.35	0.36	0.36	0.77	0.36	0.44	0.3	0.38	0.3	0.67	0.34	0.96	1.12
Russian Federation	0.42	0.47	0.35	0.35	0.36	0.36	0.77	0.36	0.44	0.3	0.38	0.3	0.67	0.34	0.96	1.12
Oceania	0.72	0.46	0	0	0.43	0.43	0.72	0.31	0.42	0.3	0.38	0.3	0.67	0.34	1.42	1.59
Near East and North Africa/Middle East	0.49	0.51	0.39	0.39	0.32	0.32	0.75	0.41	0.34	0.46	0.46	0.46	0.67	0.34	1.27	1.42
South Asia	0.51	0.63	0.57	0.57	0.32	0.32	0.74	0.37	0.34	0.46	0.46	0.46	0.67	0.34	1.48	1.47

Note: EX-ACT uses proxies from Eastern Europe for the Russian Federation.

Source: Authors' own elaboration based on Table 10.19 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A106. Default emission factors for direct N_2O emissions from manure management, in kg N_2O-N/kg N excreted, for cattle, poultry and pigs

IPCC climate zones	Lagoon	Liquid/slurry	Solid storage	Dry lot	PRP	Daily spread	Digester	Burned for fuel	Pit<1 month	Pit>1 month	Poultry manure	Deep bedding	Composting	Aerobic treatment
Cool temperate moist	0	**0.003**	**0.0075**	0.02	0.004	0	0.0006		0.002	0.002	0.001	**0.04**	**0.0065**	**0.0075**
Cool temperate dry	0	**0.003**	**0.0075**	0.02	0.002	0	0.0006		0.002	0.002	0.001	**0.04**	**0.0065**	**0.0075**
Boreal moist	0	**0.003**	**0.0075**	0.02	0.004	0	0.0006		0.002	0.002	0.001	**0.04**	**0.0065**	**0.0075**
Boreal dry	0	**0.003**	**0.0075**	0.02	0.002	0	0.0006		0.002	0.002	0.001	**0.04**	**0.0065**	**0.0075**
Warm temperate moist	0	**0.003**	**0.0075**	0.02	0.004	0	0.0006		0.002	0.002	0.001	**0.04**	**0.0065**	**0.0075**
Warm temperate dry	0	**0.003**	**0.0075**	0.02	0.002	0	0.0006		0.002	0.002	0.001	**0.04**	**0.0065**	**0.0075**
Tropical montane	0	**0.003**	**0.0075**	0.02	0.002	0	0.0006		0.002	0.002	0.001	**0.04**	**0.0065**	**0.0075**
Tropical wet	0	**0.003**	**0.0075**	0.02	0.006	0	0.0006		0.002	0.002	0.001	**0.04**	**0.0065**	**0.0075**
Tropical moist	0	**0.003**	**0.0075**	0.02	0.004	0	0.0006		0.002	0.002	0.001	**0.04**	**0.0065**	**0.0075**
Tropical dry	0	**0.003**	**0.0075**	0.02	0.002	0	0.0006		0.002	0.002	0.001	**0.04**	**0.0065**	**0.0075**

Notes: IPCC = Intergovernmental Panel on Climate Change. Bold values are the average over a given range. PRP stands for pasture, range and paddock.

Source: Authors' own elaboration based on Tables 10.21 and 11.1 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A107. Default emission factors for direct N_2O emissions from manure management, in kg N_2O-N/kg N excreted, for sheep and others livestock categories

IPCC climate zones	Lagoon	Liquid/slurry	Solid storage	Dry lot	PRP	Daily spread	Digester	Burned for fuel	Pit<1 month	Pit>1 month	Poultry manure	Deep bedding	Composting	Aerobic treatment
Cool temperate moist	0	**0.003**	**0.01**	0.02	0.003	0	0.0006		0.002	0.002	0.001	**0.04**	**0.0065**	**0.03**
Cool temperate dry	0	**0.003**	**0.01**	0.02	0.003	0	0.0006		0.002	0.002	0.001	**0.04**	**0.0065**	**0.03**
Boreal moist	0	**0.003**	**0.01**	0.02	0.003	0	0.0006		0.002	0.002	0.001	**0.04**	**0.0065**	**0.03**

IPCC climate zones	Lagoon	Liquid/slurry	Solid storage	Dry lot	PRP	Daily spread	Digester	Burned for fuel	Pit<1 month	Pit>1 month	Poultry manure	Deep bedding	Composting	Aerobic treatment
Boreal dry	0	**0.003**	**0.01**	0.02	0.003	0	0.0006		0.002	0.002	0.001	**0.04**	**0.0065**	**0.03**
Warm temperate moist	0	**0.003**	**0.01**	0.02	0.003	0	0.0006		0.002	0.002	0.001	**0.04**	**0.0065**	**0.03**
Warm temperate dry	0	**0.003**	**0.01**	0.02	0.003	0	0.0006		0.002	0.002	0.001	**0.04**	**0.0065**	**0.03**
Tropical montane	0	**0.003**	**0.01**	0.02	0.003	0	0.0006		0.002	0.002	0.001	**0.04**	**0.0065**	**0.03**
Tropical wet	0	**0.003**	**0.01**	0.02	0.003	0	0.0006		0.002	0.002	0.001	**0.04**	**0.0065**	**0.03**
Tropical moist	0	**0.003**	**0.01**	0.02	0.003	0	0.0006		0.002	0.002	0.001	**0.04**	**0.0065**	**0.03**
Tropical dry	0	**0.003**	**0.01**	0.02	0.003	0	0.0006		0.002	0.002	0.001	**0.04**	**0.0065**	**0.03**

Notes: IPCC = Intergovernmental Panel on Climate Change. Bold values are the average over a given range. PRP stands for pasture, range and paddock.

Source: Authors' own elaboration based on Tables 10.21 and 11.1 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A108. Default values for nitrogen loss fraction due to volatilisation of NH_3 and NO_x of nitrogen from manure management (FracGasMS) from swine (growing and breeding pigs)

IPCC climate zones	Lagoon	Liquid/slurry	Solid storage	Dry lot	PRP	Daily spread	Digester	Burned for fuel	Pit<1 month	Pit>1 month	Poultry manure	Deep bedding	Composting	Aerobic treatment
Cool temperate moist	0.40	0.29	0.36	0.45	0.21	0.07	0.28	0.00	0.25	0.25	0.00	0.4	0.625	0.85
Cool temperate dry	0.40	0.29	0.36	0.45	0.21	0.07	0.28	0.00	0.25	0.25	0.00	0.4	0.625	0.85
Boreal moist	0.40	0.29	0.36	0.45	0.21	0.07	0.28	0.00	0.25	0.25	0.00	0.4	0.625	0.85
Boreal dry	0.40	0.29	0.36	0.45	0.21	0.07	0.28	0.00	0.25	0.25	0.00	0.4	0.625	0.85
Warm temperate moist	0.40	0.29	0.36	0.45	0.21	0.07	0.28	0.00	0.25	0.25	0.00	0.4	0.625	0.85
Warm temperate dry	0.40	0.29	0.36	0.45	0.21	0.07	0.28	0.00	0.25	0.25	0.00	0.4	0.625	0.85
Tropical montane	0.40	0.29	0.36	0.45	0.21	0.07	0.28	0.00	0.25	0.25	0.00	0.4	0.625	0.85
Tropical wet	0.40	0.29	0.36	0.45	0.21	0.07	0.28	0.00	0.25	0.25	0.00	0.4	0.625	0.85
Tropical moist	0.40	0.29	0.36	0.45	0.21	0.07	0.28	0.00	0.25	0.25	0.00	0.4	0.625	0.85
Tropical dry	0.40	0.29	0.36	0.45	0.21	0.07	0.28	0.00	0.25	0.25	0.00	0.4	0.625	0.85

Notes: IPCC = Intergovernmental Panel on Climate Change. Bold values are the average over a given range. PRP stands for pasture/range/paddock.

Source: Authors' own elaboration based on Table 10.22 in Chapter 10 and Table 11.3 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A109. Default values for nitrogen loss fraction due to volatilisation of NH_3 and NO_x of nitrogen from manure management (FracGasMS) from dairy cow

IPCC climate zones	Lagoon	Liquid/slurry	Solid storage	Dry lot	PRP	Daily spread	Digester	Burned for fuel	Pit<1 month	Pit>1 month	Poultry manure	Deep bedding	Composting	Aerobic treatment
Cool temperate moist	0.35	0.29	0.23	0.30	0.21	0.07	0.28	0.00	0.28	0.28	0.00	0.25	0.475	0.85
Cool temperate dry	0.35	0.29	0.23	0.30	0.21	0.07	0.28	0.00	0.28	0.28	0.00	0.25	0.475	0.85
Boreal moist	0.35	0.29	0.23	0.30	0.21	0.07	0.28	0.00	0.28	0.28	0.00	0.25	0.475	0.85
Boreal dry	0.35	0.29	0.23	0.30	0.21	0.07	0.28	0.00	0.28	0.28	0.00	0.25	0.475	0.85
Warm temperate moist	0.35	0.29	0.23	0.30	0.21	0.07	0.28	0.00	0.28	0.28	0.00	0.25	0.475	0.85
Warm temperate dry	0.35	0.29	0.23	0.30	0.21	0.07	0.28	0.00	0.28	0.28	0.00	0.25	0.475	0.85
Tropical montane	0.35	0.29	0.23	0.30	0.21	0.07	0.28	0.00	0.28	0.28	0.00	0.25	0.475	0.85
Tropical wet	0.35	0.29	0.23	0.30	0.21	0.07	0.28	0.00	0.28	0.28	0.00	0.25	0.475	0.85
Tropical moist	0.35	0.29	0.23	0.30	0.21	0.07	0.28	0.00	0.28	0.28	0.00	0.25	0.475	0.85
Tropical dry	0.35	0.29	0.23	0.30	0.21	0.07	0.28	0.00	0.28	0.28	0.00	0.25	0.475	0.85

Notes: IPCC = Intergovernmental Panel on Climate Change. Bold values are the average over a given range. PRP stands for pasture/range/paddock.

Source: Authors' own elaboration based on Table 10.22 in Chapter 10 and Table 11.3 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A110. Default values for nitrogen loss fraction due to volatilisation of NH_3 and NO_x of nitrogen from manure management (FracGasMS) from poultry (chicken layer and chicken broiler)

IPCC climate zones	Lagoon	Liquid/slurry	Solid storage	Dry lot	PRP	Daily spread	Digester	Burned for fuel	Pit<1 month	Pit>1 month	Poultry manure	Deep bedding	Composting	Aerobic treatment
Cool temperate moist	0.40	0.40	0.33	0.00	0.21	0.07	0.28	0.00	0.28	0.28	0.44	0.3	0.625	0
Cool temperate dry	0.40	0.40	0.33	0.00	0.21	0.07	0.28	0.00	0.28	0.28	0.44	0.3	0.625	0
Boreal moist	0.40	0.40	0.33	0.00	0.21	0.07	0.28	0.00	0.28	0.28	0.44	0.3	0.625	0
Boreal dry	0.40	0.40	0.33	0.00	0.21	0.07	0.28	0.00	0.28	0.28	0.44	0.3	0.625	0
Warm temperate moist	0.40	0.40	0.33	0.00	0.21	0.07	0.28	0.00	0.28	0.28	0.44	0.3	0.625	0
Warm temperate dry	0.40	0.40	0.33	0.00	0.21	0.07	0.28	0.00	0.28	0.28	0.44	0.3	0.625	0
Tropical montane	0.40	0.40	0.33	0.00	0.21	0.07	0.28	0.00	0.28	0.28	0.44	0.3	0.625	0
Tropical wet	0.40	0.40	0.33	0.00	0.21	0.07	0.28	0.00	0.28	0.28	0.44	0.3	0.625	0
Tropical moist	0.40	0.40	0.33	0.00	0.21	0.07	0.28	0.00	0.28	0.28	0.44	0.3	0.625	0
Tropical dry	0.40	0.40	0.33	0.00	0.21	0.07	0.28	0.00	0.28	0.28	0.44	0.3	0.625	0

Notes: IPCC = Intergovernmental Panel on Climate Change. Bold values are the average over a given range. PRP stands for pasture/range/paddock.

Source: Authors' own elaboration based on Table 10.22 in Chapter 10 and Table 11.3 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A111. Default values for nitrogen loss fraction due to volatilisation of NH_3 and NO_x of nitrogen from manure management (FracGasMS) from other cattle

IPCC climate zones	Lagoon	Liquid/slurry	Solid storage	Dry lot	PRP	Daily spread	Digester	Burned for fuel	Pit<1 month	Pit>1 month	Poultry manure	Deep bedding	Composting	Aerobic treatment
Cool temperate moist	0.35	0.29	0.36	0.30	0.21	0.07	0.28	0.00	0.25	0.25	0.00	0.25	0.625	0.85
Cool temperate dry	0.35	0.29	0.36	0.30	0.21	0.07	0.28	0.00	0.25	0.25	0.00	0.25	0.625	0.85
Boreal moist	0.35	0.29	0.36	0.30	0.21	0.07	0.28	0.00	0.25	0.25	0.00	0.25	0.625	0.85
Boreal dry	0.35	0.29	0.36	0.30	0.21	0.07	0.28	0.00	0.25	0.25	0.00	0.25	0.625	0.85
Warm temperate moist	0.35	0.29	0.36	0.30	0.21	0.07	0.28	0.00	0.25	0.25	0.00	0.25	0.625	0.85
Warm temperate dry	0.35	0.29	0.36	0.30	0.21	0.07	0.28	0.00	0.25	0.25	0.00	0.25	0.625	0.85
Tropical montane	0.35	0.29	0.36	0.30	0.21	0.07	0.28	0.00	0.25	0.25	0.00	0.25	0.625	0.85
Tropical wet	0.35	0.29	0.36	0.30	0.21	0.07	0.28	0.00	0.25	0.25	0.00	0.25	0.625	0.85
Tropical moist	0.35	0.29	0.36	0.30	0.21	0.07	0.28	0.00	0.25	0.25	0.00	0.25	0.625	0.85
Tropical dry	0.35	0.29	0.36	0.30	0.21	0.07	0.28	0.00	0.25	0.25	0.00	0.25	0.625	0.85

Notes: IPCC = Intergovernmental Panel on Climate Change. Bold values are the average over a given range. PRP stands for pasture/range/paddock.

Source: Authors' own elaboration based on Table 10.22 in Chapter 10 and Table 11.3 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A112. Default values for nitrogen loss fraction due to volatilisation of NH_3 and NO_x from manure management (FracGasMS) from other animal categories

IPCC climate zones	Lagoon	Liquid/slurry	Solid storage	Dry lot	PRP	Daily spread	Digester	Burned for fuel	Pit<1 month	Pit>1 month	Poultry manure	Deep bedding	Composting	Aerobic treatment
Cool temperate moist	0.35	0.09	0.09	0.30	0.21	0.07	0.28	0.00	0.25	0.25	0.00	0.4	0.19	0.27
Cool temperate dry	0.35	0.09	0.09	0.30	0.21	0.07	0.28	0.00	0.25	0.25	0.00	0.4	0.19	0.27
Boreal moist	0.35	0.09	0.09	0.30	0.21	0.07	0.28	0.00	0.25	0.25	0.00	0.4	0.19	0.27
Boreal dry	0.35	0.09	0.09	0.30	0.21	0.07	0.28	0.00	0.25	0.25	0.00	0.4	0.19	0.27
Warm temperate moist	0.35	0.09	0.09	0.30	0.21	0.07	0.28	0.00	0.25	0.25	0.00	0.4	0.19	0.27
Warm temperate dry	0.35	0.09	0.09	0.30	0.21	0.07	0.28	0.00	0.25	0.25	0.00	0.4	0.19	0.27
Tropical montane	0.35	0.09	0.09	0.30	0.21	0.07	0.28	0.00	0.25	0.25	0.00	0.4	0.19	0.27
Tropical wet	0.35	0.09	0.09	0.30	0.21	0.07	0.28	0.00	0.25	0.25	0.00	0.4	0.19	0.27
Tropical moist	0.35	0.09	0.09	0.30	0.21	0.07	0.28	0.00	0.25	0.25	0.00	0.4	0.19	0.27
Tropical dry	0.35	0.09	0.09	0.30	0.21	0.07	0.28	0.00	0.25	0.25	0.00	0.4	0.19	0.27

Notes: IPCC = Intergovernmental Panel on Climate Change. Bold values are the average over a given range. PRP stands for pasture/range/paddock.

Source: Authors' own elaboration based on Table 10.22 in Chapter 10 and Table 11.3 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A113. Default values for nitrogen loss fraction due to leaching of nitrogen from manure management (FracL.MS) from all animal categories

IPCC climate zones	Lagoon	Liquid/slurry	Solid storage	Dry lot	PRP	Daily spread	Digester	Burned for fuel	Pit<1 month	Pit>1 month	Poultry manure	Deep bedding	Composting	Aerobic treatment
Cool temperate moist	0.00	0.00	0.02	0.035	0.12	0.00	0.00	0.00	0.00	0.00	0.00	0.035	0.04	0
Cool temperate dry	0.00	0.00	0.02	0.035	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.035	0.04	0
Boreal moist	0.00	0.00	0.02	0.035	0.12	0.00	0.00	0.00	0.00	0.00	0.00	0.035	0.04	0
Boreal dry	0.00	0.00	0.02	0.035	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.035	0.04	0
Warm temperate moist	0.00	0.00	0.02	0.035	0.12	0.00	0.00	0.00	0.00	0.00	0.00	0.035	0.04	0
Warm temperate dry	0.00	0.00	0.02	0.035	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.035	0.04	0
Tropical montane	0.00	0.00	0.02	0.035	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.035	0.04	0
Tropical wet	0.00	0.00	0.02	0.035	0.24	0.00	0.00	0.00	0.00	0.00	0.00	0.035	0.04	0
Tropical moist	0.00	0.00	0.02	0.035	0.12	0.00	0.00	0.00	0.00	0.00	0.00	0.035	0.04	0
Tropical dry	0.00	0.00	0.02	0.035	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.035	0.04	0

Notes: IPCC = Intergovernmental Panel on Climate Change. Bold values are the average over a given range. PRP stands for pasture/range/paddock.

Source: Authors' own elaboration based on Table 10.22 in Chapter 10 and Table 11.3 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A114. CO emission removal factors, in tCO_2-C/ha/yr, for drained organic nutrient-poor soil

IPCC climate zone	Forest	Plantation	Annual	Perennial	Flooded rice	Grassland	Peat extraction
Boreal dry	0.31	0.31	7.9	7.9	0	5.7	2.8
Boreal moist	0.31	0.31	7.9	7.9	0	5.7	2.8
Cool temperate dry	2.6	2.6	7.9	7.9	0	5.3	2.8
Cool temperate moist	2.6	2.6	7.9	7.9	0	5.3	2.8
Warm temperate dry	2.6	2.6	7.9	7.9	0	5.3	2.8
Warm temperate moist	2.6	2.6	7.9	7.9	0	5.3	2.8
Tropical dry	5.3	15.3	14.0	14.0	9.4	9.6	2.0
Tropical moist	5.3	15.3	14.0	14.0	9.4	9.6	2.0
Tropical wet	5.3	15.3	14.0	14.0	9.4	9.6	2.0
Tropical montane	5.3	15.3	14.0	14.0	9.4	9.6	2.0

Note: IPCC = Intergovernmental Panel on Climate Change.

Source: Authors' own elaboration based on Table 2.1 of IPCC. 2014. *2013 Supplement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories: Wetlands*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/wetlands/index.html

Table A115. CO_2 emission removal factors, in tCO_2-C/ha/yr, for drained organic nutrient-rich soil

IPCC climate zone	Forest	Plantation	Annual	Perennial	Flooded rice	Grassland	Peat extraction
Boreal dry	0.93	0.93	7.9	7.9	0	5.7	2.8
Boreal moist	0.93	0.93	7.9	7.9	0	5.7	2.8
Cool temperate dry	2.6	2.6	7.9	7.9	0	4.85	2.8
Cool temperate moist	2.6	2.6	7.9	7.9	0	4.85	2.8
Warm temperate dry	2.6	2.6	7.9	7.9	0	4.85	2.8
Warm temperate moist	2.6	2.6	7.9	7.9	0	4.85	2.8
Tropical dry	5.3	15.3	14.0	14.0	9.4	9.6	2.0
Tropical moist	5.3	15.3	14.0	14.0	9.4	9.6	2.0
Tropical wet	5.3	15.3	14.0	14.0	9.4	9.6	2.0
Tropical montane	5.3	15.3	14.0	14.0	9.4	9.6	2.0

Note: IPCC = Intergovernmental Panel on Climate Change*;* CO_2 = carbon dioxide; tCO_2-C/ha/yr = tonnes of carbon (as CO_2-C) per hectare per year, used to express annual carbon stock changes or emissions per unit of land.

Source: Authors' own elaboration based on Table 2.1 of IPCC. 2014. *2013 Supplement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories: Wetlands*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/wetlands/index.html

Table A116. Emission factors for dissolved organic carbon (DOC), in tC/ha/yr, on drained organic soils

IPCC climate zone	Forest	Plantation	Annual	Perennial	Flooded rice	Grassland	Peat extraction
Boreal dry	0.12	0.12	0.12	0.12	0.12	0.12	0.12
Boreal moist	0.12	0.12	0.12	0.12	0.12	0.12	0.12
Cool temperate dry	0.31	0.31	0.31	0.31	0.31	0.31	0.31
Cool temperate moist	0.31	0.31	0.31	0.31	0.31	0.31	0.31
Warm temperate dry	0.31	0.31	0.31	0.31	0.31	0.31	0.31
Warm temperate moist	0.31	0.31	0.31	0.31	0.31	0.31	0.31
Tropical dry	0.82	0.82	0.82	0.82	0.82	0.82	0.82
Tropical moist	0.82	0.82	0.82	0.82	0.82	0.82	0.82
Tropical wet	0.82	0.82	0.82	0.82	0.82	0.82	0.82
Tropical montane	0.82	0.82	0.82	0.82	0.82	0.82	0.82

Note: IPCC = Intergovernmental Panel on Climate Change.

Source: Authors' own elaboration based on Table 2.2 of IPCC. 2014. *2013 Supplement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories: Wetlands*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/wetlands/index.html

Table A117. Methane emission/removal factors, in kg CH_4/ha/yr, on drained nutrient-poor organic soil

IPCC climate zone	Forest	Plantation	Annual	Perennial	Flooded rice	Grassland	Peat extraction
Boreal dry	7	7	0	0	0	1.4	6.1
Boreal moist	7	7	0	0	0	1.4	6.1
Cool temperate dry	2.5	2.5	0	0	0	1.8	6.1
Cool temperate moist	2.5	2.5	0	0	0	1.8	6.1
Warm temperate dry	2.5	2.5	0	0	0	1.8	6.1
Warm temperate moist	2.5	2.5	0	0	0	1.8	6.1
Tropical dry	4.9	9.6	7.0	7.0	143.5	7.0	0
Tropical moist	4.9	9.6	7.0	7.0	143.5	7.0	0
Tropical wet	4.9	9.6	7.0	7.0	143.5	7.0	0
Tropical montane	4.9	9.6	7.0	7.0	143.5	7.0	0

Note: IPCC = Intergovernmental Panel on Climate Change.

Source: Authors' own elaboration based on Table 2.3 of IPCC. 2014. *2013 Supplement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories: Wetlands*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/wetlands/index.html

Table A118. Methane emission/removal factors, in kg CH_4/ha/yr, on drained nutrient-rich organic soil

IPCC climate zone	Forest	Plantation	Annual	Perennial	Flooded rice	Grassland	Peat extraction
Boreal dry	2	2	0	0	0	1.4	6.1
Boreal moist	2	2	0	0	0	1.4	6.1
Cool temperate dry	2.5	2.5	0	0	0	27.5	6.1
Cool temperate moist	2.5	2.5	0	0	0	27.5	6.1
Warm temperate dry	2.5	2.5	0	0	0	27.5	6.1
Warm temperate moist	2.5	2.5	0	0	0	27.5	6.1
Tropical dry	4.9	9.6	7	7	143.5	7	0
Tropical moist	4.9	9.6	7	7	143.5	7	0
Tropical wet	4.9	9.6	7	7	143.5	7	0
Tropical montane	4.9	9.6	7	7	143.5	7	0

Note: IPCC = Intergovernmental Panel on Climate Change.

Source: Authors' own elaboration based on Table 2.3 of IPCC. 2014. *2013 Supplement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories: Wetlands*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/wetlands/index.html

Table A119. Methane emission/removal factors, in kg CH_4/ha/yr, in drainage ditches

IPCC climate zone	Forest	Plantation	Annual	Perennial	Flooded rice	Grassland	Peat extraction
Boreal dry	217	217	1 165	1 165	0	527	542
Boreal moist	217	217	1 165	1 165	0	527	542
Cool temperate dry	217	217	1 165	1 165	0	527	542
Cool temperate moist	217	217	1 165	1 165	0	527	542
Warm temperate dry	217	217	1 165	1 165	0	527	542
Warm temperate moist	217	217	1 165	1 165	0	527	542
Tropical dry	2 259	2 259	2 259	2 259	2 259	2 259	0
Tropical moist	2 259	2 259	2 259	2 259	2 259	2 259	0
Tropical wet	2 259	2 259	2 259	2 259	2 259	2 259	0
Tropical montane	2 259	2 259	2 259	2 259	2 259	2 259	0

Note: IPCC = Intergovernmental Panel on Climate Change.

Source: Authors' own elaboration based on Table 2.4 of IPCC. 2014. *2013 Supplement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories: Wetlands*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/wetlands/index.html

Table A120. Nitrous oxide emission/removal factors, in kgN_2O-N/ha/yr, on drained nutrient-poor organic soil

IPCC climate zone	Forest	Plantation	Annual	Perennial	Flooded rice	Grassland	Peat extraction
Boreal dry	0.22	0.22	13	13	0	9.5	0.3
Boreal moist	0.22	0.22	13	13	0	9.5	0.3
Cool temperate dry	2.8	2.8	13	13	0	4.3	0.3
Cool temperate moist	2.8	2.8	13	13	0	4.3	0.3
Warm temperate dry	2.8	2.8	13	13	0	4.3	0.3
Warm temperate moist	2.8	2.8	13	13	0	4.3	0.3
Tropical dry	2.4	2.25	5	5	0.4	5	3.6
Tropical moist	2.4	2.25	5	5	0.4	5	3.6
Tropical wet	2.4	2.25	5	5	0.4	5	3.6
Tropical montane	2.4	2.25	5	5	0.4	5	3.6

Note: IPCC = Intergovernmental Panel on Climate Change.

Source: Authors' own elaboration based on Table 2.5 of IPCC. 2014. *2013 Supplement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories: Wetlands*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/wetlands/index.html

Table A121. Nitrous oxide emission/removal factors, in kgN_2O-N/ha/yr, on drained nutrient-rich organic soil

IPCC climate zone	Forest	Plantation	Annual	Perennial	Flooded rice	Grassland	Peat extraction
Boreal dry	3.2	3.2	13	13	0	9.5	0.3
Boreal moist	3.2	3.2	13	13	0	9.5	0.3
Cool temperate dry	2.8	2.8	13	13	0	4.9	0.3
Cool temperate moist	2.8	2.8	13	13	0	4.9	0.3
Warm temperate dry	2.8	2.8	13	13	0	4.9	0.3
Warm temperate moist	2.8	2.8	13	13	0	4.9	0.3
Tropical dry	2.4	2.25	5	5	0.4	5	3.6
Tropical moist	2.4	2.25	5	5	0.4	5	3.6
Tropical wet	2.4	2.25	5	5	0.4	5	3.6
Tropical montane	2.4	2.25	5	5	0.4	5	3.6

Note: IPCC = Intergovernmental Panel on Climate Change.

Source: Authors' own elaboration based on Table 2.5 of IPCC. 2014. *2013 Supplement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories: Wetlands*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/wetlands/index.html

Table A122. Emission factors for CO_2, in tCO_2-C/ha/yr, for rewetted inland organic on nutrient-poor soil

IPCC climate zone	Forest	Plantation	Annual	Perennial	Flooded rice	Grassland
Boreal dry	-0.34	-0.34	-0.34	-0.34	-0.34	-0.34
Boreal moist	-0.34	-0.34	-0.34	-0.34	-0.34	-0.34
Cool temperate dry	-0.23	-0.23	-0.23	-0.23	-0.23	-0.23
Cool temperate moist	-0.23	-0.23	-0.23	-0.23	-0.23	-0.23
Warm temperate dry	-0.23	-0.23	-0.23	-0.23	-0.23	-0.23
Warm temperate moist	-0.23	-0.23	-0.23	-0.23	-0.23	-0.23
Tropical dry	0	0	0	0	0	0
Tropical moist	0	0	0	0	0	0
Tropical wet	0	0	0	0	0	0
Tropical montane	0	0	0	0	0	0

Note: IPCC = Intergovernmental Panel on Climate Change.

Source: Authors' own elaboration based on Table 3.1 of IPCC. 2014. *2013 Supplement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories: Wetlands*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/wetlands/index.html

Table A123. Emission factors for CO_2, in tCO_2-C/ha/yr, for rewetted inland organic on nutrient-rich soil

IPCC climate zone	Forest	Plantation	Annual	Perennial	Flooded rice	Grassland
Boreal dry	-0.55	-0.55	-0.55	-0.55	-0.55	-0.55
Boreal moist	-0.55	-0.55	-0.55	-0.55	-0.55	-0.55
Cool temperate dry	0.5	0.5	0.5	0.5	0.5	0.5
Cool temperate moist	0.5	0.5	0.5	0.5	0.5	0.5
Warm temperate dry	0.5	0.5	0.5	0.5	0.5	0.5
Warm temperate moist	0.5	0.5	0.5	0.5	0.5	0.5
Tropical dry	0	0	0	0	0	0
Tropical moist	0	0	0	0	0	0
Tropical wet	0	0	0	0	0	0
Tropical montane	0	0	0	0	0	0

Note: IPCC = Intergovernmental Panel on Climate Change; N/A = not applicable; tCO_2-C/ha/yr = tonnes of carbon (as CO_2-C) per hectare per year, expressing carbon stock changes or emissions on a land area basis.

Source: Authors' own elaboration based on Table 3.1 of IPCC. 2014. *2013 Supplement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories: Wetlands*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/wetlands/index.html

Table A124. Emission factors for dissolved organic carbon, in tC/ha/yr, for rewetted inland organic soil

IPCC climate zone	Forest	Plantation	Annual	Perennial	Flooded rice	Grassland
Boreal dry	0.08	0.08	0.08	0.08	0.08	0.08
Boreal moist	0.08	0.08	0.08	0.08	0.08	0.08
Cool temperate dry	0.24	0.24	0.24	0.24	0.24	0.24
Cool temperate moist	0.24	0.24	0.24	0.24	0.24	0.24
Warm temperate dry	0.24	0.24	0.24	0.24	0.24	0.24
Warm temperate moist	0.24	0.24	0.24	0.24	0.24	0.24
Tropical dry	0.51	0.51	0.51	0.51	0.51	0.51
Tropical moist	0.51	0.51	0.51	0.51	0.51	0.51
Tropical wet	0.51	0.51	0.51	0.51	0.51	0.51
Tropical montane	0.51	0.51	0.51	0.51	0.51	0.51

Note: IPCC = Intergovernmental Panel on Climate Change.

Source: Authors' own elaboration based on Table 3.2 of IPCC. 2014. *2013 Supplement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories: Wetlands*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/wetlands/index.html

Table A125. Emission factors for CH_4, in kg CH_4-C/ha/yr, for rewetted inland organic nutrient-poor soil

IPCC climate zone	Forest	Plantation	Annual	Perennial	Flooded rice	Grassland
Boreal dry	41	41	41	41	41	41
Boreal moist	41	41	41	41	41	41
Cool temperate dry	92	92	92	92	92	92
Cool temperate moist	92	92	92	92	92	92
Warm temperate dry	92	92	92	92	92	92
Warm temperate moist	92	92	92	92	92	92
Tropical dry	41	41	41	41	41	41
Tropical moist	41	41	41	41	41	41
Tropical wet	41	41	41	41	41	41
Tropical montane	41	41	41	41	41	41

Note: IPCC = Intergovernmental Panel on Climate Change.

Source: Authors' own elaboration based on Table 3.3 of IPCC. 2014. *2013 Supplement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories: Wetlands*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/wetlands/index.html

Table A126. Emission factors for CH_4, in kg CH_4-C/ha/yr, for rewetted inland organic nutrient-rich soil

IPCC climate zone	Forest	Plantation	Annual	Perennial	Flooded rice	Grassland
Boreal dry	137	137	137	137	137	137
Boreal moist	137	137	137	137	137	137
Cool temperate dry	216	216	216	216	216	216
Cool temperate moist	216	216	216	216	216	216
Warm temperate dry	216	216	216	216	216	216
Warm temperate moist	216	216	216	216	216	216
Tropical dry	41	41	41	41	41	41
Tropical moist	41	41	41	41	41	41
Tropical wet	41	41	41	41	41	41
Tropical montane	41	41	41	41	41	41

Note: IPCC = Intergovernmental Panel on Climate Change.

Source: Authors' own elaboration based on Table 3.3 of IPCC. 2014. *2013 Supplement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories: Wetlands*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/wetlands/index.html

Table A127. Mass of fuel available for combustion, in tonne of dry matter/hectare

IPCC climate zones	Wildfire undrained peat	Wildfire drained peat	Prescribed fire (i.e. land management)
Boreal dry	66	336	0
Boreal moist	66	336	0
Cool temperate dry	66	336	0
Cool temperate moist	66	336	0
Warm temperate dry	66	336	0
Warm temperate moist	66	336	0
Tropical dry	0	353	155
Tropical moist	0	353	155
Tropical wet	0	353	155
Tropical montane	0	353	155

Source: Authors' own elaboration based on Table 2.6 of IPCC. 2014. *2013 Supplement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories: Wetlands*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/wetlands/index.html

Table A128. Emission factors for CO_2, CO and CH_4 for fire on organic soil, in g/kg of dry matter burned

IPCC climate zones	CO_2	CO	CH_4
Boreal dry	362	207	9
Boreal moist	362	207	9
Cool temperate dry	362	207	9
Cool temperate moist	362	207	9
Warm temperate dry	362	207	9
Warm temperate moist	362	207	9
Tropical dry	464	210	21
Tropical moist	464	210	21
Tropical wet	464	210	21
Tropical montane	464	210	21

Note: IPCC = Intergovernmental Panel on Climate Change, CO_2 = carbon dioxide, CH_4 = methane, N_2O = nitrous oxide.

Source: Authors' own elaboration based on Table 2.7 of IPCC. 2014. *2013 Supplement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories: Wetlands*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/wetlands/index.html

Table A129. Methane emission factors for other constructed waterbodies, in kg CH_4/ha/yr

IPCC climate zones	Freshwater pond	Brackish water pond	Canals and ditches	Saline pond
Cool temperate moist	183	183	416	30
Cool temperate dry	183	183	416	30
Boreal moist	183	183	416	30
Boreal dry	183	183	416	30
Warm temperate moist	183	183	416	30
Warm temperate dry	183	183	416	30
Tropical montane	183	183	416	30
Tropical wet	183	183	416	30
Tropical moist	183	183	416	30
Tropical dry	183	183	416	30

Note: IPCC = Intergovernmental Panel on Climate Change.

Source: Authors' own elaboration based on Table 7.12 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A130. Trophic state adjustment factor, dimensionless, and range mean annual chlorophyll-a concentration in µg/L

Trophic state	Trophic state adjustment factor	Range of the mean annual chlorophyll-a concentration
Default	1	N/A
Oligotrophic	**0.7**	0–2.6
Mesotrophic	**3.0**	2.6–20
Eutrophic	**9.9**	20–56
Hypereutrophic	**26.9**	56–>155

Notes: Bold values are the average of a given range. The range of mean annual chlo-a concentration is provided for ease of reference.

Source: Authors' own elaboration based on Table 7.11 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A131. Salinity-based definitions for water bodies

Common description	Salinity (particle per thousand [ppt])*
Tidal fresh water	< 0.5
Brackish water	0.5–18
Saline water	> 18

Note: * ppt is part per thousands (percent) and is roughly equivalent to grams of salt per litre of water.

Source: Annex 4A.1 of IPCC. 2014. *2013 Supplement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories: Wetlands*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/wetlands/index.html

Table A132. Default values for above-ground biomass, in t d.m./ha, in coastal wetlands

IPCC climate zones	Mangrove	Tidal marsh	Seagrass
Boreal dry	N/A	0	0
Boreal moist	N/A	0	0
Cool temperate dry	N/A	0	0
Cool temperate moist	N/A	0	0
Warm temperate dry	75	0	0
Warm temperate moist	75	0	0
Tropical dry	92	0	0
Tropical moist	192	0	0
Tropical wet	192	0	0

Notes: IPCC = Intergovernmental Panel on Climate Change, N/A = not applicable. Proxies from tropical wet are used for tropical moist.

Source: Authors' own elaboration based on Table 4.3 of IPCC. 2014. *2013 Supplement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories: Wetlands*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/wetlands/index.html

Table A133. Default values for the root to shoot ratio, in t root d.m./t shoot d.m., in coastal wetlands

IPCC climate zones	Mangrove	Tidal marsh	Seagrass	Tidal marsh Mediterranean
Boreal dry	N/A	0	0	N/A
Boreal moist	N/A	0	0	N/A
Cool temperate dry	N/A	2.10	1.30	N/A
Cool temperate moist	N/A	2.10	1.30	N/A
Warm temperate dry	0.96	3.65	2.40	3.63
Warm temperate moist	0.96	3.65	2.40	3.63
Tropical dry	0.29	3.65	1.70	3.63
Tropical moist	0.49	3.65	1.70	N/A
Tropical wet	0.49	3.65	1.70	N/A

Notes: IPCC = Intergovernmental Panel on Climate Change, N/A = not applicable. Proxies from tropical wet are used for tropical moist.

Source: Authors' own elaboration based on Table 4.5, Table 4.9 and Table 4.10 of IPCC. 2014. *2013 Supplement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories: Wetlands*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/wetlands/index.html

Table A134. Default values for litter, in tC/ha, in coastal wetlands

IPCC climate zones	Mangrove	Tidal marsh	Seagrass
Boreal dry	N/A	0	0
Boreal moist	N/A	0	0
Cool temperate dry	N/A	0	0
Cool temperate moist	N/A	0	0
Warm temperate dry	0.7	0	0
Warm temperate moist	0.7	0	0
Tropical dry	0.7	0	0
Tropical moist	0.7	0	0
Tropical wet	0.7	0	0

Note: IPCC = Intergovernmental Panel on Climate Change; N/A = not applicable; tC/ha = tonnes of carbon per hectare.

Source: Authors' own elaboration based on Table 4.7 of IPCC. 2014. *2013 Supplement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories: Wetlands*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/wetlands/index.html

Table A135. Default values for deadwood, in tC/ha, in coastal wetlands

IPCC climate zones	Mangrove	Tidal marsh	Seagrass
Boreal dry	N/A	0	0
Boreal moist	N/A	0	0
Cool temperate dry	N/A	0	0
Cool temperate moist	N/A	0	0
Warm temperate dry	10.7	0	0
Warm temperate moist	10.7	0	0
Tropical dry	10.7	0	0
Tropical moist	10.7	0	0
Tropical wet	10.7	0	0

Note: IPCC = Intergovernmental Panel on Climate Change, N/A = not applicable.

Source: Authors' own elaboration based on Table 4.7 of IPCC. 2014. *2013 Supplement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories: Wetlands*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/wetlands/index.html

Table A136. Default carbon stock for mineral soil for the first top metre depth, in tC/ha, in coastal wetlands

IPCC climate zones	Mangrove	Tidal marsh	Seagrass
Boreal dry	N/A	226	108
Boreal moist	N/A	226	108
Cool temperate dry	N/A	226	108
Cool temperate moist	N/A	226	108
Warm temperate dry	286	226	108
Warm temperate moist	286	226	108
Tropical dry	286	226	108
Tropical moist	286	226	108
Tropical wet	286	226	108

Note: IPCC = Intergovernmental Panel on Climate Change, N/A = not applicable.

Source: Authors' own elaboration based on Table 4.11 of IPCC. 2014. *2013 Supplement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories: Wetlands*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/wetlands/index.html

Table A137. Default carbon stock for organic soil for the first top metre depth, in tC/ha, in coastal wetlands

IPCC climate zones	Mangrove	Tidal marsh	Seagrass
Boreal dry	N/A	340	0
Boreal moist	N/A	340	0
Cool temperate dry	N/A	340	0
Cool temperate moist	N/A	340	0
Warm temperate dry	471	340	0
Warm temperate moist	471	340	0
Tropical dry	471	340	0
Tropical moist	471	340	0
Tropical wet	471	340	0

Note: IPCC = Intergovernmental Panel on Climate Change, N/A = not applicable.

Source: Authors' own elaboration based on Table 4.11 of IPCC. 2014. *2013 Supplement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories: Wetlands*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/wetlands/index.html

Table A138. Default carbon stock for aggregated soil for the first top metre depth, in tC/ha, in coastal wetlands

IPCC climate zones	Mangrove	Tidal marsh	Seagrass
Boreal dry	N/A	255	0
Boreal moist	N/A	255	0
Cool temperate dry	N/A	255	0
Cool temperate moist	N/A	255	0
Warm temperate dry	386	255	0
Warm temperate moist	386	255	0
Tropical dry	386	255	0
Tropical moist	386	255	0
Tropical wet	386	255	0

Note: IPCC = Intergovernmental Panel on Climate Change, N/A = not applicable.

Source: Authors' own elaboration based on Table 4.11 of IPCC. 2014. *2013 Supplement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories: Wetlands*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/wetlands/index.html

Table A139. Mangrove total area in 2014, in km^2, averaged soil carbon stock for the first top metre depth and standard deviation, in tC/ha

Country	No.	Area	Soil carbon stock	sd
Angola	0	213	283	
Antigua and Barbuda	0	2	283	
Australia	269	3 315	249	154
Bahamas	2	37	81	15
Bangladesh	11	1 773	118	29
Barbados	0	0	283	
Belize	13	302	647	70
Benin	3	7	235	52
Bermuda	0	0	283	

Country	No.	Area	Soil carbon stock	sd
Indonesia	40	23 143	359	181
Iran (Islamic Republic of)	0	0	283	
Jamaica	0	44	283	
Japan	47	8	274	136
Kenya	48	230	446	422
Liberia	0	47	283	
Madagascar	7	850	167	75
Malaysia	35	4 692	424	400
Maldives	0	0	283	

Country	No.	Area	Soil carbon stock	sd
Brazil	41	7 663	308	131
Brunei Darussalam	0	103	283	
Cambodia	0	320	283	
Cameroon	6	1 113	597	348
China	34	31	186	67
Colombia	4	1 672	449	164
Comoros	0	1	283	
Congo	3	70	470	385
Costa Rica	0	335	283	
Côte d'Ivoire	0	23	283	
Cuba	0	1 624	283	
Democratic Republic of the Congo	3	146	936	85
Djibouti	0	0	283	
Dominican Republic	9	101	419	88
Ecuador	12	935	327	61
Egypt	43	0	185	55
El Salvador	0	236	283	
Equatorial Guinea	0	155	283	
Eritrea	0	1	283	
Fiji	0	401	283	
French Guiana	31	698	185	129
Gabon	3	1 081	736	192
Gambia	0	48	283	
Ghana	3	23	245	40
Grenada	0	1	283	
Guadeloupe	31	698	185	129
Guatemala	0	253	283	
Guinea	0	820	283	
Guinea-Bissau	2	745	126	56
Guyana	0	188	283	
Haiti	0	42	283	
Honduras	18	524	397	142
India	160	791	253	91

Country	No.	Area	Soil carbon stock	sd
Mexico	38	2 985	370	176
Micronesia (Federated States of)	27	7	453	244
Morocco	0	0	283	
Mozambique	6	1 223	153	7
Myanmar	0	2 508	283	
New Caledonia	7	98	401	76
New Zealand	92	77	209	52
Nicaragua	0	552	283	
Nigeria	11	2 653	235	44
Oman	0	0	283	
Pakistan	3	12	181	7
Palau	7	48	482	84
Panama	10	1 323	299	46
Papua New Guinea	6	4 169	140	25
Peru	0	14	283	
Philippines	8	2 060	227	162
Puerto Rico	0	46	283	
Saudi Arabia	31	0	71	53
Senegal	2	155	139	71
Seychelles	0	5	283	
Sierra Leone	0	655	283	
Singapore	1	2	330	
Solomon Islands	0	393	283	
Somalia	0	8	283	
South Africa	7	7	235	47
Sri Lanka	6	35	503	163
Saint Vincent and the Grenadines	0	0	283	
Sudan	0	0	283	
Suriname	0	510	283	
Taiwan	3	1	122	31
United Republic of Tanzania	30	478	243	83
Thailand	14	1 876	246	82

Notes: no. is the number of observations. Values in bold are countries with missing soil carbon data. For these countries total carbon stocks were estimated using the average global soil carbon stock per unit area of 283 tC/ha.

Source: Authors' own elaboration based on Table S1 of Atwood, T.B., Connolly, R.M., Almahasheer, H., Carell, P.E., Duarte, C.M., Ewers Lewis, C.J., Irigoien, X., Kelleway, J.J., Lavery, P.S., Macreadie, P.I., Serrano, O.S., Sanders, C.J., Santos, I., Steven, A.D.L. & Lovelock, C.E. 2017. Global patterns in mangrove soil carbon stocks and losses. *Nature Climate Change*, 7: 523–528. https://doi.org/10.1038/nclimate3326

Table A140. Emission factor for carbon associated with rewetting and revegetation in coastal wetlands, in tC/ha/yr

IPCC climate zones	Mangrove	Tidal marsh	Seagrass
Boreal dry	N/A	-0.91	-0.43
Boreal moist	N/A	-0.91	-0.43
Cool temperate dry	N/A	-0.91	-0.43
Cool temperate moist	N/A	-0.91	-0.43
Warm temperate dry	-1.62	-0.91	-0.43
Warm temperate moist	-1.62	-0.91	-0.43
Tropical dry	-1.62	-0.91	-0.43
Tropical moist	-1.62	-0.91	-0.43
Tropical wet	-1.62	-0.91	-0.43

Note: IPCC = Intergovernmental Panel on Climate Change, N/A = not applicable.

Source: Authors' own elaboration based on Table 4.12 of IPCC. 2014. *2013 Supplement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories: Wetlands*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/wetlands/index.html

Table A141. Small scale fisheries Tier 1 fuel use intensities, in L/t catch

Gear	Marine	Inland	Unmotorized
Trawls	671	N/A	0
Gillnets	336	247	0
Hooks and lines	607	N/A	0
Multiple	1415	465	0
Pots and traps	884	N/A	0
Surrounding nets	73	N/A	0
Dredges	880	N/A	0
Not specified	695	356	0

Note: N/A = not applicable.

Source: WorldFish, FAO & Duke University. 2018. *Illuminating Hidden Harvests: The Contribution Of Small-Scale Fisheries To Sustainable Development*.http://pubs.iclarm.net/resource_centre/Program-Brief-Illuminating-hidden-harvests.pdf

Table A142. Large scale fisheries Tier 1 fuel use intensities, in L/t catch

Species group	Bottom trawl	Gillnets and entangling nets	Hooks and lines	Midwater trawls	Traps and lift nets	Surrounding nets	Dredges	Mobile seine	Other or mixed	Not specified
Abalones, winkles and conchs	N/A	2 162	N/A	N/A	484	N/A	N/A	N/A	839	**791**
Clams, cockles and ark shells	N/A	N/A	N/A	N/A	1 430	N/A	835	N/A	430	**787**
Cods, hakes and haddocks	490	511	702	138	N/A	512	N/A	N/A	436	**463**
Flounders, halibuts and soles	2 903	502	570	982	N/A	694	N/A	N/A	1 094	**1 822**
Herrings, sardines and anchovies	91	602	0	118	559	68	N/A	146	287	**145**
Jacks, mullets and sauries	492	496	1 836	489	N/A	434	445	N/A	497	**641**
Krill and plankton	N/A	N/A	N/A	121	N/A	N/A	N/A	N/A	N/A	**121**
Lobsters and spiny-rock lobsters	5 200	N/A	N/A	1 667	3 564	N/A	N/A	N/A	1 028	**3 937**
Mackerels, snoeks and cutlassfish	922	N/A	322	167	0	39	N/A	N/A	106	**208**
Miscellaneous marine crustaceans	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	2 230	**2 230**
Miscellaneous marine fishes	171	N/A	N/A	350	N/A	N/A	N/A	N/A	1 903	**637**
Miscellaneous marine molluscs	2 904	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0	**2 904**
Mussels	N/A	N/A	N/A	N/A	N/A	N/A	20	N/A	13	**20**
Redfish, basses and congers	754	304	472	232	921	254	N/A	N/A	912	**467**
Salmons, trots and smelts	N/A	886	835	N/A	N/A	304	N/A	N/A	N/A	**622**
Scallops and pectens	1 405	N/A	N/A	1 787	N/A	N/A	577	N/A	834	**697**
Sea-spiders and crabs	N/A	N/A	1 031	N/A	1074	N/A	N/A	N/A	1 062	**1 069**
Shads	N/A	N/A	N/A	16	N/A	N/A	N/A	N/A	N/A	**16**
Sharks and rays	982	661	365	494	153	118	N/A	N/A	253	**606**
Shrimps and prawns	2 550	630	N/A	3 074	N/A	N/A	N/A	N/A	1 224	**2 569**
Squat lobsters	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	165	**165**
Squids, cuttlefish and octopuses	313	N/A	N/A	406	N/A	N/A	N/A	N/A	N/A	**325**
Tunas, bonitos and billfishes	N/A	683	1 554	1 649	N/A	433	N/A	N/A	915	**1 295**
Not specified	**1 475**	**744**	N/A	**854**	N/A	**779**	N/A	N/A	**1 169**	**317**

Notes: IPCC = Intergovernmental Panel on Climate Change; N/A = not applicable. When gear class or species is not known, users can select "not specified", highlighted in bold and calculated from the other available values. For gear class, "not specified" values are the weighted average of all other gear class for each species group. For species, "not specified" FUI values are the median FUI values by each gear class.

Source: Parker, R.W.R. & Tyedmers, P.H. 2015. *Fuel consumption of global fishing fleets: current understanding and knowledge gaps. Fish and Fisheries,* 16(4), 684-696. https://doi.org/10.1111/faf.12087

Table A143. Direct N_2O emission factors for fertilizer on managed soils, in kg N_2O-N/kg N

IPCC climate region	Synthetic N-fertilizer	Other N inputs
Dry	0.005	0.005
Wet	0.016	0.006
Moist	0.0105	0.0055
Tropical montane	0.005	0.005

Notes: IPCC = Intergovernmental Panel on Climate Change, N/A = not applicable, N = nitrogen. The average of wet and dry is used as a proxy for moist climate.

Source: Authors' own elaboration based on Table 11.1 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A144. Emission factors from production, transportation, storage and transfer of agricultural inputs, in tCO_2eq/t of product

Type of input	Emission coefficient used by EX-ACT
Nitrogen-fertilizers	4.77
Phosphorus-fertilizer	0.73
Potassium-fertilizer	0.55
Lime and dolomite	0.59
Herbicides	23.10
Insecticides	18.70
Fungicides	14.30

Note: emission factors were originally in tonne of carbon equivalent and were converted in CO2eq using the factor 44/12.

Source: Authors' own elaboration based on Table 5 of Lal, R. 2004. Carbon Emission from Farm Operations. *Environment International,* 30: 981–990.

Table A145. Electricity grid emission factor, in tCO_2eq/MWh

UN country list	UN region	Operating margin (tCO_2eq/MWh)	Combined margin (tCO_2eq/MWh)
Afghanistan	Southern Asia	0.414	0.193
Åland Islands	Europe	N/A	0.256
Albania	Europe	N/A	0.256
Algeria	Northern Africa	0.528	0.397
American Samoa	Oceania	0.753	0.516
Andorra	Europe	0.188	0.070
Angola	Central Africa	1.476	0.748
Anguilla	Caribbean	0.753	0.472
Antigua and Barbuda	Caribbean	0.753	0.489
Argentina	South America	0.478	0.288
Armenia	Western Asia	0.390	0.205
Aruba	Caribbean	0.753	0.421
Australia	Oceania	0.808	0.421
Austria	Europe	0.242	0.113
Azerbaijan	Western Asia	0.534	0.384
Bahamas	Caribbean	0.753	0.441
Bahrain	Western Asia	0.726	0.454
Bangladesh	Southern Asia	0.528	0.412
Barbados	Caribbean	0.749	0.484

UN country list	UN region	Operating margin (tCO_2eq/MWh)	Combined margin (tCO_2eq/MWh)
Belarus	Europe	0.400	0.292
Belize	Central America	0.403	0.183
Belgium	Europe	0.252	0.124
Benin	Western Africa	0.745	0.576
Bermuda	North America	0.753	0.342
Bhutan	Southern Asia	N/A	0.388
Bolivia (Plurinational State of)	South America	0.604	0.393
Bonaire, Sint Eustatius and Saba	Caribbean	0.753	0.400
Bosnia and Herzegovina	Europe	1.197	0.739
Botswana	Southern Africa	1.486	1.070
Bouvet Island	South America	N/A	0.290
Brazil	South America	0.284	0.139
British Indian Ocean Territory	Eastern Africa	N/A	0.418
British Virgin Islands	Caribbean	N/A	0.420
Brunei Darussalam	South-Eastern Asia	0.681	0.407
Bulgaria	Europe	0.911	0.495
Burkina Faso	Western Africa	0.753	0.539
Burundi	Eastern Africa	0.414	0.197
Cabo Verde	Western Africa	0.753	0.505
Cambodia	South-Eastern Asia	1.046	0.588
Cameroon	Central Africa	0.659	0.354
Canada	North America	0.372	0.156
Cayman Islands	Caribbean	0.753	0.373
Central African Republic	Central Africa	0.188	0.077
Chad	Central Africa	0.753	0.581
Chile	South America	0.657	0.371
China	Eastern Asia	0.899	0.547
China, Hong Kong SAR	Eastern Asia	0.899	0.547
China, Macao SAR	Eastern Asia	N/A	0.512
Christmas Island	Oceania	N/A	0.451
Cocos (Keeling) Islands	Oceania	N/A	0.451
Colombia	South America	0.410	0.208
Comoros	Eastern Africa	0.753	0.589
Congo	Central Africa	0.659	0.405
Cook Islands	Oceania	0.753	0.422
Costa Rica	Central America	0.108	0.039
Côte d'Ivoire	Western Africa	0.466	0.314
Croatia	Europe	0.294	0.168
Cuba	Caribbean	0.559	0.391
Curaçao	Caribbean	0.876	0.506
Cyprus	Western Asia	0.751	0.438
Czechia	Europe	0.902	0.461
Democratic People's Republic of Korea	Eastern Asia	0.754	0.359
Democratic Republic of the Congo	Central Africa	N/A	0.0004

UN country list	UN region	Operating margin (tCO_2eq/MWh)	Combined margin (tCO_2eq/MWh)
Denmark	Europe	0.362	0.155
Djibouti	Eastern Africa	0.753	0.575
Dominica	Caribbean	0.753	0.433
Dominican Republic	Caribbean	0.601	0.426
Ecuador	South America	0.560	0.280
Egypt	Northern Africa	0.554	0.406
El Salvador	Central America	0.547	0.275
Equatorial Guinea	Central Africa	0.632	0.361
Eritrea	Eastern Africa	0.915	0.704
Estonia	Europe	1.057	0.625
Eswatini	Southern Africa	N/A	0.652
Ethiopia	Eastern Africa	N/A	0.00014
Falkland Islands (Malvinas)	South America	0.753	0.316
Faroe Islands	Europe	0.753	0.320
Fiji	Oceania	0.640	0.334
Finland	Europe	0.267	0.114
France	Europe	0.158	0.068
French Guiana	South America	0.423	0.200
French Polynesia	Oceania	0.753	0.412
French Southern Territories	Eastern Africa	N/A	0.418
Gabon	Central Africa	0.946	0.533
Gambia	Western Africa	0.753	0.591
Georgia	Western Asia	0.289	0.135
Germany	Europe	0.650	0.313
Ghana	Western Africa	0.495	0.276
Gibraltar	Europe	0.779	0.369
Greece	Europe	0.507	0.346
Greenland	North America	0.264	0.105
Grenada	Caribbean	0.753	0.523
Guadeloupe	Caribbean	0.753	0.433
Guam	Oceania	0.753	0.428
Guatemala	Central America	0.798	0.427
Guernsey	Europe	N/A	0.256
Guinea	Western Africa	0.753	0.460
Guinea-Bissau	Western Africa	0.753	0.577
Guyana	South America	0.847	0.616
Haiti	Caribbean	1.048	0.765
Heard Island and McDonald Islands	Oceania	N/A	0.451
Holy See	Europe	N/A	0.256
Honduras	Central America	0.662	0.359
Hungary	Europe	0.296	0.191
Iceland	Europe	N/A	0.000
India	Southern Asia	0.951	0.661
Indonesia	South-Eastern Asia	0.783	0.599

UN country list	UN region	Operating margin (tCO_2eq/MWh)	Combined margin (tCO_2eq/MWh)
Iran (Islamic Republic of)	Southern Asia	0.592	0.421
Iraq	Western Asia	1.080	0.788
Ireland	Europe	0.380	0.189
Isle of Man	Europe	0.436	0.204
Israel	Western Asia	0.394	0.258
Italy	Europe	0.414	0.224
Jamaica	Caribbean	0.711	0.498
Japan	Eastern Asia	0.471	0.286
Jersey	Europe	N/A	0.256
Jordan	Western Asia	0.529	0.382
Kazakhstan	Central Asia	0.797	0.532
Kenya	Eastern Africa	0.574	0.274
Kiribati	Oceania	0.753	0.530
Kuwait	Western Asia	0.675	0.400
Kyrgyzstan	Central Asia	0.217	0.098
Lao People's Democratic Republic	South-Eastern Asia	1.069	0.555
Latvia	Europe	0.240	0.117
Lebanon	Western Asia	0.794	0.567
Lesotho	Southern Africa	N/A	0.652
Liberia	Western Africa	0.677	0.374
Libya	Northern Africa	0.668	0.493
Liechtenstein	Europe	0.151	0.052
Lithuania	Europe	0.211	0.102
Luxembourg	Europe	0.220	0.095
Madagascar	Eastern Africa	0.876	0.567
Malawi	Eastern Africa	0.489	0.243
Malaysia	South-Eastern Asia	0.551	0.436
Maldives	Southern Asia	0.753	0.524
Mali	Western Africa	1.076	0.623
Malta	Europe	0.520	0.295
Marshall Islands	Oceania	0.753	0.561
Martinique	Caribbean	0.753	0.406
Mauritania	Western Africa	0.753	0.513
Mauritius	Eastern Africa	0.700	0.543
Mayotte	Eastern Africa	N/A	0.512
Mexico	Central America	0.531	0.360
Micronesia (Federated States of)	Oceania	0.753	0.557
Monaco	Europe	0.158	0.068
Mongolia	Eastern Asia	1.366	1.002
Montenegro	Europe	0.899	0.471
Montserrat	Caribbean	0.753	0.517
Morocco	Northern Africa	0.729	0.547
Mozambique	Eastern Africa	0.234	0.111
Myanmar	South-Eastern Asia	0.719	0.407

UN country list	UN region	Operating margin (tCO_2eq/MWh)	Combined margin (tCO_2eq/MWh)
Namibia	Southern Africa	0.355	0.139
Naoero	Oceania	0.753	0.521
Nepal	Southern Asia	N/A	0.000
Netherlands (Kingdom of the)	Europe	0.326	0.203
New Caledonia	Oceania	0.779	0.445
New Zealand	Oceania	0.246	0.108
Nicaragua	Central America	0.675	0.372
Niger	Western Africa	0.772	0.718
Nigeria	Western Africa	0.526	0.358
Niue	Oceania	0.753	0.459
Norfolk Island	Oceania	N/A	0.451
North Macedonia	Europe	0.851	0.563
Northern Mariana Islands	Oceania	0.753	0.416
Norway	Europe	0.047	0.017
Oman	Western Asia	0.479	0.320
Pakistan	Southern Asia	0.592	0.386
Palau	Oceania	0.753	0.497
Palestine	Western Asia	N/A	0.517
Panama	Central America	0.477	0.230
Papua New Guinea	Oceania	0.597	0.315
Paraguay	South America	N/A	0.000
Peru	South America	0.473	0.252
Philippines	South-Eastern Asia	0.672	0.525
Pitcairn	Oceania	N/A	0.451
Poland	Europe	0.828	0.532
Portugal	Europe	0.389	0.228
Puerto Rico	Caribbean	0.596	0.362
Qatar	Western Asia	0.503	0.258
Republic of Korea	Eastern Asia	0.555	0.330
Republic of Moldova	Europe	0.541	0.399
Réunion	Eastern Africa	0.772	0.421
Romania	Europe	0.489	0.289
Russian Federation	Europe	0.476	0.294
Rwanda	Eastern Africa	0.712	0.416
Saint Barthélemy	Caribbean	N/A	0.461
Saint Helena	Western Africa	0.753	0.456
Saint Kitts and Nevis	Caribbean	0.753	0.477
Saint Lucia	Caribbean	0.753	0.521
Saint Martin (French part)	Caribbean	0.753	0.484
Saint Pierre and Miquelon	North America	0.753	0.415
Saint Vincent and the Grenadines	Caribbean	0.753	0.499
Samoa	Oceania	0.753	0.434
San Marino	Europe	0.414	0.224
Sao Tome and Principe	Central Africa	0.753	0.565

UN country list	UN region	Operating margin (tCO_2eq/MWh)	Combined margin (tCO_2eq/MWh)
Saudi Arabia	Western Asia	0.592	0.374
Senegal	Western Africa	0.870	0.656
Serbia	Europe	1.086	0.678
Seychelles	Eastern Africa	0.753	0.479
Sierra Leone	Western Africa	0.489	0.246
Singapore	South-Eastern Asia	0.379	0.200
Sint Maarten (Dutch part)	Caribbean	0.753	0.463
Slovakia	Europe	0.332	0.164
Slovenia	Europe	0.620	0.285
Solomon Islands	Oceania	0.753	0.563
Somalia	Eastern Africa	0.753	0.582
South Africa	Southern Africa	1.070	0.747
South Georgia and the South Sandwich Islands	South America	N/A	0.290
South Sudan	Eastern Africa	0.890	0.704
Spain	Europe	0.402	0.209
Sri Lanka	Southern Asia	0.731	0.506
Sudan	Northern Africa	0.736	0.398
Suriname	South America	1.029	0.565
Svalbard and Jan Mayen Islands	Europe	N/A	0.256
Sweden	Europe	0.068	0.025
Switzerland	Europe	0.048	0.020
Syrian Arab Republic	Western Asia	0.713	0.546
Tajikistan	Central Asia	0.255	0.106
Thailand	South-Eastern Asia	0.450	0.351
Timor-Leste	South-Eastern Asia	0.753	0.589
Togo	Western Africa	0.859	0.597
Tokelau	Oceania	N/A	0.451
Tonga	Oceania	0.753	0.533
Trinidad and Tobago	Caribbean	0.559	0.370
Tunisia	Northern Africa	0.468	0.348
Türkiye	Western Asia	0.376	0.309
Turkmenistan	Central Asia	0.927	0.676
Turks and Caicos Islands	Caribbean	0.753	0.451
Tuvalu	Oceania	0.753	0.497
Uganda	Eastern Africa	0.279	0.116
Ukraine	Europe	0.768	0.435
United Arab Emirates	Western Asia	0.556	0.310
United Kingdom of Great Britain and Northern Ireland	Europe	0.380	0.187
United Republic of Tanzania	Eastern Africa	0.531	0.336
United States Minor Outlying Islands	Oceania	N/A	0.451
United States of America	North America	0.416	0.220
United States Virgin Islands	Caribbean	0.650	0.373

UN country list	UN region	Operating margin (tCO_2eq/MWh)	Combined margin (tCO_2eq/MWh)
Uruguay	South America	0.174	0.065
Uzbekistan	Central Asia	0.612	0.467
Vanuatu	Oceania	0.753	0.504
Venezuela (Bolivarian Republic of)	South America	0.711	0.368
Viet Nam	South-Eastern Asia	0.560	0.381
Wallis and Futuna Islands	Oceania	N/A	0.451
Western Sahara	Northern Africa	N/A	0.432
Yemen	Western Asia	0.807	0.615
Zambia	Eastern Africa	0.416	0.197
Zimbabwe	Eastern Africa	1.575	0.880

Notes: N/A = not applicable, tCO_2eq/MWh = tonnes of carbon dioxide equivalent emitted per megawatt-hour of energy produced or consumed. Values in red correspond to regional averages and replace missing or zero values for Combined Margin EFs. N/A values correspond to missing or zero values for Operating Margin EFs, for which representative regional estimates could not be identified.

Source: Authors' own elaboration based on IFI. 2021. *Methodological approach for the common fault grid emissions factors dataset*. Technical Working Group on Greenhouse Gas Accounting. IFI TWG- AHG-001. Version 3.0. December 2021. Cited 3 February 2022. https://unfccc.int/climate-action/sect oral-engagement/ifis-harmonization-of-standards-for-ghg-accounting/ifi-twg-list-of-methodologies

Table A146. Default net calorific values, in TJ/Gg, emission factors for stationary and mobile combustion, in kg GHG/TJ and tCO_2eq/m^3, and fuel density, in kg/m^3

Fuel and solid biofuels type	EF fuel	Net calorific values	EF CO_2	EF CH_4	EF N_2O	Density
Stationary - motor gasoline	2.289	44.3	69 300	10	0.6	740.7
Stationary - gasoil/diesel oil	2.706	43.0	74 100	10	0.6	843.9
Stationary - waste oil/lubricants	2.817	40.2	73 300	10	0.6	950
Stationary – Liquefied petroleum gas (LPG)	1.563	47.3	63 100	5	0.1	522.2
Stationary - natural gas	0.002	48.0	56 100	5	0.1	0.678
Mobile - motor gasoline	2.334	44.3	69 300	33	3.2	740.7
Mobile - gasoil/diesel oil	2.731	43.0	74 100	3.9	3.9	843.9
Mobile - natural gas	0.002	48.0	56 100	92	3	0.678
Mobile - LPG	1.606	47.3	63 100	62	0.2	522.2
Mobile - ethanol (cars)	0.044	27.0	1 508	18	N/A	789
Mobile - ethanol (trucks)	0.430	27.0	1 508	260	41	789
Off-road diesel	2.968	43.0	74 100	4.15	28.6	843.9
Off-road gasoline (2-stroke)	2.415	44.3	69 300	140	0.4	740.7
Off-road gasoline (4-stroke)	2.370	44.3	69 300	80	2	740.7
Wood	1.895 [0.148]	15.6	112 000	300	4	
Peat	1.131	9.8	106 000	300	1.4	
Charcoal	3.489 [0.185]	29.5	112 000	200	1	

Notes: kg GHG/TJ = kilograms of greenhouse gas emissions per terajoule of energy; tCO_2eq/m^3 = tonnes of CO_2 equivalent emissions per cubic meter of fuel; kg/m^3 = kilograms per cubic meter, expressing fuel density; EF fuel = emission factor per unit of fuel (typically kg CO_2eq per unit of fuel); EF CO_2 = carbon dioxide emission factor (kg/TJ or similar); EF CH_4 = methane emission factor; EF N_2O = nitrous oxide emission factor; Density = mass per unit volume (kg/m^3), used for converting between volume and mass. Numbers in bracket are the default emission factor for solid biofuels excluding CO_2 emissions. EFs were calculated using the IPCC AR5's GWPs without climate-carbon feedback. Users can select different GWP sets in the EX-ACT app or modify EFs in Tier 2.

Source: Authors' own elaboration based on Table 1.2 (net calorific value) Table 2.5 (GHG emission factors for stationary combustion), Tables 3.2.1 and 3.2.2 (GHG emission factors for mobile combustion), Table 3.3.1 (Emission factors for off-road mobile sources and machinery) in Volume 2 of IPCC. 2006. *2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Kanagawa, Japan. . Densities are from IEA. 2004. *Energy Statistics Manual*, OECD Publishing, Paris, https://doi.org/10.1787/9789264033986-en. Densities in bold: Waste oil/lubricants, are from expert judgment, Ethanol is from Haynes, W.M. (ed.). CRC Handbook of Chemistry and Physics. 95th Edition. CRC Press LLC, Boca Raton: FL 2014-2015, p. 3-256

Table A147. Emission factor for installation of irrigation systems

Irrigation systems	Definition	Installation energy ($kgCO_2eq/ha$)
Surface without irrigation runoff return system (IRRS)	Surface irrigation without an irrigation runoff return system	34.47
Surface with IRRS	Surface irrigation with an irrigation runoff return system	90.20
Solid set sprinkle	Irrigation with vertical tubes that are stationary for the duration of the irrigation season	444.77
Permanent sprinkle	Irrigation with vertical tubes that cannot be moved from the moment they are installed	130.17
Hand moved sprinkle	Irrigation with sprinklers that can be displaced throughout the irrigation season. This kind of sprinkler is usually used for smaller crop extensions as the continuous movement implies a high rate of labour	59.77
Solid roll sprinkle	A sprinkler line, mounted on wheels, irrigating in place. A mover unit powered by a small engine is located in the centre of the line. The sprinkler line is connected to a supply or mainline by a flexible hose	85.43
Centre-pivot sprinkle	Overhead sprinkler irrigation consisting of several segments of pipe, with sprinklers positioned along their length, joined together and supported by trusses, and mounted on wheeled towers. The machine moves in a circular pattern and is fed with water from the pivot point at the centre of the circle	79.20
Traveller sprinkler	Fully movable irrigation system. It does not entail the construction of any stable irrigation infrastructure on the field	61.97
Trickle	High precision, high efficiency drip irrigation	311.30

Note: IRRS = surface without irrigation runoff return system; $kgCO_2eq/ha$ = kilograms of carbon dioxide equivalent emitted or sequestered per hectare of land area.

Source: Authors' own elaboration based on Table 3 of Lal, R. 2004. Carbon Emission from Farm Operations. *Environment International,* 30: 981– 990.

Table A148. Emission factor for construction of buildings and roads

Type	Emission factor (tCO_2eq/m^2)
Housing (concrete)	436
Agricultural buildings (concrete)	656
Agricultural buildings (metal)	220
Industrial buildings (concrete)	825
Industrial buildings (metal)	275
Garage (concrete)	656
Garage (metal)	220
Offices (concrete)	650
Offices (metal)	158
Road (bitumen)	18
Road (asphalt)	73
Road (reinforced concrete)	86
Road rehabilitated (pavement)	9
Food sales (retail or wholesale)	515.9
Food service (restaurants)	517.4
Warehouse and storage	1 568.3
Education	440
Health care (metal)	440

Source: Authors' own elaboration based on ADEME. 2020. *Resource centre for greenhouse gas accounting*. Angers, France. Cited 26 June 2020. www.bilans-ges.ademe.fr; Bresinger, M. 2012. *Greenhouse Gas Assessment Emissions Methodology*. Technical Note IDB–TN–455. Washington, DC, Inter–American Development Bank.

Table A149. Global warming potential values

Global Warming Potential (100-year time horizon)	AR SAR	AR4	AR5 – without CC feedback	AR5 – with CC feedback	AR6
CO_2	1	1	1	1	1
CH_4 fossil origin	21	25	30	36	29.8
CH_4 non-fossil origin			28	34	27
N_2O	280	298	265	298	273
HCFC-22 (R22)	1 500	1 810	1 760	1 760	1 960
HFC-23	11 700	14 800	12 400	12 400	14 600
HFC-32	650	675	677	677	771
HFC-125	2 800	3 500	3 170	3 170	3 740
HFC-134a	1 300	1 430	1 300	1 550	1 530
HFC-143a	3 800	4 470	4 800	4 800	5 810
HFC-152a	140	124	138	138	164
HFC-227ea	2 900	3 220	3 350	3 350	3 600
HFC-236fa	6 300	9 810	8 060	8 060	8 690
HFC-245fa	-	1 030	858	858	962
HFC-365mfc	-	794	804	804	914
HFC-43-10mee	1 300	1 640	1 650	1 650	1 600
HFO-1234yf	-	4.00	0.50	0.50	0.50
HFO-1234ze(E)	-	7.00	1.37	1.37	1.37
HFO-1336mzz(Z)	-	9.00	2.00	2.00	2.08

Global Warming Potential (100-year time horizon)	AR SAR	AR4	AR5 – without CC feedback	AR5 – with CC feedback	AR6
HFO-1336mzz(E)	-	-	7.00	7.00	17.90
HCFO-1233zd(E)	-	4.50	1.00	1.00	3.88
HCFO-1224yd(Z)	-	-	1.00	1.00	-

Note: AR SAR = Second Assessment Report (IPCC, 1995); AR4 = Fourth Assessment Report (IPCC, 2007); AR5 = Fifth Assessment Report (IPCC, 2013), with and without climate–carbon feedback; AR6 = Sixth Assessment Report (IPCC, 2021); CO_2 = carbon dioxide (reference gas); CH_4 = methane (fossil and non-fossil); N_2O = nitrous oxide; HCFC = hydrochlorofluorocarbons (e.g. R22); HFC = hydrofluorocarbons; HFO = hydrofluoroolefins; HCFO = hydrochlorofluoroolefins.

Source: California Air Resources Board. 2016. Global Warming Potential Values (16 February 2016). Sacramento, CA, USA. Available at: https://ww2.arb.ca.gov/sites/default/files/2018-06/Global-Warming-Potential-Values%20%28Feb%2016%202016%29_1.pdf; Greenhouse Gas Protocol. 2024. Global Warming Potential Values (Version 2.0, 7 August 2024). Available at: https://ghgprotocol.org/sites/default/files/2024-08/Global-Warming-Potential-Values%20%28August%202024%29.pdf.

Table A150. Emission factors for type of packaging

Type of packaging	Emission factor (tCO_2eq/tonnes of packaging)
Wood	0.4
Paper	2.1
Aluminium	8.5
Plastic (mixed)	3.6
Plastic (LDPE)	2.5

Note: LDPE = low-density polyethylene, a type of plastic commonly used in agricultural applications such as mulching films, greenhouse covers, and irrigation tubing; tCO_2eq/tonnes of packaging = tonnes of carbon dioxide equivalent emitted per tonne of packaging material, expressing the greenhouse gas intensity of packaging production or use.

Source: Tables 15 and 17 from: Berneers-Lee, M. & Hoolohan, C. 2012. The Greenhouse Gas Footprint of Booths. Booths GHG report final, Small World Consulting Ltd. Lancaster, UK, Lancaster University.

Table A151. Relative stock change factors (FLU, FMG, FI) for settlement systems, dimensionless

Settlement system	IPCC climate zones	FLU	FMG	FI
Infrastructure on existing land (no paving)	All	1	1	1
Paved settlement	All	0.8	1	1
Wooded area	All	1	1	1
Turfgrass	Boreal	1	1.14	1
Turfgrass	Temperate	1	1.14	1
Turfgrass	Tropical	1	1.17	1
Turfgrass	Tropical Montane	1	1.16	1
Cultivated soil	Boreal Dry	0.77	1.03	1
Cultivated soil	Boreal Moist	0.7	1.09	1
Cultivated soil	Cool Temperate Dry	0.77	1.03	1
Cultivated soil	Cool Temperate Moist	0.7	1.09	1
Cultivated soil	Warm Temperate Dry	0.76	1.04	1
Cultivated soil	Warm Temperate Moist	0.69	1.1	1
Cultivated soil	Tropical Dry	0.92	1.04	1
Cultivated soil	Tropical Moist	0.83	1.1	1
Cultivated soil	Tropical Wet	0.83	1.1	1

Note: Values in bold are from or based on chapter 8.3.3.2 of IPCC (2006).

Source: Authors' own elaboration based on Table 5.5 and 6.2 of IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*. Geneva, Switzerland. www.ipcc-nggip.iges.or.jp/public/2019rf/index.html

Table A152. Mapping of greenhouse gases covered by Environmental eXternalities ACcounting Tool in each accounting category

Accounting category in EX-ACT	Activity represented through the accounting category	GHG coverage
Forest management	Afforestation	CO_2 fluxes from soil organic carbon stock changes CO_2 fluxes from biomass changes CH_4 from wildfires or fire use during conversion N_2O from wildfires or fire use during conversion N_2O from soil mineralisation CO_2 fluxes from harvested wood product (HWP) GHGs from the initial land use prior to afforestation depend on the type of land use selected.
	Deforestation	CO_2 fluxes from soil organic carbon stock changes CO_2 fluxes from biomass changes CH_4 from fire use during conversion N_2O from fire use during conversion N_2O from soil mineralisation CO_2 fluxes from harvested wood product (HWP) GHGs from the final land use following deforestation depend on the type of land use selected.
	Rotational Planting	CO_2 fluxes from biomass changes CO_2 fluxes from soil organic carbon stock changes (Tier 2) N_2O from soil mineralisation (Tier 2) CO_2 fluxes from harvested wood product (HWP) CO_2 from wood use for energy CH_4 from wood use for energy N_2O from wood use for energy
	Timber harvesting	CO_2 fluxes from biomass changes CO_2 fluxes from soil organic carbon stock changes (Tier 2) N_2O from soil mineralisation (Tier 2) CO_2 fluxes from harvested wood product (HWP) CO_2 from wood use for energy CH_4 from wood use for energy N_2O from wood use for energy
	Disturbance Management	CO_2 fluxes from biomass changes CH_4 from wildfires N_2O from wildfires CO_2 fluxes from soil organic carbon stock changes (Tier 2) N_2O from soil mineralisation (Tier 2) CO_2 fluxes from harvested wood product (HWP)
Annual cropland	All practices related to the management of annual and pluriannual crops	CO_2 fluxes from soil organic carbon stock changes CH_4 from residue management through burning or retention N_2O from residue management through burning or retention N_2O from soil mineralisation CO_2 fluxes from biomass changes (LUC) CH_4 from fire use during conversion (if applicable) N_2O from fire use during conversion (if applicable)
Perennial cropland	Management of woody perennials and agroforestry systems.	CO_2 fluxes from soil organic carbon stock changes CO_2 fluxes from biomass changes CH_4 from residue management through burning or retention N_2O from residue management through burning or retention N_2O from soil mineralisation CH_4 from fire use during conversion (if applicable)

Accounting category in EX-ACT	Activity represented through the accounting category	GHG coverage
		N_2O from fire use during conversion (if applicable)
Flooded rice	All practices related to the management of rice crop under partial or total flooding	CH_4 emissions from water management in flooded rice CH_4 from residue management practices N_2O from residue management practices CO_2 fluxes from soil organic carbon stock changes (Tier 2) CO_2 fluxes from biomass changes (LUC) N_2O from soil mineralisation CH_4 from fire use during conversion (if applicable) N_2O from fire use during conversion (if applicable)
Set aside land	Temporary set-aside of annual cropland or other idle cropland.	CO_2 fluxes from soil organic carbon stock changes (Tier 2) CO_2 fluxes from biomass changes (LUC) N_2O from soil mineralisation CH_4 from fire use during conversion (if applicable) N_2O from fire use during conversion (if applicable)
Grassland management	All practices related to the management of soil stocks of grasslands and pastures	CO_2 fluxes from soil organic carbon stock changes CO_2 fluxes from biomass changes (LUC) N_2O from soil mineralisation CH_4 from fires N_2O from fires CH_4 from fire use during conversion (if applicable) N_2O from fire use during conversion (if applicable)
Coastal wetland management	Extraction and excavation	CO_2 fluxes from soil organic carbon stock changes CO_2 fluxes from biomass changes
	Drainage	CO_2 from drainage
	Rewetting	CO_2 from rewetting CH_4 from rewetting
	Revegetation	CO_2 fluxes from biomass changes
Organic soils	Drainage	CO_2 from drainage (on site) CH_4 from drainage (on site) N_2O from drainage (on site) CO_2 from drainage (through DOC – off site) CH_4 from drainage (off site)
	Rewetting	CO_2 from rewetting (on site) CH_4 from rewetting (on site) N_2O from rewetting (on site) CO_2 from rewetting (through DOC – off site) CH_4 from rewetting (off site)
	Soil fire management	CO_2 from soil fires (direct CO_2 and through CO) CH_4 from soil fires
	Peat extraction (and accompanying drainage)	CO_2 from extraction (on site) CH_4 from extraction (on site) N_2O from extraction (on site) CO_2 from extraction (through DOC – off site) CH_4 from extraction (off site)
Waterbodies management	All practices related to the management of ponds, canals, ditches and other constructed water bodies.	CH_4 emissions from waterbodies
	Settlement	CO_2 fluxes from soil organic carbon stock changes (Tier 2)

Accounting category in EX-ACT	Activity represented through the accounting category	GHG coverage
Settlement and infrastructure		CO_2 fluxes from biomass changes (LUC) N_2O from soil mineralisation CH_4 from fire use during conversion (if applicable) N_2O from fire use during conversion (if applicable)
	Infrastructure creation	Other GHGs linked to installation of the new infrastructure
Other land	All practices related to the management of bare soil, rock, ice and land areas that do not fall under any of the other five categories	CO_2 fluxes from soil organic carbon stock changes (Tier 2) CO_2 fluxes from biomass changes (LUC) N_2O from soil mineralisation CH_4 from fire use during conversion (if applicable) N_2O from fire use during conversion (if applicable)
Livestock management	All practices related to the management of livestock, its productivity and its manure.	CH_4 from enteric fermentation CH_4 from manure management N_2O from manure management (including leaching and volatization)
Small fisheries	Practices linked to energy use for the fishing, refrigeration and landing of catch in small vessels.	Other GHGs linked to fuel use on vessels Other GHGs linked to electricity production for icemaking HCFC linked to refrigeration leakage
Large fisheries	Practices linked to energy use for the fishing, refrigeration and landing of catch in small vessels.	Other GHGs linked to fuel use on vessels Other GHGs linked to electricity production for icemaking HCFC linked to refrigeration leakage
Aquaculture	Practices linked to fish rearing in aquaculture ponds	N_2O emissions from fish excreta
Inputs	Practices linked to production and field use of fertilizers, pesticides and animal feed.	N_2O emissions from field use CO_2 emissions from field use Other GHGs linked to production, transportation and storage
Energy	Practices linked to the use of solid and liquid fuels and electricity	CO_2 from liquid and solid fuel use CH_4 from liquid and solid fuel use N_2O from liquid and solid fuel use Other GHGs linked to electricity use
Irrigation	New irrigation systems	Other GHGs linked to installation of the new irrigation system
	Operational phase of irrigation	Other GHGs linked to fuel and electricity use for pumping in the irrigation system
Value chain: transport	Practices linked to transporting produce and materials.	CO_2 from liquid and solid fuel use CH_4 from liquid and solid fuel use N_2O from liquid and solid fuel use Other GHGs linked to electricity use
Value chain: processing	Practices linked to processing of produce.	CO_2 from liquid and solid fuel use CH_4 from liquid and solid fuel use N_2O from liquid and solid fuel use Other GHGs linked to electricity use
Value chain: packaging	Practices linked to packaging of produce and the production of packaging materials	CO_2 from liquid and solid fuel use CH_4 from liquid and solid fuel use N_2O from liquid and solid fuel use Other GHGs linked to electricity use Other GHGs linked to production of packaging materials
Value chain: Storage	Practices linked to (refrigerated) storage of produce.	CO_2 from liquid and solid fuel use CH_4 from liquid and solid fuel use

Accounting category in EX-ACT	Activity represented through the accounting category	GHG coverage
		N_2O from liquid and solid fuel use Other GHGs linked to electricity use HCFC, HFC, HFO, HCFO linked to refrigeration leakage

Note: GHG = greenhouse gas; EX-ACT = Environemental eXternalities ACcounting Tool; CO_2 = carbon dioxide; CH_4 = methane; N_2O = nitrous oxide; HCFC = hydrochlorofluorocarbons; HFC = hydrofluorocarbons; HFO = hydrofluoroolefins; HCFO = hydrochlorofluoroolefins; LUC = land-use change; DOC = dissolved organic carbon.

Source: Author’s own elaboration

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